

Further Education and Training Certificate: Electronics

ID 83446 LEVEL 5 – 135 CREDITS

INTERGRATED LEARNER GUIDE

SAQA – 386099

Demonstrate knowledge of applied electronics

SAQA – 386093

**Inspect and test electronic equipment, components and
assemblies using measuring and testing equipment**

SAQA- 386098

**Diagnose and locate faulty circuit, component and
equipment**

SAQA: 119256

Inspect and test electrical circuits

Learner Information:

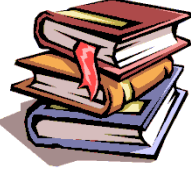
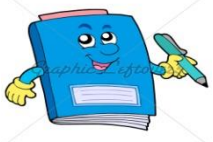
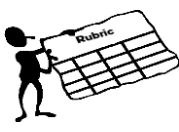
Details	Please Complete this Section
Name & Surname:	
Organisation:	
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Date Started:	
Date of Completion:	

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Key to Icons

The following icons may be used in this Learner Guide to indicate specific functions:

 Books	This icon means that other books are available for further information on a particular topic/subject.
 References	This icon refers to any examples, handouts, checklists, etc...
 Important	This icon represents important information related to a specific topic or section of the guide.
 Activities	This icon helps you to be prepared for the learning to follow or assist you to demonstrate understanding of module content. Shows transference of knowledge and skill.
 Exercises	This icon represents any exercise to be completed on a specific topic at home by you or in a group.
 Tasks/Projects	An important aspect of the assessment process is proof of competence. This can be achieved by observation or a portfolio of evidence should be submitted in this regard.

 Workplace Activities	An important aspect of learning is through workplace experience. Activities with this icon can only be completed once a learner is in the workplace
 Tips	This icon indicates practical tips you can adopt in the future.
 Notes	This icon represents important notes you must remember as part of the learning process.

Learner Guide Introduction

About the Learner Guide...	This Learner Guide provides a comprehensive overview of the Demonstrate knowledge of applied electronics, and forms part of a series of Learner Guides that have been developed for Further Education and Training Certificate: Electronics ID 83446 LEVEL 5 – 135 CREDITS. The series of Learner Guides are conceptualized in modular's format and developed for Further Education and Training Certificate: Electronics ID 83446 LEVEL 5 – 135 CREDITS. They are designed to improve the skills and knowledge of learners, and thus enabling them to effectively and efficiently complete specific tasks. Learners are required to attend training workshops as a group or as specified by their organization. These workshops are presented in modules and conducted by a qualified facilitator.
Purpose	The purpose of this Learner Guide is to provide learners with the necessary knowledge related to Demonstrate knowledge of applied electronics.
Outcomes	At the end of this module, you will be able to: • Demonstrate knowledge of digital system building blocks. • Demonstrate knowledge of analogue electronic components. • Design a specific project using applied electronics.
Assessment Criteria	The only way to establish whether a learner is competent and has accomplished the specific outcomes is through an assessment process. Assessment involves collecting and interpreting evidence about the learner's ability to perform a task. This guide may include assessments in the form of activities, assignments, tasks or projects, as well as workplace practical tasks. Learners are required to perform tasks on the job to collect enough and appropriate evidence for their portfolio of evidence, proof signed by their supervisor that the tasks were performed successfully.
To qualify	To qualify and receive credits towards the learning program, a registered assessor will conduct an evaluation and assessment of the learner's portfolio of evidence and competency
Range of Learning	This describes the situation and circumstance in which competence must be demonstrated and the parameters in which learners operate

Responsibility	The responsibility of learning rest with the learner, so: • Be proactive and ask questions, • Seek assistance and help from your facilitators, if required.
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Demonstrate knowledge of applied electronics

Learning Unit 1

UNIT STANDARD NUMBER	:	386099
LEVEL ON THE NQF	:	5
CREDITS	:	6
FIELD	:	Field 10 - Physical, Mathematical, Computer and Life Sciences
SUB FIELD	:	Information Technology and Computer Sciences

PURPOSE:	This unit standard is intended for learners who wish to gain electronics circuitry composition at intermediate and advanced levels, covering areas of analogue and digital electronics.
LEARNING ASSUMED TO BE IN PLACE:	
• Communication at NQF Level 4. • Mathematics at NQF Level 4. • Physical Science at NQF Level 4.	

SESSION 1.

Demonstrate knowledge of applied electronics

Learning Outcomes

- Digital electronics building blocks are explained in terms of their components and interrelationships.
- Digital system building blocks are analysed in terms of their characteristics and concepts in accordance with industry practice.
- A switch mode DC power supply is constructed by using specified components.
- A timing circuit is constructed using digital electronic components.

Demonstrate knowledge of applied electronics

1.LOGIC GATES – BUILDING BLOCKS OF DIGITAL CIRCUITS

A Logic gate is an electronic circuit which makes logical decisions. The AND, OR and NOT are the basic logic gates.

The NAND and NOR gates are derived from these. Any computing problem in digital circuitry can be expressed by a boolean equation or boolean expression. A Boolean expression can be minimized to Minterms or Maxterms reducing the expression to a two-level implementation which can be designed with the NAND or NOR gates. That is why, NAND and NOR gates are called the Universal gates. The exclusive – OR gates is another logic gate which can be constructed using basic gates such as OR and NOT gates.

The logic gates can have two or more inputs and only one output except for the NOT gate, which has only one input. The output signal from a logic gate appears only for certain combinations of the input signals as the number of combinations is always limited by the number of Boolean variables (where each Boolean variable represent an input channel).

So, basically, the logic gates are the building blocks of a digital circuit that can perform basic manipulation of the binary information. Any digital IC contains nothing but interconnected network of logic gates. Each gate is represented by a distinct logic symbol and its operation can be described by means of an

algebraic function. The relationship between the input and output variables of each gate can be represented by means of a truth table and signal response of any logic gate can be represented by the timing diagram. So, let us learn about various logic gates.

OR Gate –

The OR gate performs logical addition, commonly known as OR function. The OR gate has two or more inputs and only one output. The operation of OR gate is such that HIGH (1) on the output is produced when any of the inputs is HIGH (1). The output is LOW (0) only when all the inputs are LOW (0). Suppose, if A and B are the input variables of an OR gate and Y is its output, then

$$Y = A + B$$

Similarly, for more than two input variables, the output of OR gate can be expressed as follow –

$$Y = A + B + C + D + \dots$$

An OR gate built using diodes is shown below in which A and B represent the inputs and Y the output. The resistance R_L is the load resistance. The Logical symbol of OR gate is also shown below –

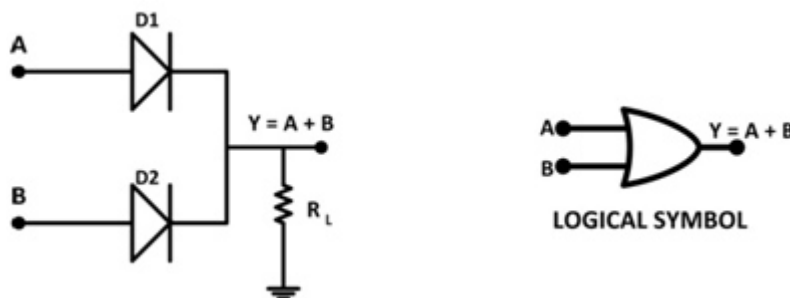


Fig. 1: Image showing OR Gate Symbol And Circuit

If $A = 0$ and $B = 0$, both the diodes will not conduct, hence the output $Y = 0$. If $A = 1$ and $B = 0$, diode D1 conducts, hence the output $Y = 1$. If $A = 0$ and $B = 1$, diode D2 conducts, hence the output $Y = 1$. If $A = 1$ and $B = 1$, both the diodes conduct and hence $Y = 1$. The output of an OR gate for two Boolean variables can be represented by the following truth table –

INPUT		OUTPUT
A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

AND Gate –

The AND gate performs logical multiplication, commonly known as AND function. The AND gate has two or more inputs and a single output. The operation of AND gate is such that the HIGH (1) is only when all the inputs are HIGH (1). Even when one of the inputs is 0, the output will be zero. Suppose, if A and B are the input variable of the AND gate and Y is its output, then

$$Y = A \cdot B$$

The dot (·) denotes the AND operation, but typically it can also be written as $Y = AB$. The two input AND gate built using the diodes is shown below in which A and B represent the inputs and Y the output. The Logical symbol of AND gate is also shown below –

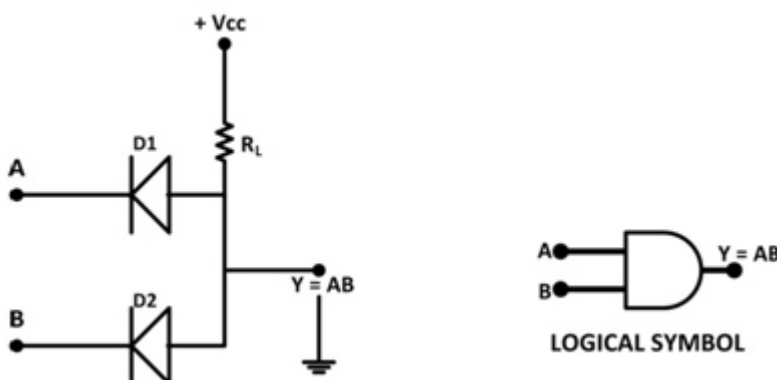


Image showing AND Gate Symbol And Circuit

If $A = 0$ and $B = 0$, both the diodes conduct as they are forward biased, hence the output $Y = 0$. If $A = 0$ and $B = 1$, the Diode D1 conducts as they are forward biased, and hence the output $Y = 0$. If $A = 1$ and $B = 0$ the diode D2 conducts as they are forward biased, and hence the output $Y = 0$. If $A = 1$ and $B = 1$ both the

diodes don't conduct as they are reverse biased, and hence the output $Y = 1$. The output of an AND gate for two Boolean variables can be represented by the following truth table –

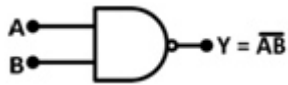


Fig. 5: Truth Table of AND Gate

The truth table for AND gate can be graphically represented by the following timing diagram –



Fig. 6: Timing Diagram of AND Gate

NOT Gate –

The NOT gate performs the basic logical function called Inversion or the complementation. The purpose of this gate is to convert the one logic level into the opposite logic level. It has one input and one output. When the High level is applied at the input the low level appears at the output and vice-versa. A NOT gate built using a transistor is shown below, where A represent the input and Y represents the output. The Logical symbol of NOT gate is also shown below –

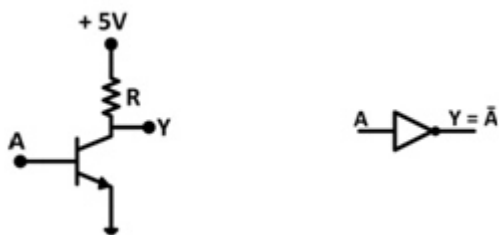


Image showing NOT Gate Symbol And Circuit

When the input is HIGH, the transistor is in the ON state and the output is LOW. When the input is LOW, the transistor is in the OFF state and the output is HIGH. The output of a NOT gate can be represented by the following truth table –

INPUT A	OUTPUT $Y = \bar{A}$
0	1
1	0

Truth Table of NOT Gate

The truth table for NOT gate can be graphically represented by the following timing diagram –

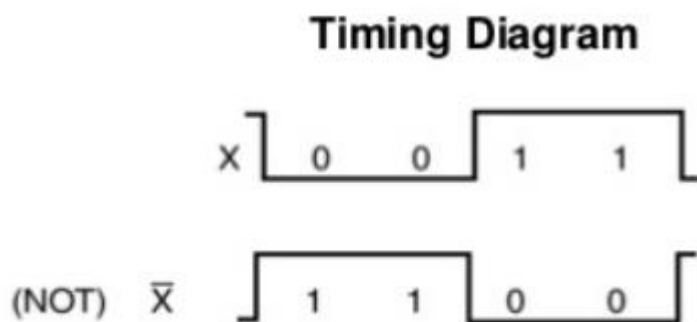


Fig. 9: Timing Diagram of NOT Gate

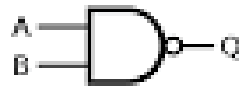
NAND Gate –

The NAND gate performs the logical function which is the serial combination of the NOT and AND Gates. It has two or more inputs and only one output. The Logical symbol for the NAND gate is shown below –



Image showing NAND Gate Symbol And Circuit

At a NAND gate, when all the inputs are HIGH, the output is LOW. When any one of the inputs is LOW, then the output is HIGH. The output of a NAND gate for two Boolean variables can be represented by the following truth table –



$$Q = A \text{ NAND } B$$

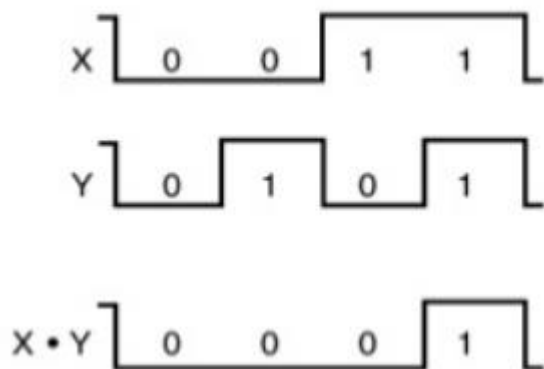
Truth Table

Input A	Input B	Output Q
0	0	1
0	1	1
1	0	1
1	1	0

Truth Table of NAND Gate

The truth table for NAND gate can be graphically represented by the following timing diagram –

Timing Diagram



Timing Diagram of NAND Gate

NOR Gate –

The NOR gate performs the logical function which is the serial combination of the NOT and OR Gates. It has two or more inputs and only one output. The Logical symbol for the NOR gate is shown below –

INPUT		OUTPUT
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Image showing NOR Gate Symbol And Circuit

At a NOR gate, when all the inputs are LOW, the output is HIGH. When any one or both of the inputs is HIGH, then the output is LOW. The output of a NOR gate for two Boolean variables can be represented by the following truth table –

INPUT		OUTPUT
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Timing Diagram of NOR Gate

EXCLUSIVE – OR (Ex – OR) Gate –

An Exclusive – OR gate is a gate with two or more inputs and outputs. The output of a two-input XOR gate assumes a HIGH state if one and only one input assumes a HIGH state i.e. the output is HIGH only if input A or input B is HIGH and the output is LOW when both the inputs are either HIGH or LOW. The logical symbol of the XOR is shown below –

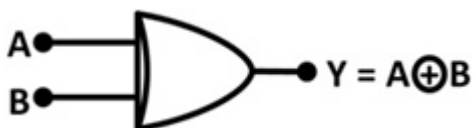


Image showing XOR Gate Symbol And Circuit

The output of an XOR gate for two Boolean variables can be represented by the following truth table –

INPUT		OUTPUT
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Truth Table of XOR Gate

The truth table of the XOR gate shows that the output is HIGH when any one, but not all of the inputs is 1 and when the both inputs are same i.e. either 0 or 1, the output is LOW. This exclusive feature eliminates a similarity to the OR gate. The truth table for XOR gate can be graphically represented by the following timing diagram –

INPUT		OUTPUT
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Timing Diagram of XOR Gate

Exclusive – NOR (XNOR) Gate –

The XNOR gate performs the logical function which is the serial combination of the NOT and XOR Gates. The output of a XNOR gate assumes a HIGH state if all the input assumes either HIGH state or LOW state. If either input has a different logic state, the output of XNOR gate is LOW. The logical symbol of the XNOR is shown below –

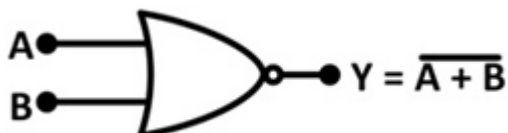


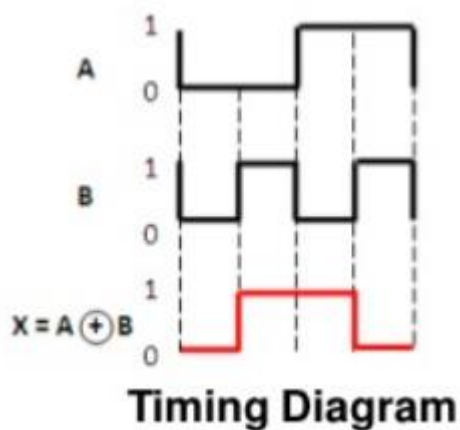
Image showing XNOR Gate Symbol And Circuit

The output of an XNOR gate for two Boolean variables can be represented by the following truth table –

INPUT		OUTPUT
A	B	$Y = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

Truth Table of XNOR Gate

The truth table for XNOR gate can be graphically represented by the following timing diagram –



Timing Diagram of XNOR Gate

These logic gates are the building blocks of any digital circuit. In the next tutorial, learn about theorems of Boolean algebra and how a Boolean expression can be minimized to Minterms and Maxterms, so, it can be implemented by two level interconnection of universal logic gates (NAND and NOR).

1. HOW TO BUILD A DC POWER SUPPLY

In this project, we will show how to build a simple DC power supply, which can be used to supply adjustable DC voltage to an electronic circuit.

By going through this project, you will gain understanding of how a DC power supply works and what parts are needed to create a simple DC power supply. If you've been doing any circuit building or electronic engineering technician work, you know how important a DC power supply is. It can be used to produce adjustable DC voltages. So instead of using a plethora of batteries for a circuit or wall warts, switching out batteries to get the precise voltage, you can simply use a DC power supply, which greatly simplifies providing DC power to a circuit. So in order to build a DC power supply, we will need the following components:

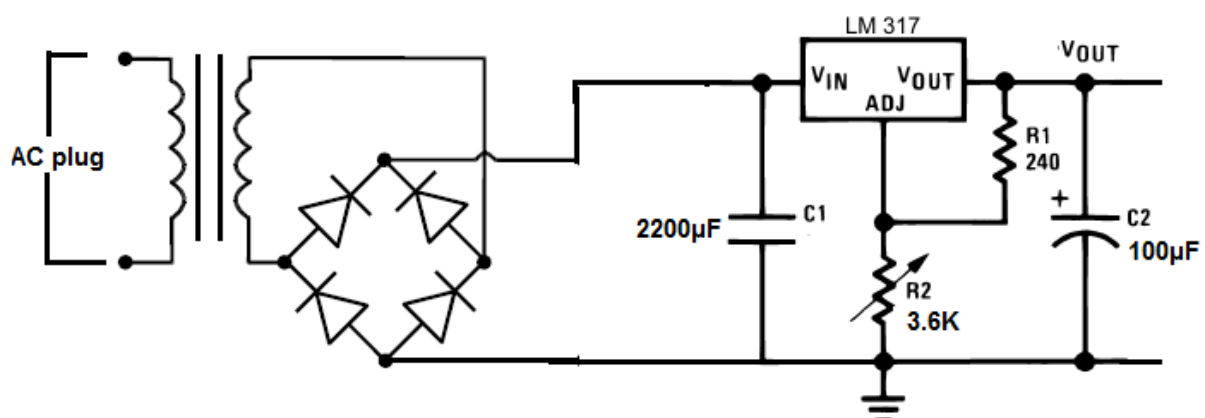
Components Needed

- ✚ AC Plug
- ✚ 24V Transformer
- ✚ Full Wave Bridge Rectifier
- ✚ LM317 Voltage Regulator
- ✚ Heat Sink
- ✚ 2200 μ F electrolytic capacitor
- ✚ 100 μ F electrolytic capacitor
- ✚ 240 Ω Resistor
- ✚ 3.6K Ω Potentiometer

DC Power Supply Schematic Diagram

Above are the parts needed to create a DC power supply?

Below is the schematic for the DC power supply, so that we can see how all the parts are connected and are brought together:



We'll now go over each part of this circuit and go over the role each component plays, so that you can know how this circuit works in its entirety.



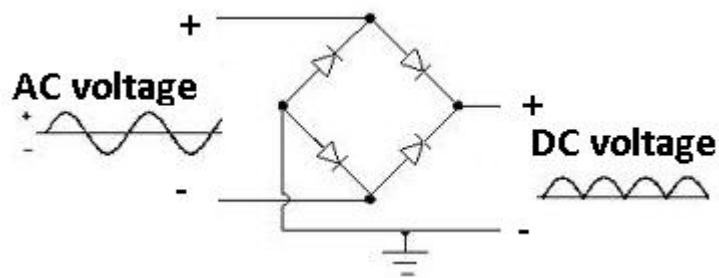
AC Plug- The first part of the circuit is the AC plug. When we create a DC power supply, it creates DC voltage from the AC mains voltage from a wall outlet. To build a DC power supply, purchase a 3-prong AC plug. It can also work with a two-prong AC plug. But having a 3-prong plug is better because ground

Transformer- after the AC plug, we need a step-down transformer. The transformer's job is to take the 120V AC voltage from the mains line and step it down to 24 volts. This is because our DC power supply will supply variable DC voltage of 1-20V. Therefore, we lower the very high voltage that we get from the mains outlet from the wall into a smaller voltage. It must still exceed the voltage of the DC which we want to output. Since we want to create up to 20VDC variable voltage output, we need a transformer that converts the mains voltage to a voltage that is higher than this 20V. A step-down transformer is a great device for lowering voltage from a mains AC voltage line. Provides better against possible electric fires.



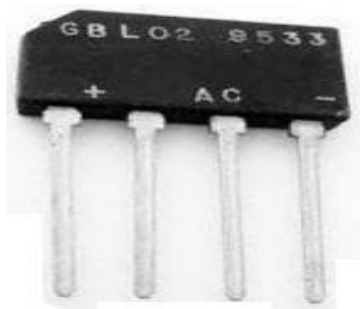
Full-wave Rectifier- The next component we need in our circuit is a full-wave rectifier. The job of the full-wave rectifier is to take the AC voltage from the transformer and rectify it so that the voltage no longer goes through a negative cycle. With the rectifier, all of the voltage is rectified positive.

Below is the how the voltage looks before and after the full-wave rectifier:



You can see how all voltage waveforms are now above the positive line. This is called pulsating DC voltage. We will later add more components so that this can be a nearly perfectly smooth DC output, which is what is desired.

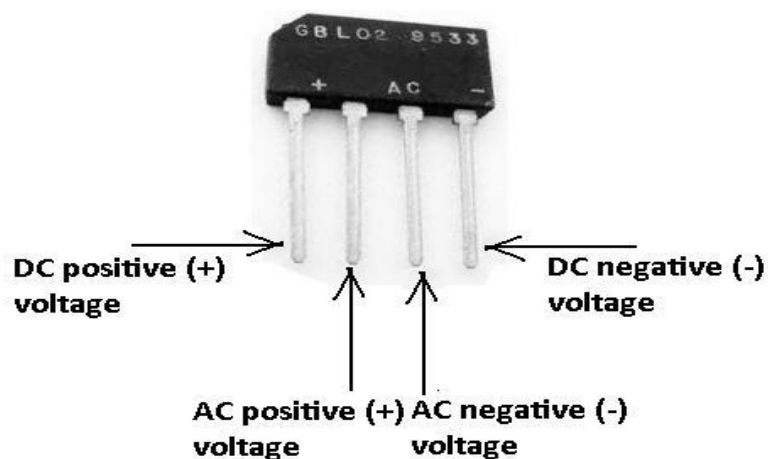
How to Connect a Full Wave Rectifier



A full wave rectifier has valuable use when a user takes an AC signal and wants DC output in exchange. A full wave rectifier converts the AC signal into DC, in the event that a device needs to be plugged into an AC outlet but needs to be powered and run on DC voltage.

Pinout

A full wave rectifier normally has 4 pins.



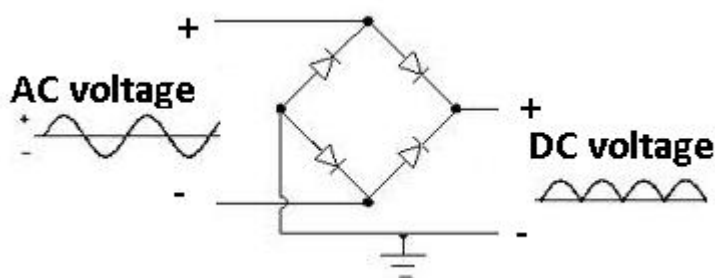
Pinout Terminals

The terminal that has the + symbol is the DC positive voltage. This is terminal that receives the converted positive DC voltage that is output by the rectifier.

The terminals that have the two AC signals ($\sim$) are the two terminals where you place the incoming AC voltage, which you want to convert into DC. It doesn't matter the polarity. The positive and negative AC voltage can be switched.

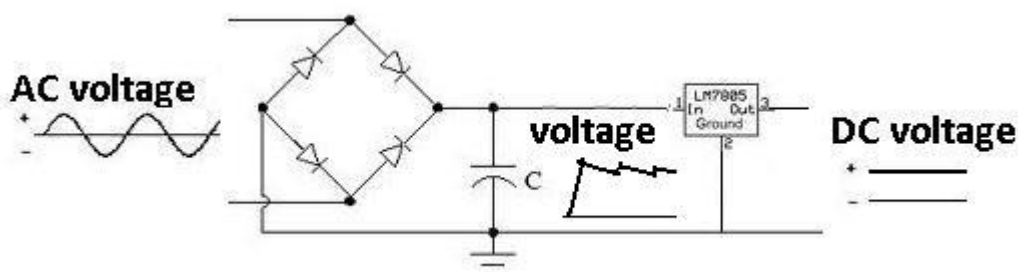
The terminal that has the - symbol is the DC negative or ground voltage. This is the terminal that receives the converted negative DC voltage that is output by the rectifier.

This is a diagram to visualize how a full wave rectifier works:



A full wave rectifier takes an AC signal input into it, such as the AC signal from a power outlet, and converts it into a DC signal. The AC is input. The DC is output. To create a smoother and steady DC signal, a smoothing capacitor is normally added in parallel to the output of the rectifier.

And to make it even smoother and steady, a voltage regulator can then be added in parallel to that to output to produce a specific DC voltage that you want as output.

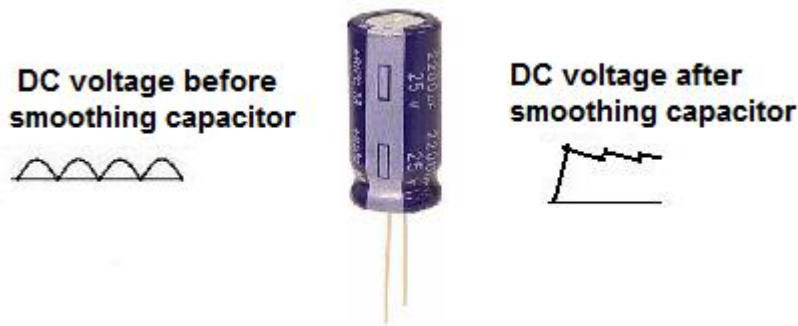


You can see after the smoothing capacitor that the voltage is pretty steady but not quite. After the voltage regulator, it should be perfectly steady and even.

Smoothing Capacitor- The next component after the full-wave rectifier is the smoothing capacitor. The smoothing capacitor acts to smooth out fluctuations in a signal, so that there is less fluctuation. As you saw in the previous

component, the rectifier creates pulsating DC signals. The smoothing capacitor, now, a 2200 μ F capacitor, acts to even out this pulsating fluctuations to create a smoother waveform.

The diagram below shows how the voltage waveform looks before and after the smoothing capacitor:



What is a Smoothing Capacitor?



A Smoothing capacitor is a capacitor that acts to smooth or even out fluctuations in a signal.

The most common and used application for smoothing capacitors is after a power supply voltage or a rectifier.

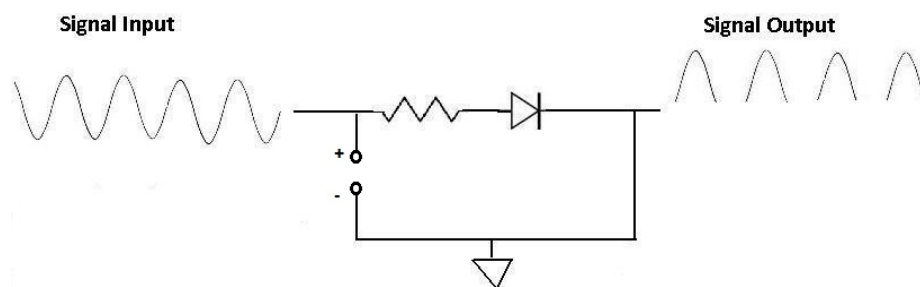
Power supply voltage can sometimes supply erratic and unsmooth voltages that fluctuate greatly.

When a steady DC signal is needed and is necessary, a smoothing capacitor is the right component needed in order to smooth out the fluctuating signal to make it steadier.

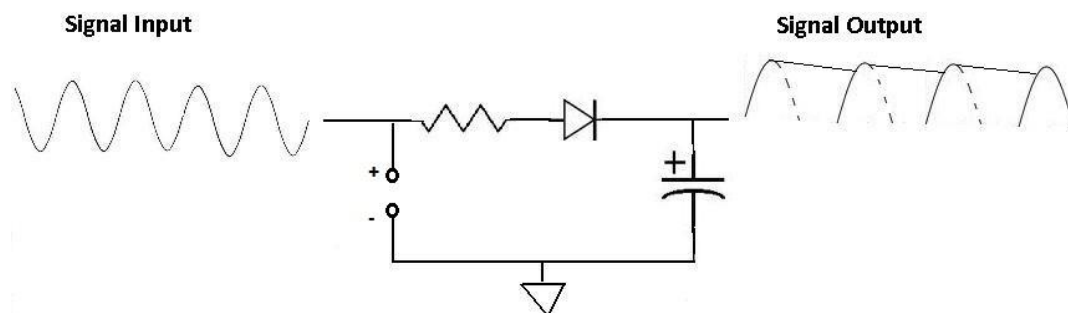
We'll go over an example of this now.

A prime example of when a smoothing capacitor is used is in conjunction with a rectifier circuit.

If you place a resistor in series with a diode and then input an AC signal into the circuit:



Now if you place a smoothing capacitor in parallel with the diode, like this, the resulting waveform will be:



You can see now how much smoother the waveform is. It no longer goes all the way down to zero and back up. The capacitor charges up from 0 to the top of the waveform and then discharges from 0 to the bottom of the waveform. This charging and discharging smooths out the waveform so that it doesn't hit the extreme ups and downs. Thus, a smoothing capacitor is extremely useful in cases of fluctuating signals that need to be more constant and steady.

Usually when choosing a smoothing capacitor, an electrolytic capacitor is used from anywhere from $10\mu\text{F}$ to a few thousand μF . The greater the amplitude of the fluctuations and the greater the waveform, the larger capacitor will be necessary. Thus, if you're smoothing a 30mV waveform, a $10\mu\text{F}$ capacitor may suffice to smooth out the signal. However, if you're dealing with a much greater signal, you will need a much larger capacitor, say, maybe $3300\mu\text{F}$ in order to smooth it out to a near DC level. Experiment with the capacitors. Check the

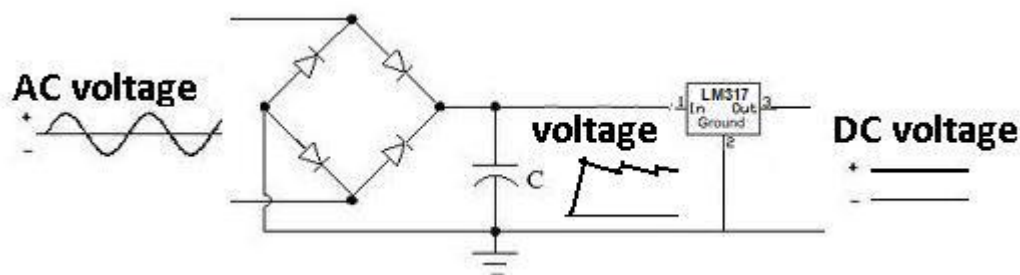
signal on an oscilloscope to see which capacitor suffices best and is best for the circuit at hand.

LM317 Voltage Regulator- After the smoothing cap is the LM317 Voltage Regulator.

This regulator serves a twofold purpose. First, it serves to further smooth out the fluctuating signal, so that it's a perfectly smoothing DC signal.

A regulator is a device that "regulates" voltage so that a perfectly smoothing DC voltage is output.

Below is a diagram of a voltage waveform after both a smoothing capacitor and a voltage regulator:



The second purpose of the LM317 regulator, since it is an adjustable voltage regulator, is to produce variable DC voltage as output. How we vary voltage is by changing the resistor R1 and the potentiometer R2.

R1 Resistor and R2 Potentiometer- The next components needed are the R1 resistor and R2 potentiometer. These two used together decide what voltage will be output by the regulator. The formula is $V_{OUT} = 1.25V (1 + R_2/R_1)$. So if R2 is turned all the way so that its resistance is near 0Ω , the voltage output will be just over 1V. When the potentiometer is turned so that its resistance is $3.6K\Omega$, the voltage output will be about 20V. This potentiometer decides how much voltage is output, just like how you turn the knob of a professional-made DC power supply. If we chose a higher voltage-valued step-down transformer, such as a 36V step-down transformer and used a higher-valued potentiometer, we could increase the amount of DC voltage output, from, say, 20V to 30V or more.

Heat sink- One thing we must do to the voltage regulator is attach a heat sink to it. This is vital for this application.



This is because when we use a regulator, we input a voltage into it and it outputs the voltage, based on the values of resistor R1 and potentiometer R2. When the potentiometer is at its highest resistance, it doesn't dissipate that much heat. Since our transformer outputs 24V, when the potentiometer is set to 3.6K Ω , the regulator outputs 20V. $24-20V=4V$. Thus, not that much wasted voltage is created. However, if the potentiometer is set to near 0 Ω , the regulator outputs approximately 1.25V. $24V-1.25V=22.75V$ of wasted, dissipated energy. This creates a lot of heat, since the voltage difference between input and output voltage is so great. Any difference appears as heat. So the greater the difference, the greater the heat. This is the reason it is vital to attach a heat sink to the regulator. When the difference between input and output voltage is great, it appears as heat. We must have a way to dissipate this heat, or else it can damage or destroy the circuitry of the power supply. The way to do this is to use a heat sink. This is the reason why professional DC power supplies always have large heat sinks in the back of them.

Attaching Heat Sinks to Voltage Regulators

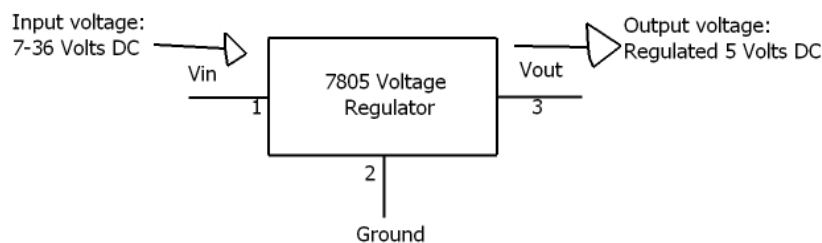


A heat sink should be attached to a voltage regulator in order to dissipate excess power that may enter into the regulator.

This prevents excess voltage from destroying and overheating the voltage regulator, since the heat sink dissipates the excess power off as heat.

Heat sinks are not always necessary with voltage regulators.

Voltage regulators work by taking in an input voltage and converting it down to its specified rated output voltage. If the input voltage is only 1-3 volts higher than its output voltage and stays steady at this value, there is only a little difference in voltage between input and output. Therefore, there is not much excess power that could overheat the regulator. For example, feeding 7.5 volts into a LM7805, which outputs 5 volts, will not overheat it, since there is only a 2.5-volt difference. However, if the input voltage fed into a regulator is much higher than its output voltage, the regulator should be attached to a heat sink. The difference between the input and output voltage appears as heat. The greater the difference between the input and output voltage, the more heat is generated. If too much heat is generated, through high input voltage, the regulator can overheat. If the regulator does not have a heat sink to dissipate this heat, it can be destroyed and malfunction.



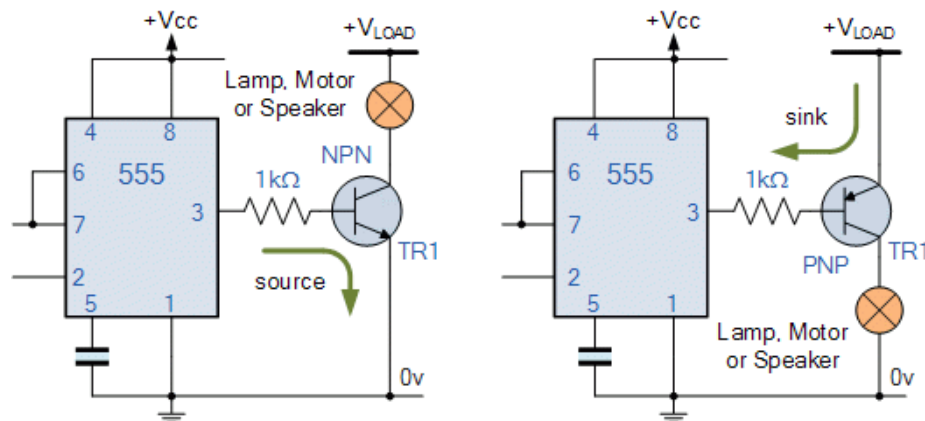
So the two options are, design your circuit so that the input voltage going into the regulator is limited to 2-3 volts above the output regulated voltage or place a heat sink in your circuit to dissipate the created heat.



Heat sinks come in many different shapes and sizes but they all have the characteristic that they're metallic so that they can dissipate heat and have a circular open where you can place a nail through to adjoin the voltage regulator.

555 Timer

The 555 Timer is a commonly used IC designed to produce a variety of output waveforms with the addition of an external RC network



We have seen that Multivibrators and CMOS Oscillators can be easily constructed from discrete components to produce relaxation oscillators for generating basic square wave output waveforms. But there are also dedicated IC's especially designed to accurately produce the required output waveform with the addition of just a few extra timing components.

One such device that has been around since the early days of IC's and has itself become something of an industry "standard" is the 555 Timer Oscillator which is more commonly called the "555 Timer".

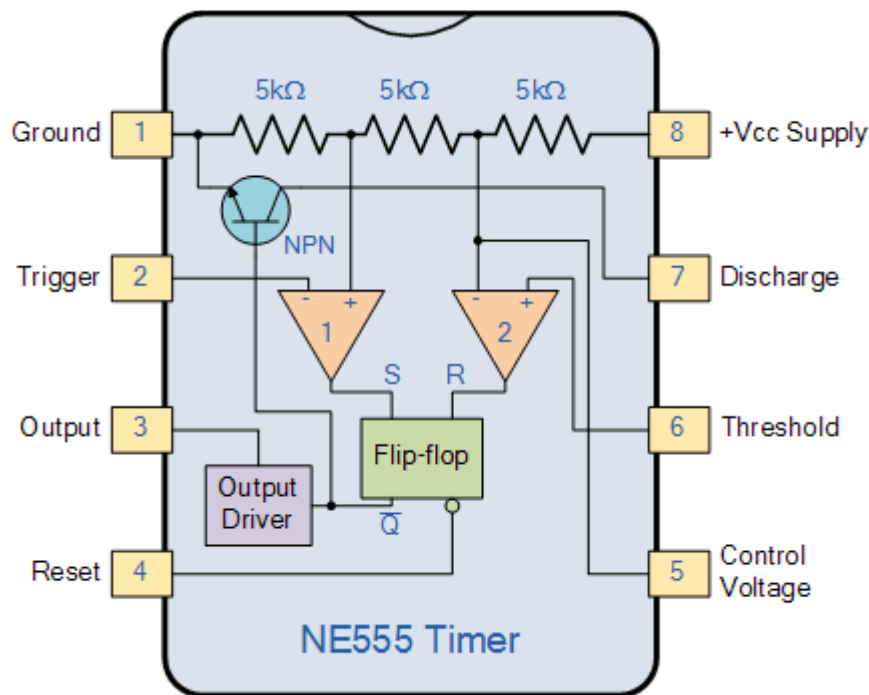
The basic 555 timer gets its name from the fact that there are three internally connected 5kΩ resistors which it uses to generate the two comparators reference voltages. The 555 timer IC is a very cheap, popular and useful precision timing device which can act as either a simple timer to generate single pulses or long time delays, or as a relaxation oscillator producing a string of stabilised waveforms of varying duty cycles from 50 to 100%.

The 555 timer chip is extremely robust and stable 8-pin device that can be operated either as a very accurate Monostable, Bistable or Astable Multivibrator to produce a variety of applications such as one-shot or delay timers, pulse generation, LED and lamp flashers, alarms and tone generation, logic clocks, frequency division, power supplies and converters etc., in fact any circuit that requires some form of time control as the list is endless.

The single 555 Timer chip in its basic form is a Bipolar 8-pin mini Dual-in-line Package (DIP) device consisting of some 25 transistors, 2 diodes and about 16 resistors arranged to form two comparators, a flip-flop and a high current output stage as shown below. As well as the 555 Timer there is also available the NE556 Timer Oscillator which combines TWO individual 555's within a single 14-pin DIP package and low power CMOS versions of the single 555 timer such as the 7555 and LMC555 which use MOSFET transistors instead.

A simplified “block diagram” representing the internal circuitry of the 555 timer is given below with a brief explanation of each of its connecting pins to help provide a clearer understanding of how it works.

555 Timer Block Diagram



- ✚ Pin 1. – Ground, The ground pin connects the 555 timer to the negative (0v) supply rail.
- ✚ Pin 2. – Trigger, The negative input to comparator No 1. A negative pulse on this pin “sets” the internal Flip-flop when the voltage drops below $1/3V_{cc}$ causing the output to switch from a “LOW” to a “HIGH” state.
- ✚ Pin 3. – Output, The output pin can drive any TTL circuit and is capable of sourcing or sinking up to 200mA of current at an output voltage equal to

approximately $V_{cc} - 1.5V$ so small speakers, LEDs or motors can be connected directly to the output.

- ✚ Pin 4. – Reset, This pin is used to “reset” the internal Flip-flop controlling the state of the output, pin 3. This is an active-low input and is generally connected to a logic “1” level when not used to prevent any unwanted resetting of the output.
- ✚ Pin 5. – Control Voltage, This pin controls the timing of the 555 by overriding the $2/3V_{cc}$ level of the voltage divider network. By applying a voltage to this pin the width of the output signal can be varied independently of the RC timing network. When not used it is connected to ground via a 10nF capacitor to eliminate any noise.
- ✚ Pin 6. – Threshold, The positive input to comparator No 2. This pin is used to reset the Flip-flop when the voltage applied to it exceeds $2/3V_{cc}$ causing the output to switch from “HIGH” to “LOW” state. This pin connects directly to the RC timing circuit.
- ✚ Pin 7. – Discharge, The discharge pin is connected directly to the Collector of an internal NPN transistor which is used to “discharge” the timing capacitor to ground when the output at pin 3 switches “LOW”.
- ✚ Pin 8. – Supply +Vcc, This is the power supply pin and for general purpose TTL 555 timers is between 4.5V and 15V.

The 555 Timers name comes from the fact that there are three 5kΩ resistors connected together internally producing a voltage divider network between the supply voltage at pin 8 and ground at pin 1. The voltage across this series resistive network holds the negative inverting input of comparator two at $2/3V_{cc}$ and the positive non-inverting input to comparator one at $1/3V_{cc}$.

The two comparators produce an output voltage dependent upon the voltage difference at their inputs which is determined by the charging and discharging action of the externally connected RC network. The outputs from both comparators are connected to the two inputs of the flip-flop which in turn produces either a “HIGH” or “LOW” level output at Q based on the states of its inputs. The output from the flip-flop is used to control a high current output

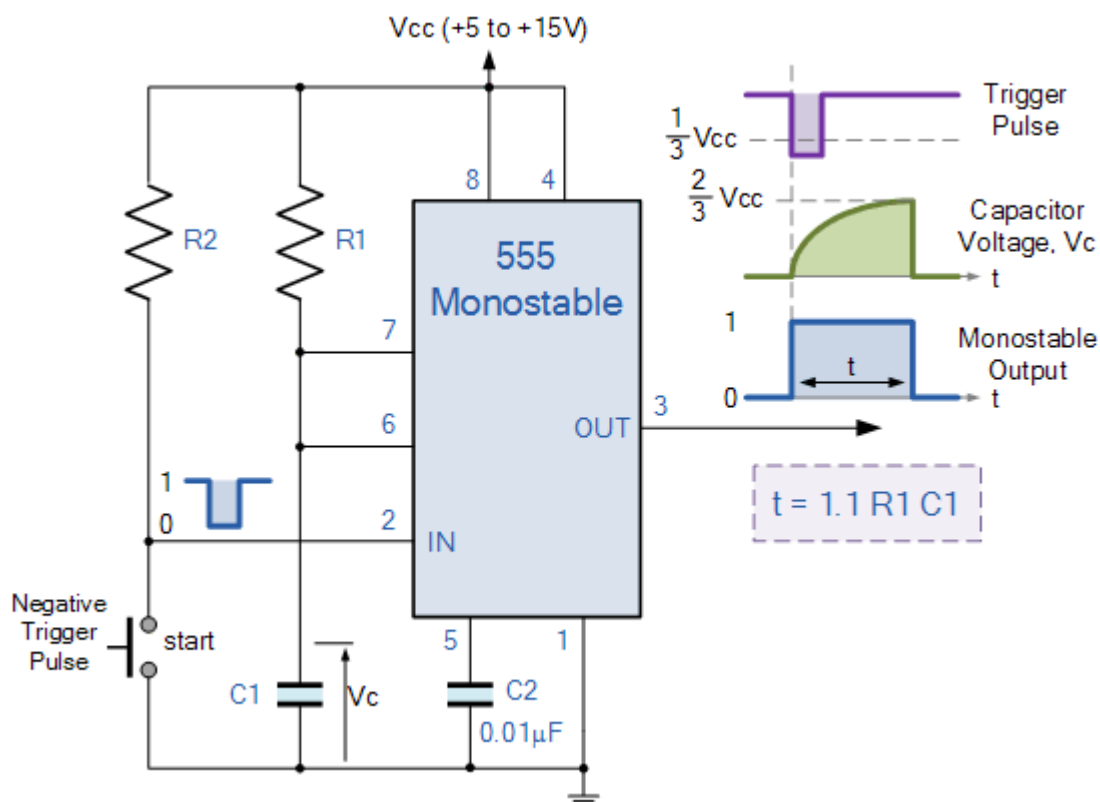
switching stage to drive the connected load producing either a “HIGH” or “LOW” voltage level at the output pin.

The most common use of the 555 timer oscillator is as a simple astable oscillator by connecting two resistors and a capacitor across its terminals to generate a fixed pulse train with a time period determined by the time constant of the RC network. But the 555 timer oscillator chip can also be connected in a variety of different ways to produce Monostable or Bistable multivibrators as well as the more common Astable Multivibrator.

The Monostable 555 Timer

The operation and output of the 555 timer monostable is exactly the same as that for the transistorised one we look at previously in the Monostable Multivibrators tutorial. The difference this time is that the two transistors have been replaced by the 555 timer device. Consider the 555 timer monostable circuit below.

Monostable 555 Timer



When a negative (0V) pulse is applied to the trigger input (pin 2) of the Monostable configured 555 Timer oscillator, the internal comparator,

(comparator No1) detects this input and “sets” the state of the flip-flop, changing the output from a “LOW” state to a “HIGH” state. This action in turn turns “OFF” the discharge transistor connected to pin 7, thereby removing the short circuit across the external timing capacitor, C1.

This action allows the timing capacitor to start to charge up through resistor, R1 until the voltage across the capacitor reaches the threshold (pin 6) voltage of $2/3V_{cc}$ set up by the internal voltage divider network. At this point the comparators output goes “HIGH” and “resets” the flip-flop back to its original state which in turn turns “ON” the transistor and discharges the capacitor to ground through pin 7. This causes the output to change its state back to the original stable “LOW” value awaiting another trigger pulse to start the timing process over again. Then as before, the Monostable Multivibrator has only “ONE” stable state.

The Monostable 555 Timer circuit triggers on a negative-going pulse applied to pin 2 and this trigger pulse must be much shorter than the output pulse width allowing time for the timing capacitor to charge and then discharge fully. Once triggered, the 555 Monostable will remain in this “HIGH” unstable output state until the time period set up by the R1 x C1 network has elapsed. The amount of time that the output voltage remains “HIGH” or at a logic “1” level, is given by the following time constant equation.

$$\tau = 1.1 R_1 C_1$$

Where, τ is in seconds, R is in Ω and C in Farads.

555 Timer Example No1

A Monostable 555 Timer is required to produce a time delay within a circuit. If a 10uF timing capacitor is used, calculate the value of the resistor required to produce a minimum output time delay of 500ms.

500ms is the same as saying 0.5s so by rearranging the formula above, we get the calculated value for the resistor, R as:

$$R = \frac{t}{1.1C} = \frac{0.5}{1.1 \times 10\mu F} = \frac{0.5}{1.1 \times 10 \times 10^{-6}} = 45.5k\Omega$$

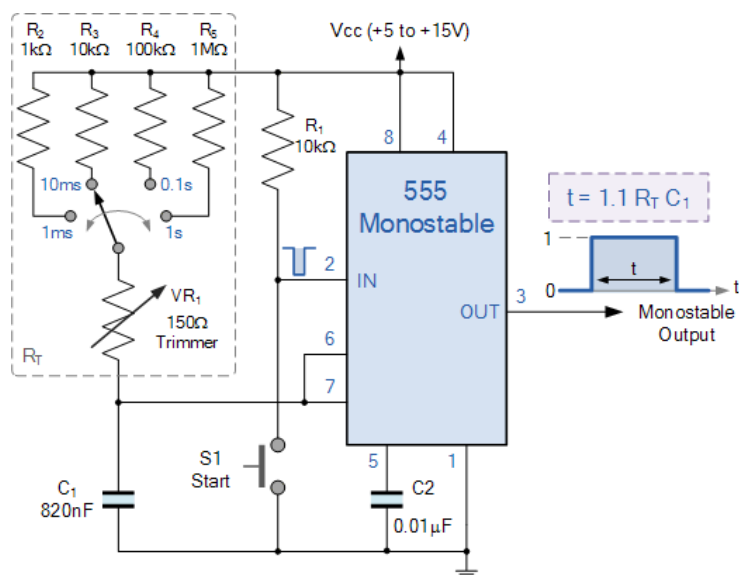
The calculated value for the timing resistor required to produce the required time constant of 500ms is therefore, 45.5K Ω . However, the resistor value of 45.5K Ω does not exist as a standard value resistor, so we would need to select the nearest preferred value resistor of 47k Ω which is available in all the standard ranges of tolerance from the E12 (10%) to the E96 (1%), giving us a new recalculated time delay of 517ms.

If this time difference of 17ms (500 – 517ms) is unacceptable instead of one single timing resistor, two different value resistor could be connected together in series to adjust the pulse width to the exact desired value, or a different timing capacitor value chosen.

We now know that the time delay or output pulse width of a monostable 555 timer is determined by the time constant of the connected RC network. If long time delays are required in the 10's of seconds, it is not always advisable to use high value timing capacitors as they can be physically large, expensive and have large value tolerances, e.g., $\pm 20\%$.

One alternative solution is to use a small value timing capacitor and a much larger value resistor up to about 20M Ω 's to produce the require time delay. Also by using one smaller value timing capacitor and different resistor values connected to it through a multi-position rotary switch, we can produce a Monostable 555 timer oscillator circuit that can produce different pulse widths at each switch rotation such as the switchable Monostable 555 timer circuit shown below.

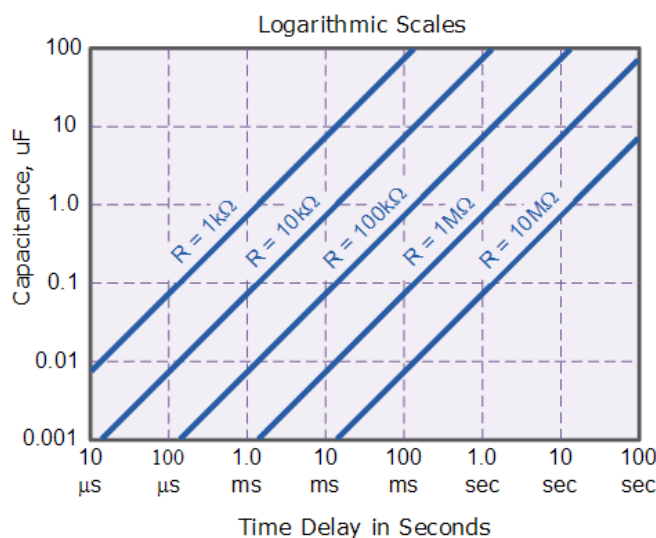
A Switchable 555 Timer



We can manually calculate the values of R and C for the individual components required as we did in the example above. However, the choice of components needed to obtain the desired time delay requires us to calculate with either kilo ohm's (KΩ), Mega ohm's (MΩ), microfarad's (μF) or picafarad's (pF) and it is very easy to end up with a time delay that is out by a factor of ten or even a hundred.

We can make our life a little easier by using a type of chart called a “Nomograph” that will help us to find the monostable multivibrators expected frequency output for different combinations or values of both the R and C. For example,

Monostable Nomograph



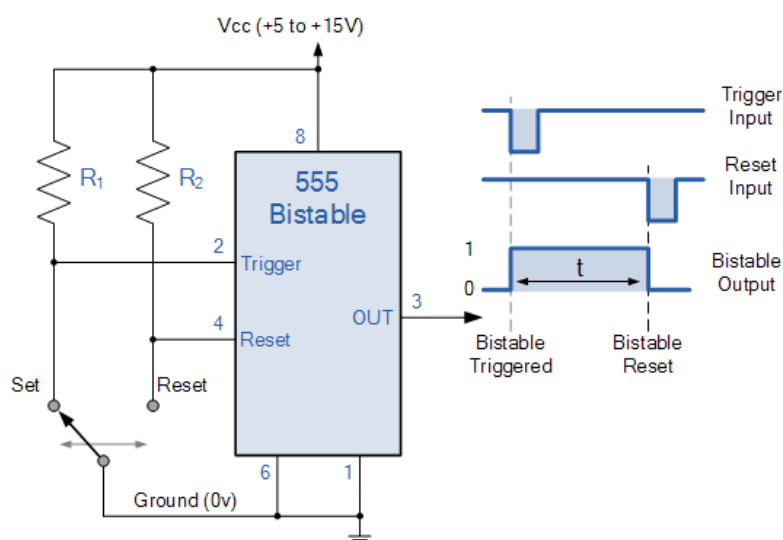
So by selecting suitable values of C and R in the ranges of 0.001uF to 100uF and 1kΩ to 10MΩ respectively, we can read the expected output frequency directly from the nomograph graph thereby eliminating any error in the calculations. In practice the value of the timing resistor for a monostable 555 timer should not be less than 1kΩ or greater than 20MΩ.

Bistable 555 Timer

As well as the one shot 555 Monostable configuration above, we can also produce a Bistable (two stable states) device with the operation and output of the 555 Bistable being similar to the transistorised one we look at previously in the Bistable Multivibrators tutorial.

The 555 Bistable is one of the simplest circuits we can build using the 555 timer oscillator chip. This bistable configuration does not use any RC timing network to produce an output waveform so no equations are required to calculate the time period of the circuit. Consider the Bistable 555 Timer circuit below.

Bistable 555 Timer (flip-flop)



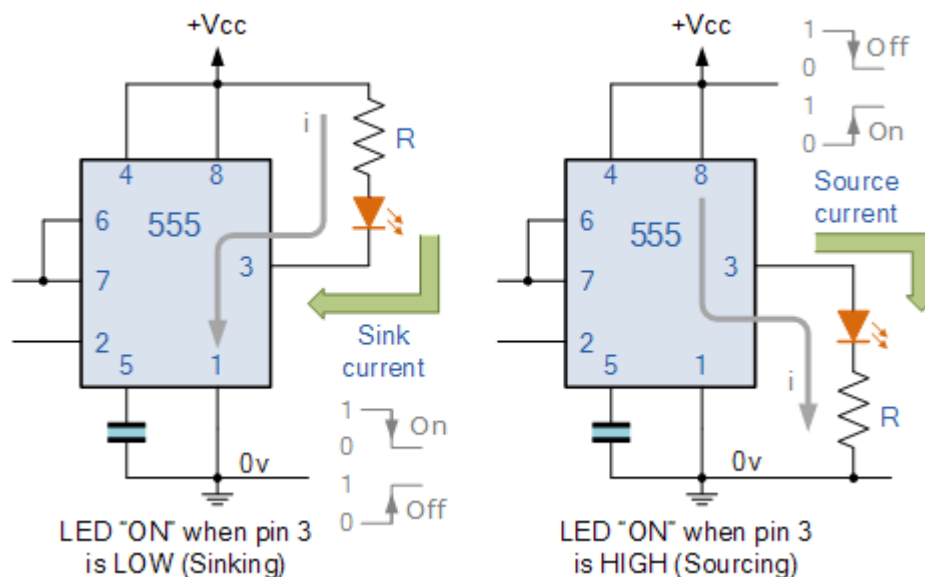
The switching of the output waveform is achieved by controlling the trigger and reset inputs of the 555 timer which are held “HIGH” by the two pull-up resistors, R₁ and R₂. By taking the trigger input (pin 2) “LOW”, switch in set position, changes the output state into the “HIGH” state and by taking the reset input (pin 4) “LOW”, switch in reset position, changes the output into the “LOW” state. This 555 timer circuit will remain in either state indefinitely and is therefore bistable. Then the Bistable 555 timer is stable in both states, “HIGH” and “LOW”.

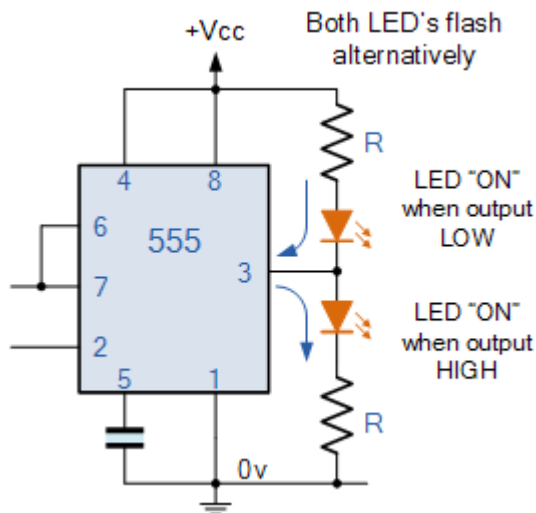
The threshold input (pin 6) is connected to ground to ensure that it cannot reset the bistable circuit as it would in a normal timing application.

555 Timer Output

We could not finish this 555 Timer tutorial without discussing something about the switching and drive capabilities of the 555 timer or indeed the dual 556 Timer IC. The output (pin 3) of the standard 555 timer or the 556 timer, has the ability to either “Sink” or “Source” a load current of up to a maximum of 200mA, which is sufficient to directly drive output transducers such as relays, filament lamps, LED's motors, or speakers etc., with the aid of series resistors or diode protection. This ability of the 555 timer to both “Sink” (absorb) and “Source” (supply) current means that the output device can be connected between the output terminal of the 555 timer and the supply to sink the load current or between the output terminal and ground to source the load current. For example.

Sinking and Sourcing the 555 Timer Output





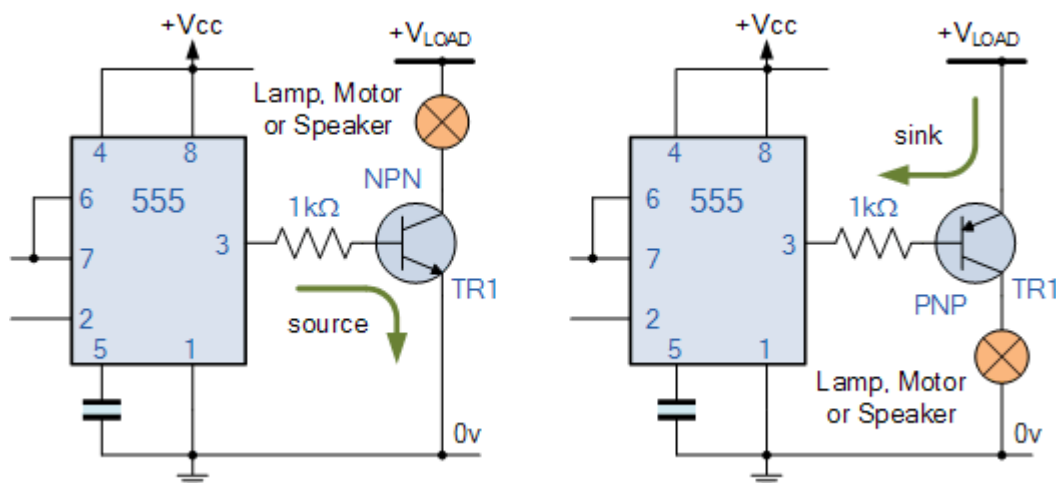
In the first circuit above, the LED is connected between the positive supply rail (+Vcc) and the output pin 3. This means that the current will “Sink” (absorb) or flow into the 555 timer output terminal and the LED will be “ON” when the output is “LOW”.

The second circuit above shows that the LED is connected between the outputs pin 3 and ground (0v). This means that the current will “Source” (supply) or flow out of the 555 timer’s output terminal and the LED will be “ON” when the output is “HIGH”.

The ability of the 555 timer to both sink and source its output load current means that both LED’s can be connected to the output terminal at the same time but only one will be switched “ON” depending whether the output state is “HIGH” or “LOW”. The circuit to the left shows an example of this. The two LED’s will be alternatively switched “ON” and “OFF” depending upon the output. Resistor, R is used to limit the LED current to below 20mA.

We said earlier that the maximum output current to either sink or source the load current via pin 3 is about 200mA at the maximum supply voltage, and this value is more than enough to drive or switch other logic IC’s, LED’s or small lamps, etc. But what if we wanted to switch or control higher power devices such as motors, electromagnets, relays or loudspeakers. Then we would need to use a Transistor to amplify the 555 timers output in order to provide a sufficiently high enough power to drive the load.

555 Timer Transistor Driver



The transistor in the two examples above, can be replaced with a Power MOSFET device or Darlington transistor if the load current is high. When using an inductive load such as a motor, relay or electromagnet, it is advisable to connect a freewheeling (or flywheel) diode directly across the load terminals to absorb any back emf voltages generated by the inductive device when it changes state.

Thus far we have look at using the 555 Timer to generate monostable and bistable output pulses. In the next tutorial about Waveform Generation we will look at connecting the 555 in an astable multivibrator configuration. When used in the astable mode both the frequency and duty cycle of the output waveform can be accurately controlled to produce a very versatile waveform generator.

SESSION 2.

Demonstrate and apply knowledge of analogue electronic components.

Learning Outcomes

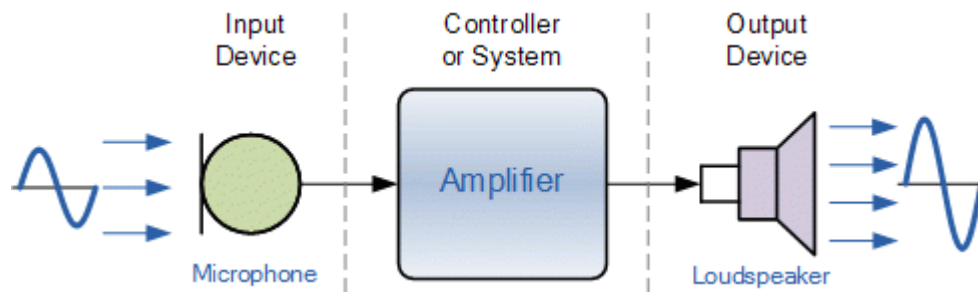
- Analogue electronic components and circuit make-up are analysed in terms of features and application.
- Related circuit is constructed using the specific electronic components for a specific task.

- Operational status of the circuit is tested and verified according to standard operation procedures.

Demonstrate and apply knowledge of analogue electronic components.

Sensors and Transducers

Simple standalone electronic circuits can be made to repeatedly flash a light or play a musical note.



But in order for an electronic circuit or system to perform any useful task or function it needs to be able to communicate with the “real world” whether this is by reading an input signal from an “ON/OFF” switch or by activating some form of output device to illuminate a single light.

In other words, an Electronic System or circuit must be able or capable to “do” something and Sensors and Transducers are the perfect components for doing this.

The word “Transducer” is the collective term used for both Sensors which can be used to sense a wide range of different energy forms such as movement, electrical signals, radiant energy, thermal or magnetic energy etc., and Actuators which can be used to switch voltages or currents.

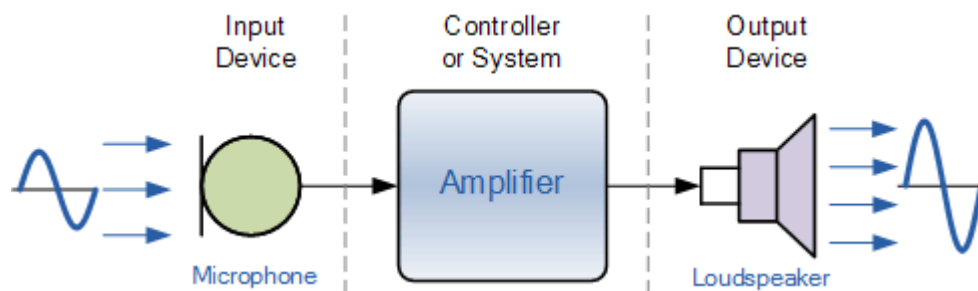
There are many different types of sensors and transducers, both analogue and digital and input and output available to choose from. The type of input or output transducer being used, really depends upon the type of signal or process being “Sensed” or “Controlled” but we can define a sensor and transducers as devices that converts one physical quantity into another.

Devices which perform an “Input” function are commonly called Sensors because they “sense” a physical change in some characteristic that changes in response to some excitation, for example heat or force and convert that into an electrical signal. Devices which perform an “Output” function are generally

called Actuators and are used to control some external device, for example movement or sound.

Electrical Transducers are used to convert energy of one kind into energy of another kind, so for example, a microphone (input device) converts sound waves into electrical signals for the amplifier to amplify (a process), and a loudspeaker (output device) converts these electrical signals back into sound waves and an example of this type of simple Input/output (I/O) system is given below.

Simple Input/output System using Sound Transducers



There are many different types of sensors and transducers available in the marketplace, and the choice of which one to use really depends upon the quantity being measured or controlled, with the more common types given in the table below:

Common Sensors and Transducers

Quantity being Measured	Input Device (Sensor)	Output Device (Actuator)
Light Level	Light Dependant Resistor (LDR) Photodiode Photo-transistor Solar Cell	Lights & Lamps LED's & Displays Fibre Optics

Temperature	Thermocouple	Heater
	Thermistor	Fan
	Thermostat	
	Resistive Temperature Detectors	
Force/Pressure	Strain Gauge	Lifts & Jacks
	Pressure Switch	Electromagnet
	Load Cells	Vibration
Position	Potentiometer	Motor
	Encoders	Solenoid
	Reflective/Slotted Opto-switch	Panel Meters
	LVDT	
Speed	Tacho-generator	AC and DC Motors
	Reflective/Slotted Opto-coupler	Stepper Motor
	Doppler Effect Sensors	Brake
Sound	Carbon Microphone	Bell
	Piezo-electric Crystal	Buzzer
		Loudspeaker

Input type transducers or sensors, produce a voltage or signal output response which is proportional to the change in the quantity that they are measuring (the

stimulus). The type or amount of the output signal depends upon the type of sensor being used. But generally, all types of sensors can be classed as two kinds, either Passive Sensors or Active Sensors.

Generally, active sensors require an external power supply to operate, called an excitation signal which is used by the sensor to produce the output signal. Active sensors are self-generating devices because their own properties change in response to an external effect producing for example, an output voltage of 1 to 10v DC or an output current such as 4 to 20mA DC. Active sensors can also produce signal amplification.

A good example of an active sensor is an LVDT sensor or a strain gauge. Strain gauges are pressure-sensitive resistive bridge networks that are external biased (excitation signal) in such a way as to produce an output voltage in proportion to the amount of force and/or strain being applied to the sensor.

Unlike an active sensor, a passive sensor does not need any additional power source or excitation voltage. Instead a passive sensor generates an output signal in response to some external stimulus. For example, a thermocouple which generates its own voltage output when exposed to heat. Then passive sensors are direct sensors which change their physical properties, such as resistance, capacitance or inductance etc.

But as well as analogue sensors, Digital Sensors produce a discrete output representing a binary number or digit such as a logic level "0" or a logic level "1".

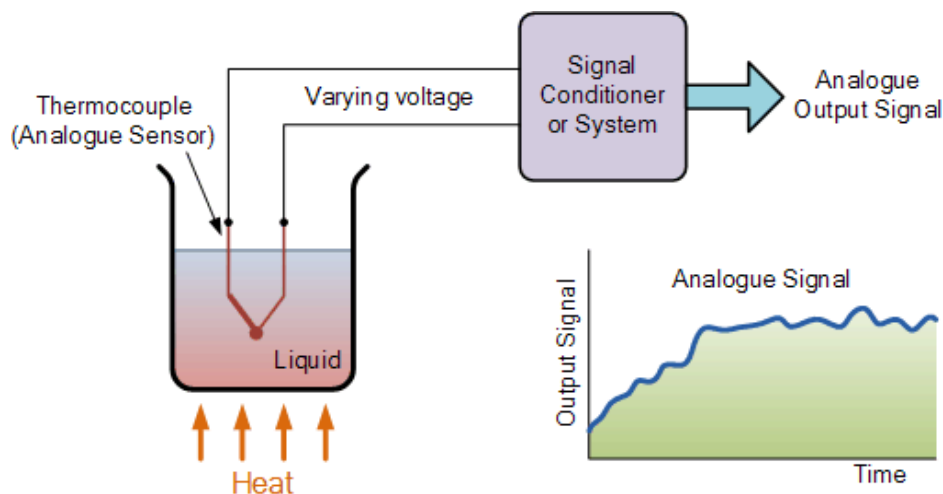
4. Analogue and Digital Sensors

Analogue Sensors

Analogue Sensors produce a continuous output signal or voltage which is generally proportional to the quantity being measured. Physical quantities such as Temperature, Speed, Pressure, Displacement, Strain etc. are all analogue quantities as they tend to be continuous in nature. For example, the temperature of a liquid can be measured using a thermometer or

thermocouple which continuously responds to temperature changes as the liquid is heated up or cooled down.

Thermocouple used to produce an Analogue Signal



Analogue sensors tend to produce output signals that are changing smoothly and continuously over time. These signals tend to be very small in value from a few micro-volts (μV) to several milli-volts (mV), so some form of amplification is required.

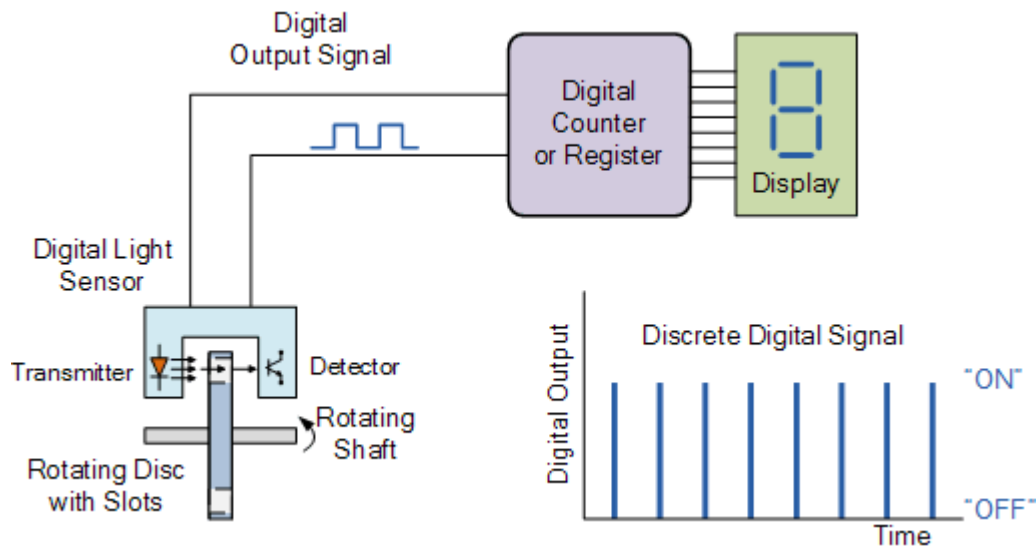
Then circuits which measure analogue signals usually have a slow response and/or low accuracy. Also analogue signals can be easily converted into digital type signals for use in micro-controller systems by the use of analogue-to-digital converters, or ADC's.

Digital Sensors

As its name implies, Digital Sensors produce a discrete digital output signals or voltages that are a digital representation of the quantity being measured.

Digital sensors produce a Binary output signal in the form of a logic "1" or a logic "0", ("ON" or "OFF"). This means then that a digital signal only produces discrete (non-continuous) values which may be outputted as a single "bit", (serial transmission) or by combining the bits to produce a single "byte" output (parallel transmission).

Light Sensor used to produce a Digital Signal



In our simple example above, the speed of the rotating shaft is measured by using a digital LED/Opt-detector sensor. The disc which is fixed to a rotating shaft (for example, from a motor or robot wheels), has a number of transparent slots within its design. As the disc rotates with the speed of the shaft, each slot passes by the sensor in turn producing an output pulse representing a logic "1" or logic "0" level.

These pulses are sent to a register of counter and finally to an output display to show the speed or revolutions of the shaft. By increasing the number of slots or "windows" within the disc more output pulses can be produced for each revolution of the shaft. The advantage of this is that a greater resolution and accuracy is achieved as fractions of a revolution can be detected. Then this type of sensor arrangement could also be used for positional control with one of the discs slots representing a reference position.

Compared to analogue signals, digital signals or quantities have very high accuracies and can be both measured and "sampled" at a very high clock speed. The accuracy of the digital signal is proportional to the number of bits used to represent the measured quantity. For example, using a processor of 8 bits, will produce an accuracy of 0.390% (1 part in 256). While using a processor of 16 bits gives an accuracy of 0.0015%, (1 part in 65,536) or 260 times more accurate. This accuracy can be maintained as digital quantities are manipulated and processed very rapidly, millions of times faster than analogue signals.

In most cases, sensors and more specifically analogue sensors generally require an external power supply and some form of additional amplification or filtering of the signal in order to produce a suitable electrical signal which is capable of being measured or used. One very good way of achieving both amplification and filtering within a single circuit is to use Operational Amplifiers as seen before.

Signal Conditioning of Sensors

As we saw in the Operational Amplifier tutorial, op-amps can be used to provide amplification of signals when connected in either inverting or non-inverting configurations.

The very small analogue signal voltages produced by a sensor such as a few milli-volts or even pico-volts can be amplified many times over by a simple op-amp circuit to produce a much larger voltage signal of say 5v or 5mA that can then be used as an input signal to a microprocessor or analogue-to-digital based system.

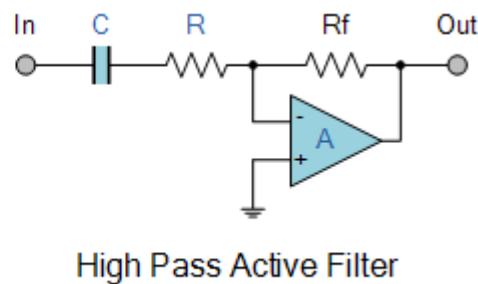
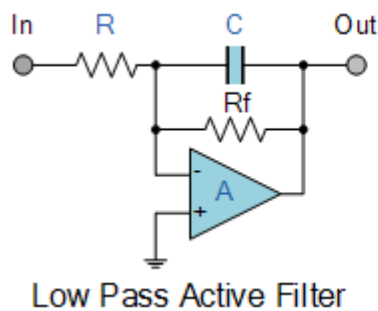
Therefore, to provide any useful signal a sensors output signal has to be amplified with an amplifier that has a voltage gain up to 10,000 and a current gain up to 1,000,000 with the amplification of the signal being linear with the output signal being an exact reproduction of the input, just changed in amplitude.

Then amplification is part of signal conditioning. So when using analogue sensors, generally some form of amplification (Gain), impedance matching, isolation between the input and output or perhaps filtering (frequency selection) may be required before the signal can be used and this is conveniently performed by Operational Amplifiers.

Also, when measuring very small physical changes the output signal of a sensor can become “contaminated” with unwanted signals or voltages that prevent the actual signal required from being measured correctly. These unwanted signals are called “Noise”. This Noise or Interference can be either greatly reduced or even eliminated by using signal conditioning or filtering techniques as we discussed in the Active Filter tutorial.

By using either a Low Pass, or a High Pass or even Band Pass filter the “bandwidth” of the noise can be reduced to leave just the output signal required. For example, many types of inputs from switches, keyboards or manual controls are not capable of changing state rapidly and so low-pass filter can be used. When the interference is at a particular frequency, for example mains frequency, narrow band reject or Notch filters can be used to produce frequency selective filters.

Typical Op-amp Filters

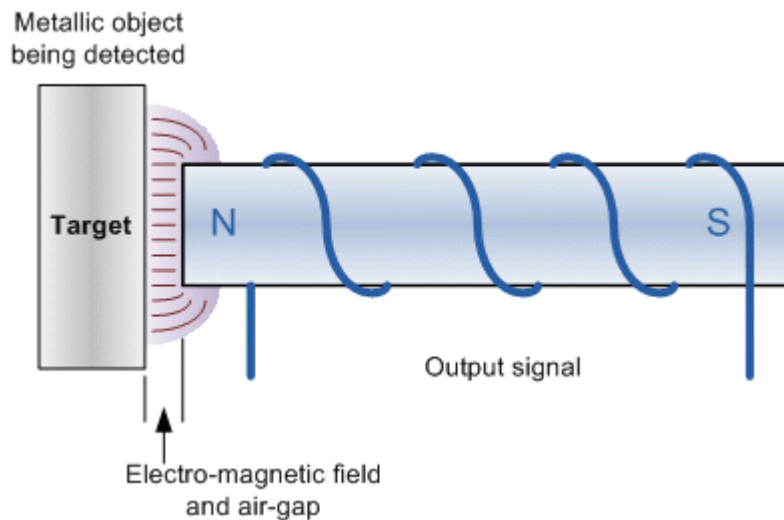


Were some random noise still remains after filtering it may be necessary to take several samples and then average them to give the final value so increasing the signal-to-noise ratio? Either way, both amplification and filtering play an important role in interfacing both sensors and transducers to microprocessor and electronics based systems in “real world” conditions.

In the next tutorial about Sensors, we will look at Positional Sensors which measure the position and/or displacement of physical objects meaning the movement from one position to another for a specific distance or angle.

Position Sensors

In this tutorial we will look at a variety of devices which are classed as Input Devices and are therefore called “Sensors” and in particular those sensors which are Positional in nature.



As their name implies, Position Sensors detect the position of something which means that they are referenced either to or from some fixed point or position. These types of sensors provide “positional” feedback.

One method of determining a position, is to use either “distance”, which could be the distance between two points such as the distance travelled or moved away from some fixed point, or by “rotation” (angular movement). For example, the rotation of a robots wheel to determine its distance travelled along the ground. Either way, Position Sensors can detect the movement of an object in a straight line using Linear Sensors or by its angular movement using Rotational Sensors.

The Potentiometer

The most commonly used of all the “Position Sensors”, is the potentiometer because it is an inexpensive and easy to use position sensor. It has a wiper contact linked to a mechanical shaft that can be either angular (rotational) or linear (slider type) in its movement, and which causes the resistance value between the wiper/slider and the two end connections to change giving an electrical signal output that has a proportional relationship between the actual wiper position on the resistive track and its resistance value. In other words, resistance is proportional to position.

Potentiometers come in a wide range of designs and sizes such as the commonly available round rotational type or the longer and flat linear slider

types. When used as a position sensor the moveable object is connected directly to the rotational shaft or slider of the potentiometer.

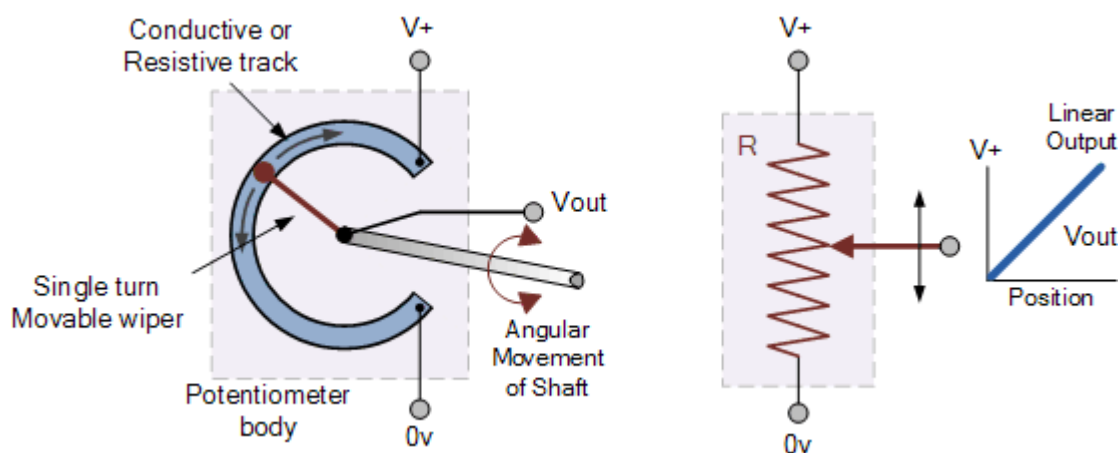


Potentiometers

A DC reference voltage is applied across the two outer fixed connections forming the resistive element. The output voltage signal is taken from the wiper terminal of the sliding contact as shown below.

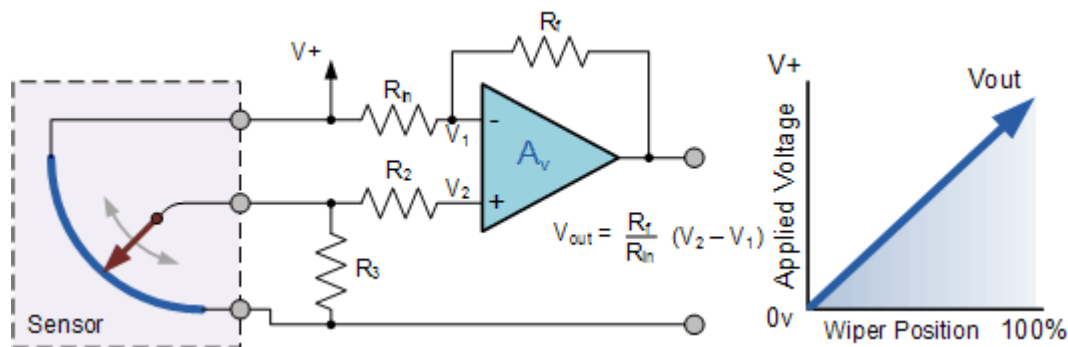
This configuration produces a potential or voltage divider type circuit output which is proportional to the shaft position. Then for example, if you apply a voltage of say 10v across the resistive element of the potentiometer the maximum output voltage would be equal to the supply voltage at 10 volts, with the minimum output voltage equal to 0 volts. Then the potentiometer wiper will vary the output signal from 0 to 10 volts, with 5 volts indicating that the wiper or slider is at its half-way or centre position.

Potentiometer Construction



The output signal (V_{out}) from the potentiometer is taken from the centre wiper connection as it moves along the resistive track and is proportional to the angular position of the shaft.

Example of a simple Positional Sensing Circuit



While resistive potentiometer position sensors have many advantages: low cost, low tech, easy to use etc., as a position sensor they also have many disadvantages: wear due to moving parts, low accuracy, low repeatability, and limited frequency response.

But there is one main disadvantage of using the potentiometer as a positional sensor. The range of movement of its wiper or slider (and hence the output signal obtained) is limited to the physical size of the potentiometer being used. For example a single turn rotational potentiometer generally only has a fixed mechanical rotation of between 0o and about 240 to 330o maximum. However, multi-turn pots of up to 3600o (10 x 360o) of mechanical rotation are also available.

Most types of potentiometers use carbon film for their resistive track, but these types are electrically noisy (the crackle on a radio volume control), and also have a short mechanical life.

Wire-wound pots also known as rheostats, in the form of either a straight wire or wound coil resistive wire can also be used, but wire wound pots suffer from resolution problems as their wiper jumps from one wire segment to the next producing a logarithmic (LOG) output resulting in errors in the output signal. These too suffer from electrical noise.

For high precision low noise applications conductive plastic resistance element type polymer film or cermet type potentiometers are now available. These pots have a smooth low friction electrically linear (LIN) resistive track giving them a low noise, long life and excellent resolution and are available as both multi-turn and single turn devices. Typical applications for this type of high accuracy

position sensor is in computer game joysticks, steering wheels, industrial and robot applications.

Inductive Position Sensors

Linear Variable Differential Transformer

One type of positional sensor that does not suffer from mechanical wear problems is the “Linear Variable Differential Transformer” or LVDT for short. This is an inductive type position sensor which works on the same principle as the AC transformer that is used to measure movement. It is a very accurate device for measuring linear displacement and whose output is proportional to the position of its moveable core.

It basically consists of three coils wound on a hollow tube former, one forming the primary coil and the other two coils forming identical secondary's connected electrically together in series but 180° out of phase either side of the primary coil.

A moveable soft iron ferromagnetic core (sometimes called an “armature”) which is connected to the object being measured, slides or moves up and down inside the tubular body of the LVDT.

A small AC reference voltage called the “excitation signal” (2 – 20V rms, 2 – 20kHz) is applied to the primary winding which in turn induces an EMF signal into the two adjacent secondary windings (transformer principles).

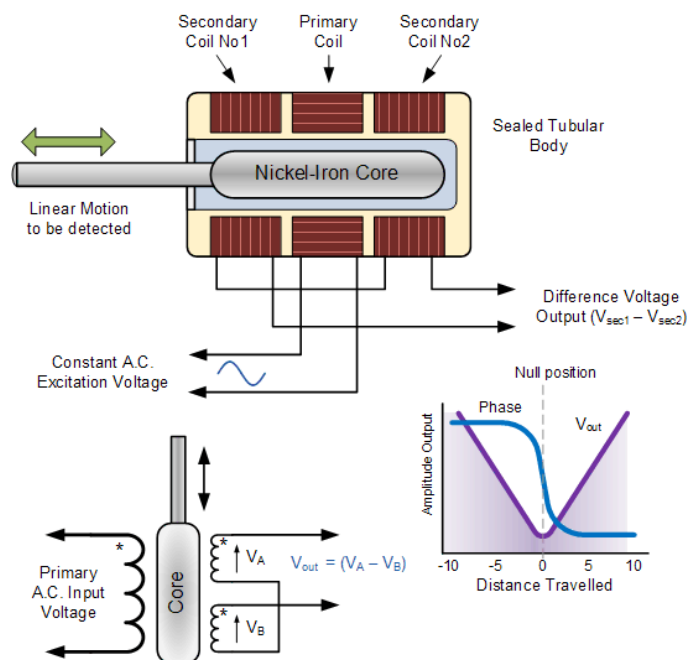
If the soft iron magnetic core armature is exactly in the centre of the tube and the windings, “null position”, the two induced emf in the two secondary windings cancel each other out as they are 180° out of phase, so the resultant output voltage is zero. As the core is displaced slightly to one side or the other from this null or zero position, the induced voltage in one of the secondary's will be become greater than that of the other secondary and an output will be produced.

The polarity of the output signal depends upon the direction and displacement of the moving core. The greater the movement of the soft iron core from its central null position the greater will be the resulting output signal. The result is a differential voltage output which varies linearly with the cores position.

Therefore, the output signal from this type of position sensor has both an amplitude that is a linear function of the core's displacement and a polarity that indicates direction of movement.

The phase of the output signal can be compared to the primary coil excitation phase enabling suitable electronic circuits such as the AD592 LVDT Sensor Amplifier to know which half of the coil the magnetic core is in and thereby know the direction of travel.

The Linear Variable Differential Transformer



When the armature is moved from one end to the other through the centre position the output voltages change from maximum to zero and back to maximum again but in the process change its phase angle by 180 deg's. This enables the LVDT to produce an output AC signal whose magnitude represents the amount of movement from the centre position and whose phase angle represents the direction of movement of the core.

A typical application of a linear variable differential transformer (LVDT) sensor would be as a pressure transducer, where the pressure being measured pushes against a diaphragm to produce a force. The force is then converted into a readable voltage signal by the sensor.

Advantages of the linear variable differential transformer, or LVDT compared to a resistive potentiometer are that its linearity that is its voltage output to

displacement is excellent, very good accuracy, good resolution, high sensitivity as well as frictionless operation. They are also sealed for use in hostile environments.

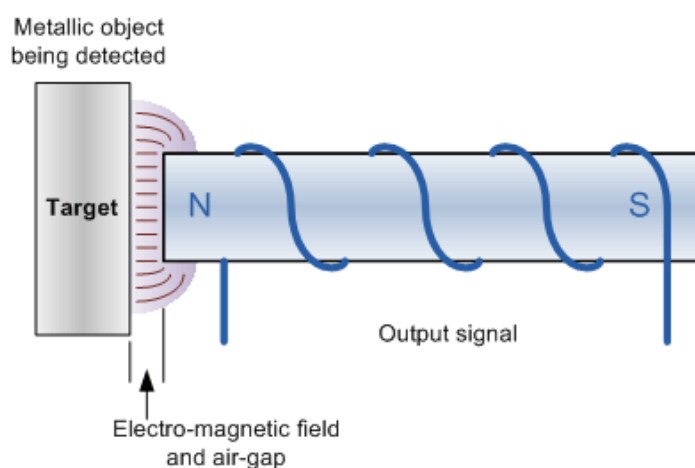
Inductive Proximity Sensors.

Another type of inductive position sensor in common use is the Inductive Proximity Sensor also called an Eddy current sensor. While they do not actually measure displacement or angular rotation they are mainly used to detect the presence of an object in front of them or within a close proximity, hence their name "proximity sensor".

Proximity sensors, are non-contact position sensors that use a magnetic field for detection with the simplest magnetic sensor being the reed switch. In an inductive sensor, a coil is wound around an iron core within an electromagnetic field to form an inductive loop.

When a ferromagnetic material is placed within the eddy current field generated around the inductive sensor, such as a ferromagnetic metal plate or metal screw, the inductance of the coil changes significantly. The proximity sensors detection circuit detects this change producing an output voltage. Therefore, inductive proximity sensors operate under the electrical principle of Faraday's Law of inductance.

Inductive Proximity Sensors



An inductive proximity sensor has four main components; The oscillator which produces the electromagnetic field, the coil which generates the magnetic field, the detection circuit which detects any change in the field when an

object enters it and the output circuit which produces the output signal, either with normally closed (NC) or normally open (NO) contacts.

Inductive proximity sensors allow for the detection of metallic objects in front of the sensor head without any physical contact of the object itself being detected. This makes them ideal for use in dirty or wet environments. The

“sensing” range of proximity sensors is very small, typically 0.1mm to 12mm.

As well as industrial applications, inductive proximity sensors are also commonly used to control the flow of traffic by changing of traffic lights at junctions and cross roads. Rectangular inductive loops of wire are buried into the tarmac road surface.



Proximity Sensor

When a car or other road vehicle passes over this inductive loop, the metallic body of the vehicle changes the loops inductance and activates the sensor thereby alerting the traffic lights controller that there is a vehicle waiting.

One main disadvantage of these types of position sensors is that they are “Omni-directional”, that is they will sense a metallic object either above, below or to the side of it. Also, they do not detect non-metallic objects although Capacitive Proximity Sensors and Ultrasonic Proximity Sensors are available.

Other commonly available magnetic positional sensors include: reed switches, Hall Effect Sensors and variable reluctance sensors.

Rotary Encoders

Rotary Encoders are another type of position sensor which resemble potentiometers mentioned earlier but are non-contact optical devices used for converting the angular position of a rotating shaft into an analogue or digital data code. In other words, they convert mechanical movement into an electrical signal (preferably digital).

All optical encoders work on the same basic principle. Light from an LED or infra-red light source is passed through a rotating high-resolution encoded disk that contains the required code patterns, either binary, grey code or BCD. Photo detectors scan the disk as it rotates and an electronic circuit processes the information into a digital form as a stream of binary output pulses that are fed to counters or controllers which determine the actual angular position of the shaft. There are two basic types of rotary optical encoders, Incremental Encoders and Absolute Position Encoders.

Incremental Encoder

Incremental Encoders, also known as quadrature encoders or relative rotary encoder, are the simplest of the two position sensors. Their output is a series of square wave pulses generated by a photocell arrangement as the coded disk, with evenly spaced transparent and dark lines called segments on its surface, moves or rotates past the light source. The encoder produces a stream of square wave pulses which, when counted, indicates the angular position of the rotating shaft.

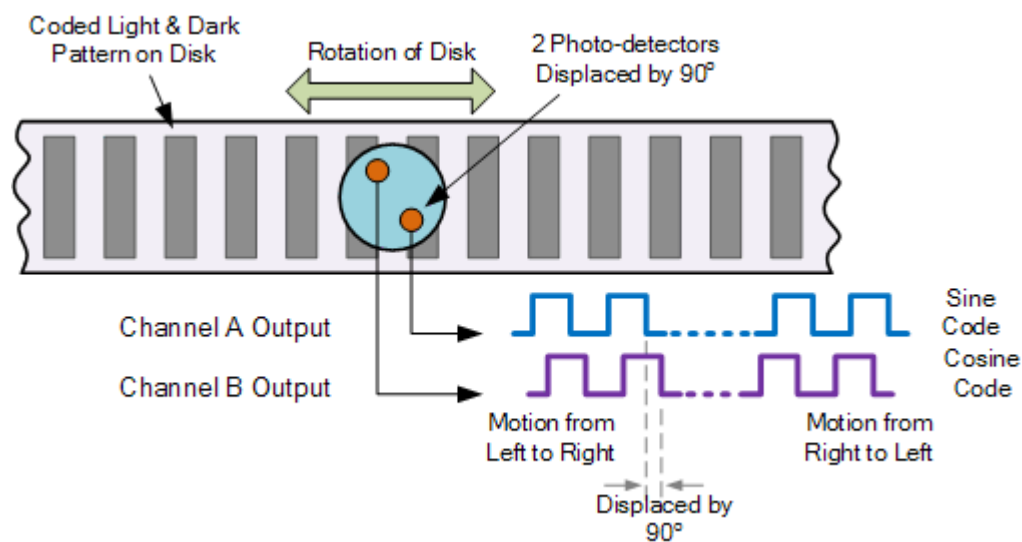


Encoder Disk

Incremental encoders have two separate outputs called “quadrature outputs”. These two outputs are displaced at 90° out of phase from each other with the direction of rotation of the shaft being determined from the output sequence. The number of transparent and dark segments or slots on the disk determines the resolution of the device and increasing the number of lines in the pattern increases the resolution per degree of rotation. Typical encoded discs have a resolution of up to 256 pulses or 8-bits per rotation.

The simplest incremental encoder is called a tachometer. It has one single square wave output and is often used in unidirectional applications where basic position or speed information only is required. The "Quadrature" or "Sine wave" encoder is the more common and has two output square waves commonly called channel A and channel B. This device uses two photo detectors, slightly offset from each other by 90° thereby producing two separate sine and cosine output signals.

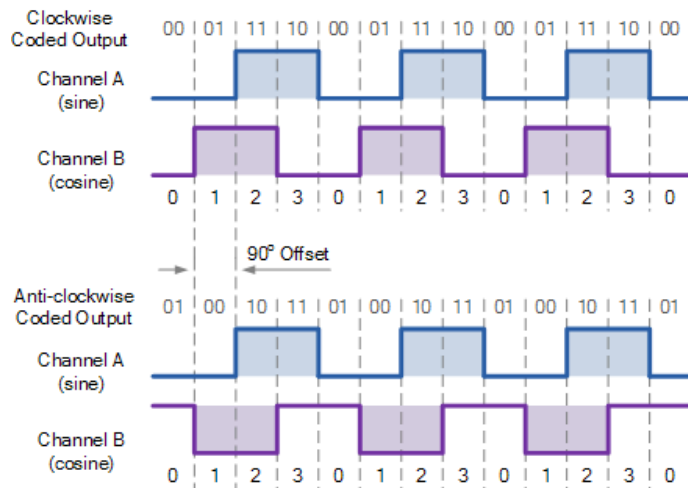
Simple Incremental Encoder



By using the Arc Tangent mathematical function the angle of the shaft in radians can be calculated. Generally, the optical disk used in rotary position encoders is circular, then the resolution of the output will be given as: $\theta = 360/n$, where n equals the number of segments on coded disk.

Then for example, the number of segments required to give an incremental encoder a resolution of 1° will be: $1^\circ = 360/n$, therefore, $n = 360$ windows, etc. Also the direction of rotation is determined by noting which channel produces an output first, either channel A or channel B giving two directions of rotation, A leads B or B leads A. This arrangement is shown below.

Incremental Encoder Output

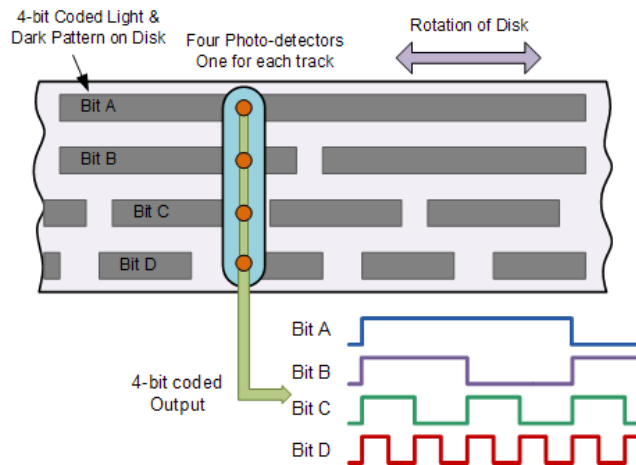


One main disadvantage of incremental encoders when used as a position sensor, is that they require external counters to determine the absolute angle of the shaft within a given rotation. If the power is momentarily shut off, or if the encoder misses a pulse due to noise or a dirty disc, the resulting angular information will produce an error. One way of overcoming this disadvantage is to use absolute position encoders.

Absolute Position Encoder

Absolute Position Encoders are more complex than quadrature encoders. They provide a unique output code for every single position of rotation indicating both position and direction. Their coded disk consists of multiple concentric "tracks" of light and dark segments. Each track is independent with its own photo detector to simultaneously read a unique coded position value for each angle of movement. The number of tracks on the disk corresponds to the binary "bit"-resolution of the encoder so a 12-bit absolute encoder would have 12 tracks and the same coded value only appears once per revolution.

4-bit Binary Coded Disc



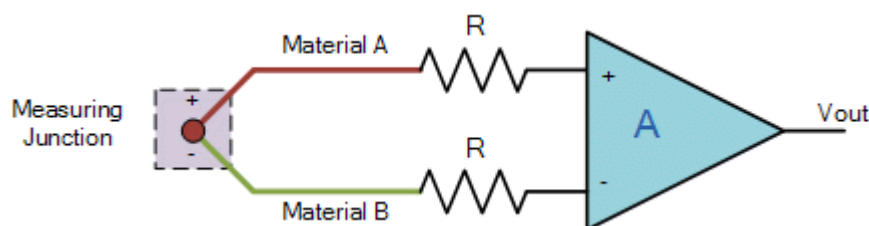
One main advantage of an absolute encoder is its non-volatile memory which retains the exact position of the encoder without the need to return to a “home” position if the power fails. Most rotary encoders are defined as “single-turn” devices, but absolute multi-turn devices are available, which obtain feedback over several revolutions by adding extra code disks.

Typical application of absolute position encoders are in computer hard drives and CD/DVD drives where the absolute position of the drives read/write heads are monitored or in printers/plotters to accurately position the printing heads over the paper.

In this tutorial about Position Sensors, we have looked at several examples of sensors that can be used to measure the position or presence of objects. In the next tutorial we will look at sensors that are used to measure temperature such as thermistors, thermostats and thermocouples, and as such are known commonly as Temperature Sensors.

Temperature Sensors

The most commonly used type of all the sensors are those types of sensors which detect Temperature or heat.



These types of temperature sensor vary from simple ON/OFF thermostatic devices which control a domestic hot water heating system to highly sensitive semiconductor types that can control complex process control furnace plants. We remember from our school science classes that the movement of molecules and atoms produces heat (kinetic energy) and the greater the movement, the more heat that is generated. Temperature Sensors measure the amount of heat energy or even coldness that is generated by an object or system, allowing us to “sense” or detect any physical change to that temperature producing either an analogue or digital output.

There are many different types of Temperature Sensor available and all have different characteristics depending upon their actual application. A temperature sensor consists of two basic physical types:

- ✚ Contact Temperature Sensor Types – These types of temperature sensor are required to be in physical contact with the object being sensed and use conduction to monitor changes in temperature. They can be used to detect solids, liquids or gases over a wide range of temperatures.
- ✚ Non-contact Temperature Sensor Types – These types of temperature sensor use convection and radiation to monitor changes in temperature. They can be used to detect liquids and gases that emit radiant energy as heat rises and cold settles to the bottom in convection currents or detect the radiant energy being transmitted from an object in the form of infra-red radiation (the sun).
- ✚ The two basic types of contact or even non-contact temperature sensors can also be sub-divided into the following three groups of sensors, Electro-mechanical, Resistive and Electronic and all three types are discussed below.

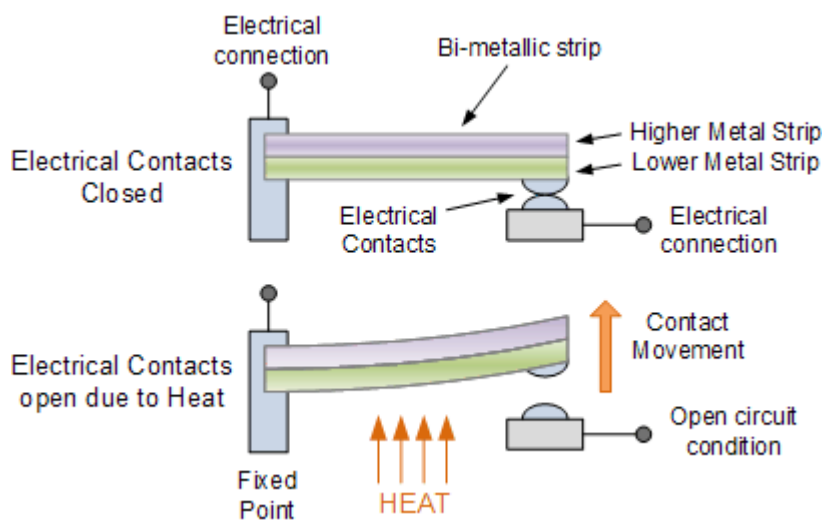
The Thermostat

The Thermostat is a contact type electro-mechanical temperature sensor or switch, that basically consists of two different metals such as nickel, copper, tungsten or aluminium etc., that are bonded together to form a Bi-metallic strip.

The different linear expansion rates of the two dissimilar metals produces a mechanical bending movement when the strip is subjected to heat.

The bi-metallic strip can be used itself as an electrical switch or as a mechanical way of operating an electrical switch in thermostatic controls and are used extensively to control hot water heating elements in boilers, furnaces, hot water storage tanks as well as in vehicle radiator cooling systems.

The Bi-metallic Thermostat



The thermostat consists of two thermally different metals stuck together back to back. When it is cold the contacts are closed and current passes through the thermostat. When it gets hot, one metal expands more than the other and the bonded bi-metallic strip bends up (or down) opening the contacts preventing the current from flowing.

There are two main types of bi-metallic strips based mainly upon their movement when subjected to temperature changes. There are the “snap-action” types that produce an instantaneous “ON/OFF” or “OFF/ON” type action on the electrical contacts at a set temperature point, and the slower “creep-action” types that gradually change their position as the temperature changes.



On/Off Thermostat

Snap-action type thermostats are commonly used in our homes for controlling the temperature set point of ovens, irons, immersion hot water tanks and they can also be found on walls to control the domestic heating system.

Creeper types generally consist of a bi-metallic coil or spiral that slowly unwinds or coils-up as the temperature changes. Generally, creeper type bi-metallic strips are more sensitive to temperature changes than the standard snap ON/OFF types as the strip is longer and thinner making them ideal for use in temperature gauges and dials etc.

Although very cheap and are available over a wide operating range, one main disadvantage of the standard snap-action type thermostats when used as a temperature sensor, is that they have a large hysteresis range from when the electrical contacts open until when they close again. For example, it may be set to 20°C but may not open until 22°C or close again until 18°C.

So the range of temperature swing can be quite high. Commercially available bi-metallic thermostats for home use do have temperature adjustment screws that allow for a more precise desired temperature set-point and hysteresis level to be pre-set.

The Thermistor

The Thermistor is another type of temperature sensor, whose name is a combination of the words THERM-ally sensitive res-ISTOR. A thermistor is a special type of resistor which changes its physical resistance when exposed to changes in temperature.

Thermistors are generally made from ceramic materials such as oxides of nickel, manganese or cobalt coated in glass which makes them easily damaged. Their

main advantage over snap-action types is their speed of response to any changes in temperature, accuracy and repeatability.

Most types of thermistor's have a Negative Temperature Coefficient of resistance or (NTC), that is their resistance value goes DOWN with an increase in the temperature, and of course there are some which have a Positive Temperature Coefficient, (PTC), in that their resistance value goes UP with an increase in temperature.



Thermistor

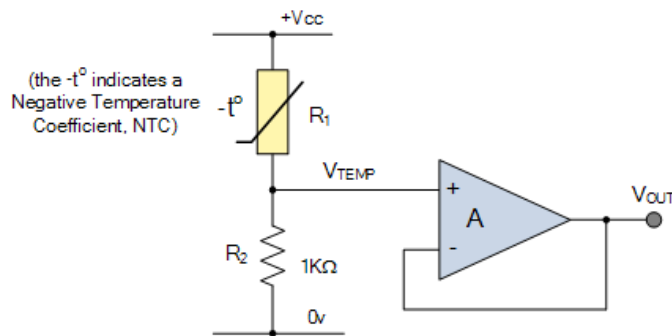
Thermistors are constructed from a ceramic type semiconductor material using metal oxide technology such as manganese, cobalt and nickel, etc. The semiconductor material is generally formed into small pressed discs or balls which are hermetically sealed to give a relatively fast response to any changes in temperature.

Thermistors are rated by their resistive value at room temperature (usually at 25°C), their time constant (the time to react to the temperature change) and their power rating with respect to the current flowing through them. Like resistors, thermistors are available with resistance values at room temperature from 10's of MΩ down to just a few Ohms, but for sensing purposes those types with values in the kilo-ohms are generally used.

Thermistors are passive resistive devices which means we need to pass a current through it to produce a measurable voltage output. Then thermistors are generally connected in series with a suitable biasing resistor to form a potential divider network and the choice of resistor gives a voltage output at some pre-determined temperature point or value for example:

Temperature Sensors Example No1

The following thermistor has a resistance value of 10K Ω at 25°C and a resistance value of 100 Ω at 100°C. Calculate the voltage drop across the thermistor and hence its output voltage (V_{out}) for both temperatures when connected in series with a 1k Ω resistor across a 12v power supply.



At 25°C

$$V_{out} = \frac{R_2}{R_1 + R_2} \times V = \frac{1000}{10000 + 1000} \times 12v = 1.09v$$

At 100°C

$$V_{out} = \frac{R_2}{R_1 + R_2} \times V = \frac{1000}{100 + 1000} \times 12v = 10.9v$$

By changing the fixed resistor value of R_2 (in our example 1k Ω) to a potentiometer or preset, a voltage output can be obtained at a predetermined temperature set point for example, 5v output at 60°C and by varying the potentiometer a particular output voltage level can be obtained over a wider temperature range.

It needs to be noted however, that thermistor's are non-linear devices and their standard resistance values at room temperature is different between different thermistor's, which is due mainly to the semiconductor materials they are made from. The Thermistor, have an exponential change with temperature and therefore have a Beta temperature constant (β) which can be used to calculate its resistance for any given temperature point.

However, when used with a series resistor such as in a voltage divider network or Wheatstone bridge type arrangement, the current obtained in response to a voltage applied to the divider/bridge network is linear with temperature. Then, the output voltage across the resistor becomes linear with temperature.

Resistive Temperature Detectors (RTD).

Another type of electrical resistance temperature sensor is the Resistance Temperature Detector or RTD. RTD's are precision temperature sensors made from high-purity conducting metals such as platinum, copper or nickel wound into a coil and whose electrical resistance changes as a function of temperature, similar to that of the thermistor. Also available are thin-film RTD's. These devices have a thin film of platinum paste is deposited onto a white ceramic substrate.

Resistive temperature detectors have positive temperature coefficients (PTC) but unlike the thermistor their output is extremely linear producing very accurate measurements of temperature.



A Resistive RTD

However, they have very poor thermal sensitivity, that is a change in temperature only produces a very small output change for example, $1\Omega/^{\circ}\text{C}$.

The more common types of RTD's are made from platinum and are called Platinum Resistance Thermometer or PRT's with the most commonly available of them all the Pt100 sensor, which has a standard resistance value of 100Ω at 0°C . The downside is that Platinum is expensive and one of the main disadvantages of this type of device is its cost.

Like the thermistor, RTD's are passive resistive devices and by passing a constant current through the temperature sensor it is possible to obtain an output voltage that increases linearly with temperature. A typical RTD has a base resistance of about 100Ω at 0°C , increasing to about 140Ω at 100°C with an operating temperature range of between -200 to $+600^{\circ}\text{C}$.

Because the RTD is a resistive device, we need to pass a current through them and monitor the resulting voltage. However, any variation in resistance due to self-heat of the resistive wires as the current flows through it, I^2R , (Ohms Law) causes an error in the readings. To avoid this, the RTD is usually connected into a

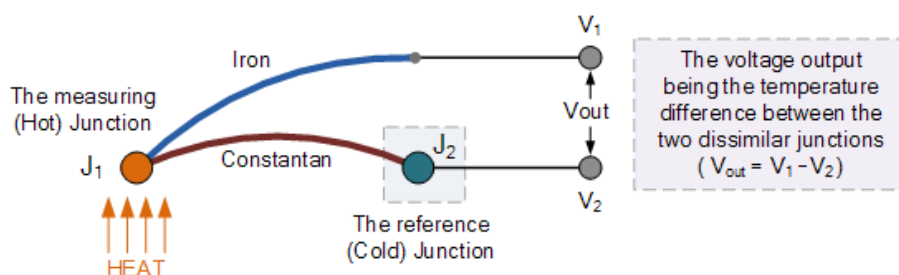
Wheatstone bridge network which has additional connecting wires for lead-compensation and/or connection to a constant current source.

The Thermocouple

The Thermocouple is by far the most commonly used type of all the temperature sensor types. Thermocouples are popular due to its simplicity, ease of use and their speed of response to changes in temperature, due mainly to their small size. Thermocouples also have the widest temperature range of all the temperature sensors from below -200°C to well over 2000°C .

Thermocouples are thermoelectric sensors that basically consists of two junctions of dissimilar metals, such as copper and constantan that are welded or crimped together. One junction is kept at a constant temperature called the reference (Cold) junction, while the other the measuring (Hot) junction. When the two junctions are at different temperatures, a voltage is developed across the junction which is used to measure the temperature sensor as shown below.

Thermocouple Construction



The operating principal of a thermocouple is very simple and basic. When fused together the junction of the two dissimilar metals such as copper and constantan produces a “thermo-electric” effect which gives a constant potential difference of only a few millivolts (mV) between them. The voltage difference between the two junctions is called the “See beck effect” as a temperature gradient is generated along the conducting wires producing an emf. Then the output voltage from a thermocouple is a function of the temperature changes.

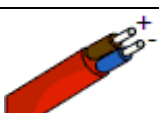
If both the junctions are at the same temperature the potential difference across the two junctions is zero in other words, no voltage output as $V_1 = V_2$. However, when the junctions are connected within a circuit and are both at

different temperatures a voltage output will be detected relative to the difference in temperature between the two junctions, $V_1 - V_2$. This difference in voltage will increase with temperature until the junction's peak voltage level is reached and this is determined by the characteristics of the two dissimilar metals used.

Thermocouples can be made from a variety of different materials enabling extreme temperatures of between -200°C to over $+2000^{\circ}\text{C}$ to be measured. With such a large choice of materials and temperature range, internationally recognised standards have been developed complete with thermocouple colour codes to allow the user to choose the correct thermocouple sensor for a particular application. The British colour code for standard thermocouples is given below.

Thermocouple Colour Codes

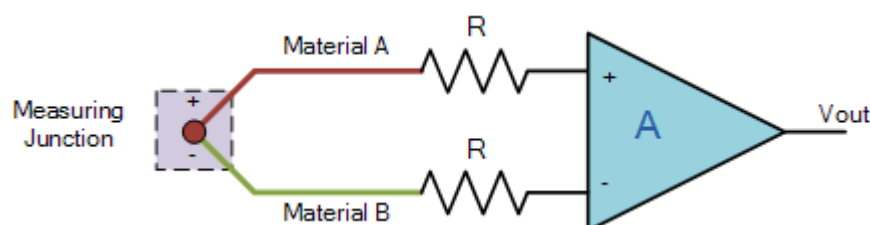
Thermocouple Sensor Colour Codes Extension and Compensating Leads

Code Type	Conductors (+/-)	Sensitivity	British BS 1843:1952
E	Nickel Chromium / Constantan	-200 to 900°C	
J	Iron / Constantan	0 to 750°C	
K	Nickel Chromium / Nickel Aluminium	-200 to 1250°C	
N	Nicrosil / Nisil	0 to 1250°C	
T	Copper / Constantan	-200 to 350°C	

U	Copper / Copper Nickel Compensating for "S" and "R"	0 to 1450°C	
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The three most common thermocouple materials used above for general temperature measurement are Iron-Constantan (Type J), Copper-Constantan (Type T), and Nickel-Chromium (Type K). The output voltage from a thermocouple is very small, only a few millivolts (mV) for a 10°C change in temperature difference and because of this small voltage output some form of amplification is generally required.

Thermocouple Amplification



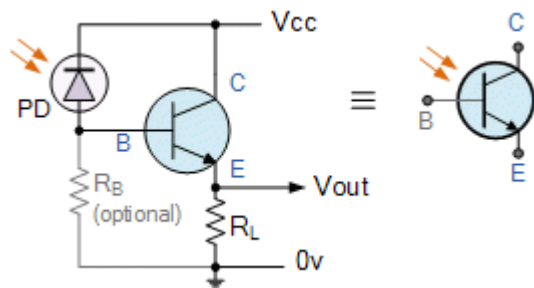
The type of amplifier, either discrete or in the form of an Operational Amplifier needs to be carefully selected, because good drift stability is required to prevent recalibration of the thermocouple at frequent intervals. This makes the chopper and instrumentation type of amplifier preferable for most temperature sensing applications.

Other Temperature Sensor Types not mentioned here include, Semiconductor Junction Sensors, Infra-red and Thermal Radiation Sensors, Medical type Thermometers, Indicators and Colour Changing Inks or Dyes.

In this tutorial about "Temperature Sensor Types", we have looked at several examples of sensors that can be used to measure changes in temperature. In the next tutorial we will look at sensors that are used to measure light quantity, such as Photodiodes, Phototransistors, Photovoltaic Cells and the Light Dependant Resistor.

Light Sensors

Light Sensors are photoelectric devices that convert light energy (photons) whether visible or infra-red light into an electrical (electrons) signal.



A Light Sensor generates an output signal indicating the intensity of light by measuring the radiant energy that exists in a very narrow range of frequencies basically called “light”, and which ranges in frequency from “Infra-red” to “Visible” up to “Ultraviolet” light spectrum.

The light sensor is a passive devices that convert this “light energy” whether visible or in the infra-red parts of the spectrum into an electrical signal output. Light sensors are more commonly known as “Photoelectric Devices” or “Photo Sensors” because the convert light energy (photons) into electricity (electrons). Photoelectric devices can be grouped into two main categories, those which generate electricity when illuminated, such as Photo-voltaics or Photo-emissives etc, and those which change their electrical properties in some way such as Photo-resistors or Photo-conductors. This leads to the following classification of devices.

- ✚ Photo-emissive Cells – These are photodevices which release free electrons from a light sensitive material such as caesium when struck by a photon of sufficient energy. The amount of energy the photons have depends on the frequency of the light and the higher the frequency, the more energy the photons have converting light energy into electrical energy.
- ✚ Photo-conductive Cells – These photodevices vary their electrical resistance when subjected to light. Photoconductivity results from light hitting a semiconductor material which controls the current flow through it. Thus, more light increase the current for a given applied voltage. The

most common photoconductive material is Cadmium Sulphide used in LDR photocells.

- ✚ Photo-voltaic Cells – These photodevices generate an emf in proportion to the radiant light energy received and is similar in effect to photoconductivity. Light energy falls on to two semiconductor materials sandwiched together creating a voltage of approximately 0.5V. The most common photovoltaic material is Selenium used in solar cells.
- ✚ Photo-junction Devices – These photodevices are mainly true semiconductor devices such as the photodiode or phototransistor which use light to control the flow of electrons and holes across their PN-junction. Photojunction devices are specifically designed for detector application and light penetration with their spectral response tuned to the wavelength of incident light.

The Photoconductive Cell

A Photoconductive light sensor does not produce electricity but simply changes its physical properties when subjected to light energy. The most common type of photoconductive device is the Photoresistor which changes its electrical resistance in response to changes in the light intensity.

Photoresistors are Semiconductor devices that use light energy to control the flow of electrons, and hence the current flowing through them. The commonly used Photoconductive Cell is called the Light Dependent Resistor or LDR.

The Photoconductive Cell

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The Light Dependent Resistor

As its name implies, the Light Dependent Resistor (LDR) is made from a piece of exposed semiconductor material such as cadmium sulphide that changes its electrical resistance from several thousand Ohms in the dark to only a few hundred Ohms when light falls upon it by creating hole-electron pairs in the material.



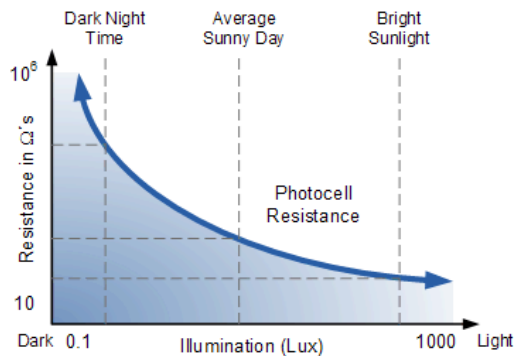
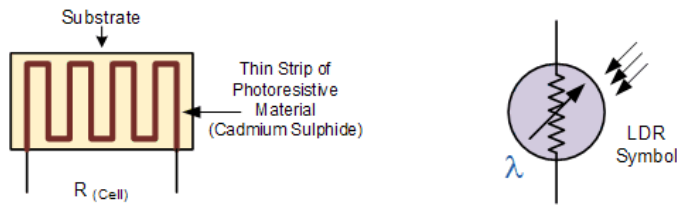
Typical LDR

The net effect is an improvement in its conductivity with a decrease in resistance for an increase in illumination. Also, photoresistive cells have a long response time requiring many seconds to respond to a change in the light intensity.

Materials used as the semiconductor substrate include, lead sulphide (PbS), lead selenide (PbSe), indium antimonite (InSb) which detect light in the infra-red range with the most commonly used of all photoresistive light sensors being Cadmium Sulphide (CdS).

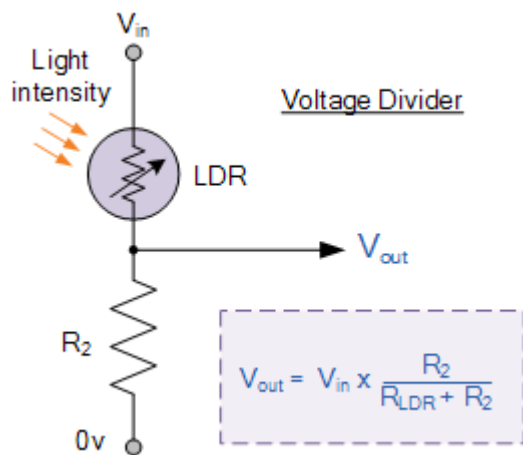
Cadmium sulphide is used in the manufacture of photoconductive cells because its spectral response curve closely matches that of the human eye and can even be controlled using a simple torch as a light source. Typically then, it has a peak sensitivity wavelength (λ_p) of about 560nm to 600nm in the visible spectral range.

The Light Dependent Resistor Cell



The most commonly used photoresistive light sensor is the ORP12 Cadmium Sulphide photoconductive cell. This light dependent resistor has a spectral response of about 610nm in the yellow to orange region of light. The resistance of the cell when unilluminated (dark resistance) is very high at about $10\text{M}\Omega$'s which falls to about 100Ω 's when fully illuminated (lit resistance).

To increase the dark resistance and therefore reduce the dark current, the resistive path forms a zigzag pattern across the ceramic substrate. The CdS photocell is a very low cost device often used in auto dimming, darkness or twilight detection for turning the street lights "ON" and "OFF", and for photographic exposure meter type applications.



LDR Switch

Connecting a light dependant resistor in series with a standard resistor like this across a single DC supply voltage has one major advantage, a different voltage will appear at their junction for different levels of light.

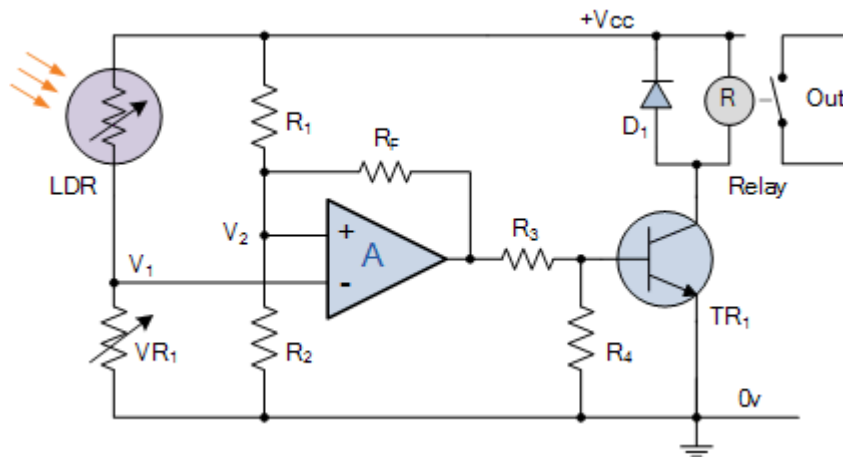
The amount of voltage drop across series resistor, R_2 is determined by the resistive value of the light dependant resistor, R_{LDR} . This ability to generate different voltages produces a very handy circuit called a "Potential Divider" or Voltage Divider Network.

As we know, the current through a series circuit is common and as the LDR changes its resistive value due to the light intensity, the voltage present at V_{OUT} will be determined by the voltage divider formula. An LDR's resistance, R_{LDR} can vary from about 100Ω in the sun light, to over $10M\Omega$ in absolute darkness with this variation of resistance being converted into a voltage variation at V_{OUT} as shown.

As the light level increases the resistance of the LDR starts to decrease causing the base bias voltage at V_1 to rise. At some point determined by the potential divider network formed with resistor R_1 , the base bias voltage is high enough to turn the transistor TR_1 "ON" and thus activate the relay which in turn is used to control some external circuitry. As the light level falls back to darkness again the resistance of the LDR increases causing the base voltage of the transistor to decrease, turning the transistor and relay "OFF" at a fixed light level determined again by the potential divider network.

By replacing the fixed resistor R_1 with a potentiometer VR_1 , the point at which the relay turns "ON" or "OFF" can be pre-set to a particular light level. This type of simple circuit shown above has a fairly low sensitivity and its switching point may not be consistent due to variations in either temperature or the supply voltage. A more sensitive precision light activated circuit can be easily made by incorporating the LDR into a "Wheatstone Bridge" arrangement and replacing the transistor with an Operational Amplifier as shown.

Light Level Sensing Circuit



In this basic dark sensing circuit, the light dependent resistor LDR1 and the potentiometer VR1 form one adjustable arm of a simple resistance bridge network, also known commonly as a Wheatstone bridge, while the two fixed resistors R1 and R2 form the other arm. Both sides of the bridge form potential divider networks across the supply voltage whose outputs V1 and V2 are connected to the non-inverting and inverting voltage inputs respectively of the operational amplifier.

The operational amplifier is configured as a Differential Amplifier also known as a voltage comparator with feedback whose output voltage condition is determined by the difference between the two input signals or voltages, V1 and V2. The resistor combination R1 and R2 form a fixed voltage reference at input V2, set by the ratio of the two resistors. The LDR – VR1 combination provides a variable voltage input V1 proportional to the light level being detected by the photoresistor.

As with the previous circuit the output from the operational amplifier is used to control a relay, which is protected by a free wheel diode, D1. When the light level sensed by the LDR and its output voltage falls below the reference voltage set at V2 the output from the op-amp changes state activating the relay and switching the connected load.

Likewise as the light level increases the output will switch back turning “OFF” the relay. The hysteresis of the two switching points is set by the feedback resistor Rf can be chosen to give any suitable voltage gain of the amplifier.

The operation of this type of light sensor circuit can also be reversed to switch the relay “ON” when the light level exceeds the reference voltage level and vice versa by reversing the positions of the light sensor LDR and the potentiometer VR1. The potentiometer can be used to “pre-set” the switching point of the differential amplifier to any particular light level making it ideal as a simple light sensor project circuit.

Photo junction Devices

Photo junction Devices are basically PN-Junction light sensors or detectors made from silicon semiconductor PN-junctions which are sensitive to light and which can detect both visible light and infra-red light levels. Photo-junction devices are specifically made for sensing light and this class of photoelectric light sensors include the Photodiode and the Phototransistor.

The Photodiode.

The construction of the Photodiode light sensor is similar to that of a conventional PN-junction diode except that the diodes outer casing is either transparent or has a clear lens to focus the light onto the PN junction for increased sensitivity. The junction will respond to light particularly longer wavelengths such as red and infra-red rather than visible light.



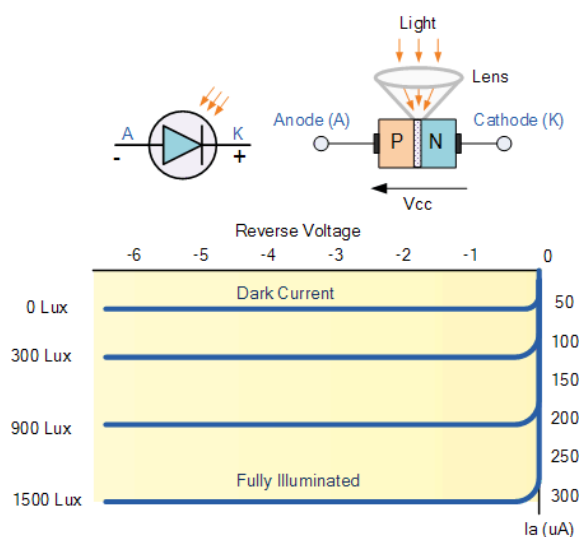
Photo-diode

This characteristic can be a problem for diodes with transparent or glass bead bodies such as the 1N4148 signal diode. LED's can also be used as photodiodes as they can both emit and detect light from their junction. All PN-junctions are light sensitive and can be used in a photo-conductive unbiased voltage mode with the PN-junction of the photodiode always “Reverse Biased” so that only the diodes leakage or dark current can flow.

The current-voltage characteristic (I/V Curves) of a photodiode with no light on its junction (dark mode) is very similar to a normal signal or rectifying diode.

When the photodiode is forward biased, there is an exponential increase in the current, the same as for a normal diode. When a reverse bias is applied, a small reverse saturation current appears which causes an increase of the depletion region, which is the sensitive part of the junction. Photodiodes can also be connected in a current mode using a fixed bias voltage across the junction. The current mode is very linear over a wide range.

Photo-diode Construction and Characteristics

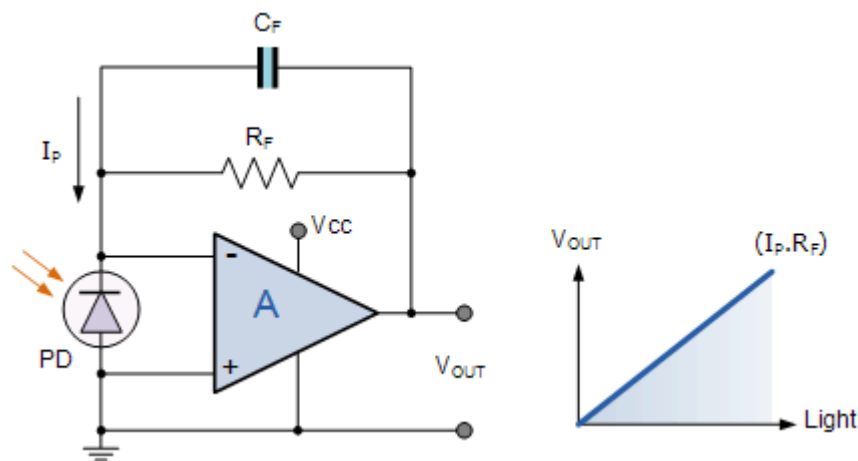


When used as a light sensor, a photodiodes dark current (0 lux) is about 10 μA for geranium and 1 μA for silicon type diodes. When light falls upon the junction more hole/electron pairs are formed and the leakage current increases. This leakage current increases as the illumination of the junction increases.

Thus, the photodiodes current is directly proportional to light intensity falling onto the PN-junction. One main advantage of photodiodes when used as light sensors is their fast response to changes in the light levels, but one disadvantage of this type of photo device is the relatively small current flow even when fully lit. The following circuit shows a photo-current-to-voltage converter circuit using an operational amplifier as the amplifying device. The output voltage (V_{out}) is given as $V_{out} = I_P \cdot R_f$ and which is proportional to the light intensity characteristics of the photodiode.

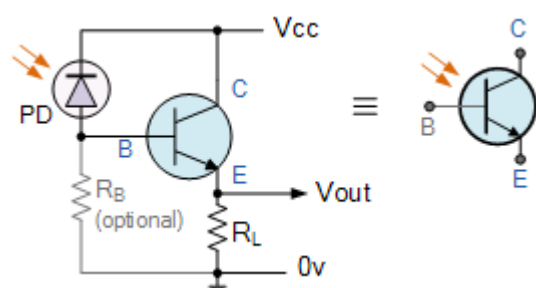
This type of circuit also utilizes the characteristics of an operational amplifier with two input terminals at about zero voltage to operate the photodiode without bias. This zero-bias op-amp configuration gives a high impedance loading to the photodiode resulting in less influence by dark current and a wider linear range of the photocurrent relative to the radiant light intensity. Capacitor C_f is used to prevent oscillation or gain peaking and to set the output bandwidth ($1/2\pi RC$).

Photo-diode Amplifier Circuit



Photodiodes are very versatile light sensors that can turn its current flow both "ON" and "OFF" in nanoseconds and are commonly used in cameras, light meters, CD and DVD-ROM drives, TV remote controls, scanners, fax machines and copiers etc., and when integrated into operational amplifier circuits as infrared spectrum detectors for fibre optic communications, burglar alarm motion detection circuits and numerous imaging, laser scanning and positioning systems etc.

The Phototransistor



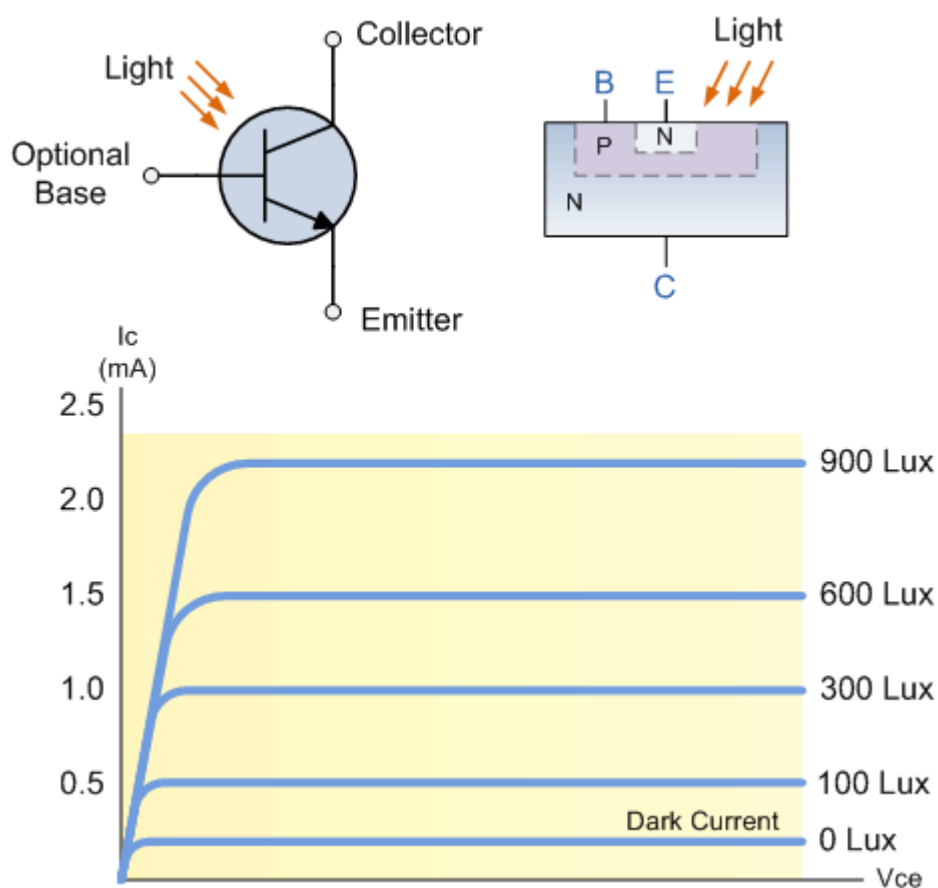
An alternative photo-junction device to the photodiode is the Phototransistor which is basically a photodiode with amplification. The Phototransistor light

sensor has its collector-base PN-junction reverse biased exposing it to the radiant light source.

Phototransistors operate the same as the photodiode except that they can provide current gain and are much more sensitive than the photodiode with currents are 50 to 100 times greater than that of the standard photodiode and any normal transistor can be easily converted into a phototransistor light sensor by connecting a photodiode between the collector and base.

Phototransistors consist mainly of a bipolar NPN Transistor with its large base region electrically unconnected, although some phototransistors allow a base connection to control the sensitivity, and which uses photons of light to generate a base current which in turn causes a collector to emitter current to flow. Most phototransistors are NPN types whose outer casing is either transparent or has a clear lens to focus the light onto the base junction for increased sensitivity.

Photo-transistor Construction and Characteristics



In the NPN transistor the collector is biased positively with respect to the emitter so that the base/collector junction is reverse biased. Therefore, with no light on the junction normal leakage or dark current flows which is very small. When light falls on the base more electron/hole pairs are formed in this region and the current produced by this action is amplified by the transistor.

Usually the sensitivity of a phototransistor is a function of the DC current gain of the transistor. Therefore, the overall sensitivity is a function of collector current and can be controlled by connecting a resistance between the base and the emitter but for very high sensitivity optocoupler type applications, Darlington phototransistors are generally used.

Photo Darlington transistors use a second bipolar NPN transistor to provide additional amplification or when higher sensitivity of a photodetector is required due to low light levels or selective sensitivity, but its response is slower than that of an ordinary NPN phototransistor.

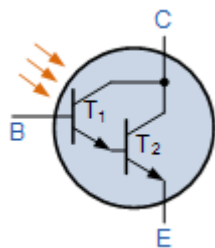


Photo-Darlington

Photo Darlington devices consist of a normal phototransistor whose emitter output is coupled to the base of a larger bipolar NPN transistor. Because a Darlington transistor configuration gives a current gain equal to a product of the current gains of two individual transistors, a photo Darlington device produces a very sensitive detector.

Typical applications of Phototransistors light sensors are in opto-isolators, slotted opto switches, light beam sensors, fibre optics and TV type remote controls, etc. Infrared filters are sometimes required when detecting visible light.

Another type of photo junction semiconductor light sensor worth a mention is the Photo-thyristor. This is a light activated thyristor or Silicon Controlled Rectifier, SCR that can be used as a light activated switch in AC applications. However

their sensitivity is usually very low compared to equivalent photodiodes or phototransistors.

To help increase their sensitivity to light, photo-thyristors are made thinner around the gate junction. The downside to this process is that it limits the amount of anode current that they can switch. Then for higher current AC applications they are used as pilot devices in opto-couplers to switch larger more conventional thyristors.

Photovoltaic Cells.

The most common type of photovoltaic light sensor is the Solar Cell. Solar cells convert light energy directly into DC electrical energy in the form of a voltage or current to a power a resistive load such as a light, battery or motor. Then photovoltaic cells are similar in many ways to a battery because they supply DC power.

However, unlike the other photo devices we have looked at above which use light intensity even from a torch to operate, photovoltaic solar cells work best using the sun's radiant energy.

Solar cells are used in many different types of applications to offer an alternative power source from conventional batteries, such as in calculators, satellites and now in homes offering a form of renewable power.

Photovoltaic cells are made from single crystal silicon PN junctions, the same as photodiodes with a very large light sensitive region but are used without the reverse bias. They have the same characteristics as a very large photodiode when in the dark.

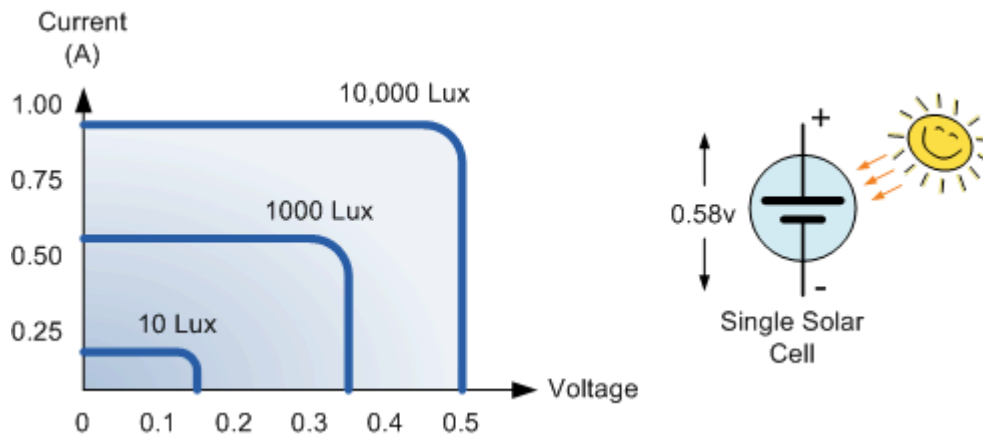


Photovoltaic Cell

When illuminated the light energy causes electrons to flow through the PN junction and an individual solar cell can generate an open circuit voltage of about 0.58v (580mV). Solar cells have a "Positive" and a "Negative" side just like a battery.

Individual solar cells can be connected together in series to form solar panels which increases the output voltage or connected together in parallel to increase the available current. Commercially available solar panels are rated in Watts, which is the product of the output voltage and current (Volts times Amps) when fully lit.

Characteristics of a typical Photovoltaic Solar Cell.



The amount of available current from a solar cell depends upon the light intensity, the size of the cell and its efficiency which is generally very low at around 15 to 20%. To increase the overall efficiency of the cell commercially available solar cells use polycrystalline silicon or amorphous silicon, which have no crystalline structure, and can generate currents of between 20 to 40mA per cm^2 .

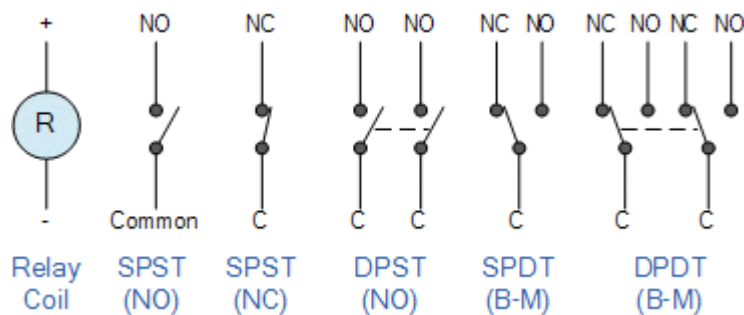
Other materials used in the construction of photovoltaic cells include Gallium Arsenide, Copper Indium Diselenide and Cadmium Telluride. These different materials each have a different spectrum band response, and so can be “tuned” to produce an output voltage at different wavelengths of light.

In this tutorial about Light Sensors, we have looked at several examples of devices that are classed as Light Sensors. This includes those with and those without PN-junctions that can be used to measure the intensity of light.

In the next tutorial we will look at output devices called Actuators. Actuators convert an electrical signal into a corresponding physical quantity such as movement, force, or sound. One such commonly used output device is the Electromagnetic Relay.

Electrical Relay

Electrical relays and contactors use a low level control signal to switch a much higher voltage or current supply using a number of different contact arrangements.



Thus far we have seen a selection of Input devices that can be used to detect or “sense” a variety of physical variables and signals and are therefore called Sensors. But there are also a variety of electrical and electronic devices which are classed as Output devices used to control or operate some external physical process. These output devices are commonly called Actuators.

Actuators convert an electrical signal into a corresponding physical quantity such as movement, force, sound etc. An actuator is also classed as a transducer because it changes one type of physical quantity into another and is usually activated or operated by a low voltage command signal. Actuators can be classed as either binary or continuous devices based upon the number of stable states their output has.

For example, a relay is a binary actuator as it has two stable states, either energised or latched or de-energised and unlatched, while a motor is a continuous actuator because it can rotate through a full 360o motion. The most common types of actuators or output devices are Electrical Relays, Lights, Motors and Loudspeakers.

We saw previously that solenoids can be used to electrically open latches, doors, open or close valves, and in a variety of robotic and mechatronic applications, etc. However, if the solenoid plunger is used to operate one or more sets of electrical contacts, we have a device called a relay that is so useful it can be used in an infinite number of different ways and in this tutorial we will look at Electrical Relays.

Electrical Relays can also be divided into mechanical action relays called “Electromechanical Relays” and those which use semiconductor transistors, thyristors, triacs, etc., as their switching device called “Solid State Relays” or SSR’s.

5. The Electromechanical Relay

The term Relay generally refers to a device that provides an electrical connection between two or more points in response to the application of a control signal. The most common and widely used type of electrical relay is the electromechanical relay or EMR.

The most fundamental control of any equipment is the ability to turn it “ON” and “OFF”. The easiest way to do this is using switches to interrupt the electrical supply. Although switches can be used to control something, they have their disadvantages. The biggest one is that they have to be manually (physically) turned “ON” or “OFF”. Also, they are relatively large, slow and only switch small electrical currents.



An Electrical Relay

Electrical Relays however, are basically electrically operated switches that come in many shapes, sizes and power ratings suitable for all types of applications. Relays can also have single or multiple contacts within a single package with the larger power relays used for mains voltage or high current switching applications being called “Contactors”.

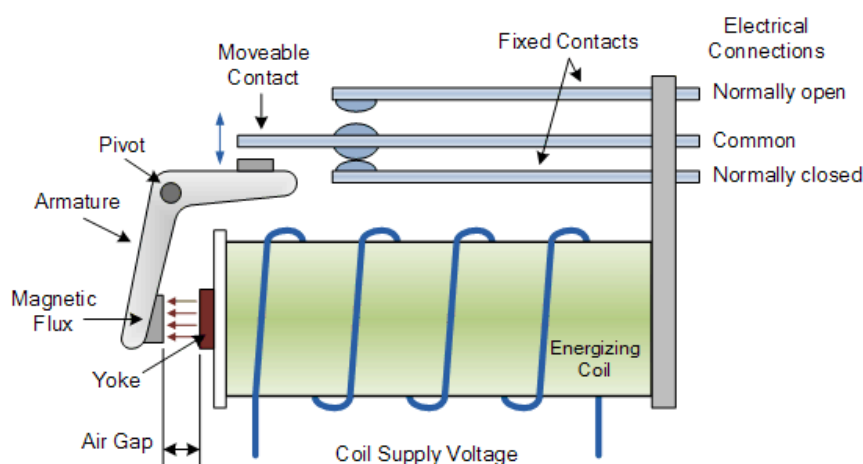
In this tutorial about electrical relays we are just concerned with the fundamental operating principles of “light duty” electromechanical relays we can use in motor control or robotic circuits. Such relays are used in general electrical and electronic control or switching circuits either mounted directly onto PCB boards or connected free standing and in which the load currents are

normally fractions of an ampere up to 20+ amperes. The relay circuit are common in Electronics applications.

As their name implies, electromechanical relays are electro-magnetic devices that convert a magnetic flux generated by the application of a low voltage electrical control signal either AC or DC across the relay terminals, into a pulling mechanical force which operates the electrical contacts within the relay. The most common form of electromechanical relay consist of an energizing coil called the “primary circuit” wound around a permeable iron core.

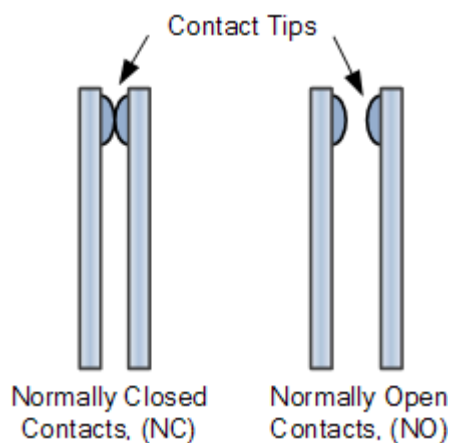
This iron core has both a fixed portion called the yoke, and a moveable spring loaded part called the armature, that completes the magnetic field circuit by closing the air gap between the fixed electrical coil and the moveable armature. The armature is hinged or pivoted allowing it to freely move within the generated magnetic field closing the electrical contacts that are attached to it. Connected between the yoke and armature is normally a spring (or springs) for the return stroke to “reset” the contacts back to their initial rest position when the relay coil is in the “de-energized” condition, i.e. turned “OFF”.

Electromechanical Relay Construction



In our simple relay above, we have two sets of electrically conductive contacts. Relays may be “Normally Open”, or “Normally Closed”. One pair of contacts are classed as Normally Open, (NO) or make contacts and another set which are classed as normally Closed, (NC) or break contacts. In the normally open position, the contacts are closed only when the field current is “ON” and the switch contacts are pulled towards the inductive coil.

In the normally closed position, the contacts are permanently closed when the field current is “OFF” as the switch contacts return to their normal position. These terms Normally Open, Normally Closed or Make and Break Contacts refer to the state of the electrical contacts when the relay coil is “de-energized”, i.e., no supply voltage connected to the relay coil. Contact elements may be of single or double make or break designs. An example of this arrangement is given below.



The relays contacts are electrically conductive pieces of metal which touch together completing a circuit and allow the circuit current to flow, just like a switch. When the contacts are open the resistance between the contacts is very high in the Mega-Ohms, producing an open circuit condition and no circuit current flows.

When the contacts are closed the contact resistance should be zero, a short circuit, but this is not always the case. All relay contacts have a certain amount of “contact resistance” when they are closed and this is called the “On-Resistance”, similar to FET's.

With a new relay and contacts this ON-resistance will be very small, generally less than 0.2Ω because the tips are new and clean, but over time the tip resistance will increase.

For example. If the contacts are passing a load current of say 10A, then the voltage drop across the contacts using Ohms Law is $0.2 \times 10 = 2$ volts, which if the supply voltage is say 12 volts then the load voltage will be only 10 volts ($12 - 2$). As the contact tips begin to wear, and if they are not properly protected

from high inductive or capacitive loads, they will start to show signs of arcing damage as the circuit current still wants to flow as the contacts begin to open when the relay coil is de-energized.

This arcing or sparking across the contacts will cause the contact resistance of the tips to increase further as the contact tips become damaged. If allowed to continue the contact tips may become so burnt and damaged to the point where they are physically closed but do not pass any or very little current.

If this arcing damage becomes too severe the contacts will eventually “weld” together producing a short circuit condition and possible damage to the circuit they are controlling. If now the contact resistance has increased due to arcing to say 1Ω the volt drop across the contacts for the same load current increases to $1 \times 10 = 10$ volts dc. This high voltage drop across the contacts may be unacceptable for the load circuit especially if operating at 12 or even 24 volts, then the faulty relay will have to be replaced.

To reduce the effects of contact arcing and high “On-resistances”, modern contact tips are made of, or coated with, a variety of silver based alloys to extend their life span as given in the following table.

Electrical Relay Contact Tip Materials

Ag (fine silver)

1. Electrical and thermal conductivity are the highest of all the metals.
2. Exhibits low contact resistance, is inexpensive and widely used.
3. Contacts tarnish easily through sulphurisation influence.

AgCu (silver copper)

1. Known as “Hard silver” contacts and have better wear resistance and less tendency to arc and weld, but slightly higher contact resistance.

AgCdO (silver cadmium oxide)

1. Very little tendency to arc and weld, good wear resistance and arc extinguishing properties.

AgW (silver tungsten)

1. Hardness and melting point are high, arc resistance is excellent.
2. Not a precious metal.

3. High contact pressure is required to reduce resistance.
4. Contact resistance is relatively high, and resistance to corrosion is poor.

AgNi (silver nickel)

1. Equals the electrical conductivity of silver, excellent arc resistance.

AgPd (silver palladium)

1. Low contact wear, greater hardness.
2. Expensive.

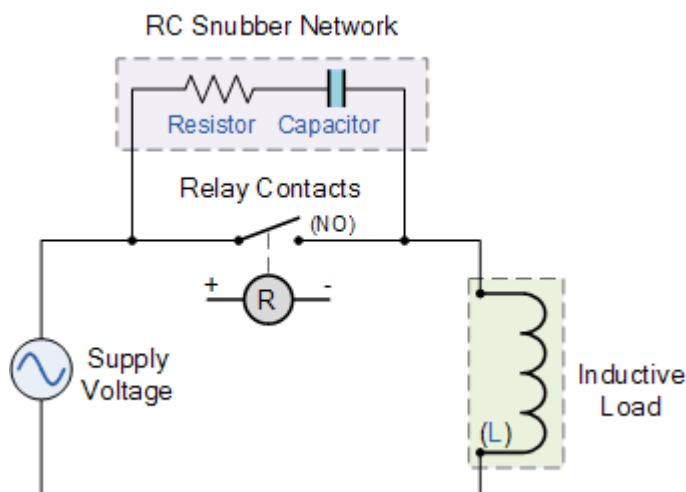
Platinum, Gold and Silver Alloys

1. Excellent corrosion resistance, used mainly for low-current circuits.

Relay manufacturers data sheets give maximum contact ratings for resistive DC loads only and this rating is greatly reduced for either AC loads or highly inductive or capacitive loads. In order to achieve long life and high reliability when switching alternating currents with inductive or capacitive loads some form of arc suppression or filtering is required across the relay contacts.

Extending the life of relay tips by reducing the amount of arcing generated as they open is achieved by connecting a Resistor-Capacitor network called an RC Snubber Network electrically in parallel with an electrical relay contact tips. The voltage peak, which occurs at the instant the contacts open, will be safely short circuited by the RC network, thus suppressing any arc generated at the contact tips. For example.

Electrical Relay Snubber Circuit



Electrical Relay Contact Types.

As well as the standard descriptions of Normally Open, (NO) and Normally Closed, (NC) used to describe how the relays contacts are connected, relay contact arrangements can also be classed by their actions. Electrical relays can be made up of one or more individual switch contacts with each “contact” being referred to as a “pole”. Each one of these contacts or poles can be connected or “thrown” together by energizing the relays coil and this gives rise to the description of the contact types as being:

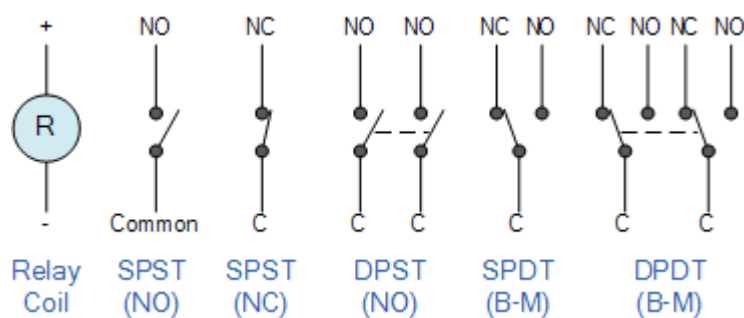
- ✚ SPST – Single Pole Single Throw
- ✚ SPDT – Single Pole Double Throw
- ✚ DPST – Double Pole Single Throw
- ✚ DPDT – Double Pole Double Throw

With the action of the contacts being described as “Make” (M) or “Break” (B). Then a simple relay with one set of contacts as shown above can have a contact description of:

“Single Pole Double Throw – (Break before Make)”, or SPDT – (B-M)

Examples of just some of the more common diagrams used for electrical relay contact types to identify relays in circuit or schematic diagrams is given below but there are many more possible configurations.

Electrical Relay Contact Configurations



Where:

C is the Common terminal

NO is the Normally Open contact

NC is the Normally Closed contact

Electromechanical relays are also denoted by the combinations of their contacts or switching elements and the number of contacts combined within a single relay. For example, a contact which is normally open in the de-energised

position of the relay is called a “Form A contact” or make contact. Whereas a contact which is normally closed in the de-energised position of the relay is called a “Form B contact” or break contact.

When both a make and a break set of contact elements are present at the same time so that the two contacts are electrically connected to produce a common point (identified by three connections), the set of contacts are referred to as “Form C contacts” or change-over contacts. If no electrical connection exists between the make and break contacts it is referred to as a double change-over contact.

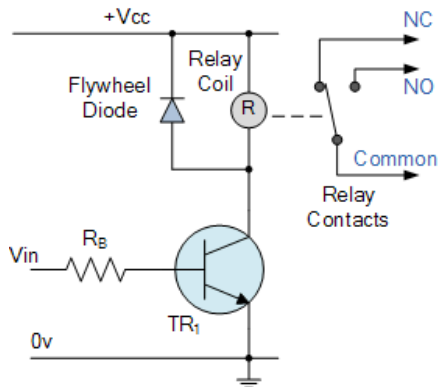
One final point to remember about using electrical relays. It is not advisable at all to connect relay contacts in parallel to handle higher load currents. For example, never attempt to supply a 10A load with two relay contacts in parallel that have 5A contact ratings each, as the mechanically operated relay contacts never close or open at exactly the same instant of time. The result is that one of the contacts will always be overloaded even for a brief instant resulting in premature failure of the relay over time.

Also, while electrical relays can be used to allow low power electronic or computer type circuits to switch relatively high currents or voltages both “ON” or “OFF”. Never mix different load voltages through adjacent contacts within the same relay such as for example, high voltage AC (240v) and low voltage DC (12v), always use separate relays for safety.

One of the more important parts of any electrical relay is its coil. This converts electrical current into an electromagnetic flux which is used to mechanically operate the relays contacts. The main problem with relay coils is that they are “highly inductive loads” as they are made from coils of wire. Any coil of wire has an impedance value made up of resistance (R) and inductance (L) in series (LR Series Circuit).

As the current flows through the coil a self-induced magnetic field is generated around it. When the current in the coil is turned “OFF”, a large back emf (electromotive force) voltage is produced as the magnetic flux collapses within the coil (transformer theory). This induced reverse voltage value may be very

high in comparison to the switching voltage, and may damage any semiconductor device such as a transistor, FET or micro-controller used to operate the relay coil.



One way of preventing damage to the transistor or any switching semiconductor device, is to connect a reverse biased diode across the relay coil.

When the current flowing through the coil is switched "OFF", an induced back emf is generated as the magnetic flux collapses in the coil.

This reverse voltage forward biases the diode which conducts and dissipates the stored energy preventing any damage to the semiconductor transistor.

When used in this type of application the diode is generally known as a Flywheel Diode, Free-wheeling Diode and even Fly-back Diode, but they all mean the same thing. Other types of inductive loads which require a flywheel diode for protection are solenoids, motors and inductive coils.

As well as using flywheel Diodes for protection of semiconductor components, other devices used for protection include RC Snubber Networks, Metal Oxide Varistors or MOV and Zener Diodes.

The Solid State Relay.

While the electromechanical relay (EMR) are inexpensive, easy to use and allow the switching of a load circuit controlled by a low power, electrically isolated input signal, one of the main disadvantages of an electromechanical relay is that it is a "mechanical device", that is it has moving parts so their switching speed (response time) due to physically movement of the metal contacts using a magnetic field is slow.

Over a period of time these moving parts will wear out and fail, or that the contact resistance through the constant arcing and erosion may make the relay unusable and shortens its life. Also, they are electrically noisy with the contacts suffering from contact bounce which may affect any electronic circuits to which they are connected.

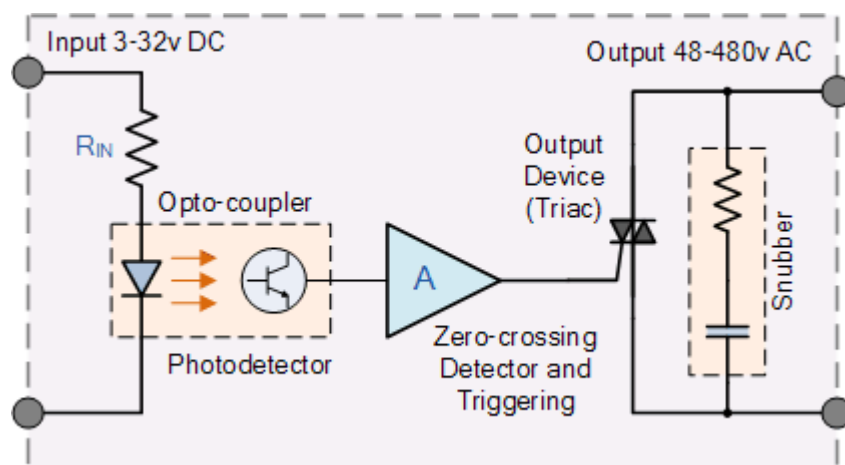
To overcome these disadvantages of the electrical relay, another type of relay called a Solid State Relay or (SSR) for short was developed which is a solid state contactless, pure electronic relay.

The solid state relay being a purely electronic device has no moving parts within its design as the mechanical contacts have been replaced by power transistors, thyristors or triac's. The electrical separation between the input control signal and the output load voltage is accomplished with the aid of an opto-coupler type Light Sensor.

The Solid State Relay provides a high degree of reliability, long life and reduced electromagnetic interference (EMI), (no arcing contacts or magnetic fields), together with a much faster almost instant response time, as compared to the conventional electromechanical relay.

Also the input control power requirements of the solid state relay are generally low enough to make them compatible with most IC logic families without the need for additional buffers, drivers or amplifiers. However, being a semiconductor device they must be mounted onto suitable heatsinks to prevent the output switching semiconductor device from overheating.

Solid State Relay



The AC type Solid State Relay turns “ON” at the zero crossing point of the AC sinusoidal waveform, prevents high inrush currents when switching inductive or capacitive loads while the inherent turn “OFF” feature of Thyristors and Triacs provides an improvement over the arcing contacts of the electromechanical relays.

Like the electromechanical relays, a Resistor-Capacitor (RC) snubber network is generally required across the output terminals of the SSR to protect the semiconductor output switching device from noise and voltage transient spikes when used to switch highly inductive or capacitive loads. In most modern SSR's this RC snubber network is built as standard into the relay itself reducing the need for additional external components.

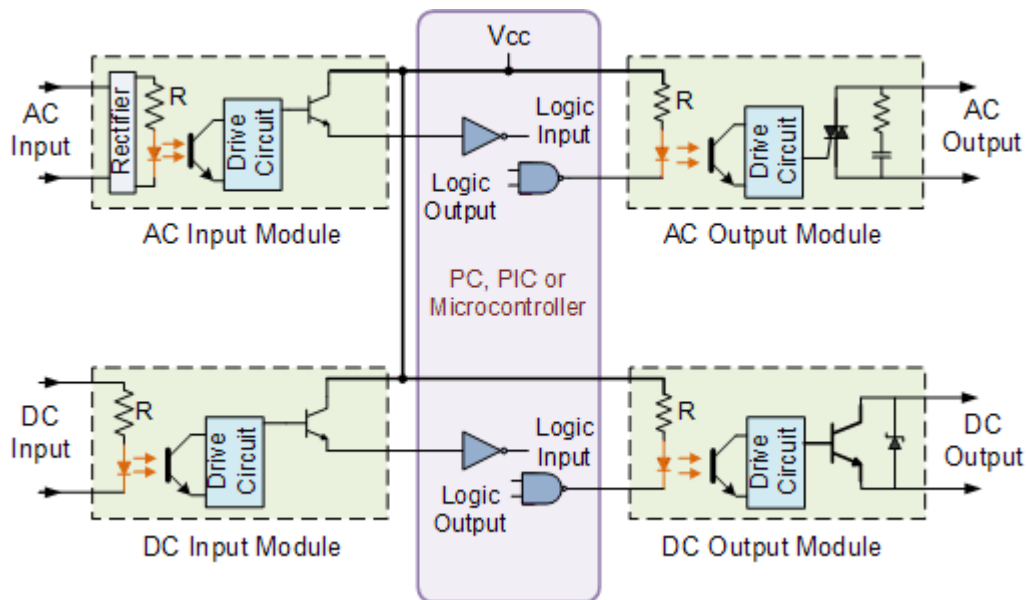
Non-zero crossing detection switching (instant “ON”) type SSR's are also available for phase controlled applications such as the dimming or fading of lights at concerts, shows, disco lighting etc., or for motor speed control type applications.

As the output switching device of a solid state relay is a semiconductor device (Transistor for DC switching applications, or a Triac/Thyristor combination for AC switching), the voltage drop across the output terminals of an SSR when “ON” is much higher than that of the electromechanical relay, typically 1.5 – 2.0 volts. If switching large currents for long periods of time an additional heat sink will be required.

Input/output Interface Modules.

Input/output Interface Modules, (I/O Modules) are another type of solid state relay designed specifically to interface computers, micro-controller or PIC's to “real world” loads and switches. There are four basic types of I/O modules available, AC or DC Input voltage to TTL or CMOS logic level output, and TTL or CMOS logic input to an AC or DC Output voltage with each module containing all the necessary circuitry to provide a complete interface and isolation within one small device. They are available as individual solid state modules or integrated into 4, 8 or 16 channel devices.

Modular Input/output Interface System.

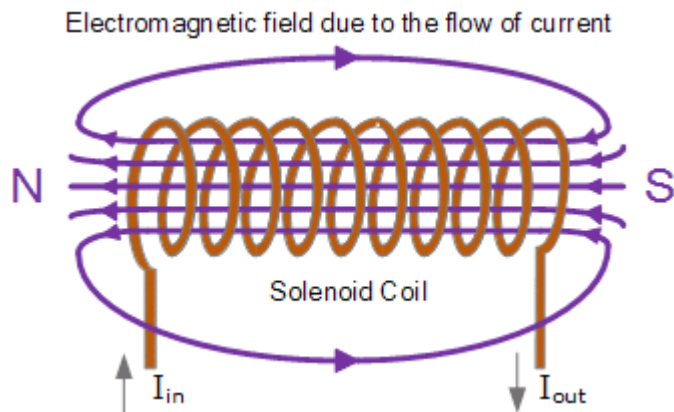


The main disadvantages of solid state relays (SSR's) compared to that of an equivalent wattage electromechanical relay is their higher costs, the fact that only single pole single throw (SPST) types are available, "OFF"-state leakage currents flow through the switching device, and a high "ON"-state voltage drop and power dissipation resulting in additional heat sinking requirements. Also they cannot switch very small load currents or high frequency signals such as audio or video signals although special Solid State Switches are available for this type of application.

In this tutorial about Electrical Relays, we have looked at both the electromechanical relay and the solid state relay which can be used as an output device (actuator) to control a physical process. In the next tutorial we will continue our look at output devices called Actuators and especially one that converts a small electrical signal into a corresponding physical movement using electromagnetism. The output device is called a Solenoid.

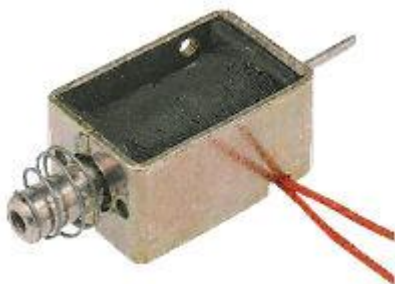
Linear Solenoid Actuator

Another type of electromagnetic actuator that converts an electrical signal into a magnetic field producing a linear motion is called the Linear Solenoid.



The linear solenoid works on the same basic principal as the electromechanical relay seen in the previous tutorial and just like relays, they can also be switched and controlled using bipolar transistors or MOSFET's. A "Linear Solenoid" is an electromagnetic device that converts electrical energy into a mechanical pushing or pulling force or motion.

Linear solenoid's basically consist of an electrical coil wound around a cylindrical tube with a Ferro-magnetic actuator or "plunger" that is free to move or slide "IN" and "OUT" of the coils body. Solenoids can be used to electrically open doors and latches, open or close valves, move and operate robotic limbs and mechanisms, and even actuate electrical switches just by energising its coil.



Linear Solenoid

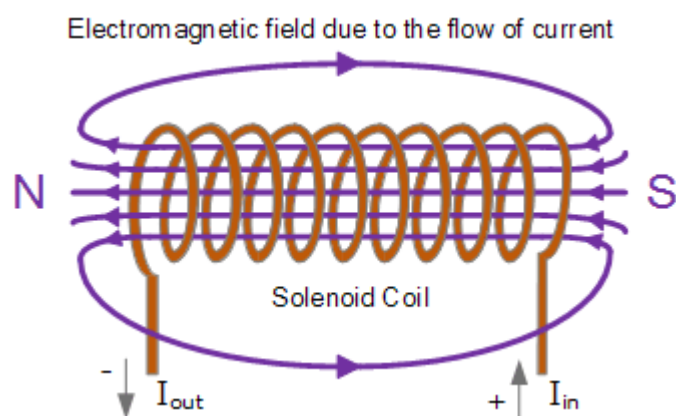
Solenoids are available in a variety of configurations and formats with the more common types being the Linear Solenoid, also known as the linear electromechanical actuator (LEMA), which as its name suggests produces a straight-line linear movement, and the Rotary Solenoid which produces a rotational movement over some fixed angle.

Both types of solenoid, linear and rotational are available as either a holding (continuously energised) or as a latching type (ON-OFF pulse) with the latching types being used in either energised or power-off applications. Linear solenoids can also be designed for proportional motion control where the plunger position is proportional to the power input.

When electrical current flows through a conductor it generates a magnetic field around itself. The direction of this magnetic field with regards to its north and south poles is determined by the direction of the current flow within the wire. Then with electric current flowing through the coil of wire it becomes an “Electromagnet” creating its own north and south poles exactly the same as that for a permanent type magnet.

The strength of this magnetic field can be increased or decreased by either controlling the amount of current flowing through the coil or by changing the number of turns or loops that the coil has. An example of an “Electromagnet” is given below.

Magnetic Field Produced by a Solenoid Coil



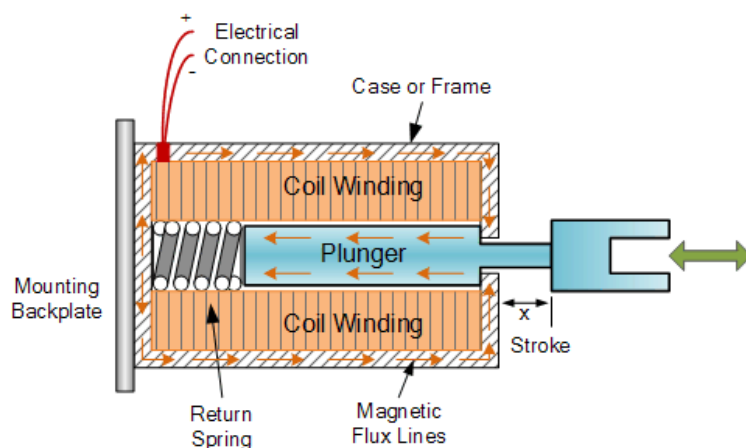
When an electrical current is passed through the coils windings, it behaves like an electromagnet and the plunger, which is located inside the coil, is attracted towards the centre of the coil by the magnetic flux setup within the coils body, which in turn compresses a small spring attached to one end of the plunger. The force and speed of the plungers movement is determined by the strength of the magnetic flux generated within the coil.

When the supply current is turned “OFF” (de-energised) the electromagnetic field generated previously by the coil collapses and the energy stored in the compressed spring forces the plunger back out to its original rest position. This back and forth movement of the plunger is known as the solenoids “Stroke”, in other words the maximum distance the plunger can travel in either an “IN” or an “OUT” direction, for example, 0 – 30mm.

Linear Solenoid Construction

This type of solenoid is commonly called a Linear Solenoid due to the linear directional movement and action of the plunger. Linear solenoids are available in two basic configurations called a “Pull-type” as it pulls the connected load towards itself when energised, and the “Push-type” that act in the opposite direction pushing it away from itself when energised. Both push and pull types are generally constructed the same with the difference being in the location of the return spring and design of the plunger.

Pull-type Linear Solenoid Construction



Linear solenoids are useful in many applications that require an open or closed (in or out) type motion such as electronically activated door locks, pneumatic or hydraulic control valves, robotics, automotive engine management, irrigation valves to water the garden and even the “Ding-Dong” door bell has one. They are available as open frame, closed frame or sealed tubular types.

Rotary Solenoid

Most electromagnetic solenoids are linear devices producing a linear back and forth force or motion. However, rotational solenoids are also available which

produce an angular or rotary motion from a neutral position in either clockwise, anti-clockwise or in both directions (bi-directional).

Rotary solenoids can be used to replace small DC motors or stepper motors where the angular movement is very small with the angle of rotation being the angle moved from the start to the end position.

Commonly available rotary solenoids have movements of 25, 35, 45, 60 and 90° as well as multiple movements to and from a certain angle such as a 2-position self-restoring or return to zero rotation, for example 0-to-90-to-0°, 3-position self-restoring, for example 0° to +45° or 0° to -45° as well as 2-position latching.



Rotary Solenoid

Rotary solenoids produce a rotational movement when either energised, de-energised, or a change in the polarity of an electromagnetic field alters the position of a permanent magnet rotor. Their construction consists of an electrical coil wound around a steel frame with a magnetic disk connected to an output shaft positioned above the coil.

When the coil is energised the electromagnetic field generates multiple north and south poles which repel the adjacent permanent magnetic poles of the disk causing it to rotate at an angle determined by the mechanical construction of the rotary solenoid.

Rotary solenoids are used in vending or gaming machines, valve control, camera shutter with special high speed, low power or variable positioning solenoids with high force or torque are available such as those used in dot matrix printers, typewriters, automatic machines or automotive applications etc.

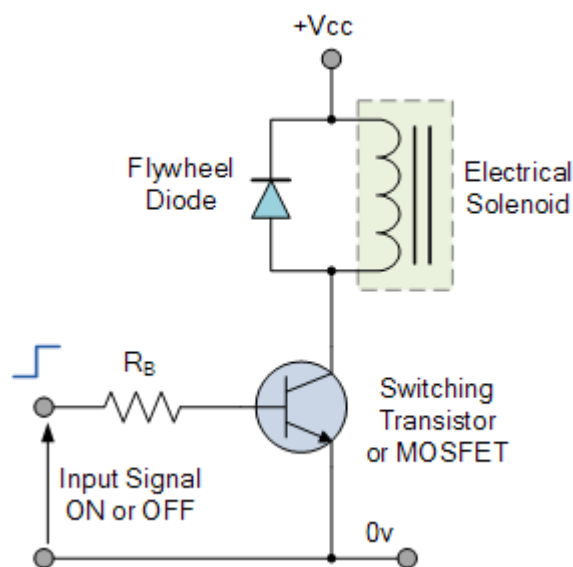
Solenoid Switching

Generally solenoids either linear or rotary operate with the application of a DC voltage, but they can also be used with AC sinusoidal voltages by using full

wave bridge rectifiers to rectify the supply which then can be used to switch the DC solenoid. Small DC type solenoids can be easily controlled using Transistor or MOSFET switches and are ideal for use in robotic applications.

However, as we saw previously with electromechanical relays, linear solenoids are “inductive” devices so some form of electrical protection is required across the solenoid coil to prevent high back emf voltages from damaging the semiconductor switching device. In this case the standard “Flywheel Diode” is used, but you could equally use a Zener diode or small value varistor.

Switching Solenoids using a Transistor



Reducing Energy Consumption

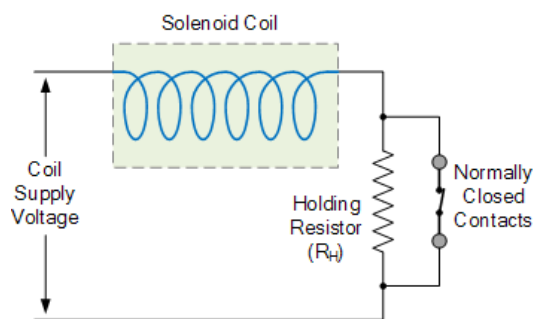
One of the main disadvantages of solenoids and especially the linear solenoid is that they are “inductive devices” made from coils of wire. This means that the solenoid coils have resistance and convert some of the electrical energy used to operate them into “HEAT” due to the I^2R heating effect of the wire.

In other words when connected for long periods of time to an electrical supply wound coils can get hot! and the longer the time that the power is applied to a solenoid coil, the hotter the coil may become. Also as the coil heats up, its electrical resistance also changes reducing both current flowing through the coil and its magnetic field strength as this depends directly upon the ampere-turns.

With a continuous voltage input applied to the coil, the solenoid coil does not have the opportunity to cool down because the input power is always on. In order to reduce this self-induced heating effect it is necessary to reduce either the amount of time the coil is energised or reduce the amount of current flowing through it.

One method of consuming less current is to apply a suitable high enough voltage to the solenoid coil so as to provide the necessary electromagnetic field to operate and seat the plunger but then once activated to reduce the coil's supply voltage to a level sufficient to maintain the plunger in its seated or latched position. One way of achieving this is to connect a suitable "holding" resistor in series with the solenoid coil, for example:

Reducing Solenoid Energy Consumption



Here, the switch contacts are closed shorting out the resistance and passing the full supply current directly to the solenoid coils windings. Once energised the contacts which can be mechanically connected to the solenoid's plunger action open connecting the holding resistor, R_H in series with the solenoid coil. This effectively connects the resistor in series with the coil.

By using this method, the solenoid can be connected to its voltage supply indefinitely (continuous duty cycle) as the power consumed by the coil and the heat generated is greatly reduced, which can be up to 85 to 90% using a suitable power resistor. However, the power consumed by the resistor will also generate a certain amount of heat, I^2R (Ohm's Law) and this also needs to be taken into account.

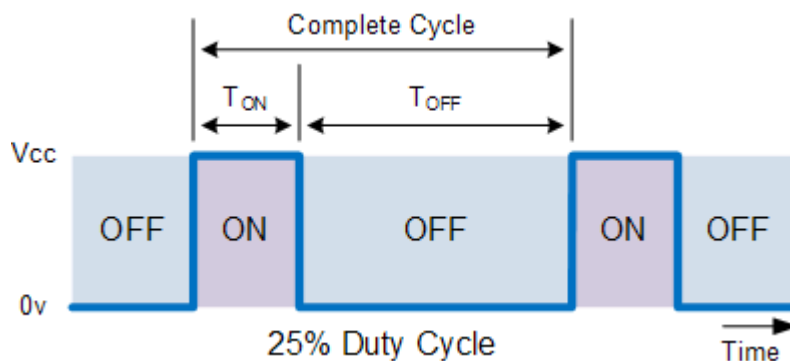
Solenoid Duty Cycle

Another more practical way of reducing the heat generated by the solenoid coil is to use an "intermittent duty cycle". An intermittent duty cycle means that

the coil is repeatedly switched “ON” and “OFF” at a suitable frequency so as to activate the plunger mechanism but not allow it to de-energise during the OFF period of the waveform. Intermittent duty cycle switching is a very effective way to reduce the total power consumed by the coil.

The Duty Cycle (%ED) of a solenoid is the portion of the “ON” time that a solenoid is energised and is the ratio of the “ON” time to the total “ON” and “OFF” time for one complete cycle of operation. In other words, the cycle time equals the switched-ON time plus the switched-OFF time. Duty cycle is expressed as a percentage, for example:

$$\text{Duty Cycle} = \frac{\text{"ON" time}}{\text{"ON" time} + \text{"OFF" time}} \times 100\%$$



Then if a solenoid is switched “ON” or energised for 30 seconds and then switched “OFF” for 90 seconds before being re-energised again, one complete cycle, the total “ON/OFF” cycle time would be 120 seconds, (30+90) so the solenoids duty cycle would be calculated as 30/120 secs or 25%. This means that you can determine the solenoids maximum switch-ON time if you know the values of duty cycle and switch-OFF time.

For example, the switch-OFF time equals 15 secs, duty cycle equals 40%, and therefore switch-ON time equals 10 secs. A solenoid with a rated Duty Cycle of 100% means that it has a continuous voltage rating and can therefore be left “ON” or continuously energised without overheating or damage.

In this tutorial about solenoids, we have looked at both the Linear Solenoid and the Rotary Solenoid as an electromechanical actuator that can be used as an output device to control a physical process. In the next tutorial we will continue our look at output devices called Actuators, and one that converts an electrical

signal into a corresponding rotational movement again using electromagnetism. The type of output device we will look at in the next tutorial is the DC Motor.

SESSION 3.

Design a specific project using applied electronics.

Learning Outcomes

- A schematic drawing is read and interpreted in order to design the given project.
- Visualisation technique is used to analyse the design process for a given project.
- Electronics components are identified and explained in terms of their functions for a given project.
- Electronics components are assembled using the required tools and technology.
- Product is tested using testing instruments to meet requirements specifications.

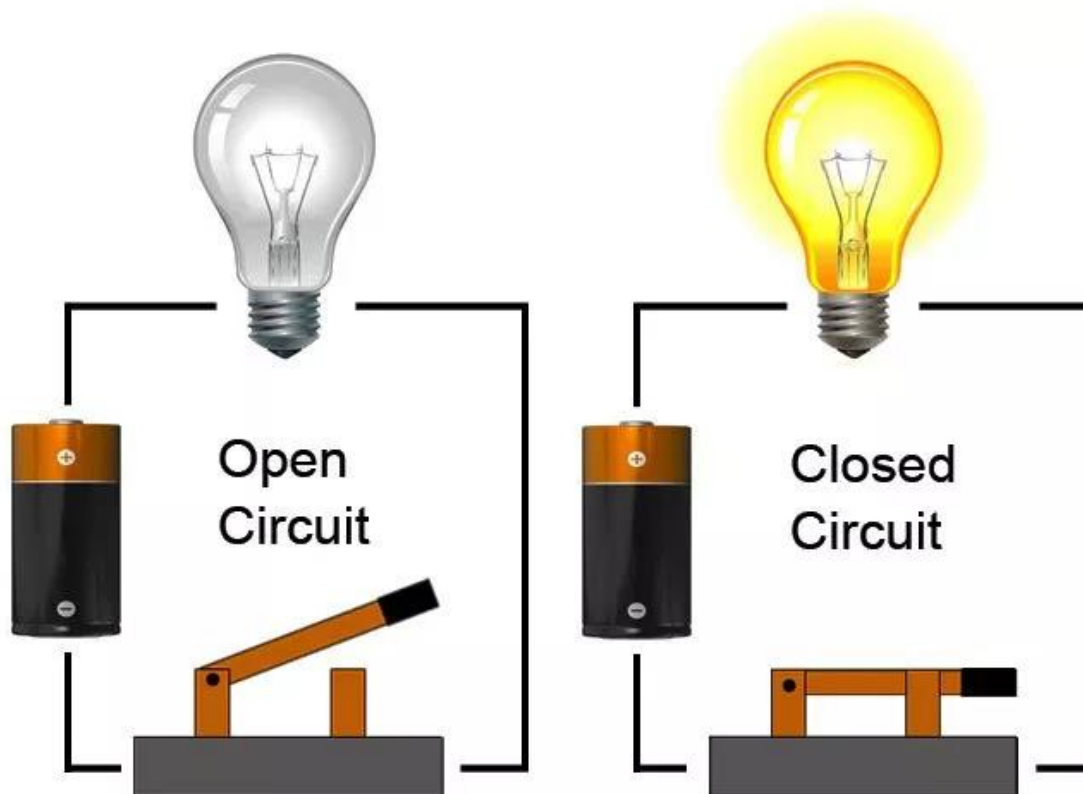
Design a specific project using applied electronics.

What Is A Circuit?

Before you design an electronic project, you need to know what a circuit is and how to create one properly.

An electronic circuit is a circular path of conductors by which electric current can flow. A closed circuit is like a circle because it starts and ends at the same point forming a complete loop. Furthermore, a closed circuit allows electricity to flow from the (+) power to the (-) ground uninterrupted.

In contrast, if there is any break in the flow of electricity, this is known as an open circuit. As shown below, a switch in a circuit can cause it to be open or closed depending on its position.



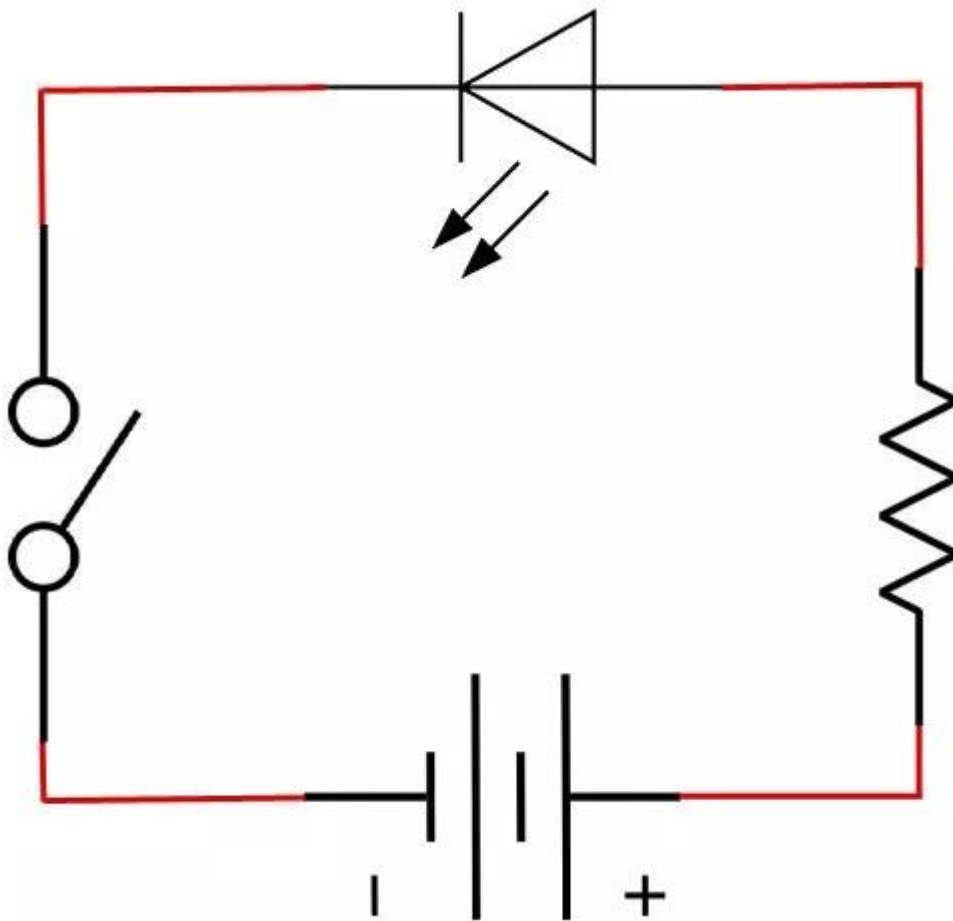
All circuits need to have three basic elements. These elements are a voltage source, conductive path and a load.

The voltage source, such as a battery, is needed in order to cause the current to flow through the circuit. In addition, there needs to be a conductive path that provides a route for the electricity to flow. Finally, a proper circuit needs a load that consumes the power. The load in the above circuit is the light bulb.

Schematic Diagram

When working with circuits, you will often find something called a schematic diagram. These diagrams use symbols to illustrate what electronic components are used and where they're placed in the circuit. These symbols are graphic representations of the actual electronic components.

Below is an example of a schematic that depicts an LED circuit that is controlled by a switch. It contains symbols for an LED, resistor, battery and a switch. By following a schematic diagram, you are able to know which components to use and where to put them. These schematics are extremely helpful for beginners when first learning circuits.



Schematic Diagram For LED Circuit

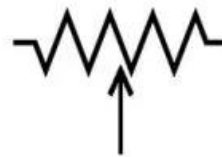
There are many types of electronic symbols and they vary slightly between countries. Below are a few of the most commonly used electronic symbols in the US.



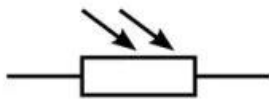
Resistor



Variable Resistor



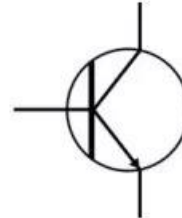
Potentiometer



Light-Dependent Resistor



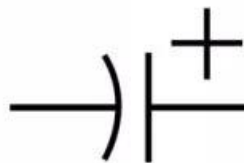
Switch



Transistor



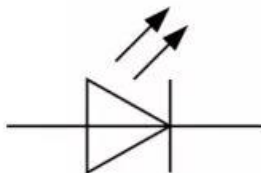
Relay



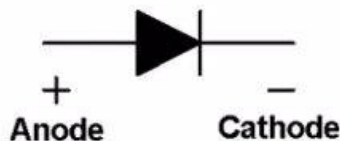
Polarized Capacitor



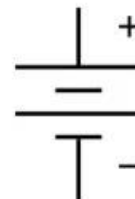
Non-Polarized Capacitor



Light-Emitting Diode



Diode



Battery

How To Determine A Resistor Size

Resistors are commonly used in electronics projects and it's important to know which size to use. To find the resistor value, you need to know the voltage and the amps for your LED and battery.

A standard LED generally needs a voltage of around 2V and a current of 20mA or .02A to operate correctly. Next, you need to find out what voltage your battery is. In this example, we will be using a 9V battery. In order to determine the resistor size, we need to use a formula known as Ohm's law as shown below.

Ohm's Law – Resistance (R) = Voltage (V) / Current (I)

- Resistance is measured in Ohms (Ω)
- Voltage is measured in volts (V)
- Current is measured in amps (A)

$$R = \frac{V_{\text{Bat}} - V_{\text{LED}}}{I_{\text{LED}}}$$

$$350 = \frac{9\text{V} - 2\text{V}}{.02\text{A}}$$

Using Ohm's law, you need to subtract the LED voltage from the battery voltage. This will give you a voltage of 7 which needs to be divided by .02 amps from the LED. This formula shows that you will need a 350 Ω resistor.

As a note, standard resistors don't come in 350 Ω but are available in 330 Ω which will work fine.

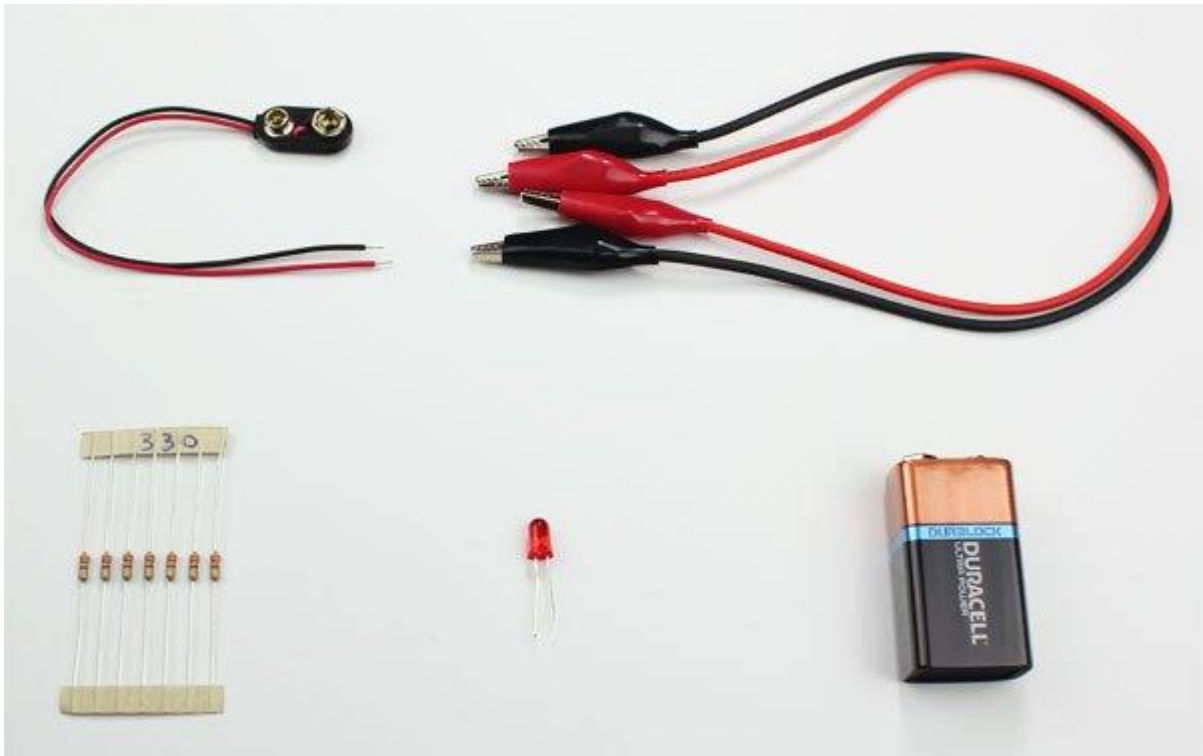
Electronics Project #1

Now it's time to combine everything you've learned and create a basic circuit. This project is a great starter project for beginners. We will be using test leads to create a temporary circuit without having to solder it together.

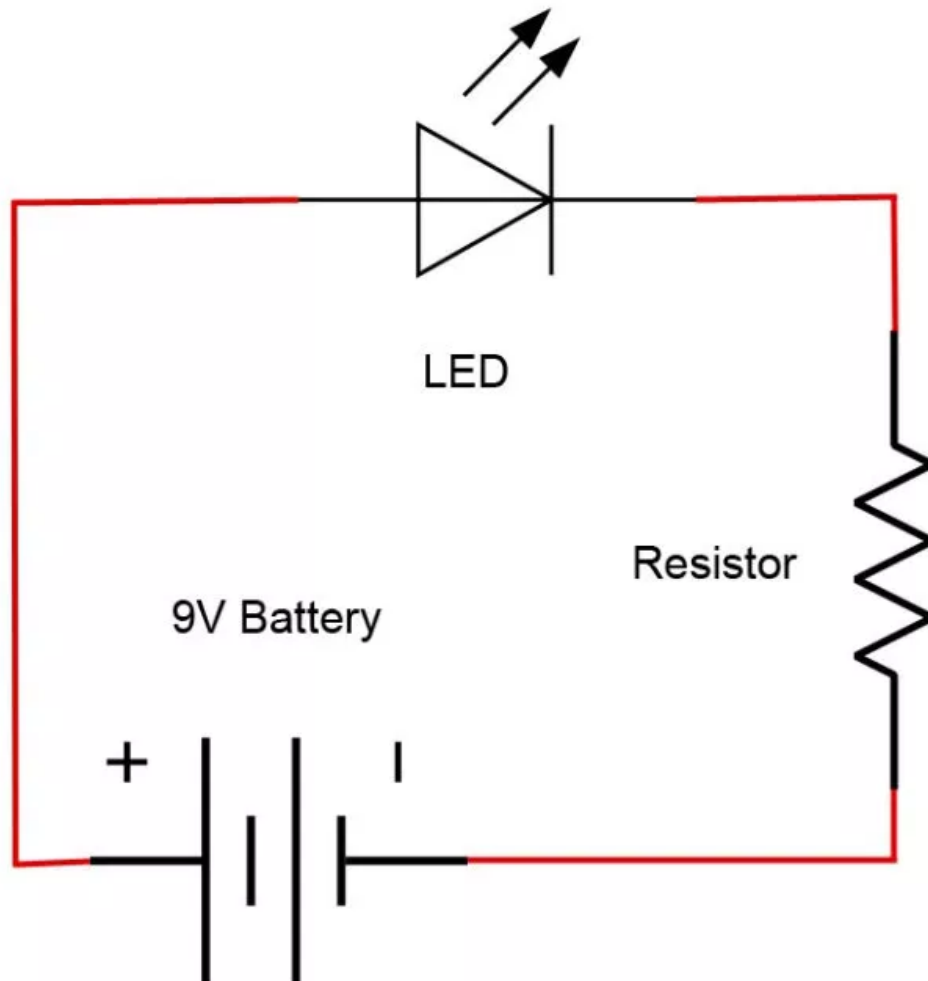
Parts Needed:

- 9V Battery

- Battery Snap-on Connector
- Test Leads w/ Alligator Clips
- 330 Ohm Resistor
- LED – Basic Red 5mm



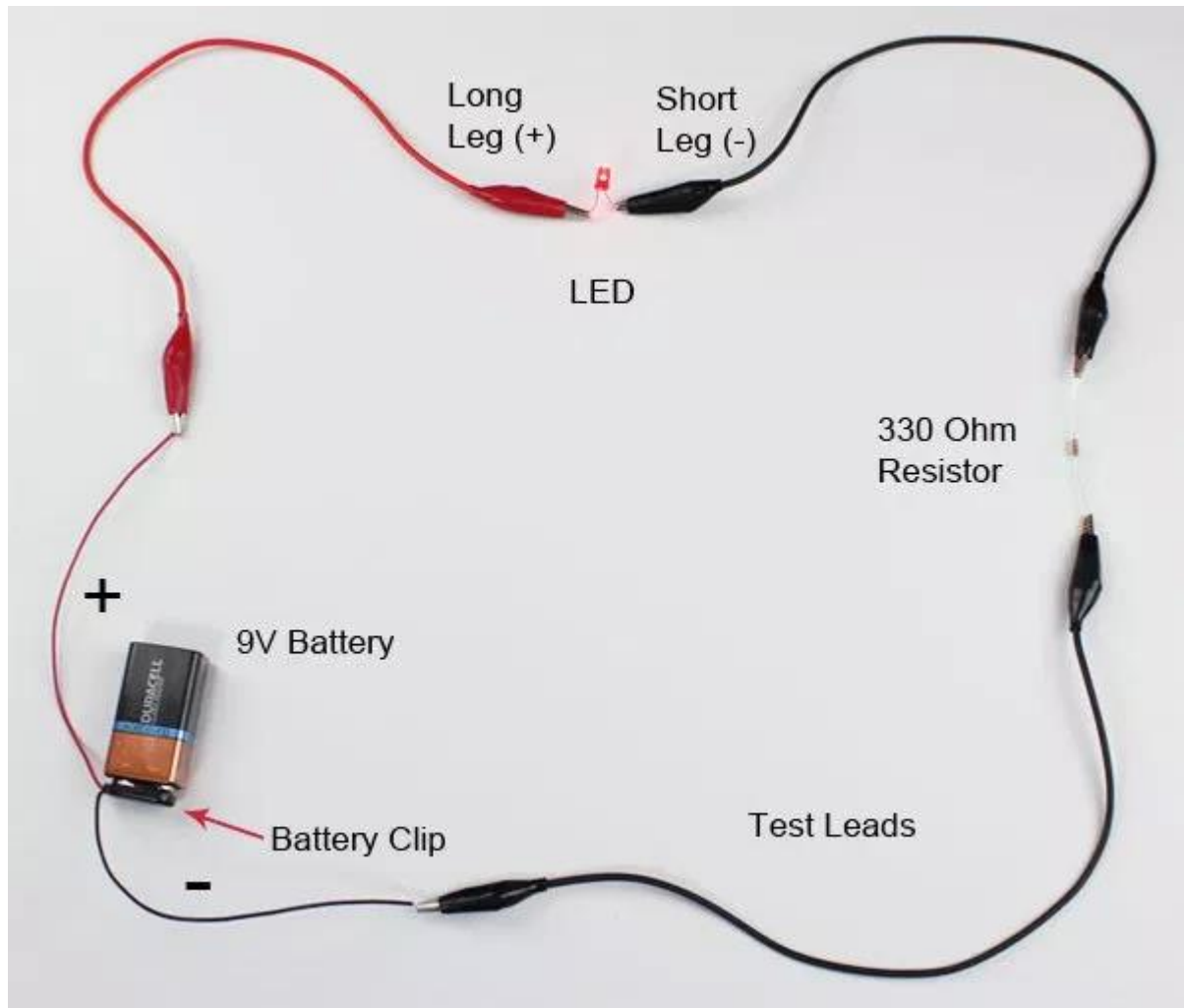
Schematic Diagram



Project Steps

1. Attach the battery clip to the top of the 9V battery.
2. Red wire from the battery clip is connected to one alligator clip on the red test lead.
3. The other end of the red test lead is connected to the long leg (+) of the LED.
4. Connect one alligator clip from black test lead to the short leg (-) of the LED.
5. The other end of the black test lead is clipped to one leg of the 330 Ω resistor.
6. Clip one side of the other black test lead to the other leg of the 330 Ω resistor.

7. The opposite end of the black test lead is connected to the black battery wire



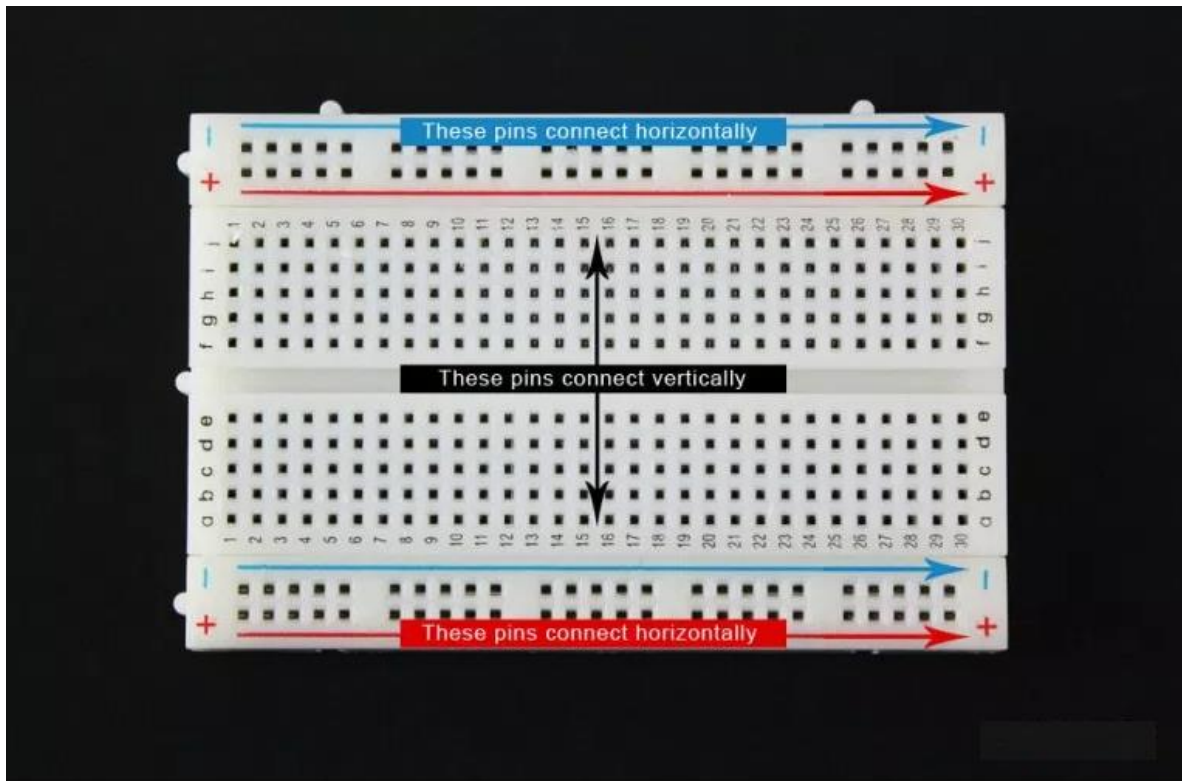
IMPORTANT – Never connect an LED directly to a 9V battery without a resistor in the circuit. Doing so will damage/destroy the LED. You can however connect an LED to a 3V or smaller battery without a resistor.

How To Use A Breadboard

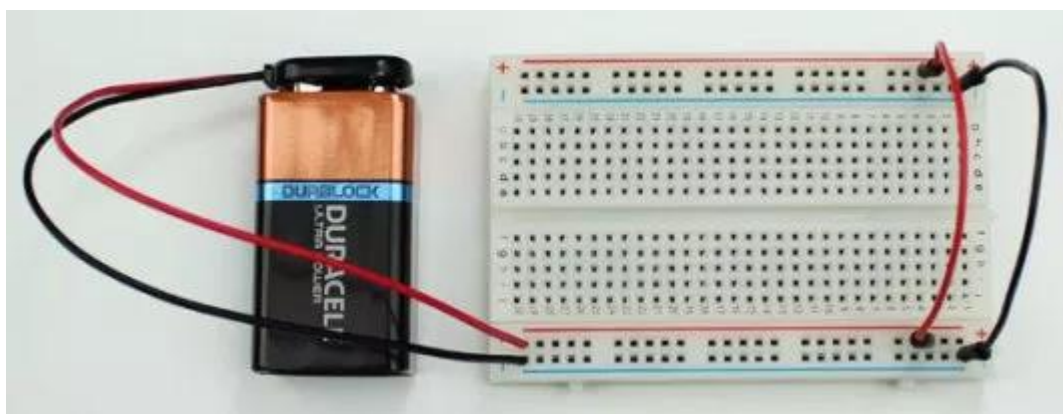
Another way to create and test a circuit is to build it on a breadboard. These boards are essential for testing and prototyping circuits because no soldering is needed. Components and wires are pushed into the holes to form a temporary circuit. Because it's not permanent, you can experiment and make changes until the desired outcome is reached.

Below the holes of each row are metal clips that connect the holes to each other. The middle rows run vertically as shown while the exterior columns are

connected horizontally. These exterior columns are called power rails and are used to receive and provide power to the board.



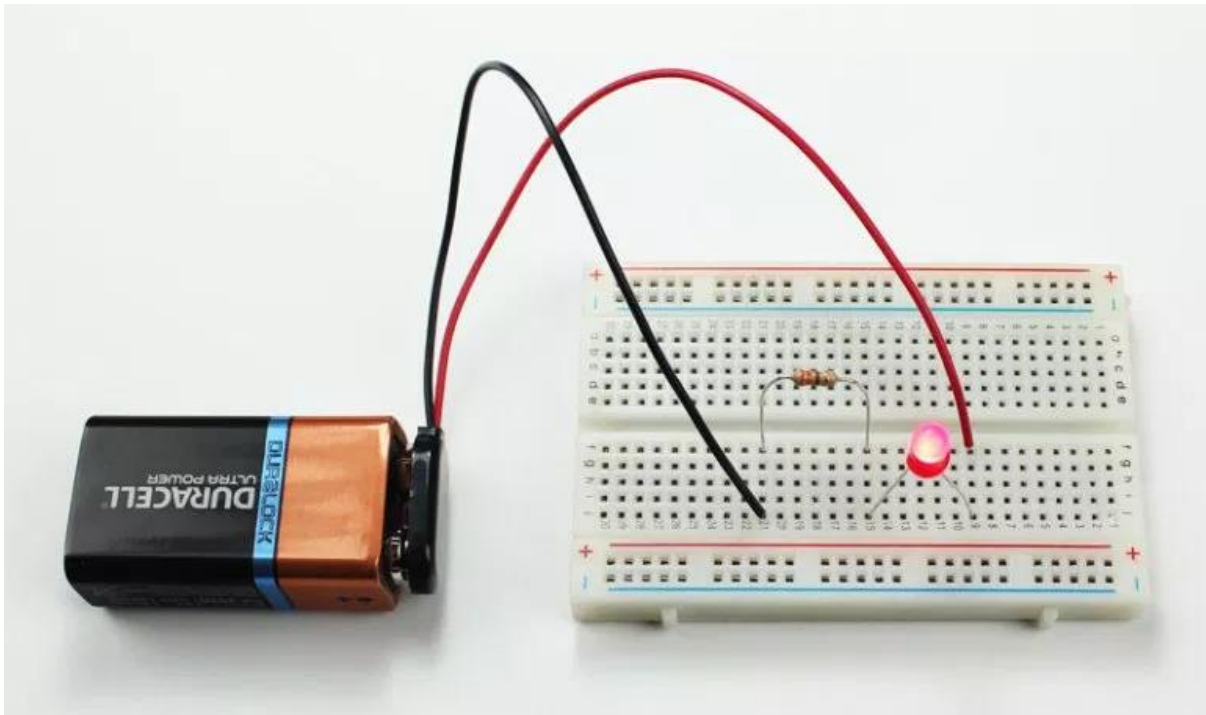
Breadboards will need to have power supplied to them and this can be done in a few ways. One of the easiest way is to plug the wires from a battery holder into the power rails. This will supply voltage to the rail it's plugged into only.



To power both rails, you will need to use a jumper wire from the (+) and (-) to the rail on the opposite side.

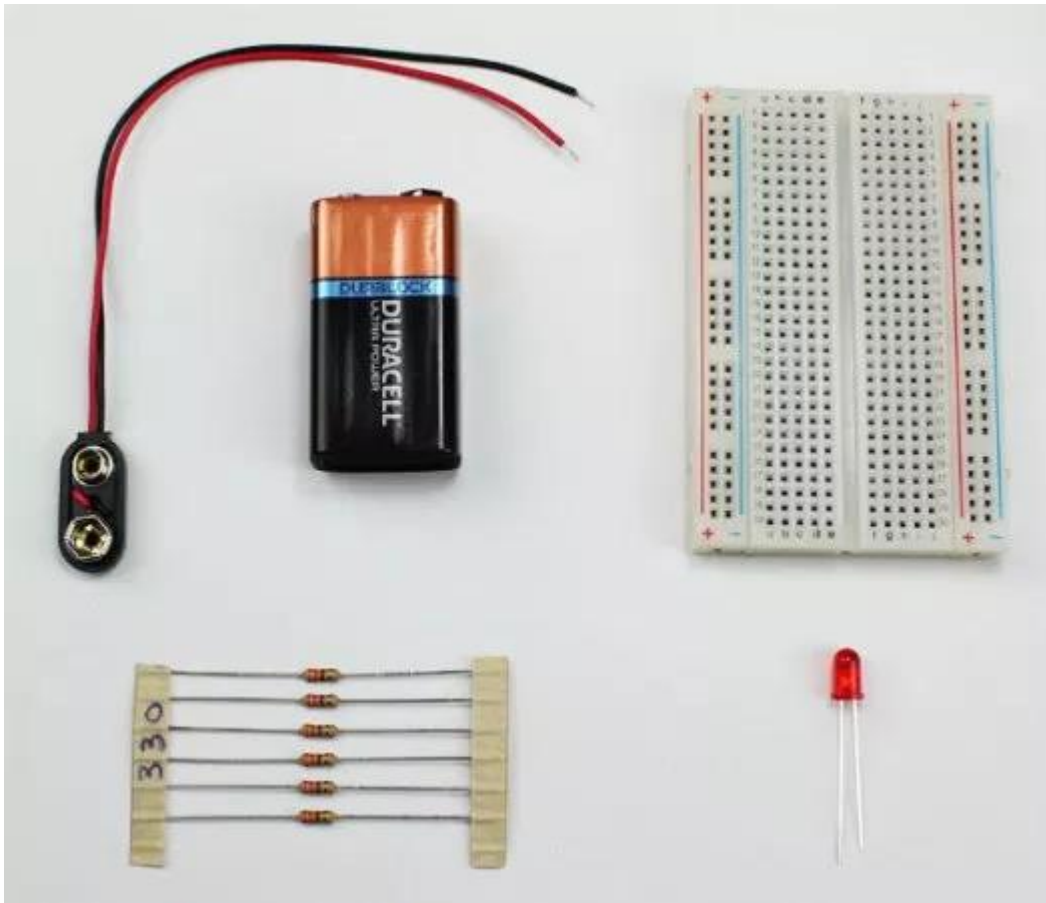
Electronics Project #2

Now we're going to learn how to create a circuit on a breadboard. This circuit is the exact same one we did earlier but we won't be using the test leads.

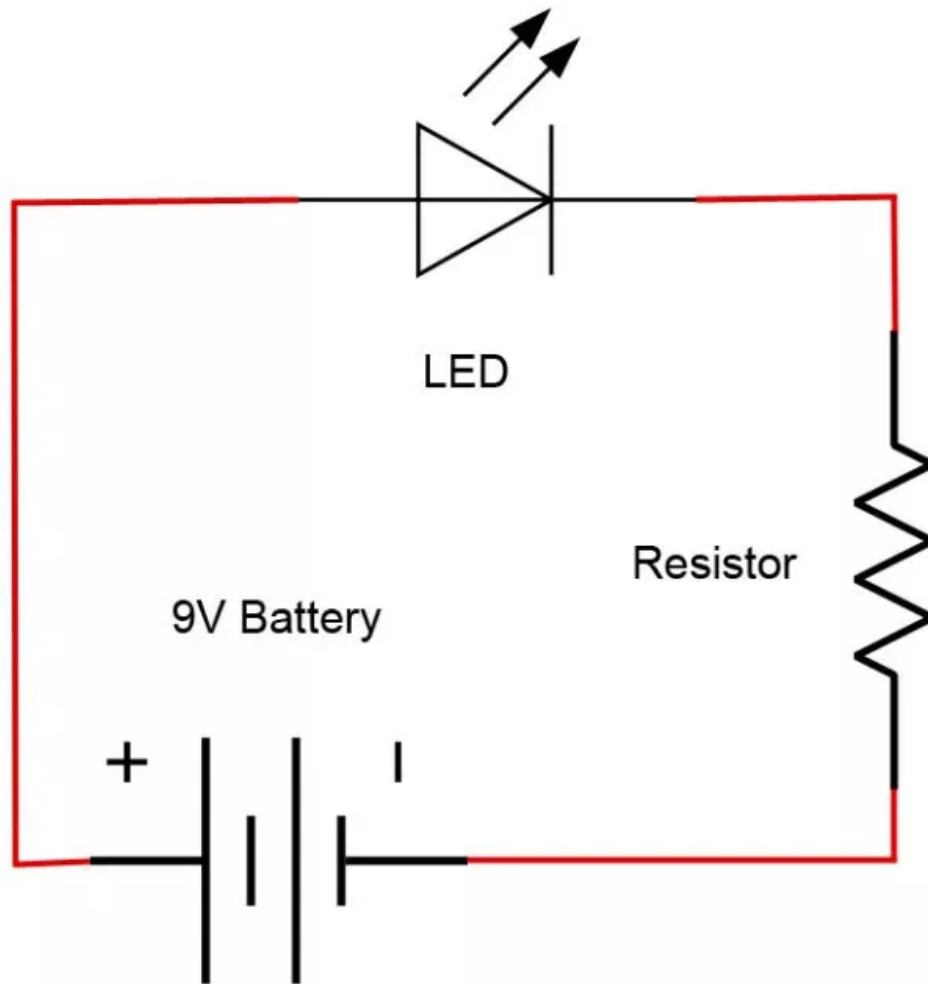


Parts Needed:

- 9V Battery
- Battery Snap-on Connector
- 330 Ohm Resistor
- LED – Basic Red 5mm
- Breadboard- Half Size

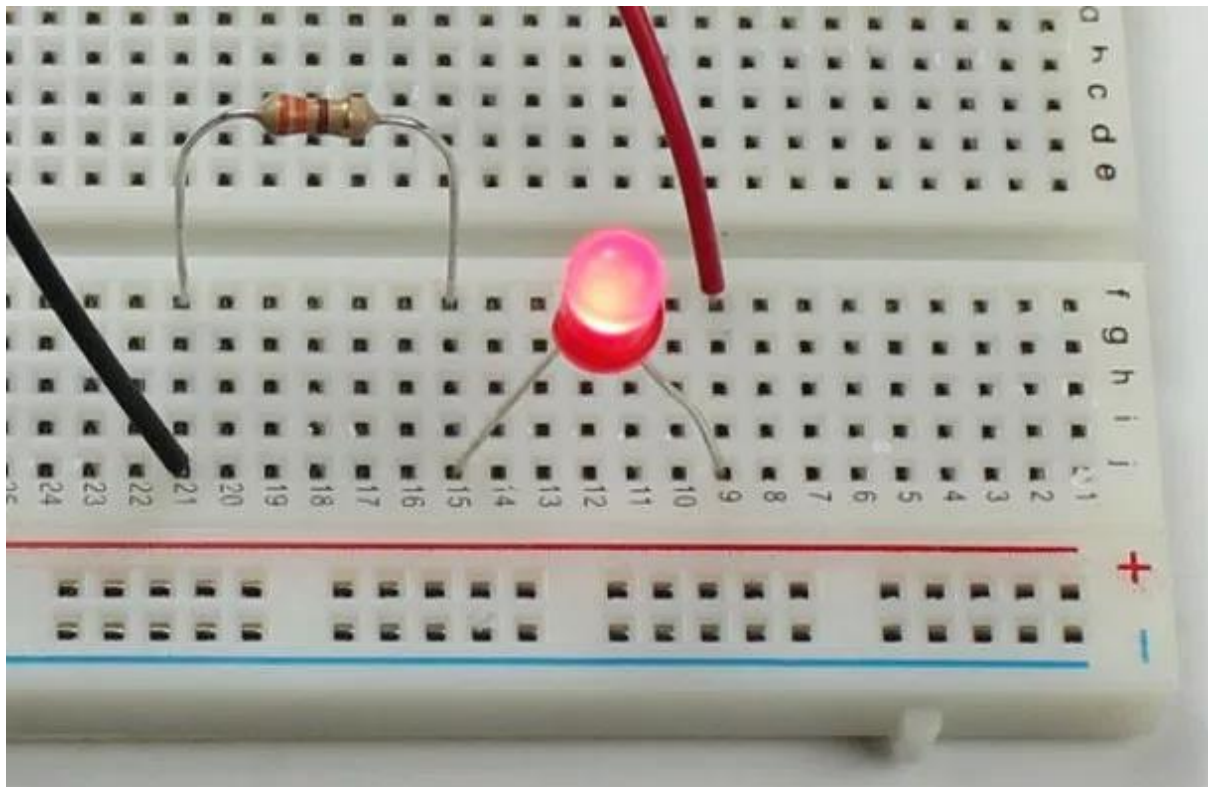


Schematic Diagram

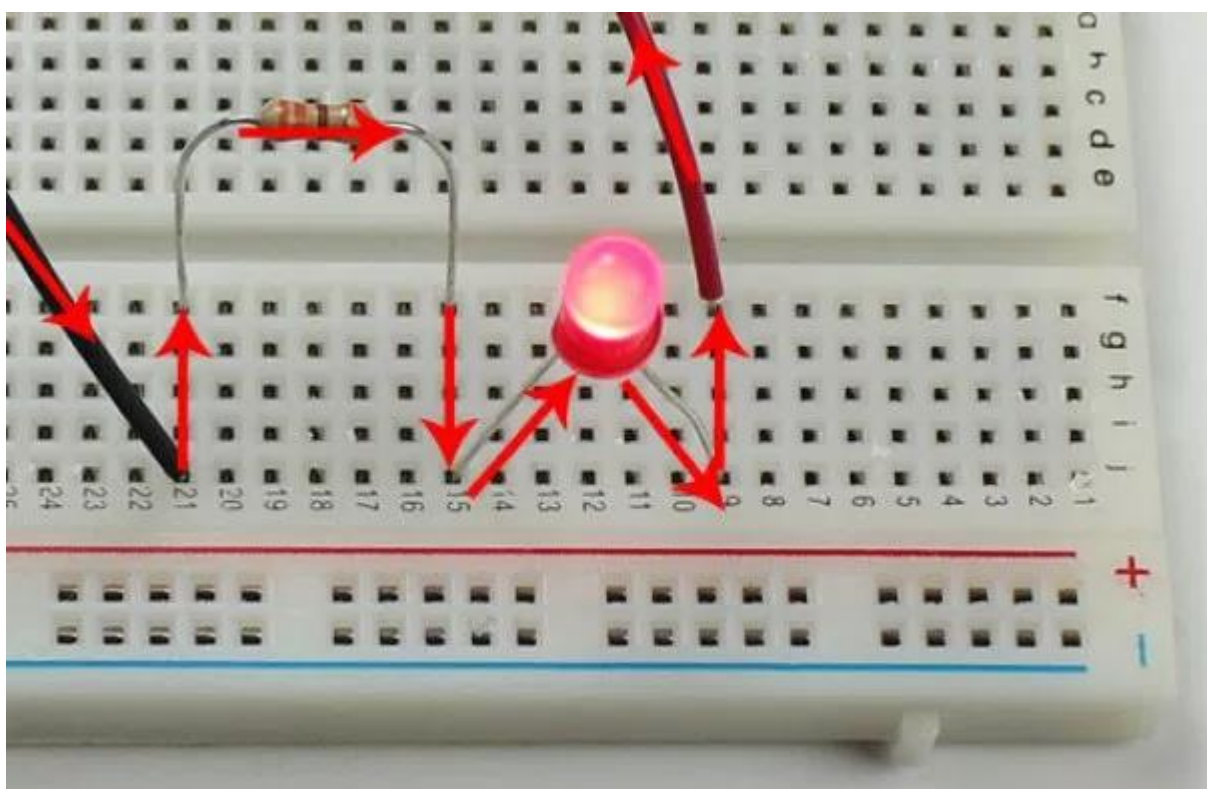


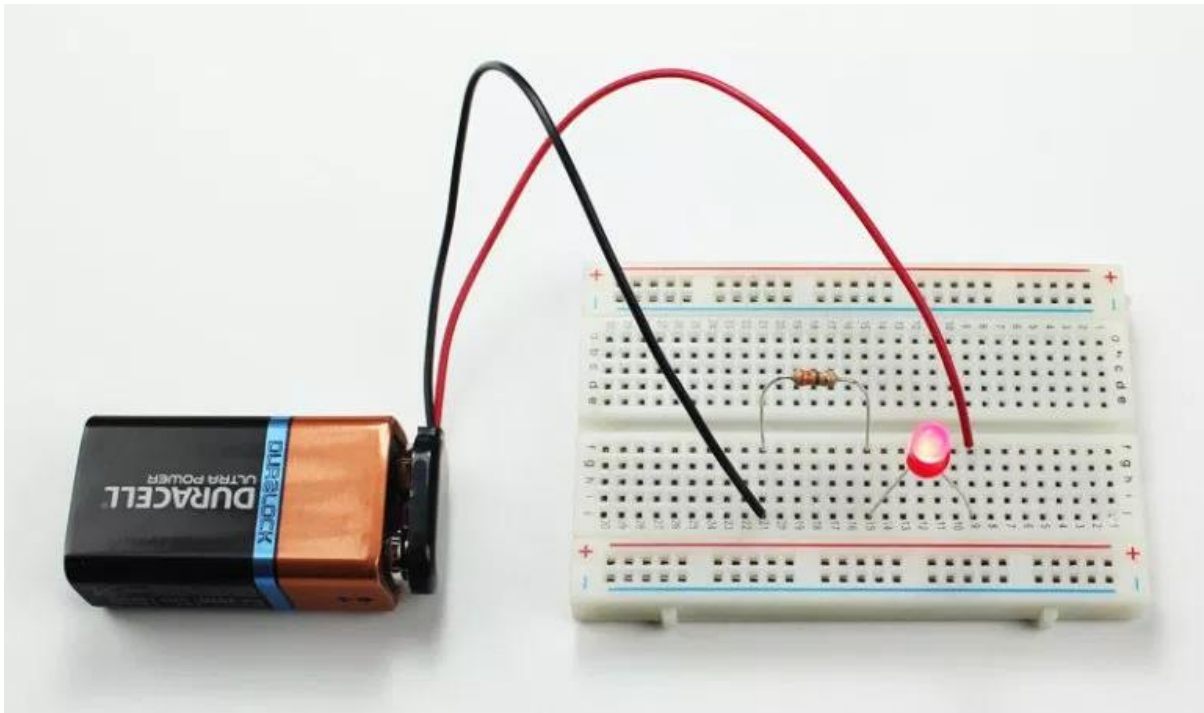
Project Steps

1. Attach the battery clip to the top of the 9V battery.
2. Place the red wire from the battery clip into F9 of the breadboard.
3. Insert the black wire from the battery clip into J21 of the breadboard.
4. Bend the legs of the 330 Ω resistor and place one leg into F21.
5. Place the other leg of the resistor into F15.
6. Insert the short leg of the LED into J15 and the long leg into J9.



The red arrows in the image below help to show how electricity is flowing in this circuit. All components are connected to each other in a circle just like when we used the test leads.





IMPORTANT – Never connect an LED directly to a 9V battery without a resistor in the circuit. Doing so will damage/destroy the LED.

How To Solder

If you want to make your circuit permanent, you'll need to solder it together. For an in-depth tutorial on soldering electronics, view our post [How To Solder](#) for a complete step-by-step guide.

Motion sensor

Motion sensor circuits are there on the Internet for quite a while now. These circuits are mainly used to drive AC load (like lights, fans) in [home automation](#) or [IoT projects](#). It is also very common in manufacturing industries, like on conveyor belts where the presence/position of a particular object has to be detected.

So, in this tutorial, we are going to use an IR sensor with a **NE555 Timer IC** to detect motion and switch the AC load according to that. The 555 timer IC is used as a switch here. Since this circuit uses a digital timer IC, so the operation of the circuit is fast and accurate with even faster detection speeds. In our previous tutorial, we learned about the **555-timer as a latching circuit**. This is the

application of that circuit, we will also demonstrate the circuit by designing it on a piece of perf-board.

Components Required to Build a Motion Detector Circuit

A list of components required to build a **motion detector circuit** is given below:

- 555 timer IC
- 220K Ω resistors*2
- 100k Ω resistor
- 1K Ω resistor
- 1uf electrolytic capacitor
- LED
- 220 Ω resistor
- SPDT Relay*2
- In4007 diode*2
- BC547 NPN transistor
- BC557 pnp transistor
- DC jack
- IR sensor module

IR Sensor Module – A Brief Introduction

IR stands for infrared radiations/light and it is the most common way of detecting motion. There are 2 types of IR sensors:

1. [PIR \(passive infrared sensor\)](#)
2. [IR sensor module](#)

Infrared comes in the place of the electromagnetic spectrum which is not visible to naked eyes. All the hot objects generate these IR radiations and by detecting these radiations, we can sense the motion. The PIR sensor does not emit any IR radiation but only detects the radiations and thus it is called “passive”. On the other hand, we have IR modules that send an IR pulse continuously and when it bounces back from an object, the module can detect it by using a photodiode. This photodiode in this case can only detect Infrared light and not visible light.

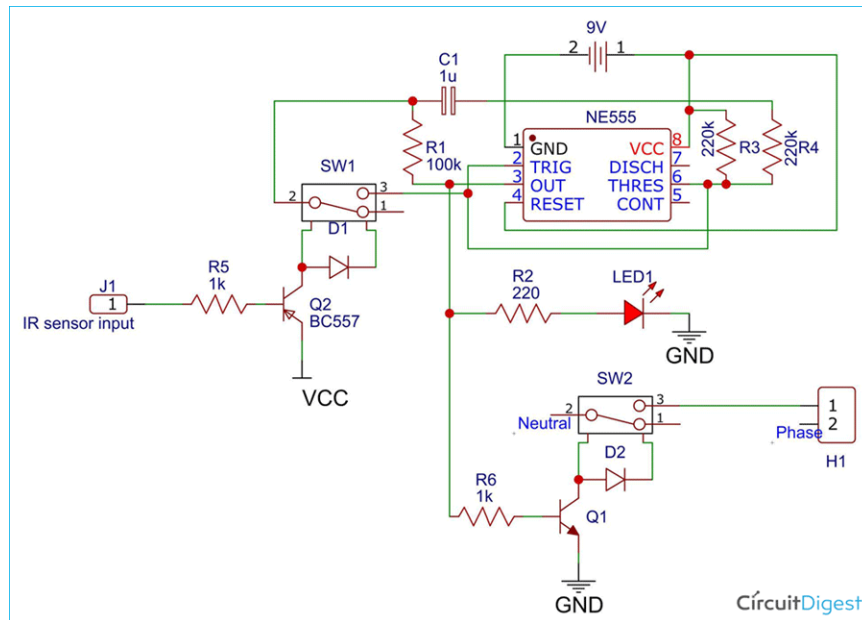
Two main components of an IR sensor module are IR LED and photodiode. LED looks exactly like a normal LED but it emits IR instead of the visible colors that we are familiar with. Photodiode is the key component that detects the bounced back radiations.

In this project, we are using an active IR sensor module as it is readily available, affordable, and easy to use.



555 Timer Motion Detector Circuit Diagram

The complete **motion detector circuit using a 555 timer** is given below.



In the circuit, pins 2 and 6 are connected, and also pins 4 and 8 are connected. The output of the voltage divider circuit is connected to pin 6 of the IC. One resistor of the voltage divider circuit is connected via a 1uF capacitor to the output pin 3 through the 100k resistor. A relay with its driver circuit is connected between pin 2 and the capacitor's positive terminal. An LED is also connected through its current limiting resistor at the output of the IC. The base of the NPN transistor used for controlling the output relay is connected to the output pin 3 of the IC via a 1K resistor. The base of the PNP transistor driving the switching relay of the IC is connected to the output of the IR sensor module.

Working of Motion Detector Circuit

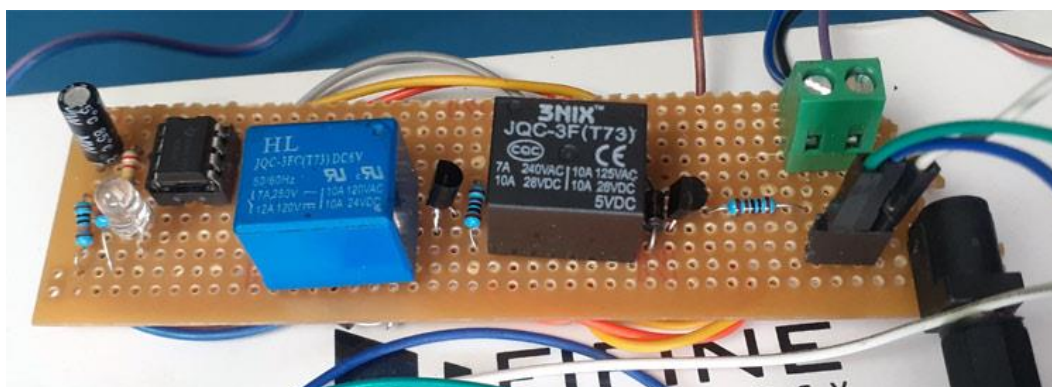
Working of the **motion sensor circuit** is given below:

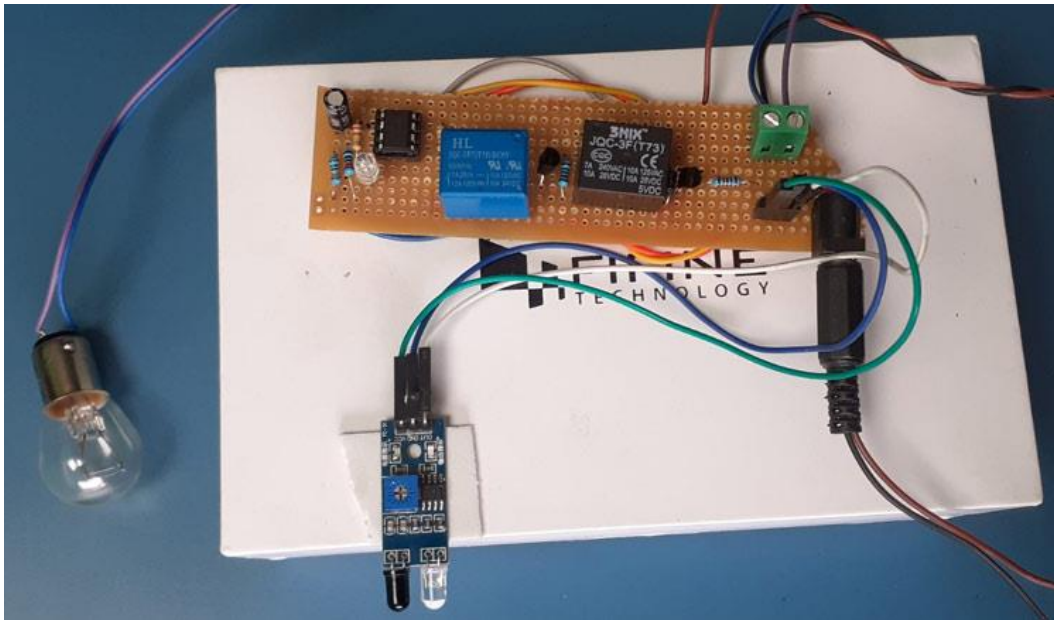
- Initially, due to the [voltage divider](#), half of the supply voltage is there on pins 2 and 6 because we are using a voltage divider with an equal value of resistors and thus the output of the IC is OFF.
- When the motion is detected by the sensor, the capacitor starts to draw current through the resistor R3 to charge, and thus the voltage drop across the resistor changes which in turn makes the voltage at pin 2 go below the 1/3 of supply voltage mark. This turns the output of the IC ON.

- Now the capacitor is fully charged to the supply voltage through the 100k Ω resistor. When the motion is detected, again pin 6 of the timer IC detects the charged capacitor which obviously is at the supply voltage and thus goes more than the 2/3 mark. This turns the output of the IC OFF.
- If you observe the circuit a little close, you can see that we can use a BJT instead of relay and you are correct but unfortunately this will not work. The reason is again the difference between the ideal and the real world. We use BJTs everywhere but they are not perfect switches and have some leakage current which in this case messes up with the circuit. We need an ideal switch in this case and thus we are using a relay.
- The transistors are used to drive the relays. We are using a PNP transistor to drive the main relay because the IR sensor module gives a constant supply voltage at its output and when it detects something, it pulls the voltage to the ground. We can use an NPN transistor to drive the output relay as the IC has active high output.

Demonstration of Motion Sensor Circuit

By following the **555 timer motion detector circuit diagram** given above, I created the circuit on perf-board, the video of which is available at the end of the article. Also, the pictures related to the circuit are given below.





Gas Leak Detection & Alarm System

Overview

In this project, we will build a Gas Leak Detection & Alarm System using IC 555 Timer and MQ-2 Gas Sensor. It works as a Home Security Alarm during Gas Leakage. Whenever the sensor detects gas, the LED & Buzzer goes high indication the audio-visual alarming.

We will use MQ-2 Gas Sensor which is one of the low cost and popular gas sensor in the market. The MQ-2 gas sensor is sensitive to LPG, i-butane, propane, methane, alcohol, Hydrogen and smoke. In the circuit, the 555 Timer IC is configured in astable mode. The output signal from this gas sensor module is used to drive a 555 timer astable multivibrator.

Components Required

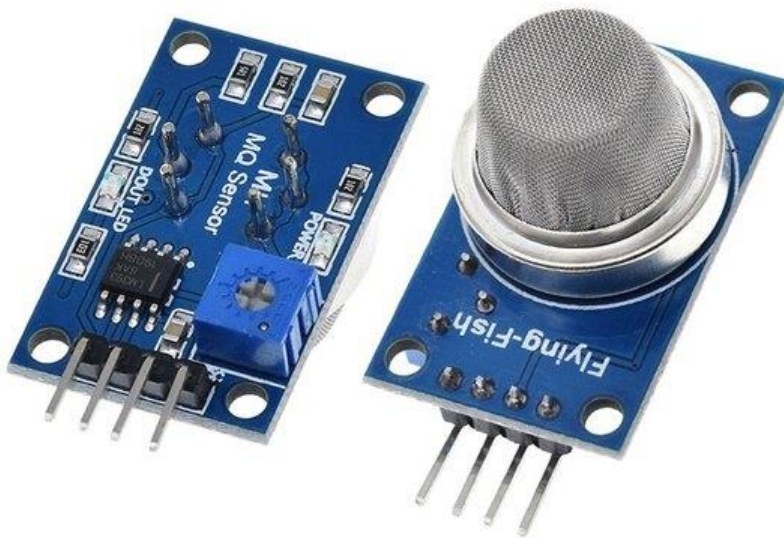
Following are the components required to build a Gas Leak Detection & Alarm using 555 Timer IC.

1. Resistors (22 Ω , 4.7k Ω , 10k Ω , 1k Ω , 560 Ω)
2. Variable Resistor 10k Ω

3. 1N4148 Zener diode
4. Capacitors 100nF – 2
5. MQ-2 Gas Sensor Module
6. 555 timer IC
7. Transistor BC547
8. 5V Buzzer
9. LED 5V Red
10. 5-9V DC Power Supply

MQ2 Gas/Smoke Sensor Module

The MQ-2 gas sensor is a versatile and cost-effective solution for detecting a variety of gases.



The MQ-2 sensor stands out as a multifunctional gas detector that can identify various gases such as alcohol, carbon monoxide, hydrogen, isobutene, liquefied petroleum gas, methane, propane, and smoke. Its affordability and user-friendly attributes make it a favorite choice for beginners in the field.

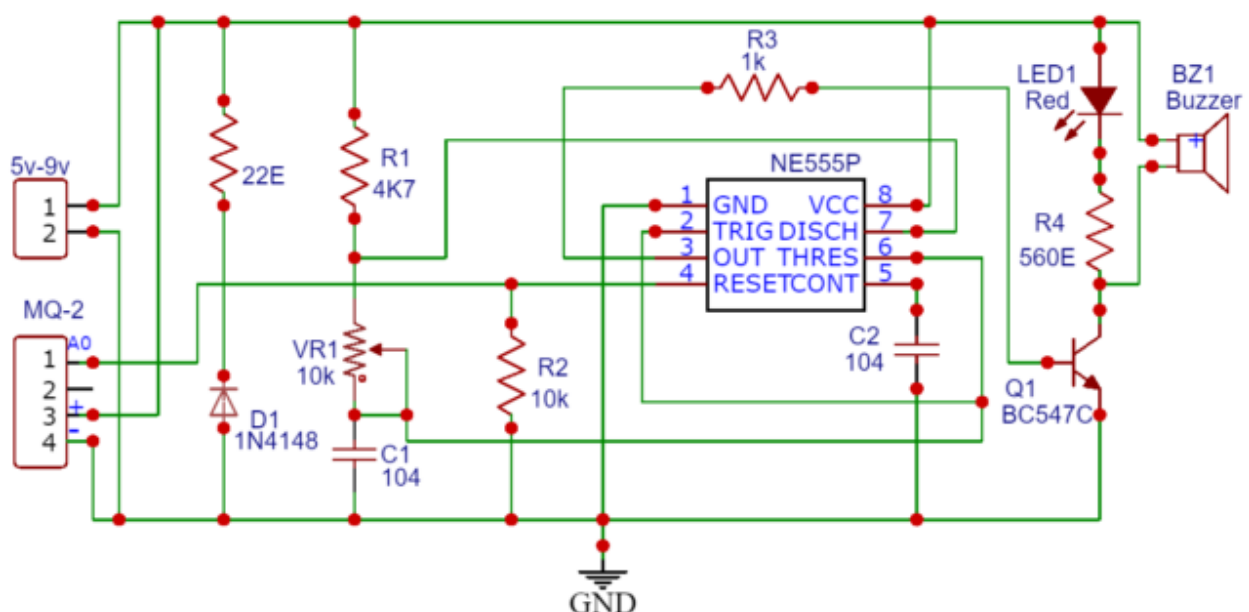
Technical Specifications

- **Target Gases:** LPG, Propane, Methane, Alcohol, Hydrogen, Smoke

- **Operating Voltage:** Typically 5V (with a preheat time of over 24 hours for initial use)
- **Load Resistance:** 20 K Ω
- **Heater Resistance:** 33 $\Omega \pm 5\%$
- **Sensing Resistance:** 10 K Ω – 60 K Ω
- **Heating consumption:** <800mw
- **Sensitivity:** High to gases in the scope of detection
- **Concentration Range:** 200 – 10000ppm
- **Preheat Duration:** 20 seconds (minimum)
- **Operating Temperature:** -10 to 50 degrees Celsius

Circuit: Gas Leak Detection & Alarm using 555 Timer

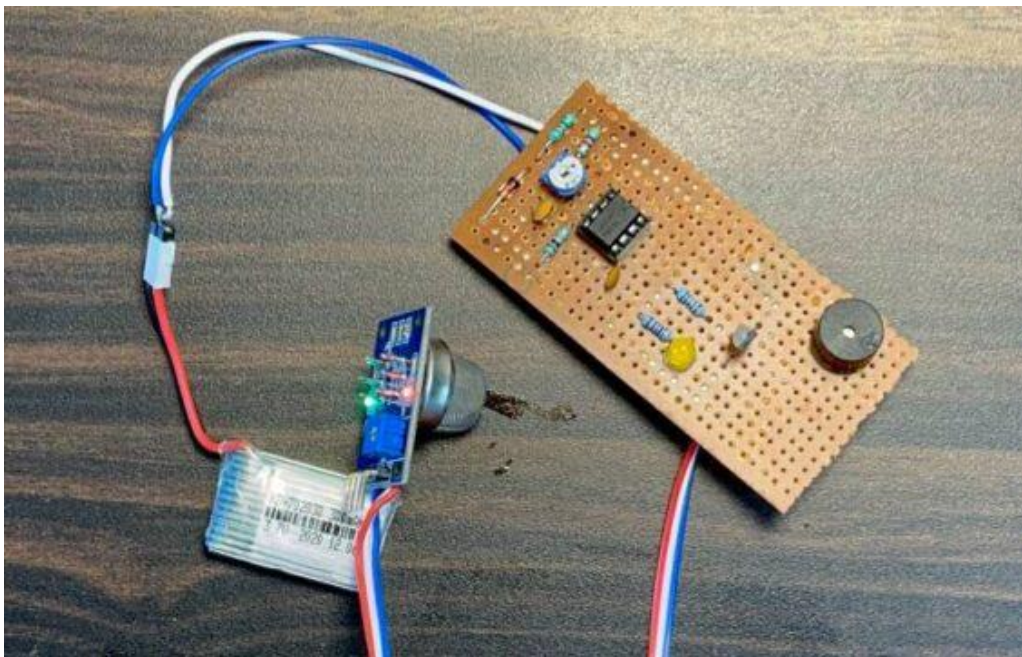
Here is the circuit diagram for Gas Leak Detection & Alarm based on 555 Timer IC and MQ-2 Gas Sensor. In this circuit the IC 555 Timer is configured in astable multivibrator mode.



The 555 timer IC (NE555P) is the core of the alarm system. It's connected as follows: pin 1 to GND, pin 8 to VCC, pin 4 to VCC for resetting, and pins 2, 6, and threshold (THR) through

a capacitor C2 (104 or 0.1 μ F). The timing components R3 (1k ohm) and C2 determine the frequency of the output oscillation on pin 3. This output drives a transistor Q1 (BC547) which in turn switches the load consisting of a red LED (LED1) and a buzzer (BZ1) in series with a resistor R4 (560 ohm) to limit the current.

The analog output of the sensor, is connected through a 22-ohm resistor to a voltage divider circuit formed by R1 (4.7K ohm) and the potentiometer VR1 (10k ohm). The sensor's output is also connected to a diode (D1, 1N4148) for protection, and this node is tied to pin 2 (Trigger) of the 555 timer IC through capacitor C1 (0.1 μ F).

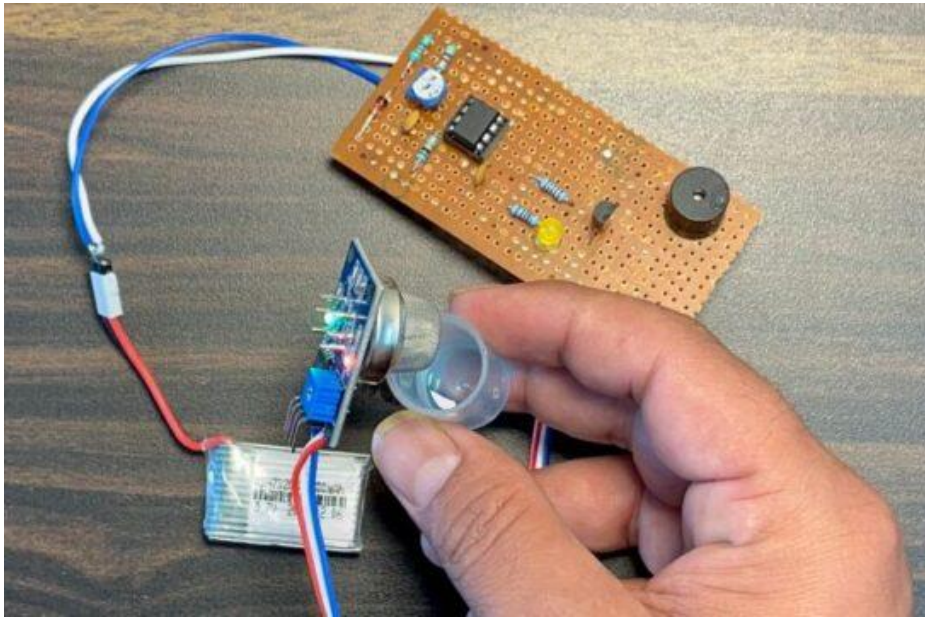


The circuit can be assembled on a Vero Board or Zero PCB. You can use a 9V Battery to Supply the power to the entire system.

Working of the Circuit

In this circuit, the purpose of the potentiometer is to set the threshold level for gas concentration. The LPG concentration of 1000 ppm (parts per million) in the area is enough to trigger the circuit. When the gas concentration exceeds this threshold, the output voltage increases to a level that triggers the 555 timer. This

output drives a transistor Q1 which in turn switches the load consisting of a red LED and a buzzer.



Using Alcohol to increase the Gas Concentration

The High output causes the LED to light up and the buzzer to sound, thus signaling the presence of gas. The LED and the buzzer will continue to alert until the gas concentration drops below the set threshold, at which point the 555 timer will reset, turning off the alarm indicators.

Further Education and Training Certificate: Electronics

ID 83446 LEVEL 5 – 135 CREDITS

LEARNER GUIDE

SAQA – 386093

**Inspect and test electronic equipment, components
and assemblies using measuring and testing equipment**

Learner Information:

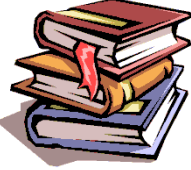
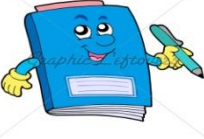
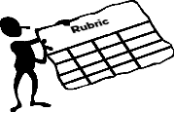
Details	Please Complete this Section
Name & Surname:	
Organisation:	
Unit/Dept:	
Facilitator Name:	
Date Started:	
Date of Completion:	

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Key to Icons

The following icons may be used in this Learner Guide to indicate specific functions:

 Books	This icon means that other books are available for further information on a particular topic/subject.
 References	This icon refers to any examples, handouts, checklists, etc...
 Important	This icon represents important information related to a specific topic or section of the guide.
 Activities	This icon helps you to be prepared for the learning to follow or assist you to demonstrate understanding of module content. Shows transference of knowledge and skill.
 Exercises	This icon represents any exercise to be completed on a specific topic at home by you or in a group.
 Tasks/Projects	An important aspect of the assessment process is proof of competence. This can be achieved by observation or a portfolio of evidence should be submitted in this regard.

 Workplace Activities	An important aspect of learning is through workplace experience. Activities with this icon can only be completed once a learner is in the workplace
 Tips	This icon indicates practical tips you can adopt in the future.
 Notes	This icon represents important notes you must remember as part of the learning process.

Learner Guide Introduction

About the Learner Guide...	This Learner Guide provides a comprehensive overview of the Inspect and test electronic equipment, components and assemblies using measuring and testing equipment, and forms part of a series of Learner Guides that have been developed for Further Education and Training Certificate: Electronics ID 83446 LEVEL 5 – 135 CREDITS. The series of Learner Guides are conceptualized in modular's format and developed for Further Education and Training Certificate: Electronics ID 83446 LEVEL 5 – 135 CREDITS. They are designed to improve the skills and knowledge of learners, and thus enabling them to effectively and efficiently complete specific tasks. Learners are required to attend training workshops as a group or as specified by their organization. These workshops are presented in modules, and conducted by a qualified facilitator.
Purpose	The purpose of this Learner Guide is to provide learners with the necessary knowledge related to Inspect and test electronic equipment, components and assemblies using measuring and testing equipment.
Outcomes	At the end of this module, you will be able to: • Conduct inspection of a product. • Test electronic equipment. • Install electronic equipment. • Disassemble and/or assemble electronic components and systems.
Assessment Criteria	The only way to establish whether a learner is competent and has accomplished the specific outcomes is through an assessment process. Assessment involves collecting and interpreting evidence about the learner's ability to perform a task. This guide may include assessments in the form of activities, assignments, tasks or projects, as well as workplace practical tasks. Learners are required to perform tasks on the job to collect enough and appropriate evidence for their portfolio of evidence, proof signed by their supervisor that the tasks were performed successfully.
To qualify	To qualify and receive credits towards the learning program, a registered assessor will conduct an evaluation and assessment of the learner's portfolio of evidence and competency

Range of Learning	This describes the situation and circumstance in which competence must be demonstrated and the parameters in which learners operate
Responsibility	The responsibility of learning rest with the learner, so: • Be proactive and ask questions, • Seek assistance and help from your facilitators, if required.

Learning Unit 1

Inspect and test electronic equipment, components and assemblies using measuring and testing equipment.

UNIT STANDARD NUMBER	:	386093
LEVEL ON THE NQF	:	5
CREDITS	:	4
FIELD	:	Field 10 - Physical, Mathematical, Computer and Life Sciences
SUB FIELD	:	Information Technology and Computer Sciences

PURPOSE:	The purpose of this unit standard is for persons who are entrusted with the responsibility to determine damages and defects in a product for corrective action. People credited with this unit standard are able to: Conduct inspection of a product. Test electronic equipment. Install electronic equipment. Disassemble and/or assemble electronic components and systems.
LEARNING ASSUMED TO BE IN PLACE:	
• Communication at NQF Level 4. • Mathematics at NQF Level 4. • Physical Science at NQF Level 4.	

SESSION 1.
Conduct inspection of a product.
Learning Outcomes
• Different forms of inspection are identified and discussed in terms of their advantages and disadvantages. • Potential causes of damages on the product are identified in terms of the operational condition of the product. • Defects are identified and recorded in order to take necessary action.

Conduct inspection of a product.

The definition of visual inspection

Visual inspection is a very basic inspection method used in manufacturing quality control and asset maintenance. It is the method of looking for flaws or imperfections using the naked eye and non-specialized inspection equipment.

Quite often, there is no empirical formula for finding defective products or inspecting a piece of equipment. Therefore, the experience of the person that is performing visual inspection plays a significant role in determining the efficacy of the process.

Looking for anomalies with the naked eye is the most common method of visual inspection. But it is not necessarily limited to that. Using other senses for inspection also falls under the broad umbrella of visual inspection. The basic tenet is that an experienced person is using his sensory capabilities to identify problems with objects they inspect.

For instance, experienced professionals can assess the quality of materials like wood and metals by judging the sound produced by them. They can just knock on the object and make a judgment based on the sound produced by it.

Similarly, a maintenance technician might *smell* a gas leak, *hear* that something is loose inside the asset, or even *feel* that a piece of equipment vibrates more than it should or radiates excessive heat. All while performing a *visual inspection*.

Visual inspections in physical asset management

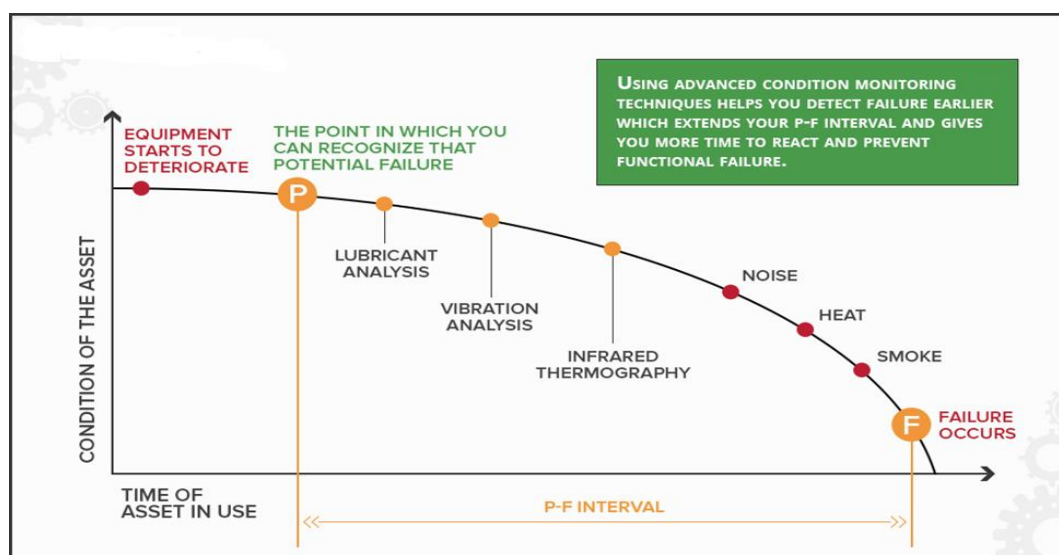
Visual inspection is an integral part of physical asset management. Maintenance professionals use it on a daily basis as part of their routine maintenance activities.

Occasionally, an organization will train machine operators to perform visual inspections instead of maintenance techs. This practice is commonly referred to as autonomous maintenance.

Regular inspections help maintenance teams spot signs of wear and tear that can lead to equipment failure. Minor issues are corrected on the spot if the technician has access to the right tools and parts. Otherwise, they will leave a note about the detected issue. The maintenance manager or supervisor will later add a new task to the preventive maintenance plan so the problem doesn't go unaddressed.

While they can be very useful, visual inspections come with a couple of downsides. Machines often need to be shut down so a technician can perform a visual inspection. As you can imagine, this reduces equipment uptime and can hurt the bottom line. The other problem is that visual inspections can sometimes spot the problem too late. If you detect noise, heat, or smoke, the failure might already be in progress.

To combat these issues, more and more organizations are using condition monitoring sensors and equipment to monitor asset health and performance in real-time. This technology allows them to notice deterioration signs much earlier than it would otherwise be possible. This is neatly represented in the graphic below.



These sensors allow organizations to run advanced maintenance strategies

like condition-based maintenance and predictive maintenance. Such strategies reduce the need for visual inspections but are far from making them obsolete. Sensors are still fairly expensive and can only detect specific deterioration signs. It means that *manual* testing methods like visual inspections are not only necessary but a more cost-effective option, especially for inexpensive assets.

Visual inspections in quality control

Visual inspection is the oldest non-destructive testing (NDT) method for quality control. In its original form, visual inspection is used to observe the surface of an object and look for cracks, scratches, misalignments, corrosion, and other physical defects.

Big brands, especially those that produce expensive products, have a lot to lose if they send out a whole line of defective products. Visual inspection is a great way to quickly filter out items with obvious defects.

For instance, automakers like Rolls-Royce give extreme importance to the fit and finish of each component that goes into their cars. Their internal research shows how **roughly 70% of all faulty components are rejected because of discrepancies found during visual inspection.**

This is not to say that visual inspections are reserved only for luxury goods. A good counter-example comes from the food and beverage industry. Here, visual inspections are performed to ensure there aren't any foreign objects in the product, that packages are uniform and properly sealed, and that the products are not mislabeled. Because of the fairly straightforward nature of these inspections, many food manufacturers are starting to adopt machine vision systems for quality control.



An example of an automated visual inspection system (Image source)

Broadly speaking, visual testing for quality control can be done in three different ways:

1. **Random sampling:** Quality checks are performed on randomly selected products. Products can be checked right at the production line for obvious visual defects. More complex inspections (like testing the interior of a product) are usually carried out at special testing areas.
2. **Full manual sampling:** All products are inspected manually. This can be a physically demanding job with a lot of repetitive actions. To maximize the health and productivity of workers performing the inspections, organizations should look to provide the necessary tools and ergonomic equipment.
3. **Automated visual inspection:** Products are inspected using cameras, image processing methods, and machine learning algorithms. More on this later in the article.

The automated systems for visual inspection seem to be gaining more traction as the private sector develops industry-specific solutions. It remains to be seen how fast will manufacturers adopt this technology.

Modern visual inspection techniques

Incremental technological improvements in camera quality and machine learning are starting to change how we think about visual inspections. Maintenance techs now have the ability to perform visual inspections at hazardous and hard-to-reach places, while being miles away from the objects they are inspecting. Quality control specialists, on the other hand, have more and more access to automated visual inspection solutions.

Remote visual inspections for asset maintenance

These days, technicians can use robots and drones to view otherwise inaccessible areas. We are talking about inspecting objects at great heights, very confined spaces, hazardous places like storage tanks, and similar.

Images or videos can be viewed from a remote location, even thousands of miles away from the location of the inspection site – hence the name remote visual inspection. The captured data can be accessed in two different ways:

1. **Live stream:** Robots or drones can relay data of the location to the base station where the inspector can watch the live feed. In this case, the robot or drone can also be controlled from the remote terminal. It can only be used in cases where there is constant wireless communication between the equipment that is being operated and the base station.
2. **Saved data:** In extremely remote or hazardous areas where there can be no wireless communication, videos and images are saved and analyzed later.

An additional advantage of these types of inspections is that the images and videos can be easily shared between technicians and other relevant team members, without having to be physically present at the inspection site. This collaboration can be extremely useful when solving complex problems or when there is a need to contact specialized experts.

Automated visual inspections (AVI) for mass production

Automated visual inspection (AVI) uses computer vision and artificial intelligence to perform visual inspections on products instead of quality inspectors. This technology aims to dramatically reduce the need for human intervention for inspection in large-scale automated factories.



Cognitive visual inspection solution from IBM (Image source)

Foxconn, one of the largest electronics manufacturers on the planet, is using AVI in their manufacturing facilities. We will use them as an example to briefly describe how this technology is supposed to work.

Suppose Foxconn is manufacturing motherboards for the latest iPhone. One ideal sample of the motherboard is made and inspected by a human visual inspector. Once it has been approved, this model is uploaded to the database and the algorithm can use it as a benchmark. Some examples of defective models are also uploaded to the server and used to teach algorithms how to recognize specific anomalies.

The rest of the motherboards will come out of the automated assembly line and pass under the camera systems for the AVI. Multiple images of the product are captured from various angles. The captured images are compared to the benchmark images stored using machine learning algorithms. Some of the factors compared could be:

- size and shape of different elements
- solder locations
- component locations
- missing components
- connectors
- color anomalies

The product passes the inspection if it is sufficiently close to the benchmark.

At face value, this technology seems like a perfect solution. After all, the main idea behind automation is to leave routine and mundane tasks to machines so that employees can focus on creative problem-solving. It is a sound idea, but there are still many limitations that need to be addressed before it becomes a standard in the industrial space.

The limitations of automated solutions

There are two main impediments to the adoption of automation in visual inspection.

The first one are technological limitations. Though AI researchers have made great strides, the inspection algorithms are far from perfect. They need a lot of examples of

good and defective products so they can learn how to detect different anomalies with enough success. Additionally, machines find it difficult to work with reflective surfaces. This would eliminate working with most metals found in a plant. Such minute differences cannot be captured by the algorithms yet, but humans can easily do that.

The second factor is cost. To implement an automated inspection system, there should be supplementary equipment like imaging machines, networking tools, computer hardware, etc. These hardware costs in addition to the software cost still make it cost-prohibitive for many organizations.

Over time, the quality and price of automated visual inspection systems should go down and we should see wider adoption.

How to standardize the visual inspection process

Standardizing any process makes it repeatable and verifiable. But a problem with the standardization of visual inspections is that a large part of the work is very subjective in nature and quite difficult to package as a checklist.

Standardization efforts in maintenance

Maintenance activities are often standardized through written standard operating procedures, safety guidelines (like LOTO), and maintenance checklists. Visual inspections, to some degree, can be tackled in the same way.

Since maintenance technicians should already know how to perform basic visual inspections, written guidelines can be created only for complex equipment where the proper inspection takes multiple steps to complete.

The process of standardization can look something like this:

- Record down all assets that are visually inspected regularly.
- Map the concerning areas to be checked.
- Note down the actions to be taken depending on the spotted deterioration sign.
- Create a flowchart based on the information above.
- Create a maintenance checklist for the cues that should be inspected.

- If possible, add the checklist and the flowchart to your CMMS system. If you have a mobile CMMS, technicians can quickly access that information on their smartphone and tablet, which can significantly speed up their workflow.
- Occasionally review and update those flowcharts and checklists.

An asset can have a million different issues but the goal of a direct visual inspection is to look for common signs of wear and tear.

Maintenance checklists should always be created by experienced technicians familiar with the inner workings of equipment at their facility. Usually, those will be the same people in charge of training new technicians. This is perfect, as new technicians will reap the most value out of these standardization efforts.

Standardization efforts for quality management

The key to standardizing visual inspections in quality management lies in the training process which should include:

- very clear guidelines on what is considered a defect
- what to do with a defective product
- when and how to use any special visual inspection tools
- who to contact for borderline cases (asking for a second opinion)
- how and where to note novel anomalies
- any other special guidelines that need to be followed on the plant floor

Even after that, there will be cases where the subjectivity of the process causes disputes. A small scratch on the product might be a deal-breaker for one inspector, while the other one would mark it as good to go. Some level of inconsistency is unavoidable but can be minimized through standardization.

WHAT ARE THE DIFFERENT TYPES OF INSPECTION IN QUALITY CONTROL?

Overview of inspections in quality control

In quality control, buyers in international trade need inspection services to verify product standards and quantity according to predetermined specifications. Companies should

partner with a trusted third-party inspection service for the best results. It makes sense to invest to ensure the highest standards for your products and services in the ever-changing global market.

Different types of QC inspections

Depending on your product and industry, you'll need a certain inspection service when searching for quality solutions. But there are some common one you should definitely know about if you're in the business of trade and manufacturing.

Below is a selection and along with their respective role of inspection in quality control explain. Types of inspection and methods of quality control inspections differ for a reason so it's good to know the details.

Sample Checking

Sample checks involve selecting a small number of items from a bigger batch or lot to check a range of their specifications such as appearance, workmanship, safety, functions, etc. prior to mass production. Basically, a sample of the products are extracted from the bunch and inspected for quality, rather than the whole lot.

Inspectors will be looking for potential defects and will compare the physical objects and products to design plans and drawings to ensure the correct measurements have been met. Style and color are other things which can be checked as well as shipping marks and packaging.

Pre-Production Inspections (PPI)

As the name of this type of inspection suggests, pre-production inspections are carried out before production begins to assess the quality and quantity of raw materials and components. This type of inspection is great for when you begin a new relationship with a supplier. You can ensure they are using the right type of materials for your production and implementing the right processes to deliver the product you expect.

Pre-Production Inspection (PPI)

- Conducted before the production process begins
- Helps to assess the quantity and quality of the raw materials and components and whether they conform to the relevant product specifications.

The PPI includes:

- ✓ Factory's production lines & capability
- ✓ Factory's facilities & equipment
- ✓ Raw materials, main components & accessories
- ✓ Semi-finished samples
- ✓ Some finished samples



But also, a pre-production inspection can help you avoid delays in your supply chain and the delivery of a production. After all, no one likes missing a deadline. Product timelines and shipping dates are as important as the product quality and communication with your supplier is key. A pre-production inspection helps you to identify issues at an early stage before it becomes too late.

During Production Inspection (DPI)

During production inspections are also sometimes known as DUPRO and these are conducted while the production is in progress. They are meant for products that are in continuous production and have strict requirements for on-time shipments and as a follow-up when quality issues are found prior to manufacturing during pre-production inspection.

During production inspection (DPI)

- Conducted while production is underway.
- Useful for products that are in continuous production that have strict requirements and when quality issues have been found prior to manufacturing during an earlier PPI.

The DPI includes:

- ✓ The production status during the manufacturing process
- ✓ Production line evaluation and timeline verification
- ✓ Random sampling of semi-finished and finished products
- ✓ Verify package and packaging material details
- ✓ Overall assessment and recommendations



Only 10-15% of units are completed when this inspection is carried out giving you a good idea of the direction of your product at an early stage. Once problems have been rectified, checks are made again to ensure everything is suitable and as it should be prior

to pre-shipment inspection. Inspectors will produce a full report together with supporting imagery to provide you with comprehensive information and data.

Pre-Shipment Inspection (PSI)

In short, the meaning of a pre-shipment inspection is that it is a method for checking the quality of goods before they are shipped. This type of inspection's purpose is to check that a production aligns with the specifications of the buyer or purchase order/letter of credit. But at what point is a pre-shipment inspection conducted exactly?

Pre-shipment inspection (PSI)

- Conducted on finished products when at least 80% of the order has been packed for shipping.
- Random samples are selected and inspected for defects against the relevant standards and procedures.

The PSI includes:

✓ Quantity verification	✓ Size specification (if required)
✓ Style and color	✓ Package details
✓ Functions and safety	✓ Shipping marks
✓ Workmanship	

A woman in a red polo shirt is standing in a warehouse, pointing at a cardboard box. She is holding a black folder or clipboard. The warehouse is filled with stacks of cardboard boxes.

They are carried out on finished products when at least 80% of the order has been packed to ship. Samples are selected at random to take part and the inspection is done according to standard acceptable quality limits or customer requirements. PSIs are especially good for overseas buyers to stay on top of differing national standards and requirements and an increase in fraudulent trade conduct.

Piece by Piece Inspection (or Sorting Inspection)

This kind of inspection involves thoroughly checking each and every item to evaluate a range of variables. These can be things such as general appearance, workmanship, function, safety, etc. and inspections can be done pre- or post-packaging. The Piece by Piece inspections are carried out on 100% of goods. Upon completion, all products which pass inspection are then sealed and certified with an HQTs sticker to ensure that every piece included in the shipment meets your specified quality requirements.

Piece by Piece Inspection



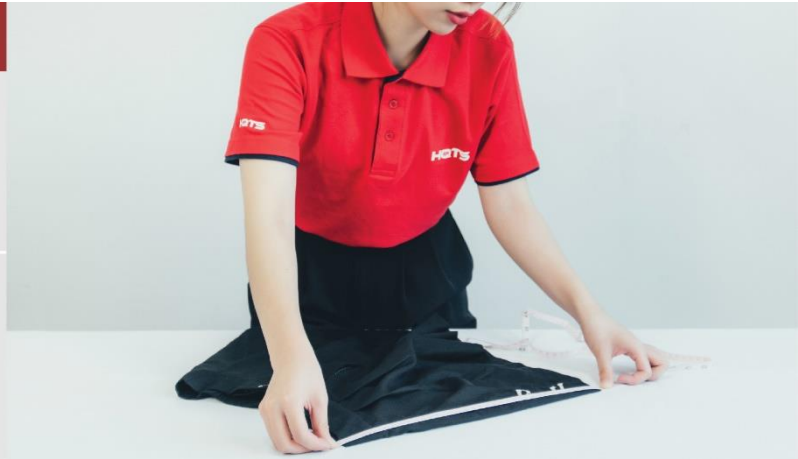
Conducted in the circumstance where the goods require particular attention to ensure compliance to specification or when the goods are high-value



Help to check each and every item to evaluate a range of variables; can be carried out either before or after the packaging inspection

The Piece by Piece includes:

- ✓ General appearance
- ✓ Workmanship
- ✓ Function
- ✓ Safety.



Metal Detection

Needles being left in garments can be a serious issue if proper metal detection procedures aren't carried out. That's why metal detection is an essential quality assurance requirement for the garment industry. Sewn items could have needle fragments or undesirable metallic substances embedded.

Textile accessories during the manufacturing and sewing process could be left in fabrics that may cause injury or harm to end consumers. The application of metal detection and X-ray detection systems are deployed at various points during the manufacturing and sewing process, to ensure detection at all potential stages of the process.

Loading and unloading supervision

This kind of inspection is the process of monitoring the entire loading and unloading of goods to ensure your products are handled professionally. Supervision of the whole container loading and unloading process to your desired location can be arranged, wherever you're shipping to and from.

Loading supervision (LS)



Container loading and unloading inspections ensure your products are loaded and unloaded correctly. The whole process will be supervised to guarantee product's safe arrival to their final destination. The procedure:

- ✓ Record the weather/arrival time of container/container No.
- ✓ Check the inner and outer condition of container
- ✓ Check the quantity of goods loaded and record the condition of outer packaging (master cartons/pallets)
- ✓ Randomly select and open sample cartons to verify compliance with customer's specifications
- ✓ Supervise the loading process to minimize breakage and maximize space utilization
- ✓ Seal the container
- ✓ Record the seal Nos. and departure time of containers



The inspection and supervision process includes an evaluation of the condition of the shipping container, verification of product information; quantities loaded and unloaded, packaging compliance and overall supervision of the loading and unloading process. It's an inspection service that's signed, sealed and delivered with quality assurance guaranteed.

Quality inspection benefits

Not only do quality inspections allow you to keep an eye on what's going on at the factory when you're not there, they also ensure your products and shipment is being prepared correctly. Poor quality can naturally be detrimental to your bottom line and brand reputation.

That's why it makes sense to increase visibility within your supply chain and ensure things run as smoothly as they can. Your duty is to keep your promise of a quality product to your customers and buyers which means inspecting your products and processes for potential flaws. Quality inspections reduce the risk of you.

5 Ways to Address Product Defects Discovered During a Quality Control Inspection

What is a quality control inspection?

A quality control inspection relates to measuring, examining or testing one or several product characteristics and comparing the results with predefined quality requirements to establish whether product compliance is achieved for each characteristic.

An individual called an inspector performs the quality control inspection. They follow a pre-established checklist based on the product specifications. Inspected products can be anything, from the components used to produce the end-product to semi-finished goods to finished goods ready for shipment to a customer. Quality inspectors rely on different types of tools to perform their tasks. For instance, they use hand-held measurement devices and electronic inspection equipment. At the end of the quality control inspection, they record the results and prepare test reports.

How to address product defects found during a quality control inspection?

The following sections below describe five approaches you can use to address unacceptable product defects in a shipment.

#1. Rework all affected products

The shipment you received doesn't pass the quality control inspection? In such a case, one of the most common measures you can take is to ask your supplier to rework the unacceptable defects in the affected products. By doing so, you will limit the number of unsellable units shipped.

Note that product rework is appropriate when the defects can be easily fixed in a short amount of time and require minimal effort. For instance, some product units have the wrong barcodes printed on the label, or the CE marking symbol doesn't have the correct dimensions. Fixing such defects can improve the quality of the affected products to an acceptable level, typically without increasing your costs or causing additional quality problems.

However, product rework is not always possible. Some types of quality defects require a different approach. For instance, there is excess glue on a toy, which may be very difficult to remove. If the factory workers of your supplier try to scrape it off, they may leave scuff marks or other abrasions on the toy's surface. Hence, affecting the salability of the product.

A good idea is always to conduct quality control inspection and product rework before shipping so you can save time and money over waiting to address product defects after you receive the order. Moreover, by sending back defective product units to your supplier, you will need to pay expensive value-added taxes and customs duties. In Addition, import restrictions may even prohibit faulty goods from entering the supplier's country. And don't even consider reworking the goods in your home country as you'll face considerably higher labour costs.

#2. Re-inspection of the defective products

Once the defective goods are reworked or replaced, you must do a re-inspection to confirm that your supplier's factory has indeed taken corrective actions and the reworked product units now fulfil your quality standards. Usually, it's done at the supplier's expense.

#3. Chargebacks or other penalties

As we said above, rework is not always a solution for you. Sometimes, you may need to take a different approach to address quality defects. Such an approach can include chargebacks or other penalties, typically set out in the agreement with your supplier. For instance, the contract may state that the supplier will bear any costs for re-inspections and pay the penalty for any substandard units shipped over a certain quantity.

Having a prior thorough agreement with your supplier will incentivize their factory early on to meet your expectations and quality standards, helping hold them accountable for product defects and other issues.

#4. Product destruction for unsellable units

If the product defects can't be corrected and render your goods unsellable, you may need to destroy all the affected units. However, carefully consider all your options before resorting to product destruction. Such an approach is appropriate if you want to protect your brand from substandard, counterfeit goods or prevent your competitors and resellers from buying and selling defective goods.

Imagine that your supplier's factory stitches your logo incorrectly on several product units. The products are now unsellable in your intended market. But you undoubtedly don't want your supplier to sell the defective units because your logo is already on them. Thus, your best course of action is product destruction. It would help if you had factory staff overseeing the destruction process to ensure all brand information is no longer legible and all defective units are entirely destroyed.

#5. Quality control inspection checklist

Ensure you update your quality control checklist with the latest inspection results to prevent the recurrence of the same types of product defects in the future. For instance, add new and relevant information such as adjustments to product dimensions, new or updated quality standards criteria, changes to product components, etc.

Providing your supplier with an up-to-date quality control checklist will help them understand your expectations better and alert them to specific defects or other issues they must take steps to prevent. By doing so, you will also improve your quality control inspector's efficiency in checking for the relevant quality issues. In Addition, you will also easily interpret future inspection reports and be sure that the inspector attends to all your quality concerns.

SESSION 2.	
Test electronic equipment.	
Learning Outcomes	
• Reasons for testing electronic equipment are discussed in line with organisational and operational requirements. • Ability to use the testing equipment is demonstrated to obtain the required results. • The equipment is tested to identify faulty or malfunctioning components. • Effects of the environment are discussed in terms of their impact on the accuracy of the results.	

Test electronic equipment.

5 good reasons to have your product tested for electrical safety:

Here are five important reasons to have your product tested for electrical safety before it goes to market.

1. Electrical safety testing is legally required for CE marking

Fulfilling the requirements of the relevant directives is a legal requirement to market products in the EU. Corresponding regulations apply outside the EU. Documentation that these requirements have been met must be kept in a technical file and made accessible for inspection during the period in which the product is marketed plus ten additional years. This technical file is required in order to prepare an EU Declaration of Conformity and to have your product CE marked.

Remember that there is always a legal obligation to ensure that your product is safe before bringing it to market. The legal requirements for electrical safety may be determined by the Low Voltage Directive, the Medical Device Directive (soon The Medical Device Regulation), the Radio Equipment Directive, the Machinery Directive (soon the Machinery Regulation), or other relevant directives. If a product is not already covered by other directives, the General Product Safety Directive (GPSD, soon GPSR) serves as a catch-all.

A common misconception is that there are no safety requirements for electronic products as long as the voltage is low enough. Indeed, some low-voltage products are

not covered by the Low Voltage Directive, but in those cases, another applicable directive will still impose electrical safety requirements. Dangerous products may not be sold.

2. It minimises the risk of damage or injury when using electric or electronic products

An electrical safety test provides insight into whether or not one's product is safely designed, such that one can avoid or reduce the risk of the product being the cause of damage or injury to people, pets and property. The product must therefore be designed and produced in such a way that a potentially dangerous condition, in which an internal or external event may result in damage - that is, a hazard - does not arise.

Safety standards address hazards that may lead to damage or injury as a result of electrical shock, energy discharge, fire, heat, mechanical conditions (e.g., sharp edges, crushing, moving parts), radiation (e.g. RF, IR, UV, laser, sound, radioactivity), or chemical conditions (e.g. toxic materials, liquids, smoke, vapour).

3. Accredited electrical safety tests lead to greater safety

If a test is accredited, that means that the testing facility is subject to quality inspections (audits) from an authority and/or professional organisation (for FORCE Technology, these are DANAK and IECEE CB Scheme). The auditor verifies that a test is performed in a methodically correct manner, and that personnel are trained to perform the test, which includes ensuring that they are capable of making the many technical assessments that are part of the standards in use.

An accredited and certified test report from a testing facility like FORCE Technology is the best proof of proper, independent testing, and is therefore more useful to you than unaccredited test reports.

As a manufacturer, an [accredited test report](#) gives you a high degree of confidence that the report will be accepted by authorities in the various EU countries. Both in and outside the EU, many countries and regions require a CB Scheme certified report. With a report certified in accordance with the CB Scheme, a manufacturer can gain access to the global market.

4. It leads to improved technical documentation

The results of an [electrical safety test](#) are recorded in a test report, which is collected in the product's technical file. The technical file must be made available for inspection by authorities while the product is available for sale, and for ten additional years after that.

A significant part of an electrical safety test is a design documentation review. Often, far less than half of the testing period is spent actually performing tests. Most of the time is spent examining the product and its documentation.

For this reason, manufacturers virtually always find that their technical documentation is significantly improved when they perform electrical safety tests. This gives you the confidence that, upon request from national authorities and in the event that any damage or injury should occur, you have taken the appropriate steps to make your product safe and secure.

5. It leads to design improvements

In the course of electrical safety testing, many businesses find that they develop new, detailed technical knowledge of their own product. Initially, the manufacturer simply becomes certain of how to make the product safe. Gradually, however, as the approval project continues on, the product developer often gains insight into their own product, creating opportunities to optimise the product's design. In this manner, a foundation for the growth of new ideas and general improvements is created, while the product is made legal and safe.



Electrical Safety: Standards and Basic Testing



Electrical safety standards

To help verify the functionality and safety of medical devices, electrical safety standards have been established in the United States, European countries, and other parts of the world. The standards differ in criteria, measurements, and protocol. The International Organization for Standardization (ISO) and the International Electro-technical Commission (IEC) organizations based in Europe provide standards worldwide in partnership with the World Trade Organization. These include standards for electro-medical equipment. There are general and specific standards for medical device electrical safety. IEC60601 AAMI/NFPA 99 The primary standard for medical devices is IEC 60601. General requirements for protection against electric shock hazards are covered in IEC 60601.1, Section 3.

In this standard, each instrument has a class:

- Class I—Live part covered by basic insulation and protective earth
- Class II—Live part covered by double or reinforced insulation
- Class IP—Internal power supply

Each patient applied part or patient lead has a type:

- Type B—Patient applied part earthed

- Type BF—Patient applied part floating (surface conductor)
- Type CF—Patient applied part floating for use in direct contact with the heart

Leakage measurement limits have been developed for equipment types and measurements. They include:

- NC—normal conditions
- SFC—single fault conditions

The terminology used in IEC 60601.1 3rd Edition includes:

- Protective earth resistance
- Earth leakage current
- Touch current (formerly enclosure leakage current)
- Patient leakage current
- Patient auxiliary current
- Mains on applied part (MAP)

Leakage current (μA)		Earth leakage current mA	Touch current (μA)	Patient leakage current AC (μA)	Patient leakage current DC (μA)	Patient leakage current mains on applied (μA)	Patient auxiliary current (μA)	Patient auxiliary current (μA)	Patient auxiliary current (μA)
Type B	NC	5	100	100	10	—	100	10	100
	SFC	10	500	500	50	—	500	50	500
Type BF	NC	5	100	100	10	—	100	10	100
	SFC	10	500	500	50	5000	500	50	500
Type CF	NC	5	100	10	10	—	10	10	10
	SFC	10	500	50	50	50	50	50	50

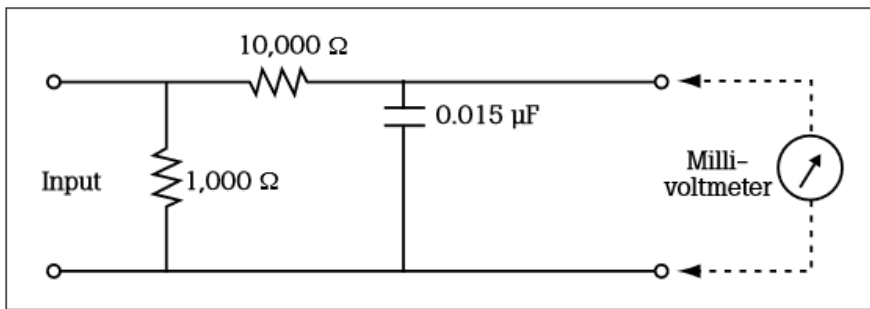


Figure 3. Impedance of a patient test load.

The figure above represents the impedance of a patient test load. Leakage current measuring devices use this impedance circuit for measurements.

Additional important points regarding IEC 60601.1 include:

- The use of up to 25 amperes ac for protective earth testing (this is a type-test and generally suitable for manufacturers)
- Leakage current is measured at 100 percent of mains voltage
- Performance of dielectric strength/insulation testing is measured at 110 percent of mains voltage.

A new IEC standard, IEC 62353, is used for medical device testing in hospitals. IEC 62353 was developed because IEC 60601.1 is a type-testing standard with no risk management criteria and is impractical for testing in the hospital environment.

IEC 62353 tests are performed on equipment prior to use on patients, during schedule periodic testing, and after repair. Thus, this standard is for field (hospital) testing and does not address equipment design. In Annex E of the document, the manufacturer is requested to provide information on testing interval and procedure based on risk, typical usage, and device history. The minimum testing requirement for life support and other critical equipment is every 24 months.

In the United States, there are several primary and secondary organizations setting standards:

1. National Fire Protection Association (NFPA): NFPA 99 Standard for Health Care Facilities is the primary standard addressing electrical safety testing required in health care institutions. Other publications include NFPA 70, National Electrical Code, and NFPA 70E, Electrical Safety in the Workplace.

2. Association for the Advancement of Medical Instrumentation (AAMI): ANSI/AAMI ES1 Safe Current Limits for Electro-medical Apparatus is another commonly-accepted standard.
3. Underwriters Laboratories (UL): UL544, Medical Equipment requirements is a standard for manufacturers, not hospitals. These standards may be referenced by accreditation, code or regulatory organizations such as the Joint Commission, Occupational Health and Safety Administration or other organizations monitoring health care institutions in the United States.
4. Canadian Standards Association (CSA): CAN/CSA C22.2 NO. 60601-1-08 Medical Electrical Equipment Part 1: General Requirements for Basic Safety and Essential Performance (Adopted IEC 60601-1:2005, third edition, 2005-12)

Global harmonization of standards has led to the development of worldwide standards. Equipment in the regions listed below must be certified to the IEC60601-1 standard or the device cannot be sold in that country.

- USA uses UL2601-1 or ANSI/AAMI ES601
- Europe uses EN60601-1
- Canada uses CAN/CSA-C22.2 No. 601.1-M90

IEC60601	AAMI/NFPA 99
Protective Earth Resistance	Ground Wire Resistance
Earth Leakage Current	Ground Wire Leakage Current
Touch or Enclosure Leakage Current	Chassis Leakage Current
Patient Leakage Current	Lead to Ground Leakage Current
Patient Auxiliary Leakage Current	Lead to Lead Leakage Current
Mains on Applied Part (MAP) Leakage Current	Isolation Leakage Current

Electrical safety testing

Testing requirements and sequence according to IEC 62353 Annex C are shown below. Only measurement equipment that meets IEC 61010-1 should be used. The sequence outlined in the figure below should be followed. For example, protective earth resistance should be measured prior to leakage current measurements

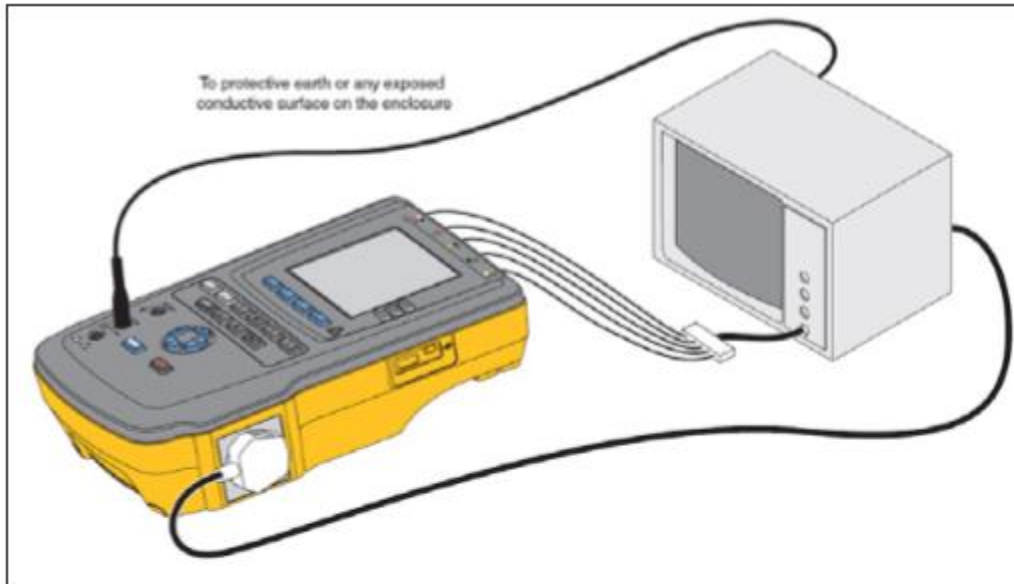


Figure 5. General connections to an electrical safety analyzer.

General connections to an electrical safety analyzer (ESA) are shown in Figure 5. Consult the operational manual for specifics for your **electrical safety analyzer**. Documentation requirements for IEC 62353 include:

- Identification of the testing group (hospital department, independent service organization, manufacturer)
- Names of the person(s), who performed the testing and evaluation(s)
- Identification of the equipment/system (e.g. type, serial number, inventory number) and the accessories tested
- Tests and measurements
- Date, type, and outcome/results of:
 - Visual inspections
 - Measurements (measured values, measuring method, measuring equipment)
 - Functional testing
- Concluding evaluation
- Date and signature of the individual who performed the evaluation.

Computerized record-keeping systems are greatly preferred for data storage, search, review, and analysis. Note the device fields must be standardized.

The ESA609 integrates all functions needed to test medical devices when patient lead



testing is not required, including:

- Line (mains) voltage
- Ground Wire (or protective earth) resistance
- Equipment current
- Ground wire (earth) leakage
- Chassis (enclosure) leakage
- Direct equipment leakage
- Point to point leakage and resistance

How do you make sure a piece of electrical equipment is safe?

What tests need to be done?

The correct tests need to be carried out on your Class I or Class II appliance in the following order:

- Visual Inspection
- Earth Continuity Test
- Insulation Resistance Test

- Leakage Test
- Further tests have to be performed on leads, RCDs etc

Firstly, before any testing can be carried out, decide what type of equipment it is eg CLASS I (earthed) or CLASS II (double insulated).

Let's look at what these words (together with a few other common terms) mean:

Class I Equipment

Safety Class I electrical equipment is electrical equipment that relies on the Protective Earth and basic or single insulation for its safety - such items usually have quite a lot of exposed metal and are often quite heavy, although this is not always the case. Some examples of Class I equipment could be dishwashers/washing machines/irons/fridges/electrical machinery/drill presses/saw benches/bench grinder etc.

- This type of appliance is often referred to as an 'earthed' appliance

Class II Equipment

Safety Class II electrical equipment relies on two layers of insulation, many times reinforced insulation will replace the two layers of insulation, but the reinforced insulation will be equivalent to the two layers of insulation.

- This type of appliance is often referred to as a 'double insulated' appliance
- It will normally be marked with the double box symbol



RCD (Residual Current Device)

An RCD is used for increasing electrical safety and is designed so that it will disconnect the power when the residual current reaches or nears a preset value. For RCDs designed to protect people this figure must be set at 30 milliamps or less.

Isolation Transformer

This is used for increasing electrical safety and is designed so that the two sets of windings (the primary and secondary) are separated from each other by at least the equivalent of that required to achieve double insulation, thereby ensuring that the electricity from the secondary (output) side is isolated from earth.

EPOD (Electrical Portable Outlet Device)

This is really a multiple outlet device and will commonly be called a multibox in New Zealand or a power board in Australia.

How to carry out the testing:

The Visual Inspection (this test is carried out on all types of equipment)

This is very important as approximately 90% of faults can be found by this inspection alone. The Visual Inspection is not just an inspection, it basically is **'everything but'** what is done during the actual electrical testing process. This part of the testing process may involve:

- Smelling
- Pulling on leads
- Tapping the appliance
- Shaking
- Turning some part/s
- Checking by trying to see that retractable guards actually retract... and all manner of other things may be done to ensure that it is safe to operate.

Then depending on whether the equipment is Class I or Class II the following tests are done:

For Class I (earthed) equipment

- Earth Continuity Test - this may be done with an Ohm Meter or a Portable Appliance Tester (PAT). The Standard AS/NZS3760: 2010 suggests a test current for PATs of 100mA to 25 Amps.
- Insulation Test/Leakage Test - a leakage test shall be done on appliances that have internal switching that is only activated when the equipment under test is powered at operating voltage. For equipment that does not have internal switching as above then a 500 Volt insulation test is an acceptable alternative.

For Class II (double insulated) equipment

- Insulation Test/Leakage Test - a leakage test shall be done on appliances that have internal switching that is only activated when the equipment under test is powered at operating voltage. For equipment that does not have internal switching as above then a 500 Volt insulation test is an acceptable alternative.

For Cord Extension Sets

- Earth Continuity Test
- Insulation Test/Leakage Test

Cord extension sets with re-wireable plugs fitted need to also have, in addition to the above tests, a

- Polarity Check

For RCDs

- Earth Continuity Test
- Insulation Test/Leakage Test
- Trip Time Test - in addition to the above tests RCDs shall have a trip time test performed on them at the time period specified in Table 4 AS/NZS3760: 2010

Electronics Testing Guide

We all own some form of electronics as they play a huge role in modern society, from information processing, telecommunication, and signal processing to the development and distribution of the very device you're reading this guide on.

Electronic devices and the components that they are comprised of have a profound effect on our lives, which is why electronics must pass stringent tests when they are produced and brought to market for functional, legal and safety purposes. This is also true for the lifecycle of an electronic product as they can gradually degrade over time, leading to failures, lower performance, and safety issues.

Where to Start with Electronics Testing?



There is a multitude of testing equipment when it comes to electronic devices and equipment, all of which are designed to ensure optimum operation and safety in any environment, from domestic dwellings to industrial warehouses.

This guide will not only tell you why electronics testing needs to be conducted but will also help you develop knowledge in the following areas:

- Phases of electronics testing
- Equipment needed for electronics testing
- Different types of electronic tests
- Strategies for testing electronics

What is Electronic Testing Equipment Used for?

The most common applications of electronic testing equipment include:

- Testing performance of the circuit and possible defects during the research & development (R&D) phase
- Identification of component level defects, faulty circuits, overloading connections, and high resistances during production testing
- Checking the overall performance of the electronic device and verifying it against the design specifications
- Verification of basic electronic parameters such as voltage, current, resistance, conductance, and capacitance. This could be applicable both in the design phase and regular maintenance of the device (if a measurement instrument)

The testing of electronics is segmented into phases before distribution and usage. This is done to troubleshoot any potential problems and eliminate them, so the product can operate at an optimum level.

If you design or manufacture any type of electronics equipment, the appropriate electronics testing procedures must be completed at the different stages of the product lifecycle, including:

- R&D testing of a proof of concept
- Design - testing a finalised design against specification
- Manufacturing or production test of each unit produced or samples thereof
- Installation and commissioning testing for specialised equipment
- Periodic calibration - if measurement instrumentation

- Troubleshooting - when equipment fails
- Periodic safety testing of hardware

Certification Testing

Once the respective stages have been completed successfully, the DUT (Device Under Test) will be suitable for use. However, it is also important to note that electronic devices must continue to be maintained under manufacturer specifications, per legislation, and in correspondence with regulatory guidelines.

Equipment Needed for Electronics Testing

So now you know why and when you need to test electronic equipment, it is important to know how to test electronic equipment and the various types of equipment used in electronics testing. This guide will cover the essential testing equipment required for different stages in the product lifecycle, from small-scale component testing to large-scale production testing.

Components Testing

The testing of electronic components is undeniably important, ensuring the circuit is built as safely as possible before being installed in a product. Considering the millions of electronic components produced each year, manufacturers can't comprehensively test every single component off the production line. Statistical process control and random sampling ensure their components are manufactured to their required specifications, focussing on four areas:

- Components that will fail early in their product life before they are added to a PCB assembly (PCBA) are identified through Highly Accelerated Life Testing (HALT) and Stress Screening (HASS). This can include vibration and shock testing, electrical stress testing, and burn-in testing
- Electrical testing verifies the components meet the rated electrical specifications as well as ensuring a PCBA meets basic electrical requirements. Automated testing equipment can check for electrical connectivity in the bare PCB and finished PCBA at a minimum. The designer needs to specify test data to the manufacturer for this stage to be carried out

- PCBAs and batches of components are put through reliability testing that tests their ability to withstand extreme mechanical loads or environments
- Functional testing is normally carried out by test engineers once a prototype is fabricated, at which point the system is tested against the predetermined performance requirements. In this case, the individual components are only tested if the board fails, and the root cause needs to be narrowed down to a specific component

Common Components Testing Equipment

Common component testing equipment includes:

- Component and IC testers – Integrated Circuit (IC) testers are battery-powered portable devices ideal for testing components such as batteries, diodes, LEDs, SCRs, and transistors. They can measure parameters such as voltage, resistance, capacitance, continuity and more
- Decade Boxes – used to substitute standard value components in a circuit, decade boxes consist of a series of resistors, capacitors, and inductors to replicate different electrical values. They allow the accuracy of test equipment to be verified before use, as well as being used for troubleshooting purposes
- Ohmmeters – Ohmmeters are used to measure the electrical resistance and continuity of an electrical circuit and its components. By passing a small current through the sample, the instrument can measure the resistance by the voltage drop that is produced when the current passes through said sample

Electronics Test Equipment

The comprehensive testing of an electronics circuit requires a variable number of different test equipment devices depending on the nature of the application. This is to ensure the DUT is adequately powered, fed with representative signals, and that outputs are assessed against the specification. This is also to enable the test engineer to gain an understanding of how the whole circuit is behaving by being able to monitor (and potentially decode signals within) any part of the circuit.

These devices can often be remote controlled and operated by test scripts to facilitate high-speed production testing.

Common test equipment for use throughout the various stages of a design's life includes:



Bench Power Supplies

To generate accurate and fully controllable DC supply at the required voltage and current.



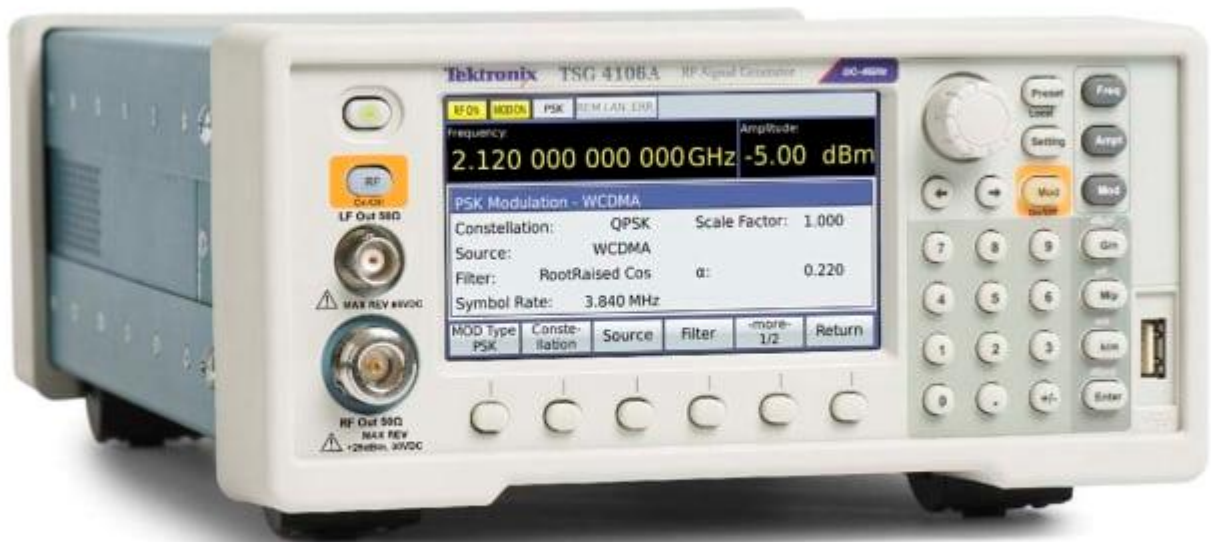
Oscilloscopes

Capture, process, display and analyse detailed information about electronic signal waveforms.



Function Generators

Produce waveforms including sine, square, triangular and sawtooth to feed into the DUT.



Signal Generators

Designed to provide highly accurate and stable signals at a given frequency and amplitude, with or without modulation applied.



Spectrum Analysers

Enable the analysis of the output of the DUT across the frequency spectrum.



Multimeters

Whether high accuracy bench digital multimeters (DMMs) or hand-held, these are effective for specific measurements of voltage, current, resistance, even sometimes capacitance and frequency. Bench DMMs can also support high numbers of measurements per second, perfect for production test situations.



Electronic Loads

Simulates an electrical load to test the power output of a DUT, whether that be for RF or high-power applications such as motors.



Logic Analysers

Allow investigation into digital signals, including timing issues, protocol decoding and related problems. This capability is now frequently available on Mixed Signal Oscilloscopes (MSO).



Data Acquisition

Monitor the DUT for extended periods, and capture key variables such as voltage, current, temperature, pressure or sound for detailed analysis.

EMC Testing

Electromagnetic Compatibility tests (EMCs) ensure that the equipment or systems can function satisfactorily in their environments without causing intolerable electromagnetic disturbances to anything else in the area. These tests also ensure that the equipment will not be adversely affected by interference from other devices or naturally occurring sources of interference (immunity testing).

EMCs are mandatory in most markets, including Europe and the United Kingdom, the United States, China, Korea, Australia, and New Zealand.

Depending on the product or device under test, there are different tests required to achieve EMC certification – called a Declaration of Conformity – including but not limited to:

- Antenna input immunity
- Antenna screening effectiveness
- Conducted disturbances immunity
- Conducted emissions (continuous and discontinuous interference)
- Electrostatic discharge (ESD) immunity
- Electrical fast transient/burst immunity
- Harmonic current emission
- Immunity from radiated fields
- Immunity to induced voltages
- On-site EMC testing
- Power interference
- Power supply conditioning
- Radiated emissions
- Radiated field immunity
- Voltage fluctuation/flicker
- Voltage surge immunity
- Voltage dips/interruptions immunity
- Magnetic field measurement (EMF)
- Electric field measurement (EMF)
- Electromagnetic field measurement (EMF)
- Induced current density due to the electric field (EMF for lightning equipment)

ESD Testing

Electrostatic Discharge (ESD) testing falls under the EMC umbrella but is perhaps one of the most well-known EMC tests and is used across a multitude of industries. ESD can occur between two electrically charged objects wherein electricity is suddenly transferred between them, an event that can be caused by direct contact, an electrical short, or a dielectric breakdown. While this discharge or visual spark is not lethal on its own, it can render devices and equipment dysfunctional, leading to replacements or repairs.

ESD events can be extremely harmful when they occur in the presence of electronics as they can cause immediate failure or slowly damage the components over time. While this is a gradual process, the latent damage will eventually lead to the component failing, causing costly replacements that can also cause downtime in the wider system.

By simulating various electrostatic impacts that may occur during transit or operation, ESD testing determines whether a product complies with the requirements of its ESD protective area and procedures. These tests can be carried out in one of two ways - either the air method or the contact method:

- The air method involves positioning a charged generator or discharge object (an ESD gun, for example) towards the device under test. Variables such as humidity and temperature can impact ESD testing, with ESD events more commonly occurring in low temperature and low humidity environments
- Contact testing eliminates the environmental variables that can affect the air method. Using this method, the ESD generator is in direct contact with the device under test and is kept in contact throughout the entire duration of the test. While this test method is typically consistent across different environmental conditions, contact ESD rarely occurs naturally and therefore the results may be considered unrealistic

Portable Appliance Testing

While tests of an electronic circuit are primarily focused on operation to specification, ensuring the complete product (including mains voltage electrical circuits) is safe for users is another key consideration.

The design must meet the relevant electrical standards for the country the product is to be supplied to and should be tested to these regulations before shipping using appropriate test equipment. This includes earth bond testers, insulation testers, and similar.

When the equipment is in use, organisations and individuals, such as employers and landlords, are legally required to create a safe working environment and PAT testing is a great way to mitigate risk.



PAT Testing

Portable Appliance Testing, or PAT Testing for short, is used to check how safe electrical appliances are for use through visual and electronic tests.

A competent person with suitable knowledge of electronics must carry out PAT Testing to be compliant with legislation such as:

- Health and Safety at Work Act 1974

- Electricity at Work Regulations 1989
- Provision and Use of Work Equipment regulations 1998
- Management of Health and Safety at Work regulations 1999

You will need to use the correct PAT testing equipment appropriate for the products under test. After the appropriate person has completed the testing and the products have passed, they will issue a PAT certificate signifying that the appropriate standards have been met and the product is safe for use.

What Do You Need to PAT Test?

There are 7 categories of appliances that require PAT testing:

- Fixed appliances
- Stationary appliances
- IT appliances
- Moveable appliances
- Portable appliances
- Cables and chargers
- Handheld appliances

Electrical appliances are mainly categorised as Class 1, 2 or 3, with Class 1 being the most dangerous if unsafe and Class 3 the least dangerous if unsafe. The class of an appliance helps determine the level of PAT testing required for the equipment so it can be used safely. Class 1 appliances need a full PAT test, Class 2 appliances need a PAT insulation test, and Class 3 appliances don't need to be PAT tested at all.

Class 1 appliances are typically made of metal, have three cables, a metal Earth pin, and a fuse in the plug. They include small and large white goods, such as microwaves, kettles, toasters, refrigerators, and microwaves.

Class 2 appliances include additional insulation and therefore do not rely on a metal Earth pin for protection, with items such as hairdryers, televisions, computers, and

photocopiers falling under this category. However, the safest way to ensure the class of your products is to look for the Class 2 appliance symbol.

Finally, **Class 3** items are low voltage appliances and are considered the safest of electrical appliances, such as laptop computers, mobile phones, and low energy lightbulbs. While the items themselves may not require PAT testing, their charging leads and plugs may require testing.

Calibration of Electronic Measurement Devices

Electrical calibration is a means of verifying or adjusting the performance of any instrument that measures or tests electrical parameters such as voltage, current, resistance, inductance, capacitance, time, and frequency.

Using precise devices that evaluate the performance of these key properties, electrical calibration can allow users to carry out performance evaluations and calibration adjustments where required, allowing for potential errors to be identified or minimised. There are two broad categories for these precision devices - calibrators/standards, and precision digital multimeters.

- Calibrators – calibrators are usually able to provide a wide range of precision output signals, such as voltage settings ranging from as little as a few microvolts to a maximum of about one kilovolt on most models. Modern calibrators commonly provide outputs for various electrical functions, including voltage, resistance, and current
- Standards – considered to be even more precise than calibrators, standards are capable of performances at least four times better than calibrators, though this is usually limited to one electrical function with one or few output settings in comparison
- Precision digital multimeters (DMMs) - DMMs provide excellent measurement performance of various electrical parameters including voltage, current, and resistance through several values. Frequency, capacitance, and other parameters can be included but are less commonly found on DMMs

How can environmental factors influence measurement accuracy?

Environmental factors can influence measurement accuracy by causing variations in the conditions under which measurements are taken.

Environmental factors such as temperature, humidity, pressure, light, and even magnetic fields can significantly influence the accuracy of measurements. For instance, in physics experiments, these factors can cause changes in the properties of the materials or equipment used, leading to variations in the results obtained.

Temperature is a key environmental factor that can affect measurement accuracy. For instance, the resistance of a conductor changes with temperature. Therefore, if a physics experiment involves measuring the resistance of a conductor, and the temperature changes during the experiment, the resistance value measured will also change, leading to inaccurate results. Similarly, the volume of a gas is directly proportional to its temperature. Hence, if the temperature changes during an experiment involving the measurement of gas volume, the results will be inaccurate.

Humidity can also influence measurement accuracy. For instance, in an experiment involving the use of electronic equipment, high humidity can cause condensation on the equipment, leading to short circuits and inaccurate measurements. On the other hand, low humidity can cause static electricity build-up, which can also affect the accuracy of measurements.

Pressure is another environmental factor that can affect measurement accuracy. For example, in an experiment involving the measurement of gas pressure, changes in atmospheric pressure can cause changes in the measured gas pressure, leading to inaccurate results.

Light can also influence measurement accuracy. For instance, in an experiment involving the use of a light meter, changes in ambient light can cause changes in the light meter readings, leading to inaccurate results. Similarly, in an experiment involving the use of a

microscope, changes in light intensity can affect the visibility of the specimen, leading to inaccurate measurements.

Magnetic fields can also affect measurement accuracy. For instance, in an experiment involving the use of a compass, changes in the Earth's magnetic field can cause changes in the compass readings, leading to inaccurate results.

Understanding the distinction between precision and accuracy is crucial in assessing the impact of environmental factors on measurements. Additionally, recognising and addressing systematic errors and random errors can further mitigate inaccuracies. Moreover, selecting the appropriate instruments for measurement is vital for enhancing the reliability of results amidst environmental challenges.

In conclusion, environmental factors can significantly influence the accuracy of measurements in physics experiments. Therefore, it is crucial to control these factors as much as possible to ensure accurate and reliable results.

SESSION 3.
Install electronic equipment.
Learning Outcomes
• Equipment for installation are identified in terms of their application. • The location of the installation is identified in terms of its suitability. • Ability to install is demonstrated in a specific working environment.

Install electronic equipment.

5 Simple Steps in the Electronic Assembly Manufacturing Process

Electronic Assembly

Electronics manufacturing refers to the process of designing, producing, and assembling electronic components and products, such as semiconductors, printed circuit boards, and consumer electronics like smartphones and computers. This industry involves technicians at various stages, including design, fabrication, assembly, quality control, and distribution of electronic devices.

1. Design

Design typically begins with the creation of a schematic diagram, illustrating the circuitry and connections between electronic components. This schematic guides the layout design, where the physical placement of components on the printed circuit board (PCB) is determined. Once the layout is finalized, a prototype is often built and tested to identify and address any issues. Following successful prototyping, the design is refined, and a detailed bill of materials (BOM) is created, specifying all necessary components. Lastly, the manufacturing files, including Gerber files, are prepped for PCB fabrication and assembly files for component placement.

2. Fabrication

Fabrication usually begins with the production of the printed circuit board (PCB), where a substrate material is coated with a layer of copper and etched to create the desired circuit pattern. Surface mount technology (SMT) is commonly employed for component placement, where automated machines precisely mount miniature components onto the PCB. Through-hole components, if used, are inserted manually or by automated insertion machines. Once components are placed, the assembly undergoes soldering, either through wave soldering for through-hole components or reflow soldering for SMT components. Inspection and testing are crucial stages to identify any defects or issues, ensuring the final product meets quality standards. The assembled electronics are then integrated into the final product, completing the fabrication process.

3. Assembly

The assembly phase of electronics assembly involves bringing together all the fabricated components to create the final electronic product. This phase of electronic assembly process begins with the careful placement of components onto the printed circuit board (PCB), whether through automated surface mount technology (SMT) processes or manual insertion for through-hole components. Following component placement, the assembly undergoes a soldering process, such as reflow soldering for SMT components or wave soldering for through-hole components, to securely attach the components to the PCB. Precision and consistency in this phase are critical to ensure proper electrical connections and the overall functionality of the electronic device. After soldering, the assembly is typically subjected to thorough inspection and testing to identify and rectify any defects. Once the quality checks are completed and the assembly is confirmed to meet specifications, it moves on to the final stages of product integration and packaging, marking the completion of the assembly phase in the electronics manufacturing process.

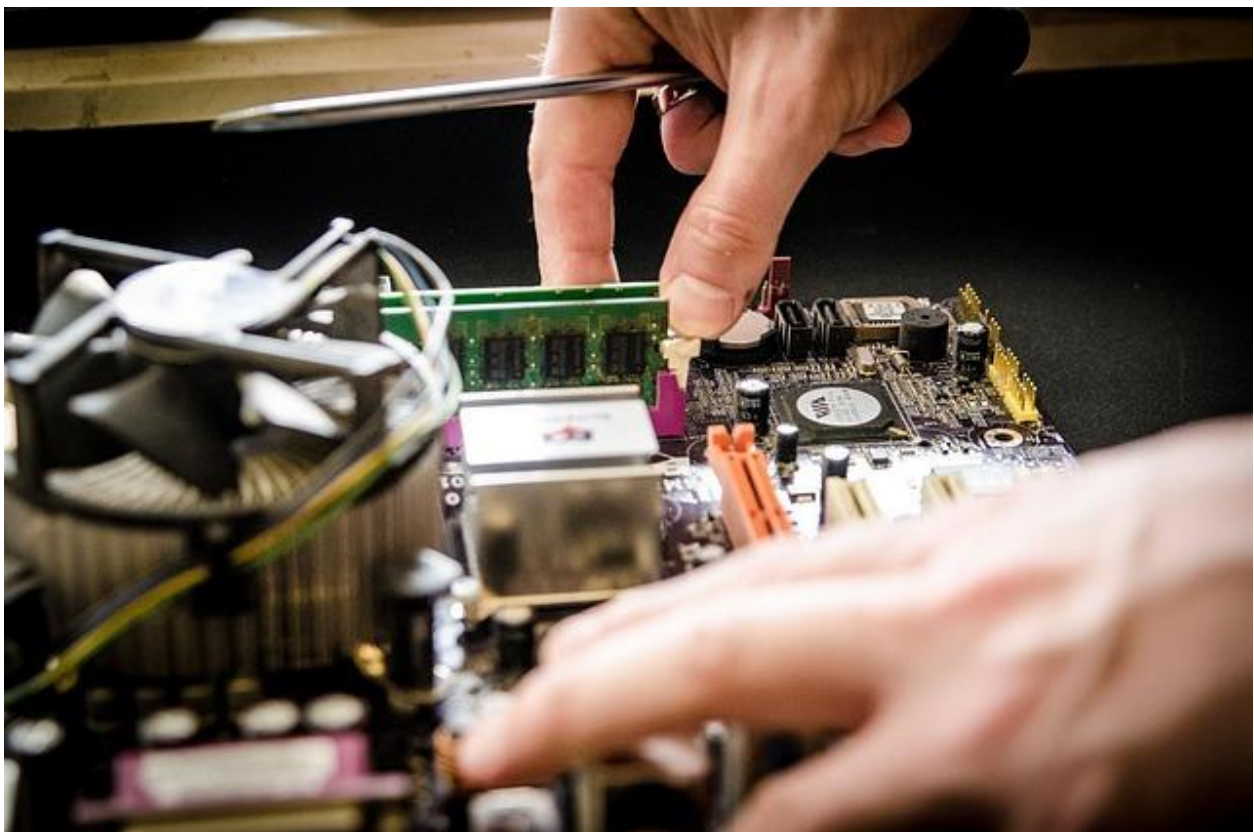
4. Quality Control

This critical stage is aimed at ensuring the reliability and functionality of the manufactured electronic devices. It involves a thorough inspection and testing process to identify any defects or deviations from design specifications. Automated inspection systems, such as optical inspection and X-ray imaging, are commonly employed to detect issues with solder joints, component placement, and overall PCB integrity. Additionally, functional testing is conducted to verify that the assembled electronics operate as intended. Quality control measures help identify and address any anomalies before the final product reaches the market, minimizing the risk of faulty or malfunctioning devices. Continuous improvement strategies may be implemented based on feedback from

quality control processes to enhance the overall manufacturing process and product quality.

5. Distribution

Involving the logistics and management processes that facilitate the movement of finished electronic products from the manufacturing facility to end-users or retailers, this phase includes packaging the assembled electronics securely for transportation, coordinating with shipping and distribution partners, and managing inventory to meet market demand. Distribution channels play a crucial role in delivering products efficiently and timely to the intended destinations. Additionally, this phase may involve tracking and traceability systems to monitor the movement of products through the supply chain. Effective distribution ensures that the electronics assembly reaches its target market in optimal condition, contributing to customer satisfaction and the success of the overall manufacturing and distribution process.



Electronic Components

Electronic components are fundamental parts used in assembling electronic devices. They include:

- **Semiconductors:** These are integrated circuits and transistors that control the flow of electrical signals.
- **Passive Components:** Resistors, capacitors, and inductors regulate and store electrical energy.
- **Connectors:** These facilitate the interconnection of components, such as USB ports or wire connectors.

- **Diodes:** Diodes control the direction of electrical current.
- **LEDs:** Light-emitting diodes emit light when current flows through them.
- **Sensors:** These detect and respond to various environmental conditions, like temperature or motion.
- **Microcontrollers and Microprocessors:** These are the brains of many electronic devices, executing instructions and processing data.
- **Power Supplies:** Components that provide and regulate electrical power to the device.

Different Ways of Circuit Board Assembly

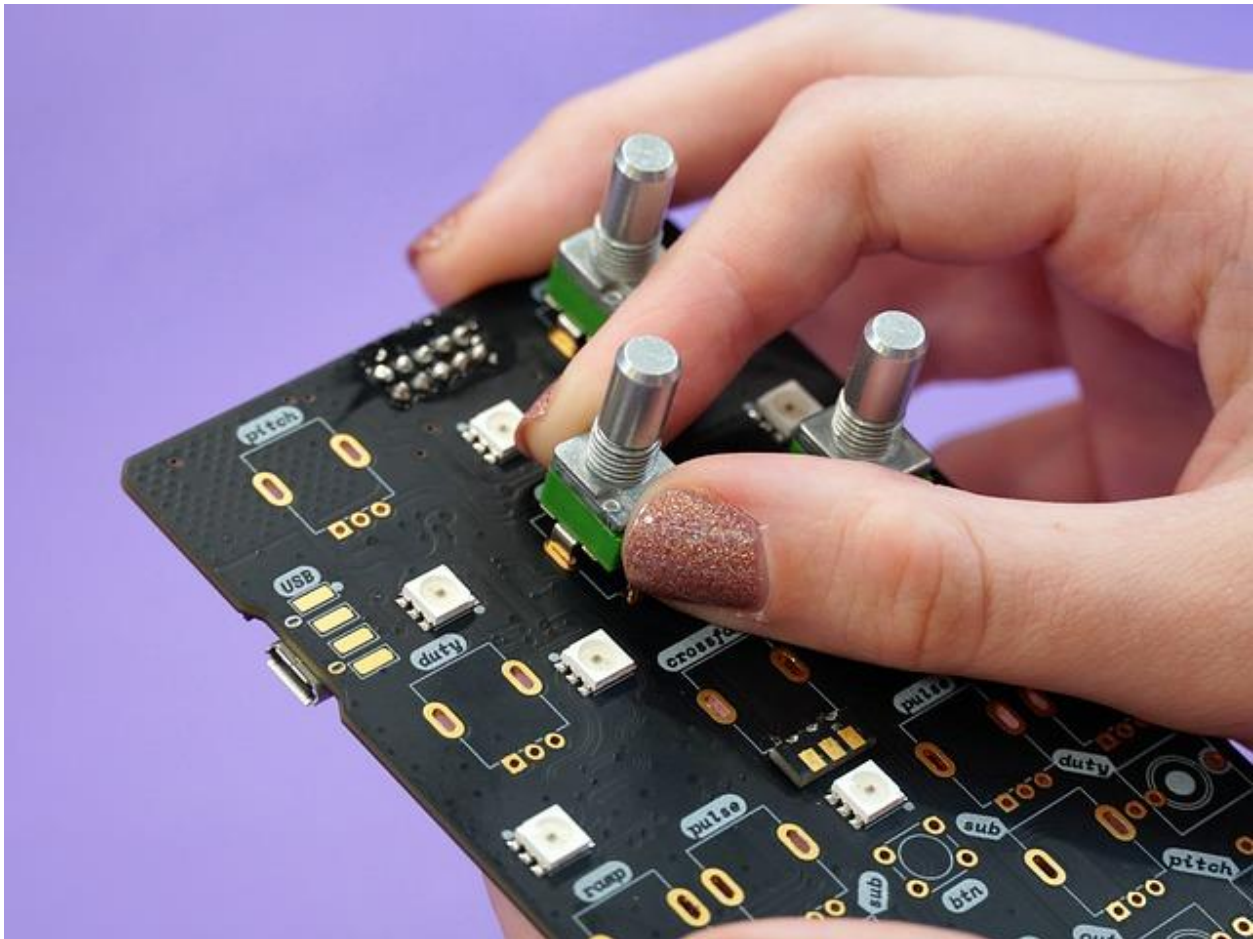
There are two primary test types of circuit board assembly: through-hole assembly and surface mount assembly.

- **Through-Hole Assembly (THA):** In through-hole assembly, components have leads that are inserted into holes on the printed circuit board (PCB). The leads are then soldered on the opposite side of the board to create the electrical connection. This method is often used for larger components and those that require extra mechanical strength, such as connectors and large capacitors.
- **Surface Mount Assembly (SMA):** Surface mount assembly involves mounting components directly onto the surface of the PCB. The components have small contacts or leads that are soldered directly onto pads on the board. This method is more space-efficient, allowing for smaller and lighter electronic devices. Surface mount technology (SMT) is widely used in modern electronics for components like resistors, capacitors, and integrated circuits.

In many cases, a combination of both through-hole and surface mount components is used, referred to as mixed-technology assembly, to leverage the strengths of each method in a single circuit board. The choice between through-hole and surface mount assembly depends on factors such as component size, board size, and the specific requirements of the electronic device being manufactured.

Printed Circuit Board Assembly (SMT + THT)

Through-Hole Technology (THT) and Surface Mount Technology (SMT) are two distinct methods used in printed circuit board (PCB) assembly, each with its advantages and applications.



Through-Hole Technology (THT):

In THT assembly, electronic components have leads that are inserted into drilled holes on the PCB. These leads protrude through the other side of the board, and solder is applied to secure the components and establish electrical connections between component leads. THT is commonly used for larger and heavier components such as connectors, switches, and electrolytic capacitors. This method provides robust mechanical connections and is often preferred for components requiring extra stability or those subject to mechanical stress.

Surface Mount Technology (SMT):

Surface Mount Technology involves mounting components directly onto the surface of the PCB. Components used in SMT have small, flat contacts or leads that are soldered directly onto pads on the PCB. SMT is widely utilized for smaller and more compact electronic devices as it allows for higher component density and is more suitable for automated assembly processes. SMT components include resistors, capacitors, integrated circuits, and other miniature devices. This method enhances manufacturing efficiency, reduces overall size and weight of electrical parts on the PCB, and is well-suited for high-volume production.

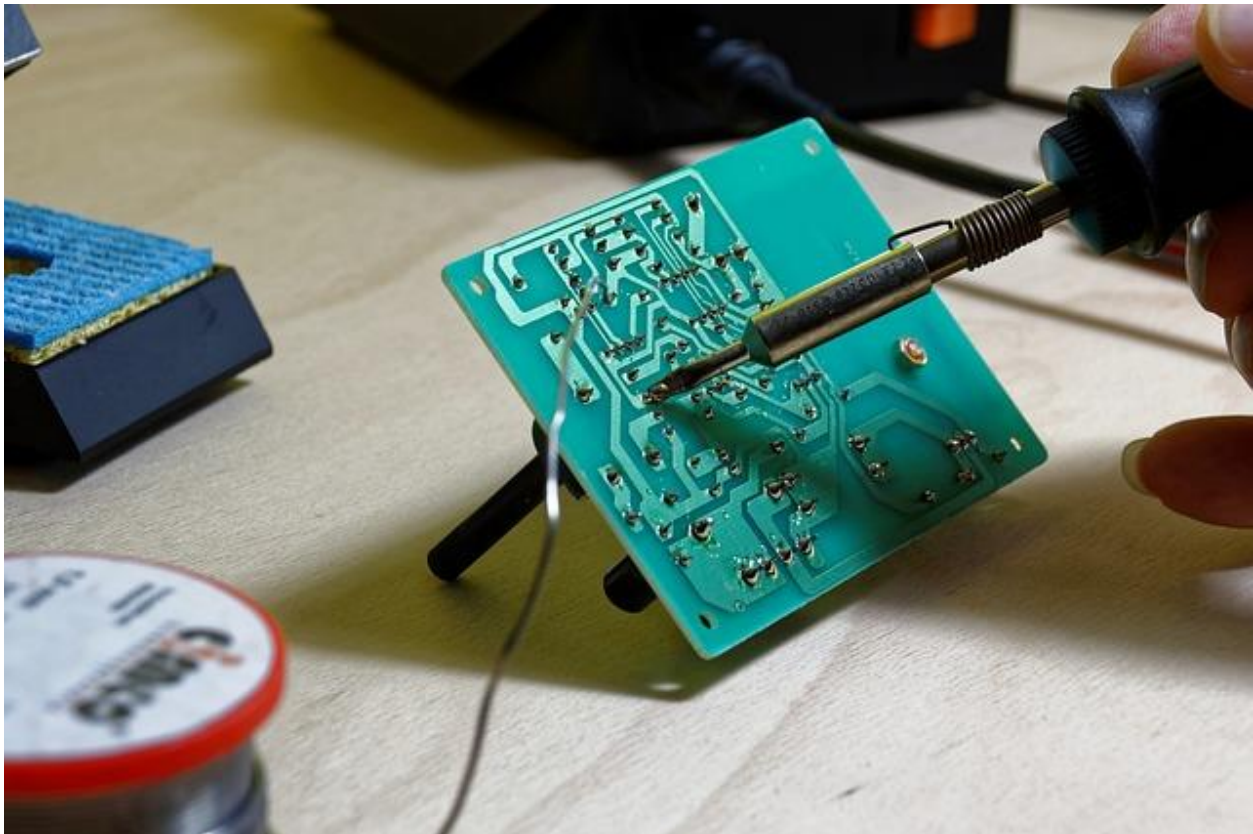
Which works best?

The choice between THT and SMT depends on factors such as the size of the components, the desired level of automation, the space constraints of the electronic device, and the

mechanical requirements of the application. THT is known for its mechanical strength and suitability for certain types of components, while SMT offers advantages in terms of miniaturization, cost-effectiveness in high-volume production, and the ability to automate assembly processes more efficiently. Many modern electronic assemblies use a combination of both THT and SMT components, known as mixed-technology assembly, to leverage the benefits of each method in the form of a single PCB.

Issues with Manual Soldering and Solder Paste

Manual soldering and the use of solder paste can present manufacturers several challenges in electronics assembly.



Manual Soldering:

Manual soldering, while a widely used technique, introduces the risk of inconsistencies and errors in the soldering process. The quality of solder joints can be highly dependent on the skill and experience of the machine operator, leading to variations in soldering temperature, duration, and technique. Inconsistent solder joints may result in issues such as poor electrical connections, insufficient mechanical strength, and the risk of cold solder joints. Additionally, manual soldering can be time-consuming and labor-intensive, making it less suitable for high-volume production where precision and efficiency are crucial.

Solder Paste Issues:

Solder paste is a crucial material in surface mount technology (SMT) assembly, where it is used to attach surface mount components to the PCB before reflow soldering. Issues with solder paste application can arise, impacting the overall quality of the solder joints. Uneven or incorrect application of solder paste can lead to problems such as solder

bridging (short circuits between adjacent pads), insufficient solder volume, or misalignment of components. Contamination, drying out, or improper storage of solder paste can also contribute to defects in the soldering process. Achieving consistent and reliable solder paste application requires careful control of the printing process and the quality of the solder paste used.



To address these issues, automated soldering techniques, such as wave soldering for through-hole components and reflow soldering for SMT components, are often employed in high-volume production settings. Automation reduces the variability introduced by manual processes, enhances precision, and improves overall solder joint quality. Additionally, advancements in solder paste formulations and application technologies continue to address challenges associated with manual soldering and solder paste, promoting more reliable and efficient electronics assembly processes.

Pick and Place

Pick and place (P&P) is a critical automation technology that significantly enhances the efficiency and accuracy of electronics assembly processes. In electronic manufacturing, P&P machines use robotic arms to precisely pick up electronic components from input trays and accurately place them onto designated locations on the printed circuit board (PCB). This automation eliminates the manual labor associated with component placement, reducing the risk of human error, improving overall production speed, and ensuring consistent component alignment. P&P machines are capable of handling a wide range of component types and sizes, making them versatile for various electronic devices. By streamlining the assembly process and enhancing precision, pick and place

technology contributes to higher production yields, improved quality, and increased throughput in electronics manufacturing.

1. What types of PCB Assemblies do you offer?

There are several types of PCB (Printed Circuit Board) assemblies, each tailored for specific applications and manufacturing processes. Here are some common types that can be sourced for you: **Through-Hole Technology (THT); Surface Mount Technology (SMT); Mixed Technology Assembly; Single-Sided PCB; Double-Sided PCB; Multilayer PCB; Rigid PCB; Flexible PCB (FPCB); Rigid-Flex PCB; High-Frequency PCB; & High-Density Interconnect (HDI) PCB.**

The choice of PCB assembly line and type depends on factors such as the application, size constraints, component types, and manufacturing requirements. These decisions are figured in the first two steps of the 5-step process.

2. Do you have the capability to place BGA and fine pitch components?

Both BGA and fine-pitch components are integral to modern electronics, enabling the development of compact and high-performance electronic devices. Their usage has become widespread in applications ranging from consumer electronics to telecommunications and industrial equipment. It's important to plan ahead in the early stages of the 5-step process. BGA packages offer advantages such as high pin count in a compact space, improved thermal performance, and enhanced signal integrity. They are commonly used in applications where miniaturization and high performance are crucial, such as in microprocessors, GPUs, and memory chips.

Fine-pitch components, on the other hand, refer to SMDs with closely spaced leads or contacts. The term "fine pitch" indicates that the distance between adjacent leads is relatively small, requiring precision during assembly. These components often have lead pitches less than or equal to 0.8 mm. Fine-pitch components include various integrated circuits, such as microcontrollers, FPGAs, and memory chips. The assembly of fine-pitch components demands careful placement and soldering techniques to ensure accurate alignment and reliable connections. Advanced manufacturing processes, including automated pick-and-place machines and precise soldering equipment, are commonly employed for the manufacture and assembly of fine-pitch components.

3. Do you Offer Electromechanical Assemblies?

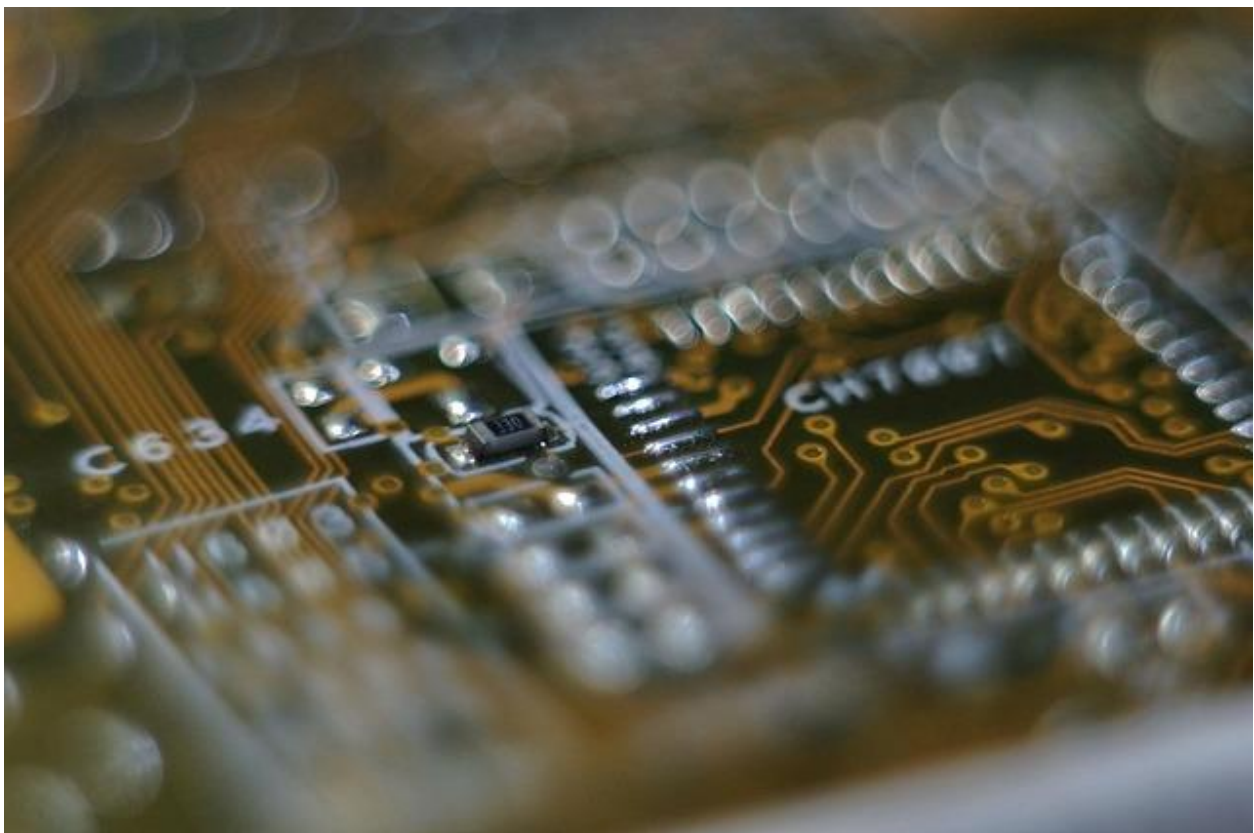
Electromechanical assemblies refer to systems or devices that integrate both electrical and mechanical components to perform a specific function. These assemblies involve the combination of electrical elements, such as circuits, sensors, or motors, with mechanical components like gears, levers, or actuators. The synergy between the electrical and mechanical aspects allows for the controlled movement, manipulation, or transformation of physical objects or signals. Examples of electromechanical assemblies include motors in appliances, robotics systems, automotive components, and various devices where electrical components and mechanical systems work together to achieve a specific outcome. The design and assembly of electromechanical systems often require interdisciplinary expertise, combining knowledge from electrical engineering and mechanical engineering.

4. Do you have conformal coat capability?

Conformal coating is a protective layer applied to the surface of printed circuit boards (PCBs) and electronic components to safeguard them from environmental factors such as moisture, dust, chemicals, and temperature extremes. The coating material is typically a liquid that, once applied, conforms to the contours of the PCB, providing a thin, protective film. Conformal coatings serve several purposes, including preventing corrosion, reducing the risk of electrical shorts due to contamination, and enhancing the overall reliability and longevity of electronic assemblies.

Common types of conformal coatings include acrylics, silicone, urethane, and epoxy. Each type offers specific properties, such as flexibility, chemical resistance, or thermal conductivity, making them suitable for different applications. The choice of conformal coating depends on the environmental conditions the electronic device is expected to face and the specific requirements of the application. The application methods include spraying, brushing, dipping, or selective coating, and the coated assemblies place components are typically cured through processes like air drying, thermal curing, or ultraviolet (UV) curing. Conformal coating is a standard practice in industries where electronic devices are exposed to challenging environmental conditions, such as automotive, aerospace, and industrial electronics.

The Future of PCB Assembly



Predicting the future of PCB assembly involves considering ongoing trends and emerging technologies. Let's highlight some directions and solutions that are gaining attention within the electronics manufacturing industry.

1. **Miniaturization and Increased Complexity:** The demand for smaller, more compact electronic devices with increased functionality has been a driving force

in PCB assembly. The trend towards miniaturization, combined with higher component density, requires advanced assembly techniques and technologies.

2. **Advanced Materials and Substrates:** The development and adoption of advanced materials, including new substrate materials and conductive inks, contribute to improvements in performance, reliability, and flexibility of PCBs. These materials are crucial for emerging applications like flexible and wearable electronics.
3. **Integration of IoT and Industry 4.0:** The integration of Internet of Things (IoT) devices and the principles of Industry 4.0 into electronic products are influencing PCB assembly. Smart manufacturing practices, including real-time monitoring, predictive maintenance, and data analytics, are becoming more prevalent in the electronics assembly process.
4. **Automation and Robotics:** The trend toward greater automation in PCB assembly is likely to continue. Automated assembly processes improve efficiency, reduce costs, and enhance precision. Robotics is being increasingly integrated into assembly lines for tasks like pick-and-place, soldering, and inspection.
5. **Environmental Considerations:** There is a growing emphasis on environmentally friendly practices in electronics manufacturing. This includes the use of lead-free solder, recyclable materials, and sustainable manufacturing processes.
6. **Advanced Inspection and Testing Technologies:** As electronic devices become more complex, the need for advanced inspection and testing technologies grows. X-ray inspection, automated optical inspection (AOI), and other advanced testing methods are becoming standard to ensure the quality and reliability of PCB assemblies.
7. **3D Printing and Additive Manufacturing:** While not yet mainstream for large-scale production, 3D printing and additive manufacturing techniques are being explored for prototyping and low-volume production of PCBs. This could offer greater design flexibility and shorter development cycles.

Six ways to achieve a culture of excellence in the workplace.

1. Communicate and understand the organization's vision.

It's not enough to simply have a stated organizational vision. To achieve a Culture of Excellence, every employee must understand not only the company's vision, but also know their own roles, responsibilities and the specific actions they need to take in order to help achieve this vision.

2. Be clear on your organization's purpose and meaning.

In a culture of excellence, employees feel that what they are working on is meaningful, significant, and purpose based. Everyone concerned is highly inspired by the common purpose, which becomes the driving force behind everything that they do.

3. Focus on high performers and developing a mastery of roles.

Companies with a culture of excellence set an expectation of high-performance organization wide. Every employee is supported and encouraged to become a master in their role and area of expertise. High performers are nurtured, rewarded, mentored

and recognized, and average performers are coached to move into the high performance category. In these cultures, there is no place for low performers, and they either move up or leave the organization.

4. Practice resilience to changes and challenges.

In most organizations, when change or challenges occur, employees become distracted and lose focus on the organization's vision and goals. In a culture of excellence, employees develop the flexibility and resilience to deal with change, challenge and uncertainty. Even when there are obstacles and challenges that may seem impossible to overcome, the motivation to achieve the organization's vision is higher than the urge to avoid the discomfort.

With that clarity of purpose and a strong desire to succeed, employees push through the barriers and move forward toward their vision. Managers support their teams in staying focused and on track, despite difficulties and challenges.

5. Create highly collaborative teams.

Most of us have worked at companies where the silo mentality reigns. Teams and individuals closely guard their expertise, projects and knowledge. Collaboration across teams is nearly non-existent unless forced. A key characteristic of a culture of excellence is highly collaborative teams— both internal and external. Because every employee and all teams are working together toward a common vision, they feel they are on the same side. And because this collaboration is encouraged and rewarded from the top down, there is no reason to protect individual roles, projects or expertise.

6. Instill a pioneer mentality.

Most companies that achieve a culture of excellence do not settle for the mediocre. They are focused on creating something that has never been created before, breaking records and achieving unprecedented results. From the outside, it may seem as if they are achieving the impossible. The resulting energy, excitement and drive, creates a certain positive tension that reverberates throughout the company. Employees and teams are encouraged to explore, nurture and co-create to achieve common goals.

How to make it happen, and create lasting change

After 18 years of experience and research in corporate development, culture and leadership, I have discovered the answer to creating a culture of excellence that both creates an immediate impact, and is sustainable over the long term.

The answer is a unique three-phrase approach:

Phase 1: Create a mindset of excellence.

The first step in building a corporate culture that will drive a high-performance organization is to create a mindset that will engage and align every employee with your vision, mission and values, and leave them speaking a common language of excellence. You will build a new capacity for growth. Your employees will think in more creative and innovate ways and will develop the tolerance to continue to move forward despite challenges, change and potential distractions.

Phase 2: Develop strategies for excellence.

With the new mindset of excellence in place, your teams will be ready to focus on achieving their specific goals and plans for performance excellence. This phase is all about growth—building the skills and competencies required to grow the business.

Phase 3: Sustain excellence.

When you reach phase 3, you should have a strong collective mindset in place, alignment around performance goals, and the skills and competencies required to deliver excellence. It's then time to focus on developing strong leadership to ensure that the new mindset and performance skills learned are sustainable, and simply, become the way things are done. This is a critical step often missed in organizations — but is required to ensure that employees don't revert back to old habits.

SESSION 4.

Disassemble and/or assemble electronic components and systems.

Learning Outcomes

- Safety measures for assembling and/or disassembling are adhered in terms manufacturer' instructions.
- Different components are analysed in terms their relations.
- Procedures for assembling and/or disassembling are followed in accordance with manufacturer' specifications.
- Parts are labelled for identification purposes during disassembling and/or assembling.

Disassemble and/or assemble electronic components and systems.

Protection Against Electrostatic Discharge (ESD)

Electronic products that have the CE marking are designed and tested for immunity to international Electrostatic Discharge (ESD) standards. While these products have been designed and determined to be compliant with standard levels for ESD, there may be situations, such as low humidity levels, that can exacerbate ESD event occurrence.

Electrostatic discharge (ESD) events can harm electronic components inside your computer. Under certain conditions, ESD may build up on your body or an object, such as a peripheral, and then discharge into another object, such as your computer. To prevent ESD damage, you should discharge static electricity from your body before you interact with any of your computer's internal electronic components, such as a memory module.

You can protect against ESD and discharge static electricity from your body by touching a metal grounded object (such as an unpainted metal surface on your computer's I/O

panel) before you interact with anything electronic. When connecting a peripheral (including handheld digital assistants) to your computer, you should always ground both yourself and the peripheral before connecting it to the computer. In addition, as you work inside the computer, periodically touch a metal-grounded object to remove any static charge your body may have accumulated.

You can also take the following steps to prevent damage from electrostatic discharge:

- When unpacking a static-sensitive component from its shipping carton, do not remove the component from the antistatic packing material until you are ready to install the component. Just before unwrapping the antistatic package, be sure to discharge static electricity from your body.
- When transporting a sensitive component, first place it in an antistatic container or packaging.
- Handle all electrostatic sensitive components in a static-safe area. If possible, use antistatic floor pads and work bench pads.



Figure 1: ESD wrist strap

General safety guidelines

Use the following safety guidelines to protect your computer from potential damage and to ensure your personal safety. Unless otherwise noted, each procedure included in this document assumes that you have read the safety information that shipped with your computer.

1. Place the equipment on a hard, level surface. Leave a 10.2 cm (4 in) minimum of clearance on all vented sides of the computer to permit the airflow required for proper ventilation. Restricting airflow can damage the computer or cause a fire.
2. Do not stack equipment or place equipment so close together that it is subject to re-circulated or preheated air.
3. Ensure that nothing rests on your equipment's cables and that the cables are not located where they can be stepped on or tripped over.
4. Ensure that all cables are connected to the appropriate connectors. Some connectors have a similar appearance and may be easily confused (for example, do not plug a telephone cable into the network connector).

5. Do not place your device in a closed-in wall unit, or on a soft, fabric surface such as a bed, sofa, carpet, or a rug.
6. Keep your device away from radiators and heat sources.
7. Keep your equipment away from extremely hot or cold temperatures to ensure that it is used within the specified operating range.
8. Do not push any objects into the air vents or openings of your equipment. Doing so can cause fire or electric shock by shorting out interior components.
9. Do not use your equipment in a wet environment, for example, near a bathtub, sink, or swimming pool or in a wet basement.
10. Do not use AC powered equipment during an electrical storm. Battery powered devices may be used if all cables have been disconnected.
11. Do not spill food or liquids on your equipment.
12. Before you clean your equipment, disconnect it from the electrical outlet. Clean your device with a soft cloth dampened with water. Do not use liquids or aerosol cleaners, which may contain flammable substances.
13. Clean the laptop LCD screen or monitor display with a soft, clean cloth and water. Apply the water to the cloth, then stroke the cloth across the display in one direction, moving from the top of the display to the bottom. Remove moisture from the display quickly and keep the display dry. Long-term exposure to moisture can damage the display. Do not use a commercial window cleaner to clean your display.

Before working inside your computer

1. To help avoid the potential hazard of electric shock, do not connect or disconnect any cables or perform maintenance or reconfiguration of your equipment during an electrical storm.
2. Save and close all open files and exit all open applications.
3. Shut down your computer. Click **Start > Power > Shut down**.

NOTE: If you are using a different operating system, see the documentation of your operating system for shut-down instructions.

4. Disconnect your computer and all attached devices from their electrical outlets.
5. Disconnect all attached network devices and peripherals, such as keyboard, mouse, and monitor from your computer.
6. Remove any media card and optical disc from your computer, if applicable.

NOTE: Before working inside your computer, read the safety information that is shipped with your computer. For more safety best practices, see the Regulatory Compliance home page at www.dell.com/regulatory_compliance.

NOTE: Disconnect all power sources before opening the computer cover or panels. After you finish working inside the computer, replace all covers, panels, and screws before connecting to the electrical outlet.

CAUTION: Before touching anything inside your computer, ground yourself by using a wrist grounding strap or by periodically touching an unpainted metal surface, such as the metal at the back of the computer. While you work, periodically touch an unpainted metal surface to dissipate static electricity, which could harm internal components.

CAUTION: To avoid damaging the computer, ensure that the work surface is flat and clean.

CAUTION: Handle components and cards with care. Do not touch the components or contacts on a card. Hold a card by its edges or by its metal mounting bracket. Hold a component such as a processor by its edges, not by its pins.

CAUTION: When you disconnect a cable, pull on its connector or on its pull tab, not on the cable itself. Some cables have connectors with locking tabs or thumb-screws that you must disengage before disconnecting the cable. When disconnecting cables, keep them evenly aligned to avoid bending any connector pins. When connecting cables, ensure that the ports and connectors are correctly oriented and aligned.

After working inside your computer

CAUTION: Leaving stray or loose screws inside your computer may severely damage your computer.

1. Replace all screws and ensure that no stray screws remain inside your computer.
2. Connect any external devices, peripherals, or cables you removed before working on your computer.
3. Replace any media cards, discs, or any other parts that you removed before working on your computer.
4. Connect your computer and all attached devices to their electrical outlets.
5. Turn on your computer.

General power safety guidelines

- Check the voltage rating before you connect the equipment to an electrical outlet to ensure that the required voltage and frequency match the available power source.
- Your device is equipped with either an internal power source or an external adapter. For internal power sources, your device is equipped with one of the following:

- **An auto-sensing voltage circuit** – Devices with an auto-sensing voltage circuit do not have a voltage selection switch on the back panel and automatically detect the correct operating voltage.
- Or **a manual voltage selection switch** – Devices with a voltage selection switch on the back panel must be manually set to operate at the correct operating voltage. Set the switch to the position that most closely matches the voltage used in your location.

CAUTION: To help avoid damaging a computer with a manual voltage selection switch, set the switch for the voltage that most closely matches the AC power available in your location.

- Ensure that the computer, monitor, and attached devices are electrically rated to operate with the AC power available in your location.
- Do not plug the equipment power cables into an electrical outlet if the power cable is damaged.
- To prevent electric shock, plug the equipment power cables into properly grounded electrical outlets. If the equipment is provided with a 3-prong power cable, do not use adapter plugs that bypass the grounding feature, or remove the grounding feature from the plug or adapter.
- If you use an extension power cable, ensure that the total ampere rating of the products plugged in to the extension power cable does not exceed the ampere rating of the extension cable.
- If you must use an extension cable or power strip, ensure the extension cable or power strip is connected to a wall power outlet and not to another extension cable or power strip. The extension cable or power strip must be designed for grounded plugs and plugged into a grounded wall outlet.
- If you are using a multiple-outlet power strip, use caution when plugging the power cable into the power strip. Some power strips may allow you to insert a plug incorrectly. Incorrect insertion of the power plug could result in permanent damage to your equipment, as well as risk of electric shock and/or fire. Ensure that the ground prong of the power plug is inserted into the mating ground contact of the power strip.
- Be sure to grasp the plug, not the cable, when disconnecting equipment from an electric socket.
- Use only the Dell provided AC adapter approved for use with this device. Use of another AC adapter may cause a fire or explosion.
- Place the AC adapter in a ventilated area, such as a desktop or on the floor, when you use it to run the computer or to charge the battery. Do not cover the AC adapter with papers or other items that will reduce cooling; also, do not use the AC adapter inside a carrying case.
- The AC adapter may become hot during normal operation of your computer. Use care when handling the adapter during or immediately after operation.

- It is recommended that you lay the adapter on the floor or desk so that the green light is visible. This will alert you if the adapter should accidentally go off due to external effects. If for any reason the green light goes off, disconnect the AC cord from the wall for a period of ten seconds, and then reconnect the power cord.
- Never use an AC adapter that shows signs of damage or excessive wear.

Desktop safety guidelines

Use the following safety guidelines to help ensure your own personal safety and to help protect your device (computer, port replicator, media base, docking station, and similar devices) and working environment from potential damage.

- Check the voltage rating before you connect the desktop to an electrical outlet to ensure that the required voltage and frequency match the available power source.
- Your desktop is equipped with either an internal power source or an external adapter. For internal power sources, your computer is equipped with an auto-sensing voltage circuit or a manual voltage selection switch.
- Turn off and unplug the desktop computer when it is going to be left unused for an extended period of time.

Laptop safety guidelines

Use the following laptop safety guidelines to help ensure your own personal safety and to help protect your device (computer, port replicator, media base, docking station, and similar devices) and working environment from potential damage.

- Do not store your laptop in a low-airflow environment, such as a carrying case or a closed briefcase, while the computer is turned on. Restricting airflow can damage the computer or cause a fire.
- Do not allow your laptop or adapter to operate with the base resting directly on exposed skin for extended periods of time. The surface temperature of the base will rise during normal operation, particularly when AC power is present. Allowing sustained contact with exposed skin can cause discomfort or a burn.
- Use only the Dell provided AC adapter approved for use with this device. Use of another AC adapter may cause a safety hazard. See your User's Guide or Owner's Manual for AC adapter electrical ratings.
- To remove all power sources from your laptop, turn the computer off, disconnect the AC adapter from the electrical outlet, and remove any battery installed in the battery bay or module bay.

NOTE: Certain laptops have a non-removable battery.

- The Laptop auto adapter allows you to power and charge your laptop in your car, boat, or in an airplane. The adapter connects your computer to a power source

via a DC power cable. The DC power cable connects via either an automobile cigarette lighter adapter or an empower-type connector. Review the operation instructions in your User's Guide before connecting and operating the auto adapter.

- Do not use your device in vehicles with high voltage (24 VDC), such as commercial transport vehicles.
- Plug the auto adapter into the computer; then with the car engine or other DC power source turned off, plug the adapter into the cigarette lighter or DC power source.
- A damaged battery may pose a risk of personal injury. Damage may include impact or shock that dents or punctures the battery, exposure to a flame, or other deformation. Do not disassemble the battery. Handle a damaged or leaking battery pack with extreme care. If the battery is damaged, electrolyte may leak from the cells or fire may result which may cause personal injury.
- Keep the battery away from children.
- Do not expose (store or place) your computer or battery pack to a heat source such as a radiator, fireplace, stove, electric heater, or other heat-generating appliance or otherwise expose it to temperatures in excess of 65°C (149°F). When heated to excessive temperatures, battery cells could vent or explode, posing risk of fire.
- Do not carry a battery pack in your pocket, purse, or other container where metal objects (such as keys) could short-circuit the battery terminals. The resulting excessive current flow can lead to extremely high temperatures and may cause damage to the battery pack or surrounding materials, or personal injury, such as burns.
- Turn off and unplug the laptop when it is going to be left unused for an extended period of time.

Monitor safety guidelines

Read and follow these instructions when connecting and using your computer monitor:

- Always be sure that your monitor is electrically rated to operate with the AC power available in your location.
- Place the monitor on a solid surface and treat it carefully. The screen is made of glass and can be damaged if dropped or hit sharply.
- Keep flat panel monitor in room temperature conditions. Excessive cold or hot conditions can have an adverse effect on the liquid crystal of the display.
- Do not subject the monitor to severe vibration or high impact conditions. For example, do not place the monitor inside a car trunk.
- To avoid electric shock, do not attempt to remove any cover or touch the inside of the monitor.

- Turn off and unplug the monitor when it is going to be left unused for an extended period of time.

Lifting equipment

CAUTION: Do not lift greater than 50 pounds (22.6 kilograms) independently. Always obtain assistance from a second person or utilize a mechanical lifting device.

1. Take your time think about the situation, plan, and follow the safety tips when lifting.
2. Get a firm balanced footing. Keep your feet apart for a stable base and point your toes out.
3. Bend your knees. Do not bend at the waist.
4. Tighten stomach muscles. Abdominal muscles support your spine when you lift, offsetting the force of the load.
5. Lift with your legs, not your back.
6. Keep the load close. The closer it is to your spine, the less force it exerts on your back.
7. Keep your back upright, whether lifting or setting down the load. Do not add the weight of your body to the load. Avoid twisting your body and back.
8. Follow the same techniques in reverse to set the load down.
9. Keep your head up when carrying equipment.
10. Always look ahead.
11. Try to move smoothly.
12. Don't handle more than what can be comfortably managed (if in doubt seek assistance).

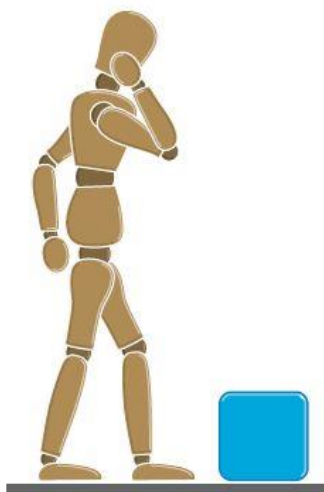


Figure 2: Take your time, think, and plan.



Figure 3: Keep your feet apart and point your toes out.



Figure 4: Bend your knees. Do not bend your waist.

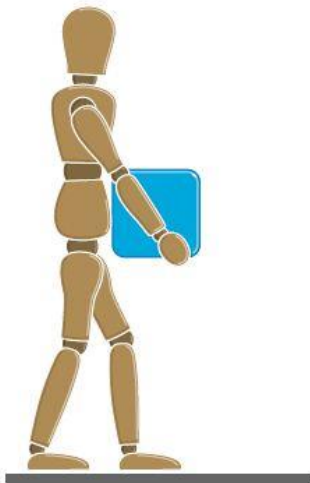


Figure 5: Keep your back in an upright position.

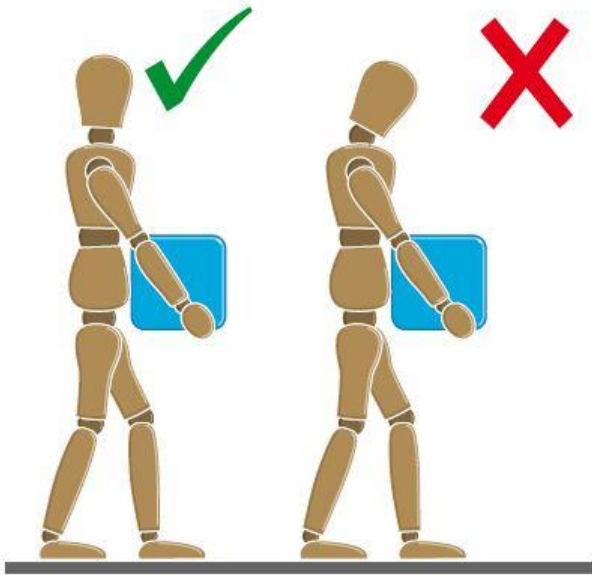


Figure 6: Keep your head when you are carrying equipment

Resistors, Capacitors, Inductors, Diodes, and Transistors

When it comes to building circuits, there are a few essential components that are used as building blocks for almost every circuit. These components are resistors, capacitors, inductors, diodes, and transistors. Each of these components plays a crucial role in the function of the circuit, and understanding their properties is essential to being able to design and troubleshoot circuits effectively. Let's dive in deeper and understand their features and functions in detail.

Resistors

Resistors are one of the most basic circuit components and are used to limit the flow of electrical current in a circuit. A resistor's resistance is measured in ohms, and the amount of current flowing through a resistor is proportional to the voltage across it and inversely proportional to its resistance.

There are different types of resistors, including carbon composition, wire-wound, and film resistors. Each of the Resistors are essential components in electronic circuits, used for limiting the flow of electrical current in a circuit. The resistance of a resistor is measured in ohms (Ω), and it determines the amount of current flowing through the resistor. The relationship between voltage, current, and resistance is defined by Ohm's law, which states that the current flowing through a resistor is proportional to the voltage across it and inversely proportional to its resistance.

There are various types of resistors available in the market, and each of them has its unique properties and advantages. Carbon composition resistors are one of the oldest and most commonly used types of resistors. They are made up of a carbon rod or tube, surrounded by an insulating material, and have leads attached to each end. These resistors have a high degree of precision and are used in applications where accuracy is crucial, such as in audio circuits.

Wire-wound resistors, on the other hand, are made by winding a resistive wire around a ceramic or fiberglass core. They are available in both fixed and variable configurations, and can handle high power and voltage levels. Wire-wound resistors are used in applications where high precision and stability are required, such as in power amplifiers and power supplies.

Film resistors are another popular type of resistor, made by depositing a thin film of resistive material onto a ceramic or fiberglass substrate. They are widely used in electronic circuits due to their high stability, low noise, and compact size. Film resistors come in both axial and surface mount configurations and are suitable for applications where high reliability and precision are required.

The choice of resistor type depends on the specific requirements of the circuit and the application. For instance, carbon composition resistors are ideal for audio applications where precision and accuracy are important, whereas wire-wound resistors are better suited for high power and voltage applications. Film resistors are commonly used in precision applications, where stability and low noise are essential.



Resistors

Capacitors

Capacitors are used to store electrical charge and release it when needed in a circuit. They are used in various applications such as filtering, timing circuits, power supplies, and many more.

Capacitance is the measure of the capacitor's ability to store electrical charge, and it is measured in farads. Capacitors come in various sizes and types, including electrolytic capacitors, ceramic capacitors, and tantalum capacitors.

Capacitors are another essential circuit component, used for storing and releasing electrical energy. A capacitor consists of two conductive plates separated by an insulating material, known as the dielectric. The capacitance of a capacitor is determined by the surface area of the plates, the distance between them, and the type of dielectric used.

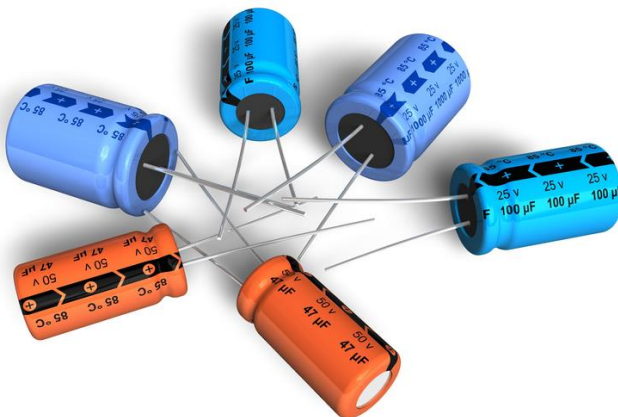
There are different types of capacitors available in the market, each with its unique properties and advantages. Some of the most common types include ceramic, electrolytic, and tantalum capacitors.

Ceramic capacitors are made up of ceramic material and have a high dielectric constant, which makes them ideal for high-frequency applications. They are also available in a wide range of capacitance values and are suitable for use in applications where stability and reliability are important.

Electrolytic capacitors are polarized capacitors, which means that they have a positive and a negative terminal. They are used in applications where high capacitance values are required, and are available in both axial and radial configurations. Electrolytic capacitors are commonly used in power supply circuits, as they can handle high voltage and current levels.

Tantalum capacitors are another popular type of capacitor, made from tantalum metal and an electrolyte. They are polarized capacitors and have a high level of stability and reliability. Tantalum capacitors are commonly used in applications where high capacitance values and high-frequency stability are required, such as in power supplies and audio circuits.

The choice of capacitor type depends on the specific requirements of the circuit and the application. For instance, ceramic capacitors are ideal for high-frequency applications, whereas electrolytic capacitors are better suited for high-capacitance applications.



Capacitors

Inductors

Inductors are an essential component in electronic circuits that play a significant role in storing energy in a magnetic field. The main function of an inductor is to resist changes in the flow of electrical current in a circuit. It achieves this by inducing a back-emf (electromotive force) when the current flowing through it changes. This back-emf opposes the change in current, which helps in stabilizing the current flow in the circuit.

The measure of an inductor's ability to store energy is called inductance. Inductance is measured in Henries and is denoted by the letter "H". In simple terms, the higher the inductance, the more energy an inductor can store in its magnetic field. The inductance

of an inductor depends on its physical properties, such as the number of turns in the coil, the size of the coil, and the type of core material used.

Inductors come in different types, each with its unique properties and applications. Air-core inductors are made up of coils of wire without any core material. They have low inductance values and are often used in high-frequency applications, where their low parasitic capacitance is beneficial.

Iron-core inductors are made up of a coil of wire wrapped around a ferromagnetic core made of iron. They have higher inductance values than air-core inductors and are often used in low-frequency applications, such as power supplies and filters.

Ferrite-core inductors are made up of a coil of wire wrapped around a core made of ferrite, a ceramic material containing iron oxide. Ferrite-core inductors have high inductance values and are used in high-frequency applications, such as inductors for switching power supplies and radio frequency (RF) circuits.



Inductor

Diodes

Diodes are crucial electronic components that are widely used in various circuits. A diode is a two-terminal device that allows the current to flow in one direction only, and blocks the current flow in the opposite direction. This characteristic of diodes makes them useful in rectifier circuits, where they are used to convert AC (alternating current) to DC (direct current) by allowing only the positive half of the AC waveform to pass through.

A diode has two terminals: the anode and the cathode. The anode is the positive terminal, and the cathode is the negative terminal. When a voltage is applied to the anode of a diode, current flows through the diode in the forward direction, from the anode to the cathode. In this state, the diode has low resistance, and current can flow easily through it.

However, if a voltage is applied to the cathode of the diode, current cannot flow through it in the reverse direction. This state of the diode is called reverse bias, and the diode has a very high resistance to current flow in this state.

There are different types of diodes, each with its unique properties and applications. Rectifier diodes are the most common type of diodes and are used in rectifier circuits to convert AC to DC. Zener diodes are used in voltage regulator circuits to maintain a constant voltage across a load. Light-emitting diodes (LEDs) are used in lighting applications and for indicating the status of electronic circuits.

Rectifier diodes are designed to handle high current and voltage levels and are used in power supply circuits. Zener diodes have a specific breakdown voltage, and they are used as voltage regulators to provide a constant voltage to a load. LEDs emit light when current flows through them and are used in a wide range of applications, including lighting, displays, and indicators.



Diode

Transistors

Transistors are one of the most essential components in electronic circuits due to their ability to amplify and switch electrical signals. They are commonly used in various applications, including audio amplifiers, radio receivers, and digital circuits. Transistors come in different types, including bipolar junction transistors (BJTs) and field-effect transistors (FETs), each of which has its own unique properties and is suited for different applications.

Bipolar junction transistors (BJTs) are three-layer semiconductor devices with two pn-junctions. They consist of a base, emitter, and collector region. BJTs are often used in analog circuits due to their high current gain and excellent linearity. The base-emitter junction of the BJT is forward-biased, allowing a small current to flow from the emitter to the collector. The amount of current that flows through the collector is proportional to the amount of current flowing into the base. This feature allows BJTs to amplify signals.

Field-effect transistors (FETs) are three-terminal devices that rely on an electric field to control the flow of current. They are often used in digital circuits due to their high input impedance, low output impedance, and fast switching speed. FETs come in two types, junction FETs (JFETs) and metal-oxide-semiconductor FETs (MOSFETs). JFETs have a simple structure, consisting of a semiconductor channel with two pn-junctions. They are commonly used in low-noise amplifier circuits due to their low noise characteristics. MOSFETs have a more complex structure and are commonly used in high-speed digital circuits due to their fast switching speed.



Transistor

Conclusion

Understanding the properties and applications of resistors, capacitors, inductors, diodes, and transistors is crucial for anyone interested in electronics and circuit design. Each of these components plays a specific role in the circuit and has its own properties that make it suitable for specific applications.

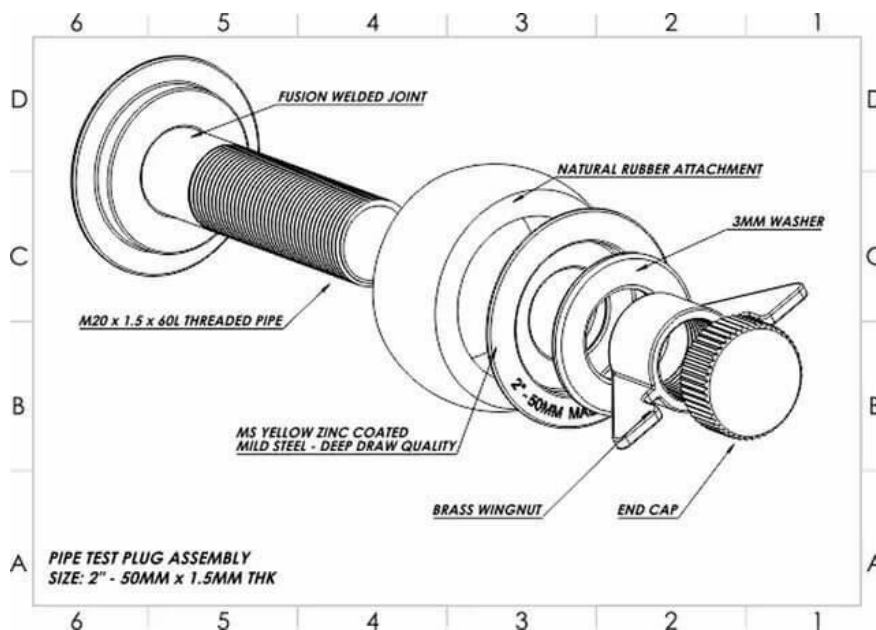
As one delves deeper into the world of electronics and circuit design, one will encounter more complex circuits that utilize these building blocks in various configurations. By understanding these fundamental components, you will be better equipped to design and troubleshoot more complex circuits effectively.

The sequence to be adopted for the dismantling/re-assembly of various types of assemblies is described:

- Assess the assembly: Before any dismantling or re-assembly begins, it is important to carefully assess the assembly to identify the different components and their connections.
- Plan the sequence: Once the assessment is complete, create a plan for the dismantling and re-assembly process. This should include a detailed sequence of steps for both activities, considering the complexity and interdependency of the components.
- Gather tools and equipment: Ensure that all necessary tools and equipment are gathered and in good working condition for the disassembly and re-assembly of the assembly. This may include wrenches, screwdrivers, pliers, and other specialized tools.
- Dismantling: Follow the planned sequence to carefully dismantle the assembly, starting with the easy-to-remove components and working towards the more complex or critical ones. Take note of the orientation and arrangement of the components for easier re-assembly.
- Inspection: Once the assembly is dismantled, thoroughly inspect each component for any damage or wear. This is also an opportunity to clean and lubricate parts as needed.
- Re-assembly: Follow the planned sequence in reverse order to re-assemble the components, ensuring that each part is correctly oriented and aligned before tightening any fasteners.
- Testing: After the re-assembly is complete, test the assembly to ensure that it functions correctly and that all components are properly connected and secure.
- Final inspection: Once the testing is complete, perform a final inspection of the assembly to ensure everything is in place and functioning as intended.
- Documentation: Document any specific procedures, measurements, or adjustments made during the dismantling and re-assembly process for future reference.
- 10. Clean up: Finally, clean up the work area and return any tools or equipment to their proper place.

Here are five types of assembly drawings and how they are used:

Although there are several types of assembly drawings, here are the five most common:



Sample of a general assembly drawing.

1. General assembly drawings

General assembly drawings identify the various components and their relationship. It contains the component's detailed drawing, the sub-assembly, and the final assembly. The drawing can be divided into four categories:

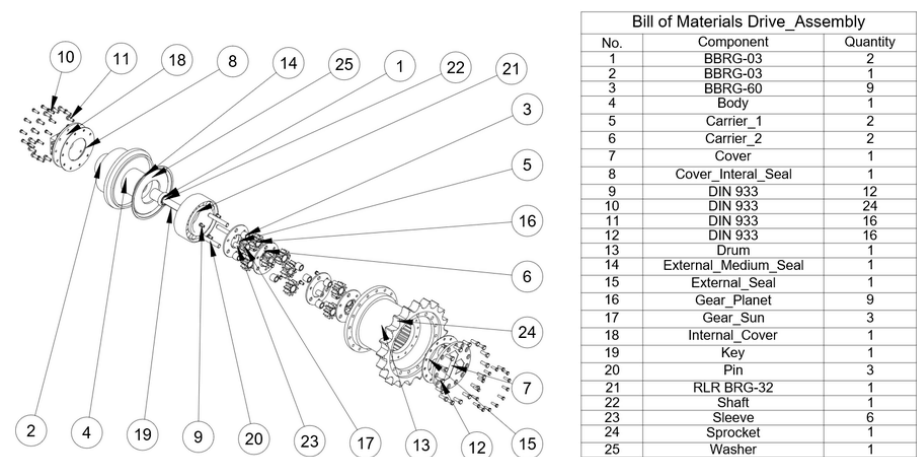
- a. Design drawings are made in the design stage and show machines, steel structures, and various constructions to be assembled from several angles. These drawings help visualize the object's workings, including its shape and the clearances of specific parts.
- b. Detailed drawings indicate how the components are assembled and provide specifics like the materials, dimensions, joining methods, etc. In addition to the regular assembly drawings, enlarged views of specific parts and how they will fit together can also be drawn up.

Some machinery and structures are made up of pre-assembled components, called sub-assemblies, and individual parts. They can be assembled and tested as a unit before they are put into their final assembly. A sub-assembly drawing will reference an assembly drawing.

- c. Assembly drawings for catalogs are unique assembly drawings prepared for company catalogs. These drawings show only the relevant details and dimensions that interest a potential buyer.

- d. Assembly drawings for instruction manuals are needed whenever a finished machine must be disassembled for shipping, reassembled, and installed at its

destination. Each component in these drawings is numbered to aid in the reassembly process.

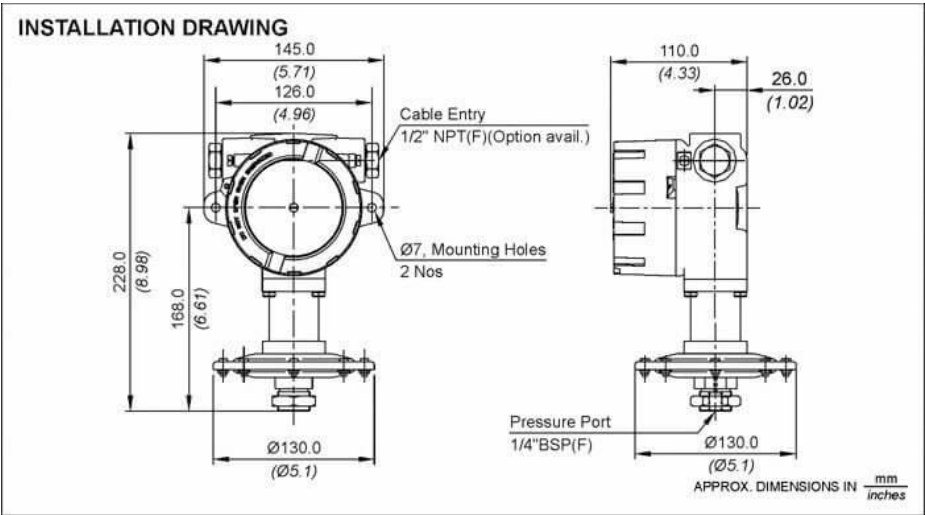


Example of exploded assembly drawing.

2. Exploded assembly drawing

These technical drawings show the components of an assembly slightly separated or suspended in space in the assembly sequence. This “exploded” view clarifies how the final product will fit together, making it easier to understand even for a layperson.

These are the drawings found in parts catalogs, assemblies, manuals, or instructional materials.

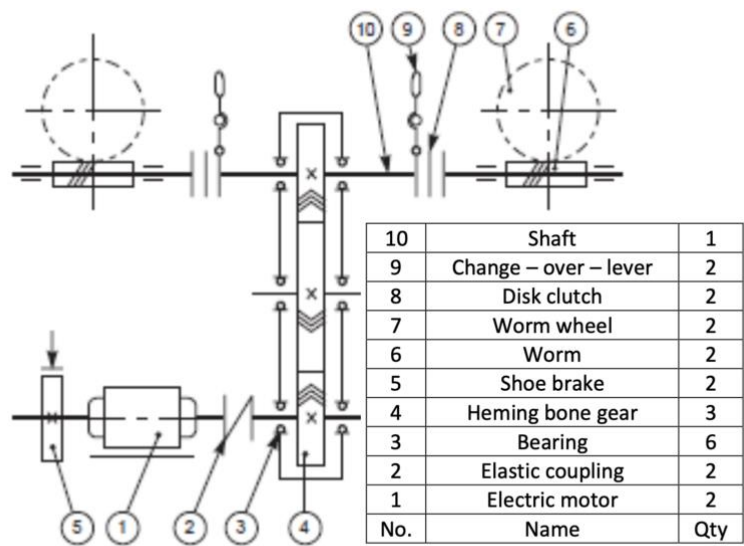


Example of installation drawing.

3. Installation drawings

Installation drawings help workers with the erection or assembly of a product. They supply information on how a component will be positioned in relation to its supporting or adjacent elements.

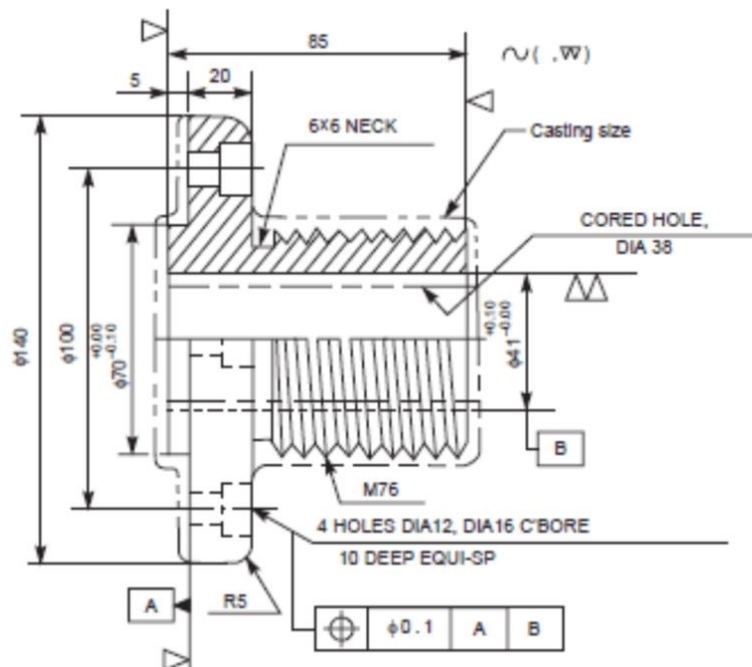
It will also show parts, dimensional data, hardware descriptions, and a general configuration.



Example of schematic assembly drawing.

4. Schematic assembly drawings

A schematic drawing is a simplified illustration of components using standardized symbols and lines. Schematic diagrams include only necessary details. For example, a schematic diagram of an electrical circuit will show how the wires and components are connected without photographs of the circuit itself.



Sample of machine shop drawing.

6. Machine shop drawings

The machine shop drawings are for the machinist or machine operator. Because this person is unconcerned about information related to the previous stages or operations, only the information about the machined part's function is included in the part drawing.

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The Importance of Assembly Drawings

Producing any mechanism on a large scale will require numerous assemblies and sub-assemblies that must be accurately manufactured. Without assembly drawings, technical information would be communicated by speech or writing rather than graphical language, resulting in expensive misunderstandings, mistakes, and wasted time. The effective use of assembly drawings has been pivotal in safely producing vehicles, aircraft, and structures on which lives might depend.

Assembly drawings typically provide enough information to enable the assembly of a component, and they will usually **include the following**:

- Two or more parts or subordinate assemblies
- Sufficient orthogonal views showing how the parts fit together
- Section views to show how pieces fit and to reveal hidden details
- Dimensions are not shown on assembly drawings unless to indicate the overall size of the assembly or for machining operations needed for assembly.
- Requirements for processes, protective finishes, and other relevant data for completing the item as an assembly

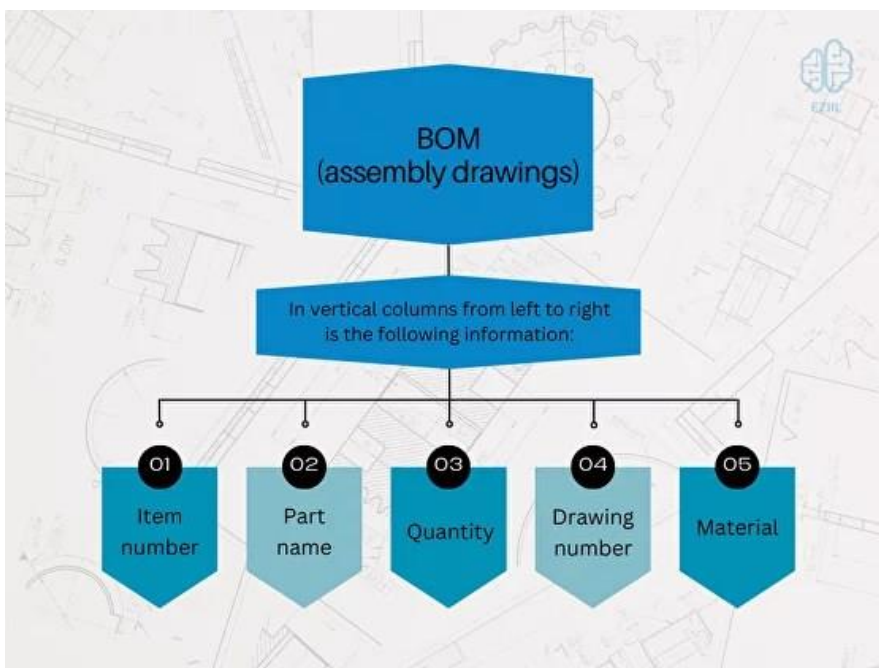
- A bill of materials (parts list) specifying a unique identifier for items that become a part of the assembly
- The revision or issue information

What is Included in the Assembly Drawing's

Bill of Materials?

A bill of materials is a complete list of the items comprising an assembly of detailed parts presented on a technical drawing. Whether included in the assembly drawing or as a separate document, the BOM provides valuable information to produce the items.

In vertical columns from left to right is the following information:



- **Item number:** The relevant item reference number is shown on the drawing and listed in sequential order
- **Part name:** Abbreviations may be used if they do not confuse. Use its traditional designation if the item describes a standard part, such as a nut or bolt.
- **Quantity:** The number of that specific part required for one complete assembly
- **Drawing number**
- **Material**

Conclusion

In conclusion, assembly drawings play an indispensable role in product design and mechanical engineering, as they provide essential visual representations of the assembly process for various parts and components. By utilizing a drawing sheet and assembly file, professionals can effectively communicate the intricacies of product

assembly to engineers, manufacturers, and assembly technicians, ensuring a smooth transition from the design phase to the manufacturing phase.

A well-prepared assembly drawing, complete with dimensions and annotations added to the drawing sheet, ensures a clear understanding of the assembly process for all stakeholders involved. The addition of dimensions, tolerances, and other relevant information not only reduces the likelihood of errors during the assembly process but also contributes to the overall efficiency and success of the project.

LEARNER GUIDE

SAQA : 386098

**Diagnose and locate faulty circuit, component and
equipment**

Learner Information:

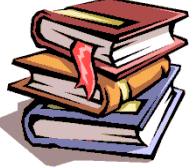
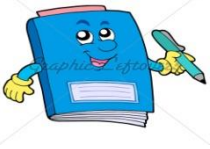
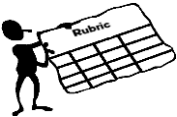
Details	Please Complete this Section
Name & Surname:	
Organisation:	
Unit/Dept:	
Facilitator Name:	
Date Started:	
Date of Completion:	

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Key to Icons

The following icons may be used in this Learner Guide to indicate specific functions:

 Books	This icon means that other books are available for further information on a particular topic/subject.
 References	This icon refers to any examples, handouts, checklists, etc...
 Important	This icon represents important information related to a specific topic or section of the guide.
 Activities	This icon helps you to be prepared for the learning to follow or assist you to demonstrate understanding of module content. Shows transference of knowledge and skill.
 Exercises	This icon represents any exercise to be completed on a specific topic at home by you or in a group.
 	An important aspect of the assessment process is proof of competence. This can be achieved by observation or a portfolio of evidence should be submitted in this regard.

Tasks/Projects	
 Workplace Activities	An important aspect of learning is through workplace experience. Activities with this icon can only be completed once a learner is in the workplace
 Tips	This icon indicates practical tips you can adopt in the future.
 Notes	This icon represents important notes you must remember as part of the learning process.

Learner Guide Introduction

About the Learner Guide...	This Learner Guide provides a comprehensive overview of the Diagnose and locate faulty circuit, component and equipment, and forms part of a series of Learner Guides that have been developed for Diagnose and locate faulty circuit, component and equipment . The series of Learner Guides are conceptualized in modular's format and developed for Diagnose and locate faulty circuit, component and equipment . They are designed to improve the skills and knowledge of learners, and thus enabling them to effectively and efficiently complete specific tasks. Learners are required to attend training workshops as a group or as specified by their organization. These workshops are presented in modules, and conducted by a qualified facilitator.
Purpose	The purpose of this Learner Guide is to provide learners with the necessary knowledge related to Diagnose and locate faulty circuit, component and equipment
Outcomes	At the end of this module. The qualifying learner is capable of: • Demonstrate knowledge of testing instruments/equipment. • Locate faulty components. • Rectify the problem.
Assessment Criteria	The only way to establish whether a learner is competent and has accomplished the specific outcomes is through an assessment process. Assessment involves collecting and interpreting evidence

	about the learner's ability to perform a task. This guide may include assessments in the form of activities, assignments, tasks or projects, as well as workplace practical tasks. Learners are required to perform tasks on the job to collect enough and appropriate evidence for their portfolio of evidence, proof signed by their supervisor that the tasks were performed successfully.
To qualify	To qualify and receive credits towards the learning program, a registered assessor will conduct an evaluation and assessment of the learner's portfolio of evidence and competency
Range of Learning	This describes the situation and circumstance in which competence must be demonstrated and the parameters in which learners operate
Responsibility	The responsibility of learning rest with the learner, so: • Be proactive and ask questions, • Seek assistance and help from your facilitators, if required.

Learning Unit 1

Diagnose and locate faulty circuit, component and equipment

UNIT STANDARD NUMBER	:	386098
LEVEL ON THE NQF	:	5
CREDITS	:	4
FIELD	:	Physical, Mathematical, Computer and Life Sciences
SUB FIELD	:	Information Technology and Computer Sciences

PURPOSE:	This unit standard will provide learners with knowledge and skills on troubleshooting techniques. People credited with this unit standard are able to: ☐ Demonstrate knowledge of testing instruments/equipment. ☐ Locate faulty components. ☐ Rectify the problem.
LEARNING ASSUMED TO BE IN PLACE:	
☐ Communication at NQF Level 4. ☐ Mathematics at NQF Level 4. ☐ Physical Science at NQF Level 4.	

SESSION 1.

Demonstrate knowledge of testing instruments/equipment.

Learning Outcomes

- Instruments used for testing are identified in terms of their functions, setting and connectivity.
- Fundamental operational principles are explained using specified/instruments.
- Data obtained is documented using troubleshooting using organisational specified methods of record keeping.

Electronic test equipment

Electronic test equipment is used to create signals and capture responses from electronic devices under test (DUTs). In this way, the proper operation of the DUT can be proven or faults in the device can be traced.

Use of electronic test equipment is essential to any serious work on electronics systems.

Practical electronics engineering and assembly requires the use of many different kinds of electronic test equipment ranging from the very simple and inexpensive (such as a test light consisting of just a light bulb and a test lead) to extremely complex and sophisticated such as automatic test equipment (ATE).

ATE often includes many of these instruments in real and simulated forms.

Generally, more advanced test gear is necessary when developing circuits and systems than is needed when doing production testing or when troubleshooting existing production units in the field.

Basic equipment

The following items are used for basic measurement of voltages, currents, and components in the circuit under test.

- Voltmeter (Measures voltage)
- Ohmmeter (Measures resistance)
- Ammeter, e.g. Galvanometer or Milliammeter (Measures current)
- Multimeter e.g., VOM (Volt-Ohm-Milliammeter) or DMM (Digital Multimeter) (Measures all of the above)
- LCR meter - inductance (L), capacitance (C) and resistance (R) meter (measure LCR values)

The following are used for stimulus of the circuit under test:

- Power supplies
- Signal generator
- Digital pattern generator
- Pulse generator

The following analyze the response of the circuit under test:

- Oscilloscope (Displays voltage as it changes over time)
- Frequency counter (Measures frequency)

And connecting it all together:

- Test probes

Advanced or less commonly used equipment

Meters

- Solenoid voltmeter (*Wiggy*)
- Clamp meter (current transducer)
- Wheatstone bridge (Precisely measures resistance)
- Capacitance meter (Measures capacitance)
- LCR meter (Measures inductance, capacitance, resistance and combinations thereof)
- EMF Meter (Measures Electric and Magnetic Fields)
- Electrometer (Measures voltages, sometimes even tiny ones, via a charge effect)

Probes

- RF probe
- Signal tracer

Analyzers

- Logic analyzer (Tests digital circuits)
- Spectrum analyzer (SA) (Measures spectral energy of signals)
- Protocol analyzer (Tests functionality, performance and conformance of protocols)
- Vector signal analyzer (VSA) (Like the SA but it can also perform many more useful digital demodulation functions)
- Time-domain reflectometer (Tests integrity of long cables)
- Semiconductor curve tracer

Signal-generating devices



Leader Instruments LSG-15 signal generator.

- Signal generator usually distinguished by frequency range (e.g., audio or radio frequencies) or waveform type (e.g., sine, square, sawtooth, ramp, sweep, modulated, ...)
- Frequency synthesiser
- Function generator
- Digital pattern generator
- Pulse generator
- Signal injector

Principles of Electronics

What are the basic principles of electronics?

Electrons, a component of atoms, and their use—known as electronics—play an important role in many pieces of household equipment. Basic electronics comprises the minimal “electronics components” that make up a part of everyday electronics equipment. These electronic components include resistors, transistors, capacitors, diodes, inductors and transformers. Powered by a battery, they are designed to work under certain physics laws and principles. Basic electronics also concerns the measurement of voltage, current (electron flow) and resistance in the assembled working "circuit."

Principles of Electronics

All electronics equipment works on a fundamental physics principle known as Ohm's law, which states that a circuit contains a voltage directly proportional to the current and resistance encountered by the current in that circuit. The circuit comprises the electronic components also known as circuit elements, linked with wires to a battery and designed to obey Ohm's law.

Power Supply

The battery produces the voltage and current that drives or powers the electronic components. The electrons flow through wires and are modulated by the arrangement of the electronics components to produce specific results. The voltage is measured in units called volts and the current in units called amperes, or amps.

Resistors

The resistor is a circuit element designed to offer resistance to the flow of electrons called current. In so doing, it produces heat and, in turn, dissipates power—a resistance multiplied by the square of the current. A heating coil is an example of an application of a resistor. Arranged in series, resistors have an additive value (that is, the circuit's total resistance equals the sum of all of the resistors' value). Arranged in parallel, their combined value diminishes. The unit of resistance is the ohm. In practice, we have kilohm and megaohm units.

Capacitors

Capacitors are circuit elements that represent the opposite of resistors: They store electrical power. Their capacity is measured in farads and practical subunits of a farad such as microfarad and picofarad. When arranged in parallel, their combined value increases. When in series, their combined value diminishes. Basic electronics circuits always include some capacitors.

Diodes and Transistors

The diode, a circuit element, encourages current to flow in only one direction. It has two terminals, unlike the transistor—which has three. In the transistor, current can flow in more than one direction. Both diodes and transistors modulate current direction and voltage.

Inductors, Transformers and RLC Circuit

Basic electronics circuits often include inductors and transformers as circuit elements. An inductor is a coil of wire that produces a desired magnetic field, unlike a capacitor, which produces a desired electric field. When coupled with a resistor and capacitor, an inductor contributes to a special “tuning” circuit, called an RLC circuit, which can be tuned to various frequencies as electric current flows through it. Transformers, another type of circuit element, can step up a voltage or step it down to desired values. All these circuit components make up what is generally called “basic electronics.”

Measuring Instruments

Basic electronics is not complete without mention of the basic instruments used in electronics measurements. These include analog and digital meters that measure voltages, current, resistance and capacitance; power supplies, which provide stable regulated voltage and current; oscilloscopes, which measure circuit waveforms from circuits; and function generators, which provide standard desired waveforms.

What are the basic principles of electronics?

- Principles of Electronics. ...
- Power Supply. ...
- Resistors. ...
- Capacitors. ...

- Diodes and Transistors. ...
- Inductors, Transformers and RLC Circuit. ...
- Measuring Instruments.

Key Technical Issues

DATA MODEL

Which information and metadata are saved for each preserved record? How are they represented? Using which data types? The “data model” is the specification that answers these questions. The ingest process builds representations of records that conform to the model and the access process finds records, deciphers the representation according to the model, and presents the results to the user. The data model is thus a key interface in the system: it is the interface between the ingest process and the access process, using a linkage provided by the storage system. It is an “interface to the future.”

The data model must be designed to evolve, because new data types and requirements will emerge over the life of the system. Thus a careful design for the data model that tries to anticipate future ERA needs will simplify the system. While it is easy to allow different models to coexist in a single archive simply by labeling each record with the identity of the data model used to store it, a proliferation of data models will result in a costly proliferation of software to interpret them.

Some Possible Elements of a Digital Record

1. *Original files: one or more digital files, the original bit stream or native form of the record represented in its native data type.* A record may consist of more than one file, e.g., a report in which each chapter is represented by a separate word-processor file.
2. *Optional derived forms: digital files obtained from the original files by a converter.* ERA policies might encourage certain kinds of derived forms to be saved, for example these:
 1. A form whose data type is chosen to simplify “presentation” or “rendering” the record into a visible form for printing or display.
 2. A form whose data type is chosen to simplify content searching.

3. *Metadata for the record.* In addition to the usual metadata normally captured for records, there is additional metadata associated with electronic records, such as:

- Data types of the digital files and derived forms, with sufficient information to allow finding documentation about the data types. Data types will usually be versioned.
- Relationship among the digital files that constitute the record.
- Integrity checks—e.g., a cryptographic hash, for the digital files and the metadata.
- Ephemeral/nonderivable metadata—i.e., properties of the context in which the record was created that are not specified in the record itself.
- Derived metadata—i.e., properties that have been extracted from the record.
- Provenance and history, such as evidence that it was transferred accurately to the archive (a form of ephemeral metadata). In the case of derived forms, metadata identify the converter used to obtain the derived form from the original.
- Unique identifier of the record.
- The data type of the metadata—i.e., the definitions of metadata elements used to construct the metadata for this record.

A separate file. Metadata that pertain to collections as a whole might be stored in yet another file, referenced in metadata for each record in the collection.

The data model must also deal with embedding and aggregation. Embedding occurs when one record is embedded within another, e.g., a spreadsheet is embedded as an attachment within an e-mail message. Aggregation occurs when several records are saved together, e.g., a series of e-mail messages is saved in a single file, even though each message is to be treated as a separate record. Another form of aggregation that may be desirable is the container—e.g., as used by the SDSC demonstration—which simply collects a group of records into a single digital file for more efficient handling by the file system.

The archive should contain complete documentation about all versions of the data model, including specifications of the data types it uses. Since metadata sets are likely to proliferate.

A container is distinct from an archivist's "collection." A collection may span several containers, and several collections might fit within a single container.

and be complex, it is essential that any use of metadata be linked to a complete definition of its terms.

Several aspects of the data model require more discussion:

Evolution. The key to smooth evolution of the data model is to carefully label each digital file that is part of the stored record with its type; when the file is read, the type identifier selects software that can interpret the file correctly. This is sometimes called "self-identifying data." Whenever a data type is chosen as part of the data model, the system designer should ask, How do I introduce a new version of this data type without disrupting existing records? Clearly, if new types are introduced too frequently, the proliferation of types will lead to overly complex software to decode them all. Note that types must be versioned—it is not enough to say "this is a Microsoft Word document." A version number and perhaps platform (e.g., PC or Macintosh) are required for unambiguous identification

Unique identifier. The ERA should assign to each record a unique identifier that can be used, both inside and outside the archive, to refer to records. Such identifiers are used inside a system to identify records without regard to their location or storage mechanism; they are used outside a system to specify a link to a specific record. A related question is how separate parts of the representation of a record (e.g., distinct original files or derived forms) are identified. The unique identifier for digital records should be harmonized with ways of identifying other records held by NARA.³ There are a number of digital identifier techniques and implementations today, and it is entirely possible that the particular scheme used by the ERA may change a few times over the lifetime of the archive. The system design should accommodate such changes.

Modifying the archive. Although the principal idea of an archive is that records, once ingested, must not be modified, the ERA must allow certain kinds of information to be added or changed. Records themselves, once ingested, are rarely, if ever, modified or removed. It may be advantageous to add new derived forms long after the record was first ingested, perhaps when improved migration tools and techniques become available. And some of the metadata surrounding records (such as conditions of use, where to find derived versions, and relation to records added at a later date) may

need to be modified. Some metadata may need to be updated—for example, to record ephemeral metadata that come to light after the original ingestion, or to note changes to access controls for a record that result from new regulations or the passage of time. Importantly, all such modifications need to be logged in such a way that it is possible later to identify causes of mistakes and untangle them. It also may be wise to design the system so that information cannot be deleted, only augmented.

NARA does have a procedure for establishing a unique identifier for each record series for its paper records. Essentially, the records hierarchy has 13 levels, from Record Group to item, that assign a unique identifier based on the specific Record Group and the level of the record. This does not go all the way down to a unique ID for each record. It would be possible to add additional data (such as a date and sequence number) to provide unique identification.

Computer science researchers have designed “write-once” file systems in which it is not possible to delete a file, only to add new files or supplant old ones (add new versions of old files). This may not be a wise approach for the ERA. Suppose, however, certain records must be expunged from the archive, perhaps as a result of a court order. Even though rare, such a modification might be required.

Commonalities with digital libraries. The issues confronting the design of a data model for the ERA are almost exactly the same as those for a digital library. Unfortunately, there are no standard data models that can simply be adopted by NARA. Nevertheless, NARA should seek to align its design with that of digital libraries, since this will increase the likelihood of providing uniform access to libraries and archives in the future.

Closely related to the specification of the data model is a set of policies used to operate the archive. For example, the data model determines how derived forms are recorded in the archive, but a policy will specify which kinds of derived forms should be generated for a collection, perhaps specified as part of the collection's profile. Policies will also apply to modifications to the archive—who may introduce modifications and under what conditions (perhaps even requiring two staffers to confirm certain kinds of changes, as is common practice in the financial services industry).

Data Types and Obsolescence

The data model confronts the most challenging problem of an archive: For records to be useful many decades after they are ingested, they must be expressed in the data model using data types that can still be decoded and interpreted at the time of access. By that time the computers and software used to create the original records may be obsolete.

The archival community has written at length on this topic. The debate over emulation vis-à-vis migration continues (see the brief discussion of emulation and migration in the preceding chapter), and neither one is firmly established as the only way to preserve digital records. In fact, some researchers in digital preservation reject the notion that emulation and migration are mutually exclusive (assuming, as we do here, that the requisite information to support each is retained) and argue that each is appropriate under particular circumstances. Consequently, the ERA should be designed to:

- Record in the archive the information necessary to support both emulation and migration when and if either becomes common. The original bit stream and careful records of the software environment in which it was created (data type and ephemeral metadata to record the fonts, operating system, application program, and other digital resources or pointers to them) are essential.
- Function adequately in the absence of either emulation or migration solutions. This requires some pragmatic choices, which are discussed here and in

NARA shares with others an interest in making one or another of these preservation strategies operational, such as building emulators for a few of the most popular computers.

Metadata

Metadata are conventionally expressed as a series of "attribute, value" pairs (also sometimes called "tag, value" pairs). A metadata set is a collection of attribute names and corresponding definitions used to express metadata. There are no standard universal metadata sets, though many popular subsets exist.

Some attempts are under way to try to relate all metadata definitions to a common ontology, but this complex approach is very risky. It seems clear that metadata sets will

change frequently over the lifetime of the ERA and that the system should be designed to accommodate such change, an observation confirmed by the experience of the SDSC demonstrations. XML offers one convenient syntax today for recording metadata, in part because it is easily extensible; it is already the representation of choice in the digital library community.

STORAGE

The storage function of the ERA is used to save files for a long time (perhaps it should be called a “long-term file system”). It thus serves as a transparent bridge between ingest and access functions: The files produced by ingest are delivered without modification to access. In many respects it is like any other file system. Some of its requirements, though not commonplace, are shared with other very large, mission-critical systems, including these:

- *Scalability.* It must be able to grow to hold many (even hundreds of) petabytes of data, in many billions or even trillions of individual files.
- *Robustness.* It must not corrupt data or fail to deliver data requested.
- *Access control.* Only authorized software, under the control of authorized staff, may store files or make other modifications to what is stored.

It is in the area of robustness over a very long period of time—the lifetime of its data is longer than that of any known data storage device or medium—that the system presents unusual requirements.

The following strategies can be used to preserve digital files for many decades:

Store redundant replicas. To survive the failure of a storage device, information is stored redundantly. But care must be taken to categorize failure modes and determine appropriate storage strategies. For example, if a file is stored on two different disks, both of which are located in the same room, a fire might destroy both copies. So some kind of geographic replication is a must—usually achieved today by saving copies of a file at one or more remote sites connected by a high-speed network. Replication is governed by policies enforced by the file system that specify how many copies should be kept, where they should be stored, and so on.

Detect errors and correct them automatically. In order to detect failed devices in a large file system, it is necessary to use a background task to read all the data in the system constantly and check for errors caused by deterioration. This process requires computing a hash or checksum of each file and comparing the value with a value computed from the same data when it was first stored. When a corrupted file is detected, one of its redundant copies is used to create a fresh copy somewhere else so as to conform to the replication policy. (And other files stored on the same device that produced errors are probably also moved, since the device as a whole may be failing. The device is then removed from service or replaced.)

Refresh media. As storage media age, copy all the files stored on them to new devices,

Calling it an archival file system risks confusing it with archival file systems offered by some computer system vendors, often as part of a hierarchical storage management product.

The robustness requirements may well not be uniform for all records. The requirements might, for example, be a property of a record that is assigned at ingest time and might be reflected in how much the storage system “invests” in preserving that record. This is a good example of how complicated some of the architectural issues for the ERA can be.

Disk and tape recording formats include checksums that are verified on each read, but these are viewed as insufficiently robust. A file-level hash is advisable as well.

The term “refresh” is preferred to “migrate,” because the second term is used to describe a conversion of data type.

Rebuild directories and indexes by scanning the archive. If sensitive data that describe the structure of a file system are lost, the entire file system can be scanned to rebuild the data. Such a design requires storing files on the disk in a way that records their structure within the file system.

Rebuild directories and indexes by scanning the archive. If sensitive data that describe the structure of a file system are lost, the entire file system can be scanned to rebuild the data. Such a design requires storing files on the disk in a way that records their structure within the file system.

These techniques are all widely used (or at least widely advocated) today. The engineering challenge for NARA is that no one has yet demonstrated that they can, together, implement a file system that will preserve files for a hundred years or more.

The scalability of the file system, as well as its ability to connect new storage devices to replace old ones, can be achieved using network-connected storage systems. When it is no longer possible or economically feasible to expand the capacity of any existing storage system, a new system is procured, attached to the network, and configured to participate in the overall file system.

The overall file system operates as a “federation” of the network-connected storage components. This technique uses a software layer that makes the collection of network-connected systems appear to be a single file system. It would provide services such as these:

- Naming of files, and mapping names to locations where files are stored;
- Routing file access requests to the appropriate network-connected store, incorporating newly added stores, and providing the necessary “drivers” for new classes of network storage services;
- Access control;
- Redundancy control, as described above (maintaining redundant copies, scanning for corrupted files, refreshing old media, removing old storage from service);
- Audit log, used to track all changes to the file system;
- Performance measurement, used to determine the load placed on the file system, its internal overhead (e.g., for integrity auditing or media refresh), occurrences of faulty media, and so on;
- Management processes, used to modify configurations when new storage is added to the federation or to direct that old storage should be evacuated and abandoned, etc.

The software layer insulates the ERA clients of the file system from the various implementations used in the federation. Implementations can be replaced by

changing at most a driver in the software. The Storage Request Broker (SRB), used in the SDSC demonstrations, is one example of such a distributed file system, but the technology is quite common.

File System Performance Requirements

The file system must be designed to meet the scale and performance requirements that the ERA will face. Examples of some of the file-system-specific performance metrics required to guide design are these:

- *Size*. What is the target size of the initial repository? How many bytes? How many files? How will each of these scale over time?
- *Bandwidth*. How much bandwidth will be necessary to support ingest, access, file scanning, and media refresh?
- *Refresh time*. How long will it take to copy or recreate the entire archive? This parameter is important, because if the file system becomes so large that it takes 5 years to copy and the refresh cycle needs to be repeated every 4 years, the design will not work.
- *Ingest rate*. At what rate will new records be incorporated into the repository? Note that the geographic redundancy requirement means that the ingest rate heavily influences the bandwidth required for communication with remote storage, as new data are copied to remote sites.
- *Response time*. Must the system support real-time queries? If so, what are reasonable response time targets?
- *Life span*. At what rate are components expected to be replaced (particularly the storage devices themselves). What is the unit of replacement? What is the impact of replacement on performance?

Performance requirements for access are harder to anticipate than those for ingest and storage. Although the conventional model for access is a user conducting a search and reading records on a computer screen, it is also likely that users in the future will want to run statistical queries, perform data mining and automated cross-referencing and correlation, and do other things that a future access system may or may not support and which could have significant bearing on an access system's performance

requirements. Moreover, it is likely that access will involve creating full-text indexes of collections, which imposes an additional load on the long-term file system.

These targets may eliminate certain technology choices (for example, access performance requirements might eliminate the possibility that tape could be used as a primary preservation medium). They are essential to guide the implementation of software that controls the federation—for example, the design of the naming system is quite sensitive to anticipated system size.

Selecting Storage Media

Presently, NARA stores most of its electronic records using off-line tape storage; this is also the approach used in the SDSC demonstrations. For new systems, disks are becoming the preferred storage choice. Instead of storing files on tape and copying and storing backups offsite, digital archives are kept on geographically separated disk replicas. Data are transferred to other locations using either a network connection or by shipping disks or servers containing the files offsite. While other storage media, such as optical disks, have been considered for long-term preservation, they suffer from many of the same drawbacks as tape.

Disks have a number of advantages compared with tape, including these:

- *Lower cost.* The overall system costs of tape and disk storage are roughly equal, but the cost per byte of disk storage is declining faster than that of tape. (Cost comparisons of disk and tape are complicated by debates about what to count, especially the human support staff required.) Projections favor disks in coming years.
- *Volumetric density.* Disks take less physical space to store the same amount of data. Densities are improving by a factor of about 2 every year.
- *Fast access.* Disks allow fast access, suitable for interactive applications. By contrast, tape processing incurs a variety of access delays. Tapes must be mounted (either by a robot or manually) and once mounted, must be read sequentially.
- *Less complexity.* Disk-only file systems are much simpler than hierarchical storage management schemes that must manage the migration of data between disk

and tape. Tapes also require large file sizes to be efficient, which results in complex file-aggregation mechanisms. If accessing a record requires reading files from several disks or tapes because they were created at different times (e.g., metadata or derived forms that were added long after the record was ingested), disk-only systems will perform far better than tape systems. A tape system could be designed to collect and rewrite in one spot disparate portions of a record, but this adds complexity.

- *Automatic error detection and recovery.* Regardless of actual disk lifetimes, the file system must be designed to save files on more than one disk and perform file integrity checks frequently enough that files corrupted due to bad disks are identified before the last replica of the file is also unreadable. When a disk is found to be bad, all files stored on it are copied elsewhere and the disk is removed from service or replaced.

SESSION 2.

Locate faulty components.

Learning Outcomes

- Diagnosis of faults is planned and prepared to ensure Occupational Health and Safety policies and procedures are followed in accordance with organisational requirements.
- Tools, equipment and testing devices needed to carry out the work are obtained in accordance with established procedures.
- Symptoms that may lead to the cause of the problem are identified to take a corrective action.
- Faulty components are checked against job requirements and worksite procedures.
- Components, circuit and equipment are checked as being isolated where necessary using testing procedures.
- Fault(s) is/are diagnosed in accordance with requirements without damage or distortion to the surrounding environment or services.

Electronic Kit Basic Fault Finding Guide

Basic Fault Finding

Here are some simple fault finding techniques that you can try.

1. Check that all components are in the correct position and orientation.
2. Check that there are no short circuits (excessively high current drain on the power supply may indicate shorts). Examine the PCB for solder splashes across pads/tracks and stray component leads/wires.
3. Carry out some basic electrical tests to ensure power is reaching all parts of the circuit.
4. Check all components are functioning correctly. Has overheating during soldering damaged any components? IC's and transistors are particularly susceptible.
5. Remove and test suspect components using appropriate test equipment. Alternatively, substitute dubious components with tried and tested ones.
6. Make a thorough check of circuit voltages/currents/signals using appropriate

test equipment.

7. Where kits are connected to other equipment check for compatibility problems by testing on alternative equipment.

By adopting a systematic approach to fault finding you will eventually discover where the problem lies. If you wish to learn more about the design and testing of electronic circuits we recommend that you explore the extensive range of books and magazines that are available on the subject.

Please note that we are unable to provide specific help or advice on design modifications or adaptations of project kits for individual customer applications.

Fault finding can be quick and easy or a case of time, effort and perseverance. It may (and will) be frustrating at times but it really is one of the best teachers, providing you with a greater knowledge and understanding of electronics. We have all been there. Have faith and keep trying! The sense of satisfaction of finally cracking the problem can be immense.

Electrical test instruments

Safety is still the first tool

Safety is paramount with electrical test instruments and great care must be taken when using equipment or understanding the codes involved.



There is a great deal of attention devoted to safe work practices during electrical construction, maintenance and repair work. Industry electrical publications regularly report on safety issues, including the use of the proper tools and equipment used for energized and de-energized work, as well as utilizing the correct personal protective equipment (PPE) for each workplace situation.

Electrical test instruments are given very little discussion, if any, in safety articles. Examples include using the wrong test instruments or improperly using them, which can have catastrophic results. Some of the most frequently used test instruments include noncontact voltage testers, multimeters, insulation testers and ground-resistance testers. The issue with using a non-contact or proximity device is that the requirement to test a circuit to ensure that it is de-energized requires the circuit to be tested phase-to-phase and phase-to-ground, which cannot be done using this type of tester.

When electrical safety is discussed, the subjects of shock, arc flash, and arc blast dominate the discussions. One question is often asked: "How do I identify when these hazards are present, or likely to be present, when I am using electrical test instruments on electrical circuits and equipment?" A review of these electrical hazards, along with requirements for assessing the workplace to identify the electrical hazards and PPE associated with using test instruments, is one way to get to the answer.

Electrical hazards

Electricity is a serious workplace hazard, exposing employees to electrical shock, electrocution, burns, fires and explosions. Employees have been killed or injured in fires and explosions caused by electricity.

Besides the electrical hazards of arc flash and arc blast, extremely high energy arcs can damage equipment and cause fragmented metal to fly in all directions. In atmospheres that contain explosive gases or vapors, or combustible dusts, even low-energy arcs can cause violent explosions. In these cases, the electric arc may be the ignition source for a much bigger explosion and fire.

Improper use of electrical test instruments can result in shock or electrocution, as well as an arc flash incident. This article addresses these issues, along with the requirements for selecting and utilizing the test instruments to verify the presence of voltage.

Selection of test instruments

Regardless of whether you are performing electrical installation work, equipment maintenance, verifying the absence of voltage for de-energized work, troubleshooting, voltage measurements or similar diagnostic work, collecting accurate and consistent information from these tests is imperative. To comply with electrical industry standards and regulations, there is a need to select and use the right test instruments according to the application.

When conducting voltage verification, for energized and de-energized work, the electrical worker must select the right test instruments and equipment applicable to the work to be performed. As a minimum, these should include the following:

- Voltage indicating instrument suitable for conditions
- Environment
- Correct CAT category I, II, III, or IV
- Continuity test instrument
- Insulation resistance test instrument.

All test instruments include specific manufacturer's operational instructions. Test instruments must be certified and display a label of an independent verification lab, such as UL, CSA, CE, ETL or TÜV. Make sure all meters, test leads and probes have an adequate category (CAT) safety rating. Sometimes, the only thing standing between an electrical worker and an unexpected spike is their meter and test leads. If you use the wrong equipment with the wrong voltage, you could be putting yourself and others at risk. So, before conducting any test, make sure your choice of instrument is correct.

Electrical standards, such as UL, ANSI, IEC, and CAN, specify protection from currents at levels well above a system's rated capacity. Without this additional protection, transient overvoltages, which are becoming increasingly common, can lead to equipment failure and serious injury or death.

Minimizing such risks requires that everyone working in electrical environments has safety equipment as required. They need properly rated gloves, eye protection and

electrical measurement test instruments that provide appropriate protection. Having the correct electrical testing and measurement instruments and using the correct procedures can improve job safety.

Use of electrical test instruments

Due to the potential electrical hazards associated with the use of electrical test instruments, only *qualified persons* are permitted to perform tasks such as testing, troubleshooting and voltage measuring when working within the Limited Approach Boundary of exposed energized electrical conductors or circuit parts operating at 50 volts or more, or where any other electrical hazard may exist. Improper use of electrical test instruments can result in shock or electrocution, as well as creating an arc flash incident.

The following additional requirements apply to test instruments, equipment, and all associated test leads, cables, power cords, probes, and connectors:

- Must be rated for circuits and equipment where they are utilized.
- Must be designed for the environment to which they will be exposed and for the manner in which they will be utilized.
- Must be visually inspected for external defects and damage before each use. If there is a defect or evidence of damage that might expose an employee to injury, the defective or damaged item shall be removed from service.

When test instruments are used for testing the absence of voltage on conductors or circuit parts operating at 50 volts or more, the operation of the test instrument must be:

- Verified on a known voltage source before an absence of voltage test is performed.
- Test for the absence of voltage on the de-energized conductor or circuit part. A zero reading might mean that no voltage is present during the testing, or it could mean that the instrument has failed.
- Verified on a known voltage source after an absence of voltage test is performed.

This verification primarily applies to conductors or circuit parts operating at 50 volts or more. However, under certain conditions (such as wet contact or immersion) even circuits operating under 50 volts can pose a shock hazard.

Only qualified persons are permitted to perform tasks such as testing, troubleshooting, and voltage measuring, due to the electrical hazards associated with energized work. All required PPE, for the associated hazards, must be utilized when performing these tasks. Test instruments must be rated for the conditions under which testing is to be performed. When selecting voltage testing instruments, an assessment must be performed to determine the proper category (CAT) rating required, based on the highest hazard exposure.

When test instruments are used for testing the absence of voltage, for de-energized work, on conductors or circuit parts operating at 50 volts or more, the operation of the test instrument must be verified on a known voltage source before and after an absence of voltage test is performed.

Electrical safety checklist

The fundamentals of electrical safety can be overlooked, especially by seasoned electricians. It's worth reviewing a few safety tips, both for the novice electrician and the veteran:

- Use a meter that meets accepted safety standards for the environment in which it will be used.
- Use a meter with fused current inputs and be sure to check the fuses before making current measurements.
- Inspect test leads for physical damage before making a measurement.
- Use the meter to check continuity of the test leads.
- Use test leads that have shrouded connectors and finger guards.
- Use meters with recessed input jacks.
- Select the proper function and range for your measurement.
- Be certain the meter is in good operating condition.

- Follow all equipment safety procedures.
- Always disconnect the "hot" (red) test lead first.
- Don't work alone.
- Use a meter that has overload protection on the ohms function.
- When measuring current without a current clamp, turn the power off before connecting into the circuit.
- Be aware of high-current and high-voltage situations and use the appropriate equipment, such as high-voltage probes and high-current clamps.

CAT ratings and their definitions

Here's a brief review of the four category (CAT) ratings and their basic definitions:

Category I

This typically covers electronic equipment. Signal level for telecommunications, electronic equipment and low-energy equipment with transient-limiting protection. The peak impulse transient range is from 600 to 4,000 volts with a 30-ohm source.

- Protected electronic equipment
- Equipment connected to (source) circuits in which measures are taken to limit transient overvoltages to an appropriately low level
- Any high-voltage-low-energy source derived from a high-winding resistance transformer, such as the high-voltage section of a copier.

Category II

Single-phase receptacle connected loads. Local level for fixed or non-fixed powered devices-everything from lighting to appliances to office equipment. Also, all outlets at more than 10 meters (30 feet) from Category III sources and all outlets at more than 20 meters (60 feet) from Category IV sources. The peak impulse transient range is from 600 to 6,000 volts with a 12-ohm source.

- Appliance, portable tools and other household and similar loads
- Outlet and long branch circuits
- Outlets at more than 10 meters from CAT III source

- Outlets at more than 20 meters from CAT IV source.

Category III

Three-phase distribution, including single-phase commercial lighting. Distribution level-fixed primary feeders or branch circuits. These circuits are usually separated from Category IV (whether utility service or other high-voltage source) by a minimum of one level of transformer isolation; for example, feeders and short branch circuits, distribution branch panels and heavy appliance outlets with "short" connections to service entrance. The peak impulse transient range is from 600 to 8,000 volts with a 2-ohm source.

- Equipment in fixed installations, such as switchgear and polyphase motors
- Bus and feeders in industrial plants
- Feeders and short branch circuits, distribution panel devices
- Lighting systems in larger buildings
- Appliance outlets with short connections to service entrance.

Category IV

Three-phase at utility connection, any outdoor conductors or primary supply level. It will cover the highest and most dangerous level of transient overvoltage you are likely to encounter-in utility service to a facility both outside and at the service entrance, as well as the service drop from the pole to the building, the overhead line to a detached building, and the underground line to a well pump. The peak impulse transient range is from 600 to 12,000 volts with a less than 1-ohm source.

- "Origin of installations," such as where low-voltage connection is made to utility power
- Electricity meters, primary overcurrent protection equipment
- Outside and service entrance, service drop from pole to building, run between meter and panel
- Overhead line to detached building, underground line to well pump.

Transistor Circuit Test & Fault Finding using a Multimeter

One of the main uses for multimeters whether they are analogue multimeters or digital multimeters, DMMs is to test and fault find circuits like those in a transistor radio. Multimeters are ideal items of test equipment for finding many faults in a transistor circuit.

However to use a multimeter to test a circuit and find faults it is necessary to have a little knowledge about the circuit, and also to adopt a logical approach in tracking down any faults that may exist.

A word of warning

Some transistor equipment may be mains powered. Only qualified persons should attempt to repair mains powered equipment or equipment that contains high or hazardous voltages. High voltages can kill so be warned!

Look for obvious faults

The first step when looking at tracing any faults and testing a transistor circuit of any sort is to look for the obvious or major faults. Fortunately most faults with electronics equipment such as transistor radios are relatively major and therefore easy to find. Accordingly the first step in any fault finding is to look for the major problems.

Check the supply to the circuit: The first steps in checking the circuit are to ensure that it has power being supplied to it. This is easily done using a multimeter set to a voltage range. Measure the voltage using a multimeter at the points where the supply enters the circuit board. If the multimeter indicates that there is no supply voltage then there can be a number of possibilities to investigate:

1. Battery could be flat if the equipment is battery powered
2. If the transistor equipment is battery powered, the battery could have been left in for many months and corroded the battery compartment. Check for this and if there are signs of leakage then clean and remove all signs of corrosion being careful not to touch any of the residue.
3. On-off switch faulty. This can be checked by disconnecting any power source and checking for continuity across the switch
4. Corroded connector. One common problem is that connectors become

corroded with time and connections can become very poor, especially if the equipment has not been used for some time. To overcome this it can help to unplug and then re-mate the connector.

5. Check for any broken wiring that would prevent the power reaching the circuit board.

Check the outputs from the board: In the same way that broken connections may exist for the power line, the same may be true of the outputs from the board. Again it is worth checking any connectors that may have corroded or oxidised with time, and check for any broken connections.

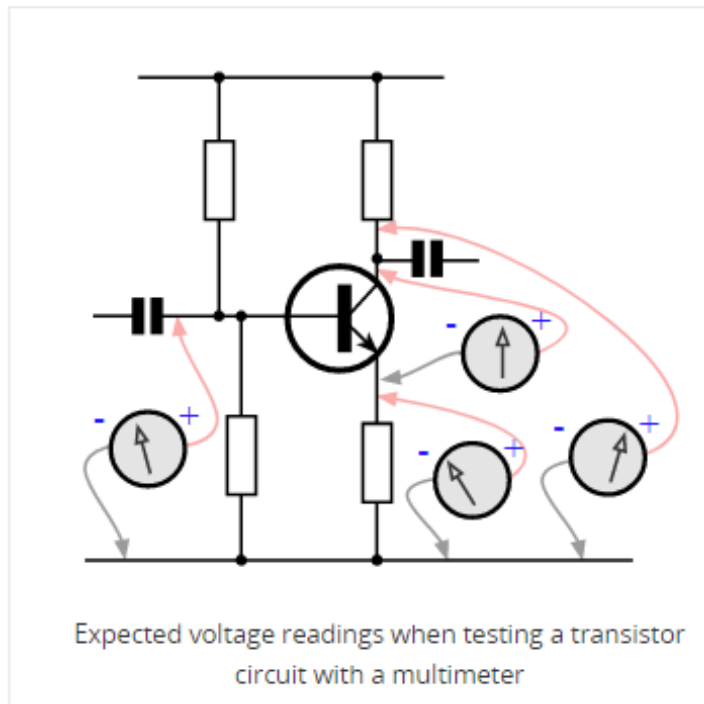
Check the inputs to the circuit: Likewise, if the signal inputs are not reaching the board then it will not be able to perform. Again any switches, and connectors along with any broken wires should be checked. Often a multimeter can be used to check the continuity of the wires, but first ensure there is no power applied to the circuit.

By using a multimeter for the fault finding, it is possible to find many of the obvious faults that can occur. If the problem cannot be found, and it appears that the correct power is reaching the transistor circuit, and the inputs are all connected and present as well as the output lines being intact, then further fault finding on the transistor circuit board itself may be needed. Again a multimeter can assist in this.

Expected voltages in a transistor circuit

If all the inputs to the board appear to be correct, further tests can be undertaken using a multimeter for the fault finding and to track down the problem. Again a systematic approach should be adopted.

When testing a particular transistor circuit, a multimeter can be used to determine if the voltages around the circuit are correct. To test and fault find a particular transistor circuit, it is necessary to have an idea what the steady voltages should be. The circuit below is a typical basic transistor circuit. Many circuits are similar to it, and it provides good starting point to explain some of the points to note.



The circuit shows several of the points where the voltage can be measured in a circuit. Most of them are measured with respect to ground. This is the easiest way to make a voltage measurement because the "common" or negative probe can be clipped to a suitable ground point (many black probes used for the negative line have a crocodile or alligator clip for this purpose). Then all the measurements can be made relative to ground.

There are normally a number of points around a transistor circuit that are easy to measure, and the expected voltages can be anticipated for the most part if a few assumptions are made:

- Assume the circuit is operating in a linear mode, i.e. it is not a switching circuit.
- Assume the circuit is operating in a common emitter mode as shown in the diagram.
- Assume the circuit has a resistive collector load.

If the assumptions above are true, then the following voltages can be expected. If not then allowances need to be made for the changes.

1. The collector voltage should sit at approximately half the rail voltage. More specifically it should sit at half the rail voltage less the emitter voltage. In this way

the greatest voltage swing can be obtained. If the transistor has an inductive load, as in the case of the intermediate frequency amplifier in a radio which may have an IF transformer in the collector circuit, then the collector should sit at virtually the same voltage as the rail voltage.

2. The emitter voltage should sit at around a volt or two. In most class A common emitter circuits an emitter resistor is included to give some DC feedback. The voltage across this resistor is typically a volt or so.
3. The base voltage should sit at the PN junction turn on voltage above the emitter. For a silicon transistor, which is the most common type, this is around 0.6 volts.

Indications of the types of voltage expected can be seen on the circuit diagram.

In addition to this, there are many other types of circuit which may need fault finding. Switching circuits are quite common these days where transistors are used to drive other elements such as relays or other devices. These do not operate in a linear mode. Instead all voltages are either on or off.

The collector voltage will either be approximately zero when the transistor is on, and approximately the rail voltage when off. The emitter will usually be connected to ground, and the base voltage will be high, i.e. approximately 0.6 volts for a silicon transistor when the transistor is on (i.e. collector near zero), and low, (zero volts) when the transistor is off and the collector is high.

Summary

A multimeter is an ideal piece of test equipment to help with fault finding an electronics transistor circuit. Often circuits like transistor radios fail after they have been used for many years, and it is useful to be able to mend them. Also when constructing equipment, circuits do not always work first time and it is necessary to fault-find these circuits. While it will not be possible to solve all problems using a multimeter, it is one of the most useful basic tools for any fault finding job.

How to Find a Fault in a Circuit Board

- Remove the cover from your equipment. Most electronic and computer covers are held on by four screws, two on each side near the bottom. Remove the screws with the screwdriver and set them aside. Lift the cover off.
- Check to see if all the circuit boards (CB) are plugged in tightly. You may have only one, but if you have more, check them all. Many CB's have multi-pin edge connectors. Moving can cause them to work loose. Push the cards gently into their connecting slots to make sure they're plugged all the way in.
- Check individual plug-in components. Some chips, "daughter boards" or "piggy back" boards fit into their own sockets, and can have a loose connection. Pull them out and inspect them for dirty or corroded pins, and then reseat by pushing them gently in place so they are properly seated. The pins should be shiny and bright if they are clean and not corroded.
- Look for water or foreign objects anywhere on the circuit board. Sometimes a paper clip dropped onto a circuit board can stop it from working.
- Check secondary wiring plugs. Daughter boards and piggy-back boards may have a cable connection that just pushes on. Some of these can work loose in moving. Pull them off, inspect them for wear or corrosion and then replace them to ensure a snug fit.
- Look for blackened components or melted or broken parts of the printed circuit traces, the little metal lines that connect components together on both sides of the board. Components can burn out and when they do they can smoke. Also look for swollen or discolored components.
- Look for broken leads on the components. Some components have tiny wire leads that can easily break off right next to the circuit board or close to the component itself. Lightly jiggle larger components with wires to see if a wire might be broken.
- Look for cracks on the circuit board. A hairline crack may not look too damaging but it can include broken circuit traces.

Swap identical components

In a system with identical or parallel subsystems, swap components between those subsystems and see whether or not the problem moves with the swapped component. If it does, you've just swapped the faulty component; if it doesn't, keep searching!

This is a powerful troubleshooting method, because it gives you both a positive and a negative indication of the swapped component's fault: when the bad part is exchanged between identical systems, the formerly broken subsystem will start working again and the formerly good subsystem will fail.

I was once able to troubleshoot an elusive problem with an automotive engine ignition system using this method: I happened to have a friend with an automobile sharing the exact same model of ignition system. We swapped parts between the engines (distributor, spark plug wires, ignition coil—one at a time) until the problem moved to the other vehicle. The problem happened to be a “weak” ignition coil, and it only manifested itself under heavy load (a condition that could not be simulated in my garage). Normally, this type of problem could only be pinpointed using an ignition system analyzer (or oscilloscope) and a dynamometer to simulate loaded driving conditions. This technique, however, confirmed the source of the problem with 100% accuracy, using no diagnostic equipment whatsoever.

Occasionally you may swap a component and find that the problem still exists, but has changed in some way. This tells you that the components you just swapped are *somehow different* (different calibration, different function), and nothing more. However, don't dismiss this information just because it doesn't lead you straight to the problem—look for other changes in the system as a whole as a result of the swap, and try to figure out what these changes tell you about the source of the problem.

An important caveat to this technique is the possibility of causing further damage. Suppose a component has failed because of another, less conspicuous failure in the system. Swapping the failed component with a good component will cause the good component to fail as well. For example, suppose that a circuit develops a short, which “blows” the protective fuse for that circuit. The blown fuse is not evident by inspection, and you don't have a meter to electrically test the fuse, so you decide to swap the suspect fuse with one of the same rating from a working circuit. As a result of this, the good fuse that you move to the shorted circuit blows as well, leaving you with two

blown fuses and two non-working circuits. At least you know for certain that the original fuse was blown, because the circuit it was moved to stopped working after the swap, but this knowledge was gained only through the loss of a good fuse and the additional "down time" of the second circuit.

Another example to illustrate this caveat is the ignition system problem previously mentioned. Suppose that the "weak" ignition coil had caused the engine to backfire, damaging the muffler. If swapping ignition system components with another vehicle causes the problem to move to the other vehicle, damage may be done to the other vehicle's muffler as well. As a general rule, the technique of swapping identical components should be used only when there is minimal chance of causing additional damage. It is an excellent technique for isolating non-destructive problems.

Example 1: *You're working on a CNC machine tool with X, Y, and Z-axis drives. The Y axis is not working, but the X and Z axes are working. All three axes share identical components (feedback encoders, servo motor drives, servo motors).*

What to do: Exchange these identical components, one at a time, Y axis and either one of the working axes (X or Z), and see after each swap whether or not the problem has moved with the swap.

Example 2: *A stereo system produces no sound on the left speaker, but the right speaker works just fine.*

What to do: Try swapping respective components between the two channels and see if the problem changes sides, from left to right. When it does, you've found the defective component. For instance, you could swap the speakers between channels: if the problem moves to the other side (i.e. the same speaker that was dead before is still dead, now that its connected to the right channel cable) then you know that speaker is bad. If the problem stays on the same side (i.e. the speaker formerly silent is now producing sound after having been moved to the other side of the room and connected to the other cable), then you know the speakers are fine, and the problem must lie somewhere else (perhaps in the cable connecting the silent speaker to the amplifier, or in the amplifier itself).

If the speakers have been verified as good, then you could check the cables using the same method. Swap the cables so that each one now connects to the other channel

of the amplifier and to the other speaker. Again, if the problem changes sides (i.e. now the right speaker is now "dead" and the left speaker now produces sound), then the cable now connected to the right speaker must be defective. If neither swap (the speakers nor the cables) causes the problem to change sides from left to right, then the problem must lie within the amplifier (i.e. the left channel output must be "dead").

Remove components

If a system is composed of several parallel or redundant components which can be removed without crippling the whole system, start removing these components (one at a time) and see if things start to work again.

Example 1: *A "star" topology communications network between several computers has failed. None of the computers are able to communicate with each other.*

What to do: Try unplugging the computers, one at a time from the network, and see if the network starts working again after one of them is unplugged. If it does, then that last unplugged computer may be the one at fault (it may have been "jamming" the network by constantly outputting data or noise).

Example 2: *A household fuse keeps blowing (or the breaker keeps tripping open) after a short amount of time.*

What to do: Unplug appliances from that circuit until the fuse or breaker quits interrupting the circuit. If you can eliminate the problem by unplugging a single appliance, then that appliance might be defective. If you find that unplugging almost any appliance solves the problem, then the circuit may simply be overloaded by too many appliances, neither of them defective.

Divide systems into sections

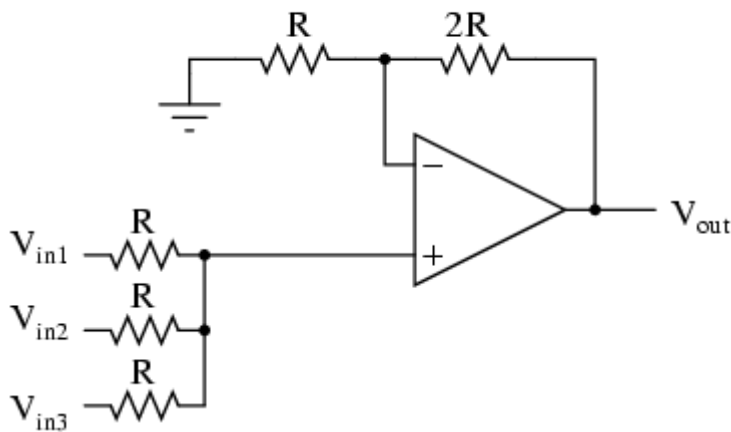
In a system with multiple sections or stages, carefully measure the variables going in and out of each stage until you find a stage where things don't look right.

Example 1: *A radio is not working (producing no sound at the speaker)*

What to do: Divide the circuitry into stages: tuning stage, mixing stages, amplifier stage, all the way through to the speaker(s). Measure signals at test points between these stages and tell whether or not a stage is working properly.

Example 2: An analog summer circuit is not functioning properly.

Analog summer circuit



What to do: test the passive averager network (the three resistors at the lower-left corner of the schematic) to see that the proper (averaged) voltage was seen at the noninverting input of the op-amp. I would then measure the voltage at the inverting input to see if it was the same as at the noninverting input (or, alternatively, measure the voltage difference between the two inputs of the op-amp, as it should be zero). Continue testing sections of the circuit (or just test points within the circuit) to see if you measure the expected voltages and currents.

Simplify and rebuild

Closely related to the strategy of dividing a system into sections, this is actually a design and fabrication technique useful for new circuits, machines, or systems. It's always easier begin the design and construction process in little steps, leading to larger and larger steps, rather than to build the whole thing at once and try to troubleshoot it as a whole.

Suppose that someone were building a custom automobile. He or she would be foolish to bolt all the parts together without checking and testing components and subsystems as they went along, expecting everything to work perfectly after its all assembled. Ideally, the builder would check the proper operation of components along the way through the construction process: start and tune the engine *before* its connected to the drivetrain, check for wiring problems *before* all the cover panels are put in place, check the brake system in the driveway *before* taking it out on the road, etc.

Countless times I've witnessed students build a complex experimental circuit and have trouble getting it to work because they didn't stop to check things along the way: test all resistors *before* plugging them into place, make sure the power supply is regulating voltage adequately *before* trying to power anything with it, etc. It is human nature to rush to completion of a project, thinking that such checks are a waste of valuable time. However, more time will be wasted in troubleshooting a malfunctioning circuit than would be spent checking the operation of subsystems throughout the process of construction.

Take the example of the analog summer circuit in the previous section for example: what if it wasn't working properly? How would you simplify it and test it in stages? Well, you could reconnect the op-amp as a basic comparator and see if its responsive to differential input voltages, and/or connect it as a voltage follower (buffer) and see if it outputs the same analog voltage as what is input. If it doesn't perform these simple functions, it will never perform its function in the summer circuit! By stripping away the complexity of the summer circuit, paring it down to an (almost) bare op-amp, you can test that component's functionality and then build from there (add resistor feedback and check for voltage amplification, then add input resistors and check for voltage summing), checking for expected results along the way.

Trap a signal

Set up instrumentation (such as a datalogger, chart recorder, or multimeter set on "record" mode) to monitor a signal over a period of time. This is especially helpful when tracking down intermittent problems, which have a way of showing up the moment you've turned your back and walked away.

This may be essential for proving what happens first in a fast-acting system. Many fast systems (especially shutdown "trip" systems) have a "first out" monitoring capability to provide this kind of data.

Example #1: *A turbine control system shuts automatically in response to an abnormal condition. By the time a technician arrives at the scene to survey the turbine's condition, however, everything is in a "down" state and its impossible to tell what signal or condition was responsible for the initial shutdown, as all operating parameters are now "abnormal."*

What to do: One technician I knew used a videocamera to record the turbine control panel, so he could see what happened (by indications on the gauges) first in an automatic-shutdown event. Simply by looking at the panel after the fact, there was no way to tell *which* signal shut the turbine down, but the videotape playback would show what happened in sequence, down to a frame-by-frame time resolution.

Example #2: *An alarm system is falsely triggering, and you suspect it may be due to a specific wire connection going bad. Unfortunately, the problem never manifests itself while you're watching it!*

What to do: Many modern digital multimeters are equipped with "record" settings, whereby they can monitor a voltage, current, or resistance over time and note whether that measurement deviates substantially from a regular value. This is an invaluable tool for use in "intermittent" electronic system failures.

FAULT DIAGNOSIS

Basic Concepts

A unit under test (UUT) fails when its observed behavior is different from its expected behavior. Diagnosis consists of locating the physical fault(s) in a structural model of the UUT. The degree of accuracy to which faults can be located is called *diagnostic resolution*. *Functionally equivalent faults* (FEF) cannot be distinguished. The partition of all faults into distinct subsets of FEF defines the *maximal fault resolution*. A test that achieves the maximal fault resolution is said to be a *complete fault-location test*.

Repairing the UUT often consists of substituting one of its *replaceable units* (RU) referred as a *faulty RU*, rather than in an accurate identification of the real fault inside an RU. We characterize this process by *RU resolution*. Suppose that the results of the test do not allow to distinguish between two suspected RUs U1 and U2. We could replace now one of these RUs, say U1 with a good RU, and return to the test experiment. If the new results are correct, the faulty RU was the replaced one; otherwise, it is the remaining one U2. This type of procedure we call *sequential diagnosis procedure*.

The diagnosis process is often *hierarchical*, carried out as a top-down process (with a system operating in the field) or bottom-up process (during the fabrication of the system).

In the *top-down approach* (system $\rightarrow$ boards $\rightarrow$ ICs) first-level diagnosis may deal with "large" RUs like boards called also *field-replaceable units*. The faulty board is then tested in a maintenance center to locate the faulty component (IC) on the board. Accurate location of faults inside a faulty IC may be also useful for improving its manufacturing process.

In the *bottom-up approach* (ICs $\rightarrow$ boards $\rightarrow$ system) a higher level is assembled only from components already tested at a lower level. This is done to minimize the cost of diagnosis and repair, which increases significantly with the level at which the faults are detected.

The rule of 10: if it costs R10 to test an IC, the cost of locating the same defective IC when mounted on a board and of repairing the board is about R100; when the defective board is plugged into a system, the cost of finding the fault and repairing the system is R1000.

In manufacturing, the most likely faults are fabrication errors affecting the interconnections between components; in the field the most likely faults are physical failures internal to components (because every UUT has been successfully tested in the past). Knowing the most likely class of faults helps in fault location.

Combinational Fault Diagnosis Methods

This approach does most of the work before the testing experiment. It uses fault simulation to determine the possible responses to a given test in the presence of faults. The database constructed in this step is called a *fault table* or a *fault dictionary*. To locate faults, one tries to match the actual results of test experiments with one of the precomputed expected results stored in the database. The result of the test experiment represents a *combination* of effects of the fault to each test pattern. That's why we call this approach *combinational fault diagnosis method*. If this look-up process is successful, the fault table (dictionary) indicates the corresponding fault(s).

Fault Table

In general, a *fault table* is a matrix $FT = \|a_{ij}\|$ where columns F_j represent faults, rows T_i represent test patterns, and $a_{ij} = 1$ if the test pattern T_i detects the fault F_j , otherwise if the test pattern T_i does not detect the fault F_j , $a_{ij} = 0$.

Denote the actual result of a given test pattern by 1 if it differs from the precomputed expected one, otherwise denote it by 0. The result of a test experiment is represented by a vector $\bar{E} = |e_i|$ where $e_i = 1$ if the actual result of the test patterns does not match with the expected result, otherwise $e_i = 0$. Each column vector f_j corresponding to a fault F_j represents a possible result of the test experiment in the case of the fault F_j .

Three cases are now possible depending on the quality of the test patterns used for carrying out the test experiment:

1. The test result $\bar{E}$ matches with a single column vector f_j in FT. This result corresponds to the case where a single fault F_j has been located. In other words, the maximum diagnostic resolution has been obtained.
2. The test result $\bar{E}$ matches with a subset of column vectors $\{f_i, f_j \dots f_k\}$ in FT. This result corresponds to the case where a subset of indistinguishable faults $\{F_i, F_j \dots F_k\}$ has been located.
3. No match for $\bar{E}$ with column vectors in FT is obtained. This result corresponds to the case where the given set of vectors does not allow to carry out fault diagnosis. The set of faults described in the fault table must be incomplete (in other words, the real existing fault is missing in the fault list considered in FT).

Example:

	F_1	F_2	F_3	F_4	F_5	F_6	F_7
T_1	0	1	1	0	0	0	0
T_2	1	0	0	1	0	0	0
T_3	1	1	0	1	0	1	0
T_4	0	1	0	0	1	0	0
T_5	0	0	1	0	1	1	0
T_6	0	0	1	0	0	1	1

E_1	E_2	E_3
0	0	1
0	1	0
0	1	0
1	0	1
1	0	1
0	0	0

Faults F_1 and F_4 are not distinguishable

Fault F_5 located

No match,
diagnosis not possible

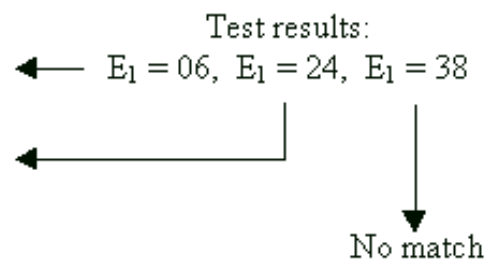
In the example the results of three test experiments E_1, E_2, E_3 are demonstrated. E_1 corresponds to the first case where a single fault is located, E_2 corresponds to the second case where a subset of two indistinguishable faults is located, and E_3 corresponds to the third case where no fault can be located because of the mismatch of E_3 with the column vectors in the fault table.

Fault Dictionary

Fault dictionaries (FD) contain the same data as the fault tables with the difference that the data is reorganized. In FD a mapping between the potential results of test experiments and the faults is represented in a more compressed and ordered form. For example, the column bit vectors can be represented by ordered decimal codes (see the example) or by some kind of compressed signature.

Example:

No	Bit vectors	Decimal numbers	Faults
1	000001	01	F ₇
2	000110	06	F ₅
3	001011	11	F ₆
4	011000	24	F ₁ , F ₄
5	100011	35	F ₃
6	101100	44	F ₂



Minimization of Diagnostic Data

To reduce large computational effort involved in building a fault dictionary, in fault simulation the detected faults are *dropped* from the set of simulated faults. Hence, all the faults detected for the first time by the same vector will produce the same column vector (signature) in the fault table, and will be included in the same equivalence class of faults. In this case the testing experiment can stop after the first failing test, because the information provided by the following tests is not used. Such a testing experiment achieves a lower diagnostic resolution. A tradeoff between computing time and diagnostic resolution can be achieved by dropping faults after $k > 1$ detections.

Example:

In the fault table produced by fault simulation with fault dropping, only 19 faults need to be simulated compared to the case of 42 faults when simulation without fault dropping is carried out (the simulated faults in the fault table are shown in shadowed boxes). As the result of the fault dropping, however, the following faults remain not distinguishable: {F₂, F₃}, {F₁, F₄}, {F₂, F₆}.

	F ₁	F ₂	F ₃	F ₄	F ₅	F ₆	F ₇
T ₁	0	1	1	0	0	0	0
T ₂	1	0	0	1	0	0	0
T ₃	0	0	0	0	0	1	0
T ₄	0	0	0	0	1	0	0
T ₅	0	0	0	0	0	0	0
T ₆	0	0	0	0	0	0	1

Fault Location by Structural Analysis

Assume a single fault in the circuit. Then there should exist a path from the site of the fault to each of the outputs where errors have been detected. Hence the fault site should belong to the intersection of cones of all failing outputs. A simple structural analysis can help to find faults that can explain all the observed errors.

Sequential Fault Diagnosis Methods

In sequential fault diagnosis the process of fault location is carried out step by step, where each step depends on the result of the diagnostic experiment at the previous step. Such a test experiment is called *adaptive testing*. Sequential experiments can be carried out either by observing only output responses of the UUT or by pinpointing by a special probe also internal control points of the UUT (*guided probing*). Sequential diagnosis procedure can be graphically represented as *diagnostic tree*.

Fault Location by Edge-Pin Testing

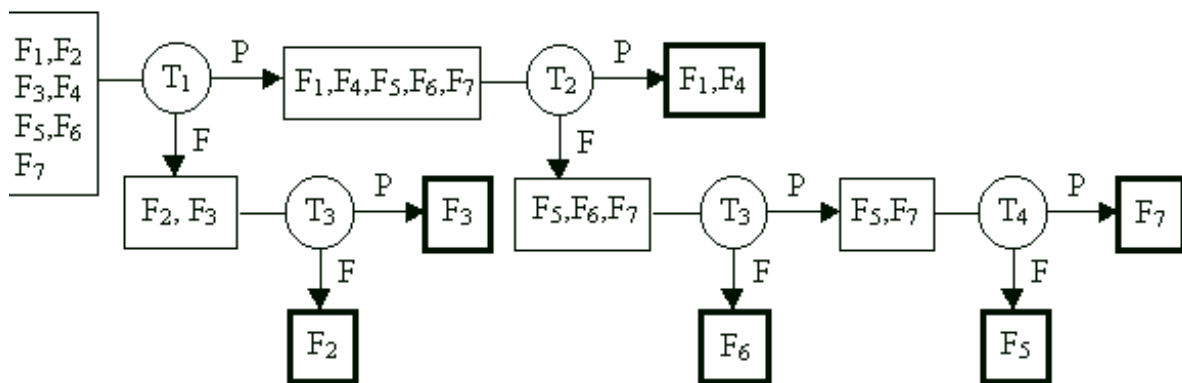
In fault diagnosis test patterns are applied to the UUT step by step. In each step, only output signals at edge-pins of the UUT are observed and their values are compared to the expected ones. The next test pattern to be applied in adaptive testing depends on the result of the previous step. The diagnostic tree of this process consists of the fault nodes FN (rectangles) and test nodes TN (circles).

A FN is labeled by a set of not yet distinguished faults. The starting fault node is labeled by the set of all faults. To each FN k a TN is linked labeled by a test pattern T_k to be applied as the next. Every test pattern distinguishes between the faults it detects and the ones it does not. The task of the test pattern T_k is to divide the faults in FN k into two groups - detected and not detected by T_k faults. Each test node has two outgoing

edges corresponding to the results of the experiment of this test pattern. The results are indicated as *passed* (P) or *failed* (F). The set of faults shown in a current fault node (rectangle) are equivalent (not distinguished) under the currently applied test set.

Example:

The diagnostic tree in the Figure below corresponds to the example considered in. We can see that most of the faults are uniquely identified, two faults F_1, F_4 remain indistinguishable. Not all test patterns used in the fault table are needed. Different faults need for identifying test sequences with different lengths. The shortest test contains two patterns the longest four patterns.



Rather than applying the entire test sequence in a fixed order as in combinational fault diagnosis, adaptive testing determines the next vector to be applied based on the results obtained by the preceding vectors. In our example, if T_1 fails, the possible faults are $\{F_2, F_3\}$. At this point applying T_2 would be wasteful, because T_2 does not distinguish among these faults. The use of adaptive testing may substantially decrease the average number of tests required to locate a fault.

Generating Tests to Distinguish Faults

To improve the fault resolution of a given test set T , it is necessary to generate tests to distinguish among faults equivalent under T .

Consider the problem of generating a test to distinguish between faults F_1 and F_2 . Such a test must detect one of these faults but not the other, or vice versa. The following cases are possible.

1. F_1 and F_2 do not influence the same set of outputs. Let $OUT(F_k)$ be the set of

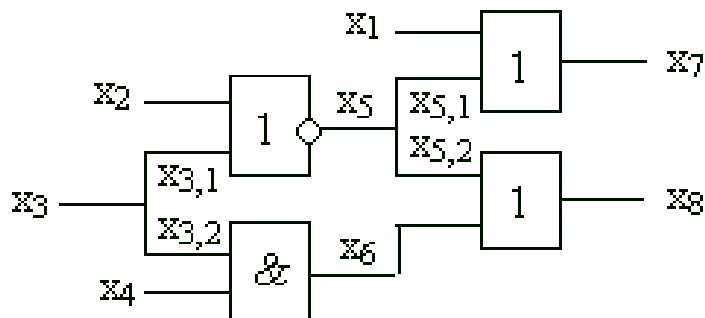
outputs influenced by the fault F_k . A test should be generated for $F1$ using only the circuit feeding the outputs $OUT(F1)$, or for $F2$ using only the circuit feeding the outputs $OUT(F2)$.

2. $F1$ and $F2$ influence the same set of outputs. A test should be generated for $F1$ without activating $F2$, or vice versa, for $F2$ without activating $F1$.

Three possibilities can be mentioned to keep a fault $F2: x_k \equiv e$ not activated, where x_k denotes a line in the circuit, and $e \in \{0,1\}$:

1. The value $\neg e$ should be assigned to the line x_k .
2. If this is not possible then the activated path from $F2$ should be blocked, so that the fault $F2$ could not propagate and influence the activated path from $F1$.
3. If the 2nd case is also not possible then the values propagated from the sites $F1$ and $F2$ and reaching the same gate G should be opposite on the inputs of G .

Example:



1. There are two faults in the circuit: $F1: x_{3,1} \equiv 0$, and $F2: x_4 \equiv 1$. The fault $F1$ may influence both outputs, the fault $F2$ may influence only the output x_8 . A test pattern 0010 activates $F1$ up to the both outputs, and $F2$ only to x_8 . If both outputs will be wrong, $F1$ is present, and if only the output x_8 will be wrong, $F2$ is present.
2. There are two faults in the circuit: $F1: x_{3,2} \equiv 0$, and $F2: x_{5,2} \equiv 1$. Both of them influence the same output of the circuit. A test pattern 0100 activates the fault $F2$. The fault $F1$ is not activated, because the line $x_{3,2}$ has the same value as it would have had if $F1$ were present.

3. There are the same two faults in the circuit: $F1: x_{3,2} \equiv 0$, and $F2: x_{5,2} \equiv 1$. Both of them influence the same output of the circuit. A test pattern 0110 activates the fault $F2$. The fault $F1$ is activated at its site but not propagated through the AND gate, because of the value $x_4 = 0$ at its input.
4. There are two faults in the circuit: $F1: x_{3,1} \equiv 1$, and $F2: x_{3,2} \equiv 1$. A test pattern 1001 consists the value $x_1 \equiv 1$ which creates the condition where both of the faults may influence only the same output x_8 . On the other hand, the test pattern 1001 activates both of the faults to the same OR gate (i.e. none of them is blocked). However, the faults produce different values at the inputs of the gate, hence they are distinguished. If the output value on x_8 will be 0, $F1$ is present. Otherwise, if the output value on x_8 will be 1, either $F2$ is present or none of the faults $F1$ and $F2$ are present.

Guided-Probe Testing

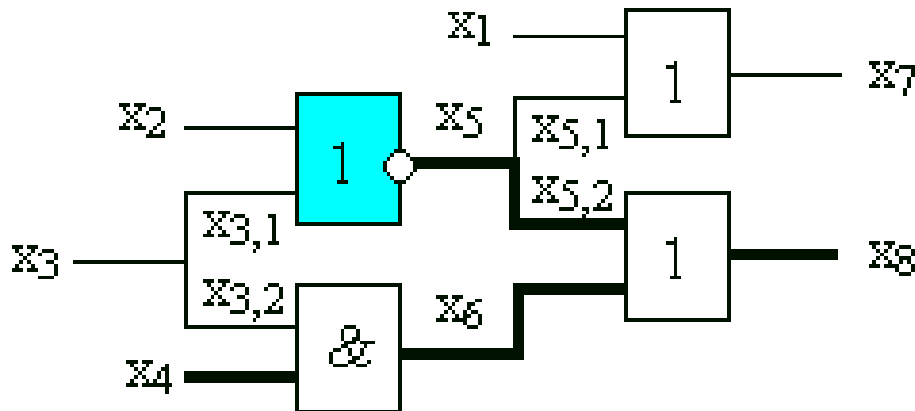
Guided-probe testing extends edge-pin testing process by monitoring internal signals in the UUT via a probe which is moved (usually by an operator) following the guidance provided by the test equipment. The principle of guided-probe testing is to backtrace an error from the primary output where it has been observed during edge-pin testing to its physical location in the UUT. Probing is carried out step-by-step. In each step an internal signal is probed and compared to the expected value. The next probing depends on the result of the previous step.

A diagnostic tree can be created for the given test pattern to control the process of probing. The tree consists of internal nodes (circles) to mark the internal lines to be probed, and of terminal nodes (rectangles) to show the possible result of diagnosis. The results of probing are indicated as *passed* (P) or *failed* (F).

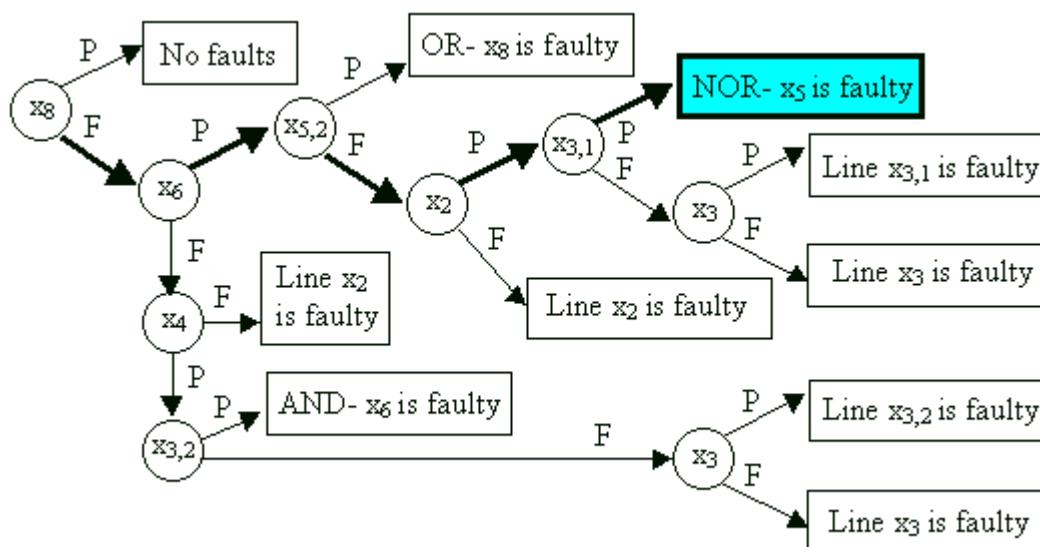
Typical faults located are opens and defective components. An open between two points A and B in a connection line is identified by a mismatch between the error observed at B and the correct value measured at A. A faulty device is identified by detecting an error at one of its outputs, while only correct values are measured at its inputs.

The most time-consuming part of guided-probe testing is moving the probe. To speed-up the fault location process, we need to reduce the number of probed lines. A lot of methods to minimize the number of probings are available.

Example:



Let have a test pattern 1010 applied to the inputs of the circuit. The diagnostic tree created for this particular test pattern is shown. On the output x_8 , instead of the expected value 0, an erroneous signal 1 is detected. By backtracing (indicated by bold arrows in the diagnostic tree) the faulty component NOR- x_5 is located.



Diagnostic tree allows to carry out optimization of the fault location procedure, for example to generate a procedure with minimum average number of probes.

Fault Location by UUT Reduction

Initially the UUT is the entire circuit and the process starts when its test fails. While the failing UUT can be partitioned, half of the UUT is disabled and the remaining half is tested. If the test passes, the fault must be in the disabled part, which then becomes the UUT. If the test fails, the tested part becomes the UUT.

SESSION 3

Rectify the problem.

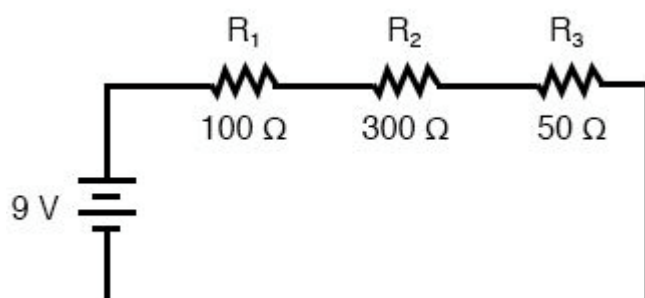
Learning Outcomes

- Possible causes of fault are isolated and removed in terms sub-system and component level.
- Faulty component or sub-system is replaced following standard assembly and disassembly procedures.

Component Failure Analysis

Analyzing Failures on a Simple Series Circuit

Let's start with a simple series circuit:



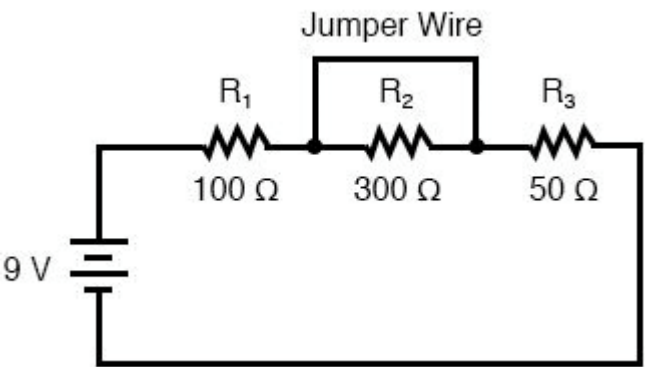
With all components in this circuit functioning at their proper values, we can mathematically determine all currents and voltage drops:

	R_1	R_2	R_3	Total	
E	2	6	1	9	Volts
I	20m	20m	20m	20m	Amps
R	100	300	50	450	Ohms

Shorted Components in a Series Circuit

Now let us suppose that R_2 fails shorted. *Shorted* means that the resistor now acts as a straight piece of wire, with little or no resistance. The circuit will behave as though a "jumper" wire were connected across R_2 (in case you were wondering, "jumper wire" is a common term for a temporary wire connection in a circuit). What causes the shorted

condition of R_2 is no matter to us in this example; we only care about its effect upon the circuit:



With R_2 shorted, either by a jumper wire or by an internal resistor failure, the total circuit resistance will *decrease*. Since the voltage output by the battery is a constant (at least in our ideal simulation here), a decrease in total circuit resistance means that total circuit current *must increase*:

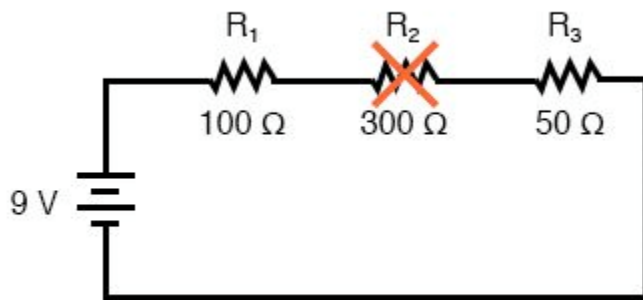
	R_1	R_2	R_3	Total	
E	6	0	3	9	Volts
I	60m	60m	60m	60m	Amps
R	100	0	50k	150	Ohms

↑
Shorted resistor

As the circuit current increases from 20 milliamps to 60 milliamps, the voltage drops across R_1 and R_3 (which haven't changed resistances) increase as well, so that the two resistors are dropping the whole 9 volts. R_2 , being bypassed by the very low resistance of the jumper wire, is effectively eliminated from the circuit, the resistance from one lead to the other having been reduced to zero. Thus, the voltage drop across R_2 , even with the increased total current, is zero volts.

Opened Components in a Series Circuit

On the other hand, if R_2 were to fail “open”—resistance increasing to nearly infinite levels—it would also create wide-reaching effects in the rest of the circuit:



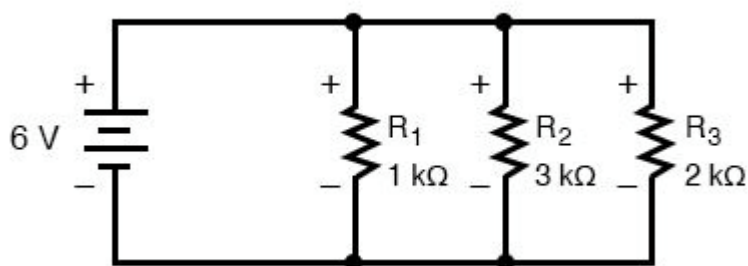
	R_1	R_2	R_3	Total	
E	0	9	0	9	Volts
I	0	0	0	0	Amps
R	100	∞	50	∞	Ohms


 Open resistor

With R_2 at infinite resistance and total resistance is the sum of all individual resistances in a series circuit, the total current decreases to zero. With zero circuit current, there is no current to produce voltage drops across R_1 or R_3 . R_2 , on the other hand, will manifest the full supply voltage across its terminals.

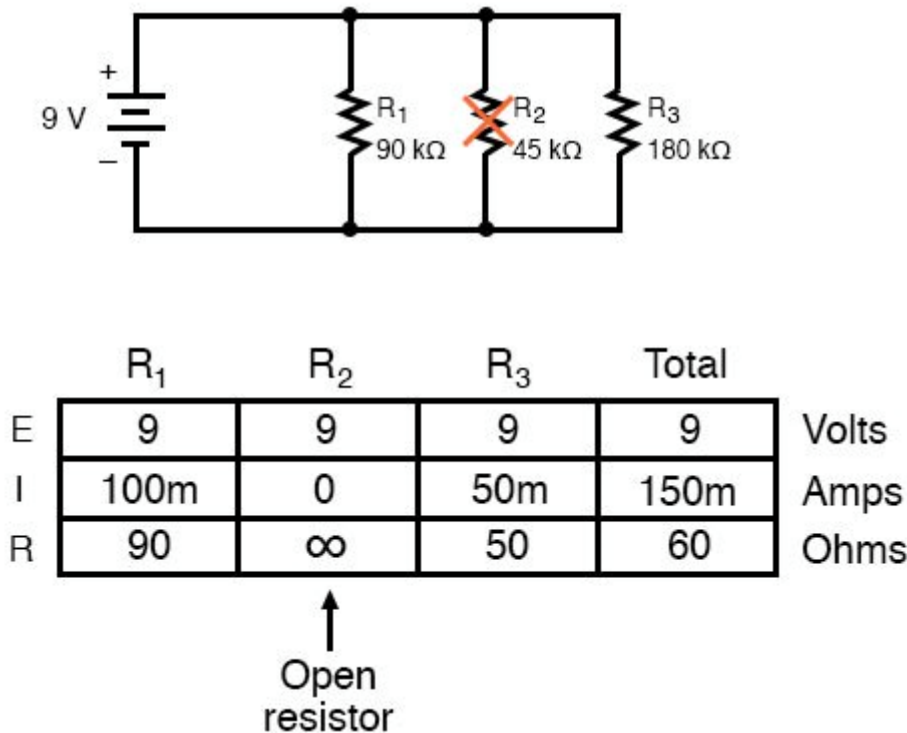
Analyzing Failures on a Simple Parallel Circuit

We can apply the same before/after analysis technique to parallel circuits as well. First, we determine what a “healthy” parallel circuit should behave like.



Opened Components in a Parallel Circuit

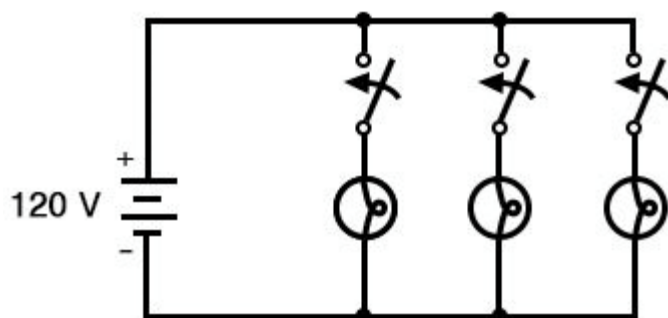
Supposing that R₂ opens in this parallel circuit, here's what the effects will be:



Notice that in this parallel circuit, an open branch only affects the current through that branch and the circuit's total current. Total voltage—being shared equally across all components in a parallel circuit, will be the same for all resistors. Due to the fact that the voltage source's tendency is to hold the voltage *constant*, its voltage will not change, and being in parallel with all the resistors, it will hold all the resistors' voltages the same as they were before: 9 volts. Being that voltage is the only common parameter in a parallel circuit, and the other resistors haven't changed resistance value, their respective branch currents remain unchanged.

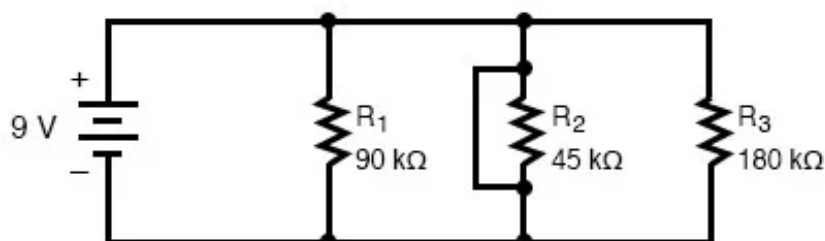
Household Lighting Application

This is what happens in a household lamp circuit: all lamps get their operating voltage from power wiring arranged in a parallel fashion. Turning one lamp on and off (one branch in that parallel circuit closing and opening) doesn't affect the operation of other lamps in the room, only the current in that one lamp (branch circuit) and the total current powering all the lamps in the room:



Shorted Components in a Parallel Circuit

In an ideal case (with perfect voltage sources and zero-resistance connecting wire), shorted resistors in a simple parallel circuit will also have no effect on what's happening in other branches of the circuit. In real life, the effect is not quite the same, and we'll see why in the following example:



R_2 "shorted" with a jumper wire

	R_1	R_2	R_3	Total	
E	9	9	9	9	Volts
I	100m	∞	50m	∞	Amps
R	90	0	180	0	Ohms

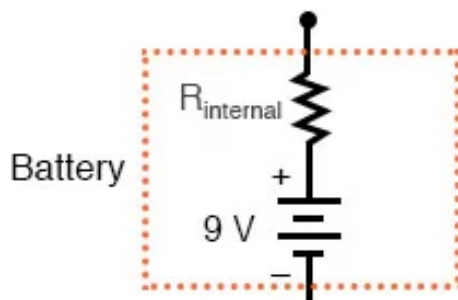
↑
 Shorted resistor

A shorted resistor (resistance of $0\ \Omega$) would theoretically draw infinite current from any finite source of voltage ($I=E/0$). In this case, the zero resistance of R_2 decreases the circuit total resistance to zero Ω as well, increasing total current to a value of infinity. As

long as the voltage source holds steady at 9 volts, however, the other branch currents (I_{R1} and I_{R3}) will remain unchanged.

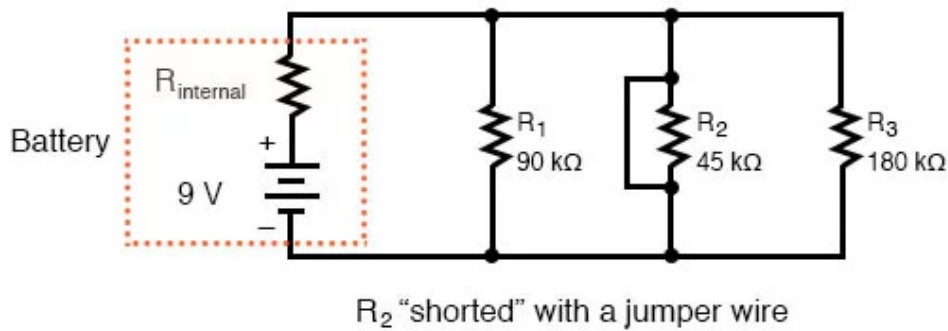
Non-ideal Assumptions in Analysis

The critical assumption in this “perfect” scheme, however, is that the voltage supply will hold steady at its rated voltage while supplying an infinite amount of current to a short-circuit load. This is simply not realistic. Even if the short has a small amount of resistance (as opposed to absolutely zero resistance), no *real* voltage source could arbitrarily supply a huge overload current and maintain the steady voltage at the same time. This is primarily due to the internal resistance intrinsic to all electrical power sources, stemming from the inescapable physical properties of the materials they’re constructed of:



These internal resistances, small as they may be, turn our simple parallel circuit into a series-parallel combination circuit. Usually, the internal resistances of voltage sources are low enough that they can be safely ignored, but when high currents resulting from shorted components are encountered, their effects become very noticeable. In this case, a shorted R_2 would result in almost all the voltage being dropped across the

internal resistance of the battery, with almost no voltage left over for resistors R_1 , R_2 , and



	R_1	R_2	R_3	Total	
E	low	low	low	low	Volts
I	low	high	low	high	Amps
R	90	0	150	0	Ohms

↑

Shorted resistor

↑

Supply voltage decrease due to voltage drop across internal resistance

R_3 :

uffice it to say, intentional direct short-circuits across the terminals of any voltage source is a bad idea. Even if the resulting high current (heat, flashes, sparks) causes no harm to people nearby, the voltage source will likely sustain damage, unless it has been specifically designed to handle short-circuits, which most voltage sources are not.

Eventually, in this book I will lead you through the analysis of circuits *without the use of any numbers*, that is, analyzing the effects of component failure in a circuit without knowing exactly how many volts the battery produces, how many ohms of resistance is in each resistor, etc. This section serves as an introductory step to that kind of analysis.

Whereas the normal application of Ohm's Law and the rules of series and parallel circuits is performed with numerical quantities ("*quantitative*"), this new kind of analysis without precise numerical figures is something I like to call *qualitative* analysis. In other words, we will be analyzing the *qualities* of the effects in a circuit rather than the precise *quantities*. The result, for you, will be a much deeper intuitive understanding of electric circuit operation.

REVIEW:

- To determine what would happen in a circuit if a component fails, re-draw that circuit with the equivalent resistance of the failed component in place and recalculate all values.
- The ability to intuitively determine what will happen to a circuit with any given component fault is a *crucial* skill for any electronics troubleshooter to develop. The best way to learn is to experiment with circuit calculations and real-life circuits, paying close attention to what changes with a fault, what remains the same, and *why*!
- A *shorted* component is one whose resistance has dramatically decreased.
- An *open* component is one whose resistance has dramatically increased. For the record, resistors tend to fail open more often than fail shorted, and they almost never fail unless physically or electrically overstressed (physically abused or overheated).

LEARNER GUIDE

SAQA: 119256

Inspect and test electrical circuits

Learner Information:

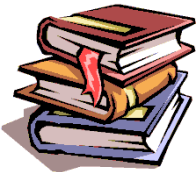
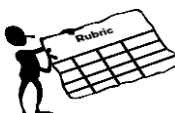
Details	Please Complete this Section
Name & Surname:	
Organization:	
Unit/Dept:	
Facilitator Name:	
Date Started:	
Date of Completion:	

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Key to Icons

The following icons may be used in this Learner Guide to indicate specific functions:

 Books	This icon means that other books are available for further information on a topic/subject.
 References	This icon refers to any examples, handouts, checklists, etc...
 Important	This icon represents important information related to a specific topic or section of the guide.
 Activities	This icon helps you to be prepared for the learning to follow or assist you to demonstrate understanding of module content. Shows transference of knowledge and skill.
 Exercises	This icon represents any exercise to be completed on a specific topic at home by you or in a group.
 Tasks/Projects	An important aspect of the assessment process is proof of competence. This can be achieved by observation or a portfolio of evidence should be submitted in this regard.



Workplace Activities

An important aspect of learning is through workplace experience. Activities with this icon can only be completed once a learner is in the workplace



Tips

This icon indicates practical tips you can adopt in the future.



Notes

This icon represents important notes you must remember as part of the learning process.

Learner Guide Introduction

About the Learner Guide...	This Learner Guide provides a comprehensive overview of the Inspect and test electrical circuits, and forms part of a series of Learner Guides that have been developed for Inspect and test electrical circuits. The series of Learner Guides are conceptualized in module's format and developed for Inspect and test electrical circuits. They are designed to improve the skills and knowledge of learners, and thus enabling them to effectively and efficiently complete specific tasks. Learners are required to attend training workshops as a group or as specified by their organization. These workshops are presented in modules and conducted by a qualified facilitator.
Purpose	The purpose of this Learner Guide is to provide learners with the necessary knowledge related to Inspect and test electrical circuits .
Outcomes	A person credited with this unit standard is able to: • Read and interpret electric circuit diagrams. • Prepare work area. • Inspect electrical circuits. • Test electrical circuits. • Complete work activities.
Assessment Criteria	The only way to establish whether a learner is competent and has accomplished the specific outcomes is through an assessment process. Assessment involves collecting and interpreting evidence about the learner's ability to perform a task. This guide may include assessments in the form of activities, assignments, tasks or projects, as well as workplace practical tasks. Learners are required to perform tasks on the job to collect enough and appropriate evidence for their portfolio of evidence, proof signed by their supervisor that the tasks were performed successfully.
To qualify	To qualify and receive credits towards the learning program, a registered assessor will conduct an evaluation and assessment of the learner's portfolio of evidence and competency
Range of Learning	This describes the situation and circumstance in which competence must be demonstrated and the parameters in which learners operate
Responsibility	The responsibility of learning rest with the learner, so: • Be proactive and ask questions, • Seek assistance and help from your facilitators, if required.

Learning Unit 1

Inspect and test electrical circuits

UNIT STANDARD NUMBER	:	119256
LEVEL ON THE NQF	:	4
CREDITS	:	6
FIELD	:	Manufacturing, Engineering and Technology
SUB FIELD	:	Engineering and Related Design

PURPOSE:	This unit standard will be useful to people who are required to inspect and test electrical circuits to ensure safety of public and personnel in accordance with the Occupational Health and Safety Act. People credited with this unit standard can: • Read and interpret electric circuit diagrams. • Prepare work area. • Inspect electrical circuits. • Test electrical circuits. • Complete work activities.
LEARNING ASSUMED TO BE IN PLACE:	
The credit calculation assumes that learners are already competent in terms of the following outcomes or areas of learning when starting to learn towards this unit standard: • Basic knowledge of electrical circuits and systems. • SABS standards and worksite safety procedures. • Applicable statutory requirements. • Working at heights and in confined places. • Use of relevant testing and measuring equipment.	

SESSION 1

Read and interpret electric circuit diagrams.

Learning Outcomes

- Electrical circuit diagrams are read and interpreted in terms of the symbols, components and functionality.
- Reasons for the existence of circuit diagrams are given in terms of the function they perform in relation to the safe electrification of a unit.
- Electrical components that require inspection and testing are identified in relation to the electrified unit.
- The basic principles of electricity pertaining to the circuit diagram are explained in accordance with electrical safety requirements.

Electrical circuit diagrams are read and interpreted in terms of the symbols, components and functionality.

Circuit diagrams

When electricity flows through a circuit it can be used to light up a globe or make an electric motor work. A circuit diagram is used to show you which parts make up the circuit and in what order.

What is a Circuit Diagram?

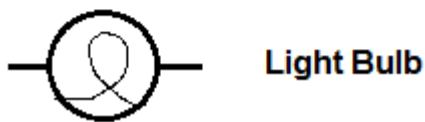
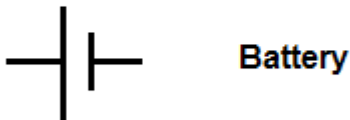
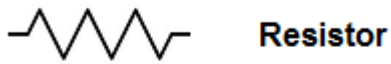
A circuit diagram is a visual display of an electrical circuit using either basic images of parts or industry standard symbols. Symbol usage depends on the audience viewing the diagram. These two different types of circuit diagrams are called pictorial (using basic images) or schematic style (using industry standard symbols). A schematic style circuit diagram is used to give a visual representation of an electrical circuit to an electrician. The pictorial style circuit diagram would be used for a broader, less technical audience.

Electric circuit diagrams use symbols to represent the parts of a circuit. They show you how to connect the circuit and make it work and can also be used to determine the current in different parts of the circuit.

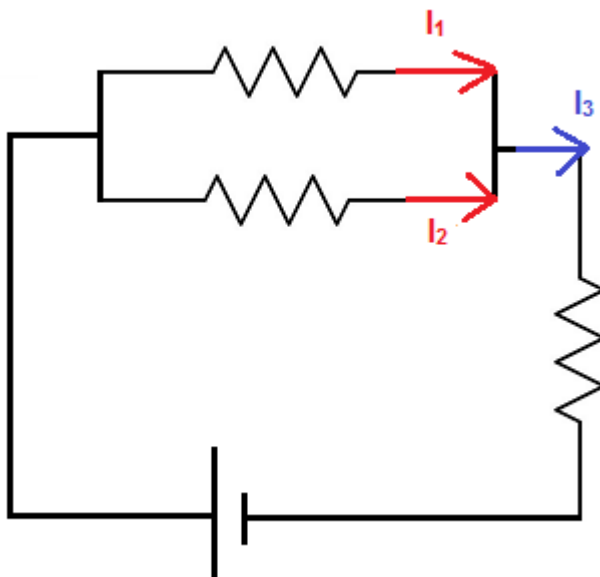
Ohm's Law

There is a simple relationship, known as Ohm's Law, between current, voltage, and resistance for many types of resistors. Using Ohm's Law, together with a circuit diagram, you can determine the current through any resistor in a circuit.

Common Symbols in Electric Circuit Diagrams:

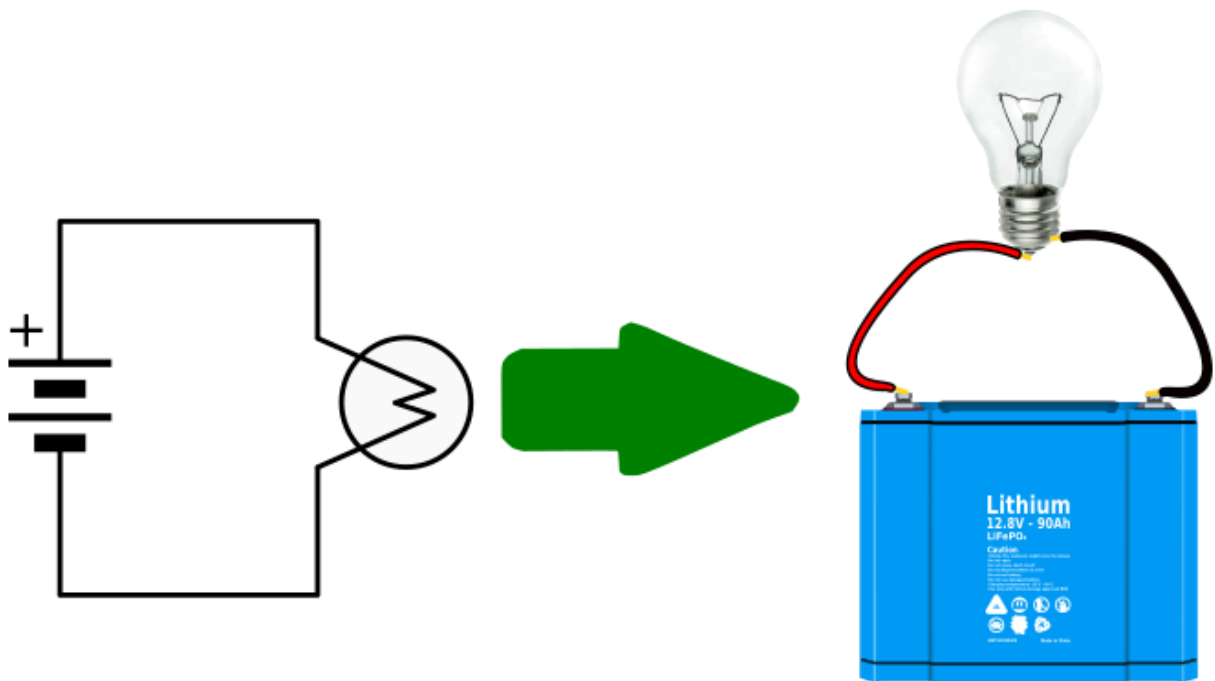


Ohm's Law:



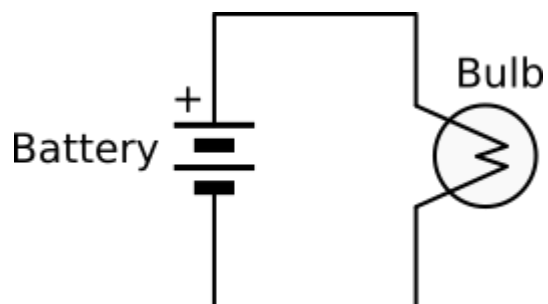
How to Read Circuit Diagrams

Circuit or schematic diagrams consist of symbols representing physical components and lines representing wires or electrical conductors. In order to learn how to read a circuit diagram, it is necessary to learn what the schematic symbol of a component looks like. It is also necessary to understand how the components are connected together in the circuit.



Battery and Light Bulb Circuit

Probably the simplest circuit that can be drawn is one that you may have seen in a school science class: a battery connected to a light bulb as shown below.



Circuit Symbols and Physical Components

Each electronic or electrical component is represented by a symbol as can be seen in this simple circuit diagram. Lines used to connect the symbols represent conductors or wires. Each symbol represents a physical component that may look as follows.

Battery Symbol



Light Bulb Symbol



Wire or Conductor



Physical Battery



Physical Light Bulb



Physical Wire



Physical Circuit

The physical circuit for the above circuit diagram may look something like the image below, although a more practical physical circuit would have a light bulb holder and clamps that connect to the battery terminals. A light bulb holder would have screw terminals to connect the wires to, and a socket to screw the light bulb into. Battery clamps would allow the wires to easily be connected between the battery and light bulb holder.



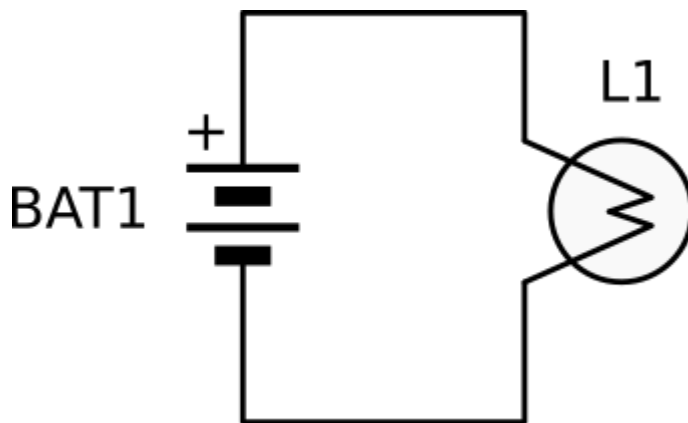
Specifying Components

Typically, the actual battery type and bulb type would be specified in a component list that accompanies the circuit diagram. More information on the bulb and battery type could also be included in the circuit as text. For example, the battery may be specified as a 12.8V 90Ah Lithium battery, or a 9V PM9 battery. The light bulb may be specified as a 12V 5W incandescent bulb, or 9V 0.5W torch bulb.

Component References

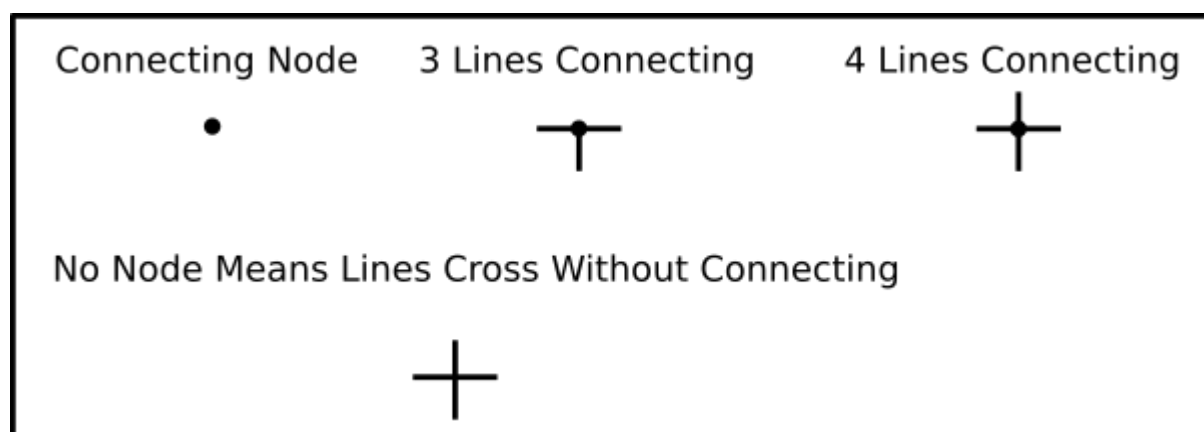
Components in a circuit should always have references, also called reference designators, used to identify the components in the circuit. This allows the components to easily be referenced in text or a component list. A battery may have the reference designator "BAT" and a light bulb may have a reference "L".

Because there could be more than one battery or light bulb in a circuit, reference designators will usually always end with a number, e.g. BAT1 and L1 as shown in the circuit below. A second light bulb in the circuit would then have the reference designator L2.



Circuit Diagram Connections

Circuit diagrams or schematic diagrams show electrical connections of wires or conductors by using a node as shown in the image below. A node is simply a filled circle or dot. When three or more lines touch each other or cross each other and a node is placed at the intersection, this represents the lines or wires being electrically connected at that point.

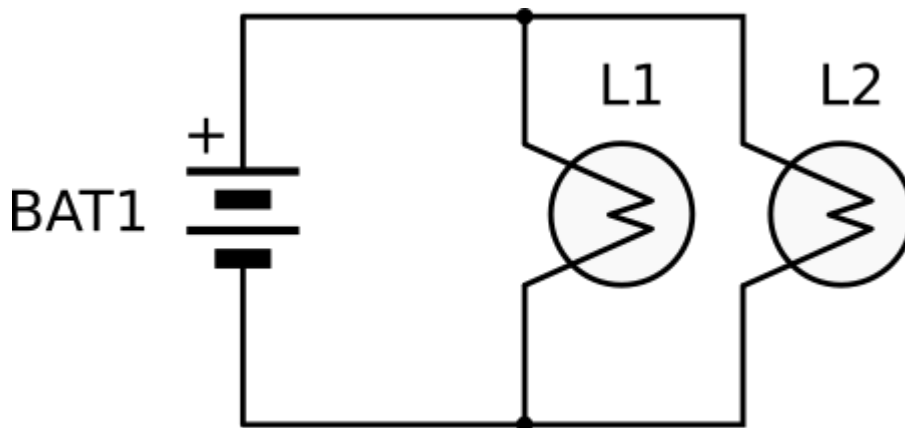


If wires or lines cross each other and there is no node, as shown at the bottom of the above image, the wires are not electrically connected. In this case the wires are crossing each other without connecting, like two insulated wires placed one on top of the other.

Parallel Circuit Example

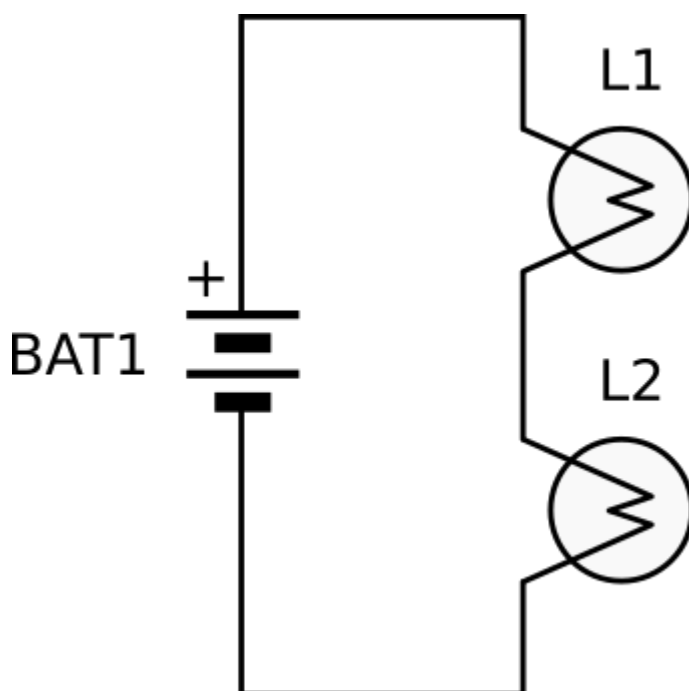
In the circuit below, two light bulbs are connected in parallel to a battery power source. It can be seen that the top terminals of the two light bulbs are connected together and to the positive terminal of the battery. We know this because the three terminals or connection points have a node where they intersect.

The bottom terminals of the bulbs are connected to each other and to the negative terminal of the battery, because the second node shows these connections.



Series Circuit Example

In the series circuit below, two light bulbs are connected in series. No nodes are necessary in this circuit to show the bulbs connecting to each other and to the battery because single wires are connecting straight to each other. Nodes are only placed if three or more wires are connected.



Some Circuit Diagram Rules

The following are general circuit diagram rules.

- Wires or lines in circuit diagrams are usually horizontal or vertical. In some cases a diagonal line may be used which is placed at 45 degrees.
- Component symbols in a circuit diagram are usually placed horizontally or vertically. On very rare occasions a component may be placed at 45 degrees, but only for a very good reason.
- Circuit diagrams are drawn as simply and neatly as possible. This means that the physical implementation of the circuit may look different to the circuit diagram, but they are electrically the same.
- Lines connecting components can be thought of as insulated wires in most cases, with only the ends of the wires being bare conductors for electrical connection.
- When lines cross each other in a circuit diagram, they can be thought of as two insulated wires crossing if there is no node where the wires intersect or cross each other.
- Three lines intersecting at a point with a node at the intersection means that the three wires are electrically connected. This connection can be thought of as three insulated wires bared at the point of intersection and soldered together.
- Two wires that cross each other with a node at the intersection of the crossing point means that the wires are electrically connected.

How to Read Electrical Drawing

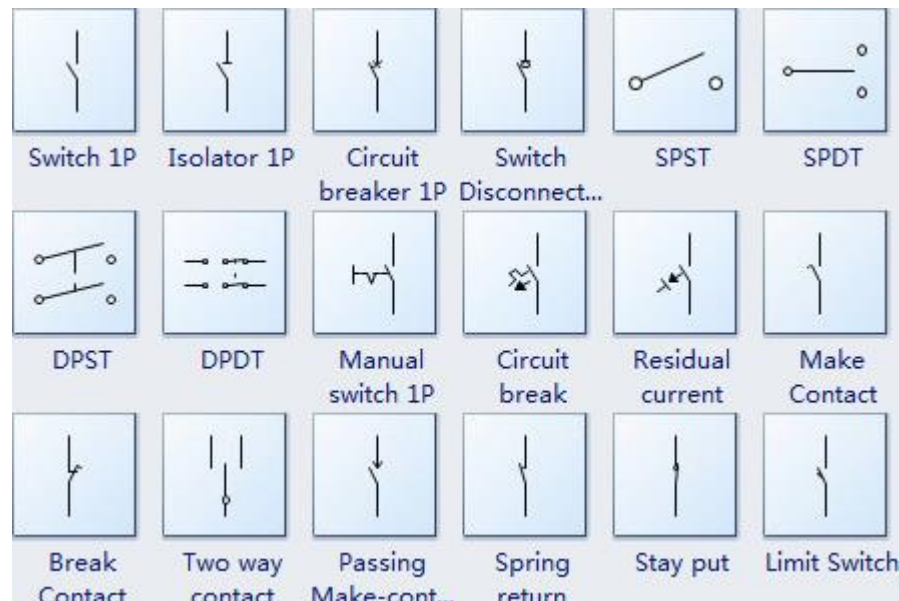
1. Familiarize with the Standardized Electrical Symbols

Knowing what the symbols in your electrical drawing mean will help you find different appliances. Symbols usually resemble specific meaning. Familiarize yourself with descriptions for electrical appliances, understanding that different symbols appear for different objects. Refer to Basic Electrical Symbols and Their Meanings for specific resources and learn these symbols visually. You'd better get to know the following basic symbols by heart.

Lights are shown as ovals with a squiggly line inside. They look like light bulbs. Different types of lights may be indicated with different symbols.

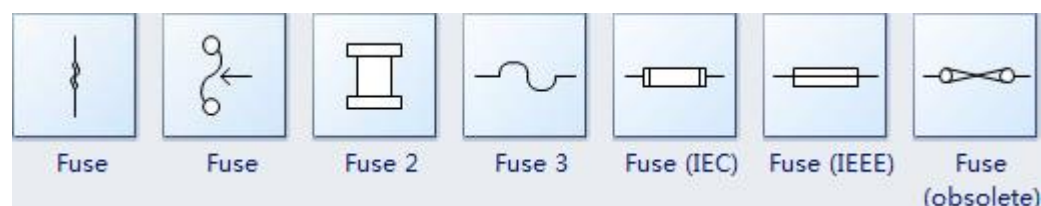


Switches are symbolized by an opening or break in the line. It looks like the flip of a light switch.

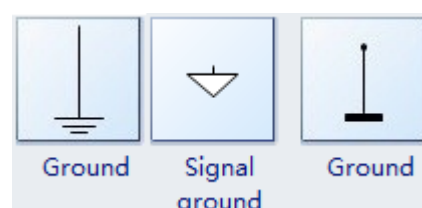


Thermostats are symbolized by an opening or break in the line, but they also have a squiggle that connects them to the line.

A fuse is represented by a slight zigzag in the line. Motors are displayed by bumps along the line. It looks like an "M" with 5 or 6 humps.



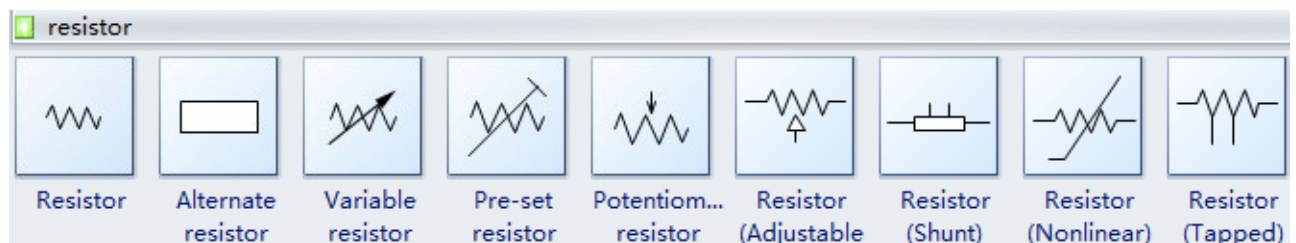
Ground is represented by either a triangle pointing down or a set of parallel lines that become shorter as they appear below each other, in effect representing the inner area of the triangle pointing down. Ground is a common reference point that schematics use to show the overall unity of the various functions of the circuit. It does not refer to the actual ground of the earth.



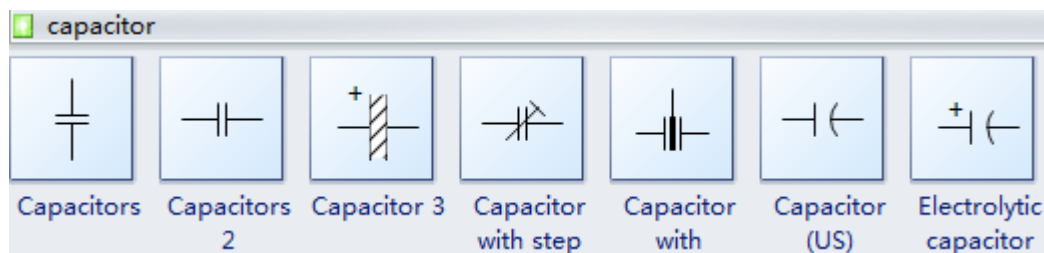
A line represents a wire. Wires are used to link the devices together. All points along the wire are identical and connected. Wires may cross each other on an electrical drawing, but that does not necessarily mean that they connect. If they do not connect, one will be shown looping around the other in a semicircle. If they do connect, they will cross and a dot will be seen at the point where the lines cross.



A zigzag shape stands for a resistor. Resistors act to impede the flow of the circuit to an extent determined by the resistance value used. They are used to scale and shape the signal.



Learn that **capacitors** are represented by two parallel lines. Capacitors are used to condition rapidly changing signals, as opposed to the static or slower changing signals that are conditioned by resistors. The traditional use of capacitors in modern circuits is to draw noise, which is inherently a rapidly changing signal, away from the signal of interest and drain it away to ground.



2. Learn Reading Pattern

Read schematics in the pattern that you would read text. With rare exceptions, schematics should be read left to right and top to bottom. The signal being generated or used by the circuit will flow in this direction. The user can follow the same path that the signal uses to understand what the signal does or how it is being modified.

3. Identify Polarity

Some components to a circuit board are polarized, meaning one side is positive and the other negative. This means you have to attach it in the specified way. For most symbols, polarity is included in the symbol. To identify the polarity of the physical part, a general rule of thumb is to find out which metal lead wire is longer. The longer part is the + side.

4. Understand Names and Values

Values help define exactly what a component is. For electrical components like resistors, capacitors, and inductors the value tells us how many ohms, farads, or henries they have. For other components, like integrated circuits, the value may just be the name of the chip. Crystals might list their oscillating frequency as their value. Basically, the value of a schematic component calls out its most important characteristic.

Component names usually consist of one or two letters and a number. The letter part of the name represents the type of component - R's for resistors, C's for capacitors, U's for integrated circuits, etc. Each component name in an electrical drawing should be unique; if you have multiple resistors in a circuit, for example, they should be named R1, R2, R3, etc.

Components names help us reference specific points in schematics. The prefixes of names are pretty well standardized. For some components, like resistors, the prefix is just the first letter of the component. Other name prefixes are not so literal; inductors, for example, are L's (because current has already taken I [but it starts with a C... electronics is a silly place]). Here's a quick table of common components and their name prefixes:

Table of Components Names

Name	Identifier Component
R	Resistors
C	Capacitors
L	Inductors
S	Switches
D	Diodes

Q	Transistors
U	Integrated Circuits
Y	Crystals and Oscillators

Reasons for the existence of circuit diagrams are given in terms of the function they perform in relation to the safe electrification of a unit.

The Importance of Electrical Wiring Diagrams

1. Electrical Wiring Diagrams Save Time

Everyone knows the feeling of certain projects taking far longer than anticipated. One of the ways to avoid this delay is to use an electrical wiring diagram. The diagram contains key areas of the structure that professionals can anticipate as causing potential problems before they even get there. Anticipating sharp corners, tight walls, or potential wet spots can help electricians save time when wiring a building because they can anticipate the problem before they get there. The professionals will have a plan for that spot when they arrive.

2. Electric Wiring Diagrams Save Money

Nobody likes to spend more money than they have to. Electrical wiring diagrams can help people plan for the exact type and length of wire that the various parts of the project will require. This means that people will never have to go back and purchase or cut additional wire which would otherwise drive up overhead costs sharply. Make sure this doesn't happen with an electrical wiring diagram. It is always better to plan ahead.

3. Electrical Wiring Diagrams Prevent Injury

Anyone with experience dealing with electricity knows that people can get hurt wiring a structure. Electrical wiring diagrams can help technicians anticipate areas of the building that might have the potential for injury. This could involve nails and sharp objects or even areas where people can get shocked putting the wiring together. If people are injured on the job, this can result in a substantial delay and an expensive hospital trip. The most effective form of medical treatment is to prevent the problem in the first place.

An electrical schematic diagram provides electrical engineers with an idea of how a circuit layout should look when it is completed, and it also aids in planning repairs. For

example, looking at an electrical schematic diagram for a circuit board indicates where wires connect between components of the system.

On a larger scale, electrical schematics help plans for wiring buildings. For safety, circuits can only carry a given amount of current. For example, if a light circuit carries 10 amperes, that information appears on the schematics. Referring to the diagram during construction reduces the likelihood of potentially dangerous wiring mistakes.

Electrical components that require inspection and testing are identified in relation to the electrified unit.

Testing Electrical and Electronics Components and Devices with Multimeter

To be a good analyzer and troubleshooter, you must know the following basic techniques and has good skills in Electrical and Electronics troubleshooting, design and analyzing electric / electronic circuits. For this purpose, we have started a multimeter tutorial where we will use DMM (Digital Multimeter) and AVO Meter (Ampere-Voltage-Resistance Meter) or Multimeter (Digital/Analog) to test different electrical / electronics devices, instruments and components to find their terminals and condition such as are they short, open, good or faulty.

We will use digital and analog multimeter to check the following electrical and electronic components, devices, tools and instruments:

- Cable and Wires
- Switch/Push Buttons
- Fuse
- Capacitors & Inductors
- Resistors & Burnt resistors
- Diodes and LED
- Battery
- Transistors
- Relays

Cable and Wires

To check if the cable and wires are in good condition or broken before sizing the proper cable & wire for Electrical Wiring installation, we perform the continuity test. For this

purpose, take AVO meter (or Digital Multimeter) and select "Resistance" (in AVO meter...Rotate the knob to the " Ω " or Resistance).

Now connect both terminals, i.e. both naked ends of the cable/wire with the AVO or Digital Multimeter Terminals. If the meter reading shown "0 Ω ", it means Cable/Wire is in "Good Condition". On the other hand, if meter reading is "Infinite", it shows the cable/wire may be defected or broken. So you need to replace it with a new one.

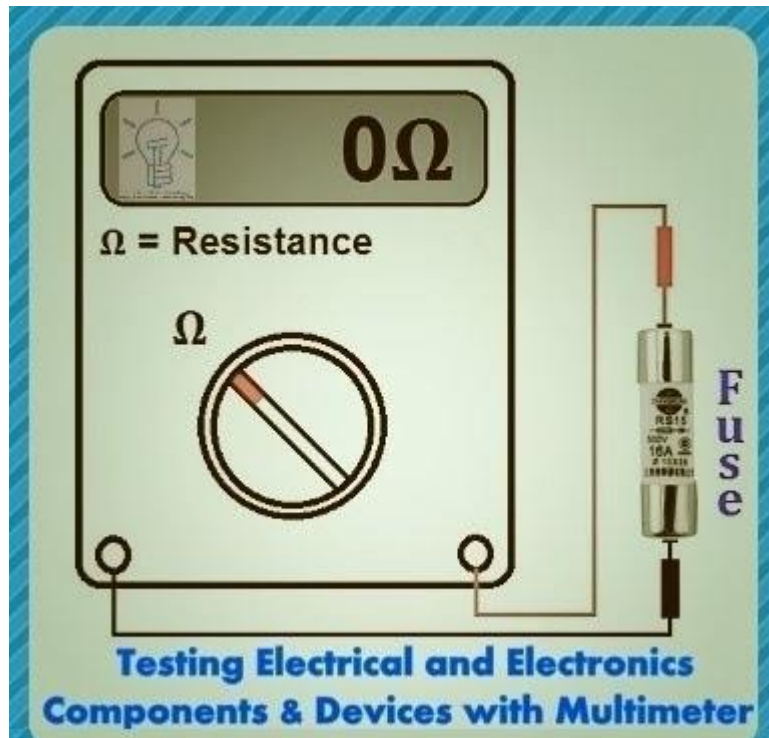
Switch/Push Buttons

Use the same method (mentioned above for checking the cable and wires)...to perform this method correctly, you will need to apply this method in both cases (ON & OFF positions) on switches and push buttons...In other words, first apply this method on switches/push buttons and then "Push" the push button and perform the same method again.

At first attempt, if meter reading is "Zero" and in the second attempt, the meter reading is infinite, it means Switch/Push button is in good condition. If Multimeter reading is "Zero" or "infinite" in both attempts, it means switch is in short circuit or continuity connection is broken and you should replace it with a new one.

Fuse

To verify Fuse condition, i.e. is "Fuse" in good condition or damaged? ... We perform the same method i.e. continuity test as mentioned above. In short, if the meter reading is "Zero" it means Fuse is in good condition. If Multimeter reading is infinite, it means Fuse continuity may be broken or blown. So you should replace it with a new one immediately.



Resistor & Burnt Resistors

To check if the Resistor is in good condition or broken, we use a multimeter. For this purpose, take AVO meter (or Digital Multimeter) and select "Resistance" (in AVO meter...Rotate the knob to the " Ω " or Resistance). Now connect both ends of resistor with the AVO or Digital Multimeter Terminals. If the meter reading shows the exact value of resistance or with a percentage tolerance, it means Resistor in "Good Condition".

For Example, $1\text{k}\Omega = 1000\Omega$ with a 5% tolerance will show the reading near about 950Ω to 1050Ω . On the other hand, if meter reading is "Infinite", it shows the Resistor may be defected or broken and open. So you need to replace it with a new one (exact value).

How to use a circuit tester

- Connect the tester clip to the negative terminal of the battery and touch the probe to the positive one.
- If the tester lamp does not light, the battery is dead (or the bulb in the tester has blown).
- If it lights, try again with the clip earthed to the car body: if the lamp fails to light, the battery negative terminal is not earthed properly.
- Earth the clip near the switch of the circuit being tested and touch the probe to the 'live' (battery) side of the switch. If the lamp does not light, the wiring between the battery and the switch is faulty, or a fuse has blown.

If the lamp lights, turn the switch on and probe its other

side: if the lamp does not light, the switch is faulty.

If the switch works, leave it on, earth the clip near the component and probe the live side of the component. If the lamp does not light, the wiring from switch to component is faulty, or a fuse has blown.

If all of the checks so far are satisfactory, transfer the clip to the live side of the battery.

Now probe the earthed side of the component (it may be earthed by its metal body being fixed to the car body). If the lamp does not light, the component is badly earthed. If it does light, the component itself is faulty.

The basic principles of electricity pertaining to the circuit diagram are explained in accordance with electrical safety requirements.

The Basic Principles of Electricity

Electricity

Electricity, simply put, is the flow of electric current along a conductor. This electric current takes the form of free electrons that transfer from one atom to the next. Thus, the more free electrons a material has, the better it conducts. There are three primary electrical parameters: the volt, the ampere and the ohm.

1. The Volt

The pressure that is put on free electrons that causes them to flow is known as electromotive force (EMF). The volt is the unit of pressure, i.e., the volt is the amount of electromotive force required to push a current of one ampere through a conductor with a resistance of one ohm.

2. The Ampere

The ampere defines the flow rate of electric current. For instance, when one coulomb (or 6×10^{18} electrons) flows past a given point on a conductor in one second, it is defined as a current of one ampere.

3. The Ohm

The ohm is the unit of resistance in a conductor. Three things determine the amount of resistance in a conductor: its size, its material, e.g., copper or aluminum, and its temperature. A conductor's resistance increases as its length increases or diameter decreases. The more conductive the materials used, the lower the conductor resistance becomes. Conversely, a rise in temperature will generally increase resistance in a conductor.

Ohm's Law

- Ohm's Law defines the correlation between electric current (I), voltage (V), and resistance (R) in a conductor.
- Ohm's Law can be expressed as: $V = I \times R$
- Where: V = volts, I = amps, R = ohms

Ampacity

Ampacity is the amount of current a conductor can handle before its temperature exceeds accepted limits. These limits are given in the National Electrical Code (NEC), the Canadian Electrical Code and in other engineering documents such as those published by the Insulated Cable Engineers Association (ICEA). It is important to know that many external factors affect the ampacity of an electrical conductor and these factors should be taken into consideration before selecting the conductor size.

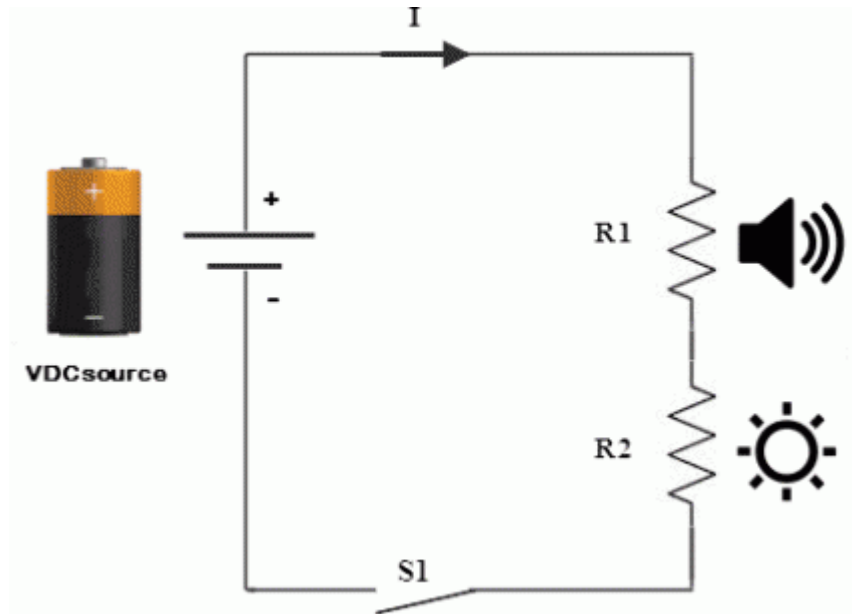
Applying Ohm's Law

Now, let's apply Ohm's law to the following circuit (for exercise purposes only, circuits are theoretical) and calculate the voltage and current supplied to each load. The schematic below shows a supply circuit for a child's bedtime toy. R1 represents the resistance value of the speaker and R2 shows the resistance value of the LEDs. R1 is equal to 430 Ohms, R2 is equal to 284 Ohms and the supply is a battery with 5VDC and 5A. What is the voltage supplied to the LEDs and to the speaker? First, we need to find the current in the loop once the button is pressed and switch 1 (S1) closes. The supply offers 5 amps of current, but the circuit will only use what is demanded by the loads. Using Ohm's law, we can reconfigure the formula to solve for current in the loop, or $I (\text{loop}) = V (\text{loop}) / R (\text{loop})$. Using the supplied values, we can calculate that $I (\text{loop current}) = 5\text{VDC} / 714\Omega = 7\text{mA}$.

Ohm's Law Applied to a Series Circuit

So now that we know the current in the loop is 7mA and in a series circuit that current is constant throughout, we can use Ohm's law to calculate the voltage supplied to the

speaker: $V(\text{speaker}) = I(\text{loop}) \times R(\text{speaker})$ or $V(\text{speaker}) = (7\text{mA}) \times (430\Omega)$ or $\sim 3\text{VDC}$. The LEDs will in turn have a supply voltage of: $V(\text{LED}) = (7\text{mA}) \times (284)$ or $\sim 2\text{VDC}$. This circuit is known as a voltage divider circuit. The supply voltage was divided among the loads in proportion to the resistance each load carries. R1 had a higher resistance and received 3VDC of the total 5VDC supply and R2 received the rest or 2VDC. It can otherwise be stated that R1 has a voltage drop of 3VDC and R2 has a voltage drop of 2VDC.

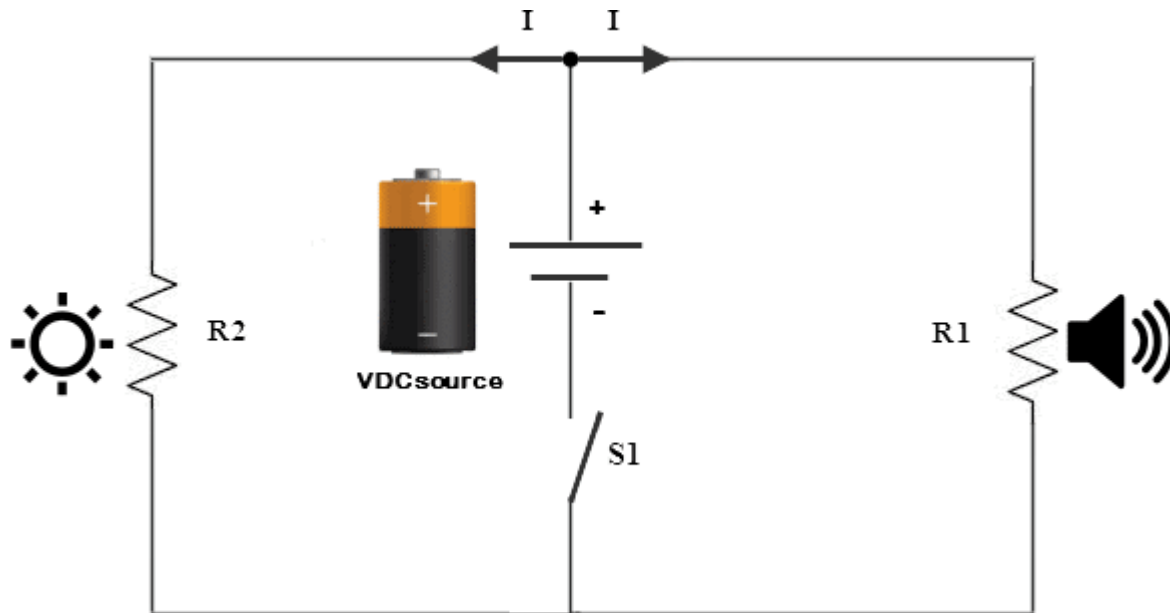


Kirchoff's Voltage Law (KVL)

This voltage drop principle leads to another important law in basic electrical engineering, Kirchoff's Voltage Law (KVL). This law states that the algebraic sum of the voltages in a closed loop is always equal to zero. If we only knew the supply potential and the voltage drop of R1, we could use KVL to find the other voltage drop. With KVL you must follow the current path and use the polarities of the components shown. If current path is unknown, you must assume one. We will use the positive to negative (clockwise) path. $V(\text{supply}) + V(1) + V(2) = 0$ or $-5\text{VDC} + (+3\text{VDC}) + (+V(2))$. Solving for $V(2)$, $V(2) = 2\text{VDC}$, which we know to be correct. KVL really comes in handy when there are multiple supplies in a loop or multiple loops.

Ohm's Law Applied to a Parallel Circuit

Now let's take the same toy and rewire it so that the speaker and LEDs are in parallel with the power source, as seen below.

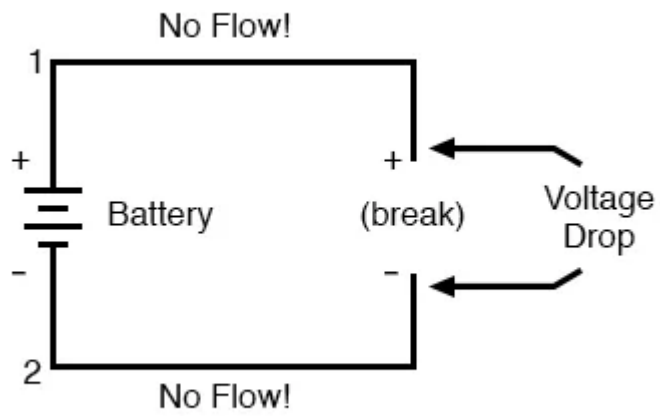


Understanding the Concept of Electric Current

As long as the battery continues to produce voltage and the continuity of the electrical path isn't broken, charge carriers will continue to flow in the circuit. Following the metaphor of water moving through a pipe, this continuous, uniform flow of charge through the circuit is called a current. So long as the voltage source keeps “pushing” in the same direction, the charge carriers will continue to move in the same direction in the circuit. This single-direction flow of current is called a Direct Current, or DC. In the second volume of this book series, electric circuits are explored where the direction of current switches back and forth: Alternating Current, or AC. But for now, we'll just concern ourselves with DC circuits.

Because electric current is composed of individual charge carriers flowing in unison through a conductor by moving along and pushing on the charge carriers ahead, just like marbles through a tube or water through a pipe, the amount of flow throughout a single circuit will be the same at any point. If we were to monitor a cross-section of the wire in a single circuit, counting the charge carriers flowing by, we would notice the exact same quantity per unit of time as in any other part of the circuit, regardless of conductor length or conductor diameter.

If we break the circuit's continuity at any point, the electric current will cease in the entire loop, and the full voltage produced by the battery will be manifested across the break, between the wire ends that used to be connected:



SESSION 2

Prepare the work area.

Learning Outcomes

- Authorisation to access the worksite is obtained according to worksite procedures, SABS safety requirements and company procedures. The consequences of not obtaining authorisation are explained in terms of work interruptions and liability.
- Work area is prepared for inspection and testing according to statutory requirements and worksite procedures. Potential hazards are identified and rectified in accordance with safety legislation.
- Defects and potentially hazardous conditions are identified, and corrective actions taken according to worksite procedures.
- Public access to the worksite is restricted in accordance with statutory requirements and worksite procedures. Consequences of not restricting public access are explained in terms of workflow and liability.

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Access & Egress

Having good access and egress to the site and to the plant and equipment operating on the site is a basic health and safety requirement that is often overlooked. Over 30% of workplace accidents are as a result of slips, trips and falls. Addressing the fundamental issues associated with access and egress can help reduce accidents, incidents and near hits in the workplace.

This written procedure provides advice and support on what to look for when undertaking a risk assessment. It is broken down into sections looking at static plant and equipment and mobile plant and considers issues such as traffic and pedestrian routing and confined space entry. It should be read in conjunction with the mobile and static plant inspection and the housekeeping modules.

What is this?

This is a written procedure which explains the control measures required to ensure that all persons have suitable access and egress in the workplace

What does the responsible manager need to do?

The responsible manager should:

- Undertake a risk assessment to identify all access and egress hazards within the workplace.
- Transfer any recommendations for improvement onto a site action plan.
- Communicate the findings of the access and egress assessment to all employees and, if applicable, contractors.
- Monitor the effectiveness of the access and egress by using the site daily inspection form.

How to combat unauthorized access?

First and foremost, you need to define how unauthorized access can occur at your company and develop a program aimed at eradicating any possible loopholes.

Different levels of security are crucial to prevent unauthorized access . Robust access control system, employee control and emergency response help prevent unsanctioned access to facilities, devices and information.

- Begin with perimeter security. Make sure you use fences, gates, guards and video surveillance around the perimeter.
- By installing motion detectors and alarm systems you can attain an additional level of security.
- Implement identification cards to verify people entering the premises, including visitors, contractors and personnel.
- Lock up areas with sensitive information. It is also advised to enforce delay control on server room doors.
- Conduct background check of employees before onboarding. If you are curious about onboarding definition, it is a process of integration of a new employee into an organization, training and orienting them.
- Make sure you have a new hire forms checklist thanks to which you can verify your new employee's work eligibility.
- Set up different access control levels. Each employee should be granted permission to enter facilities depending on their role within an organization.
- Use cable locks for computers in order to prevent theft of electronic devices.
- Desktop locks are aimed at protecting computer equipment from theft
- Lock up sensitive files or USB drives in safes or drawers.

- Develop an emergency plan and train employees to troubleshoot possible security issues and inform of any suspicious behavior they notice.

It is important to remember that security starts at a physical level. By implementing measures aimed at preventing unauthorized access you can protect your assets, information and personnel from internal and external security threats which otherwise might have a detrimental impact on your business.

SAFETY & SECURITY

SABS is a leading business services provider to organizations worldwide, offering a range of services for management system certification, product testing and certification, and standardization.

This includes the standardization in the field of manufacturing, conversion and testing in Safety & Security, including all aspects of their effects on man and his environment relating to activities in the following fields: security systems, including electric fences for agricultural purposes, intruder alarms and perimeter protection, access control, vehicle alarms (Dormant), closed circuit television, electric fences, security systems, steel and aluminium structures, security management systems, safes and strong room doors, perimeter protection, locks, security barriers, firearms, commercial ammunition, ranges and body armor, devices or systems relating to vehicle monitoring or tracking and stolen vehicle recovery, life jackets, flotation aids, buoyancy aids for swimming, fire prevention and fire protection.

Growing environmental awareness and the increasing awareness of quality implies the need for standardized requirements and test methods.

Work area is prepared for inspection and testing according to statutory requirements and worksite procedures. Potential hazards are identified and rectified in accordance with safety legislation.

Workplace Inspection Procedure

Safe work practices are generally written methods that define how tasks are performed while minimizing risks to people, equipment, materials, environment, and processes. Safe Work Procedures are documented procedures for performing tasks.

What is the purpose of Workplace safety procedures and instructions?

Safe Work Procedures are documented procedures for performing tasks. The purpose of a safe work procedure is to reduce the risk to health and safety in the workplace and reduce the likelihood of an injury by ensuring that employees know how to work safely when carrying out the tasks involved in their jobs. Safe work procedures may also be called safe work method statements (SWMS).

The purpose of a safe work procedure is to reduce the risk to health and safety in the workplace and reduce the likelihood of an injury through improving employees know how to work safely when carrying out the tasks involved in their jobs. Another term for safe work procedures is safe work method statements (SWMS).

What are the components or elements of Workplace safety procedures and instructions?

The following steps should be followed to ensure a sound safe work procedure is developed:

- Observe the task/activities: It is important to observe the task/activity being performed the preferred way to ensure safest method is documented.
- Review associated legislative requirements: Some task/activities are governed by legislative requirements. These must be considered when developing a safe work procedure to ensure any legal requirements are included.
- Record the sequence of basic job steps: write down the steps that make up the task/activity.
- Record potential hazards of each step: Next to each step identify what may have potential to cause injury or disease
- Identify ways of eliminating and controlling the hazards: list the measures that need to be put in place to eliminate or control any likely risk.
- Test the procedure: Observe staff/student following the safe work procedure
- Obtain approval: Before the safe work procedure can be used it must be approved by each approver nominated.
- Monitor and review: Make sure the activity is supervised to ensure the documented process is being followed.

Hazard identification and risk control – Identifying any hazard as a source of potential harm to people, or damage to property, and reducing risk.

Occupational Health and Safety (OHS) – OHS concerns the health and safety of all people in a workplace, including employers, employees and visitors

Safe work practices and procedures – These provide practical guidance to business on how to fulfil their duty to provide a safe and healthy workplace.

Defects and potentially hazardous conditions are identified, and corrective actions taken according to worksite procedures.

What types of hazards do we look for in a workplace?

Types of workplace hazards include:

- Safety hazards such as those caused by inadequate machine guards, unsafe workplace conditions, unsafe work practices.
- Biological hazards caused by organisms such as viruses, bacteria, fungi and parasites.
- Chemical hazards caused by a solid, liquid, vapour, gas, dust, fume or mist.
- Ergonomic hazards caused by physiological and psychological demands on the worker, such as repetitive and forceful movements, awkward postures arising from improper work methods, and improperly designed workstations, tools, and equipment.
- Physical hazards caused by noise, vibration, energy, weather, heat, cold, electricity, radiation and pressure.
- Psychosocial hazards that can affect mental health or well-being such as overwork, stress, bullying, or violence.

Determine which products are used in the workplace and whether safety data sheets are available. Find out if all sources of exposure are properly controlled. Make sure that all workers have received education and training in how to safely use, handle and store the products they work with. Check that all hazardous products are labelled appropriately according to Workplace Hazardous Materials Information System (WHMIS) requirements.

Inspection Principles

When conducting inspections, follow these basic principles:

- Draw attention to the presence of any immediate danger - other items can await the final report.

- Shut down and "lock out" any hazardous items that cannot be brought to a safe operating standard until repaired.
- Do not operate equipment. Ask the operator for a demonstration. If the operator of any piece of equipment does not know what dangers may be present, this is cause for concern. Never ignore any item because you do not have knowledge to make an accurate judgement of safety.
- Look up, down, around and inside. Be methodical and thorough. Do not spoil the inspection with a "once-over-lightly" approach.
- Clearly describe each hazard and its exact location in your rough notes. Allow "on-the-spot" recording of all findings before they are forgotten. Record what you have or have not examined in case the inspection is interrupted.
- Ask questions, but do not unnecessarily disrupt work activities. This interruption may interfere with efficient assessment of the job function and may also create a potentially hazardous situation.
- Consider the static (stop position) and dynamic (in motion) conditions of the item you are inspecting. If a machine is shut down, consider postponing the inspection until it is functioning again.
- Consider factors such as how the work is organized or the pace of work and how these factors impact safety.
- Discuss as a group, "Can any problem, hazard or accident generate from this situation when looking at the equipment, the process or the environment?" Determine what corrections or controls are appropriate.
- Do not try to detect all hazards simply by relying on your senses or by looking at them during the inspection. You may have to monitor equipment to measure the levels of exposure to chemicals, noise, radiation or biological agents.
- Take a photograph if you are unable to clearly describe or sketch a particular situation.

Common Electrical Hazards in the Workplace

On the jobsite, working around electricity can be very safe when workers properly identify and control hazards. But, inadequate training, lack of experience and failure to recognize potential hazards could result in electric shock or death.

Who is at Risk?

Engineers, electricians, and overhead line workers are at the top of the list of professionals who are most exposed to electrical hazards. Common tasks that put these workers at risk

include electrical installation and repairs, testing of fixtures and equipment and inspection and maintenance activities. However, people who are indirectly working with electricity like office workers are also exposed to electrical hazards.

Here are 7 of the most common electrical hazards in the workplace and tips on what you can do to mitigate these risks.

Hazard #1: Overhead Power Lines

Overhead powered and energized electrical lines have high voltages which can cause major burns and electrocution to workers. Remember to maintain a minimum distance of 10 feet from overhead power lines and nearby equipment. Conduct site surveys to ensure that nothing is stored under overhead power lines. Also, safety barriers and signs must be installed to warn nearby non-electrical workers of the hazards present in the area.

Hazard #2: Damaged Tools and Equipment

Exposure to damaged electrical tools and equipment can be very dangerous. Do not fix anything unless you are qualified to do so. Thoroughly check for cracks, cuts or abrasions on cables, wires and cords. In case of any defects, have them repaired or replaced. Lock Out Tag Out (LOTO) procedures should be performed at all times before commencing electrical maintenance and repairs. LOTO procedures are there to protect all workers on a worksite.

Hazard #3: Inadequate Wiring and Overloaded Circuits

Using wires with inappropriate size for the current can cause overheating and fires to occur. Use the correct wire suitable for the operation and the electrical load to work on. Use the correct extension cord designed for heavy duty use. Also, do not overload an outlet and use proper circuit breakers. Perform regular fire risk assessments to identify areas at risk of bad wiring and circuits.

Hazard #4: Exposed Electrical Parts

Examples of exposed electrical parts include temporary lighting, open power distribution units, and detached insulation parts on electrical cords. These hazards can cause potential shocks and burns. Secure these items with proper guarding mechanisms and always check for any exposed parts to be repaired immediately.

Hazard #5: Improper Grounding

The most common OSHA electrical violation is improper grounding of equipment. Proper grounding can eliminate unwanted voltage and reduce the risk of electrocution. Never remove the metallic ground pin as it is responsible for returning unwanted voltage to the ground.

Hazard #6: Damaged Insulation

Defective or inadequate insulation is a hazard. Be aware of damaged insulation and report it immediately. Turn off all power sources before replacing damaged insulation and never attempt to cover them with electrical tape.

Hazard #7: Wet Conditions

Never operate electrical equipment in wet locations. Water greatly increases the risk of electrocution especially if the equipment has damaged insulation. Have a qualified electrician inspect electrical equipment that has gotten wet before energizing it.

Knowing your limits and applying best electrical safety practices can help reduce the risk of electrical shock and death. It is safer to work within your scope of expertise instead of taking the risk of working beyond your capacity. If you are not confident to do the job, don't hesitate to call for help from an authorized person.

Public access to the worksite is restricted in accordance with statutory requirements and worksite procedures. Consequences of not restricting public access are explained in terms of workflow and liability.

Protecting the public

The law says you must conduct your business without putting members of the public at risk. This includes the public and other workers who may be affected by your work.

The project client should provide information about:

- boundaries
- adjacent land usage
- access; and
- measures to exclude unauthorized persons

This will influence the measures contractors take.

Managing site access

Site boundaries: You need to define boundaries physically, where necessary, by suitable fencing. The type of fencing should reflect the nature of the site and its surroundings.

Determining the boundary is an important aspect of managing public risk. You need to:

- plan what form the perimeter will take;
- provide the fencing; and
- maintain the fencing.

Questions you need to ask yourself include:

- What is the nature and type of the construction work?
- How heavily populated is the area is?
- Who will need to visit the site during the work?
- Will the site attract children?
- What are the site characteristics (eg existing site boundaries, location, proximity to other buildings).

Typically, in populated areas, this will mean a two-metre high small mesh fence or hoarding around the site.

Authorisation: The principal contractor must take reasonable steps to prevent unauthorised people accessing the site.

- People may be authorised to access the whole site or be restricted to certain areas;
- You must explain relevant site rules to authorised people and undertake any necessary site induction;
- You may need to supervise or accompany some authorised visitors while they are on site or visiting specific areas.

Hazards causing risk to the public

Many hazards have the potential to injure members of the public and visitors. Consider if they exist on your project and how you will manage them.

Falling objects - You must make sure objects cannot fall outside the site boundary. On scaffolds you can achieve this using toe-boards, brick guards and netting. You may also need fans and/or covered walkways.

Delivery and other site vehicles - Make sure pedestrians cannot be struck by vehicles entering or leaving the site. Obstructing the pavement during deliveries may force pedestrians into the road, where they can be struck by other vehicles.

Scaffolding and other access equipment - Prevent people outside the boundary being struck while they are erecting, dismantling and using scaffolding and other access equipment.

Storing and stacking materials - You can reduce the risks associated with the storage of materials by storing materials within the site perimeter, preferably in secure compounds or away from the perimeter fencing.

Openings and excavations - People can be injured if they fall into excavations, manholes, stairwells or from open floor edges. You'll need to put up barriers or covers.

Other hazards include -

- slips, trips and falls within pedestrian areas;
- plant, machinery and equipment;
- hazardous substances;
- electricity and other energy sources;
- dust, noise and vibration; and
- road works.

Vulnerable groups

The elderly, children and people with certain disabilities may need special attention. Work in premises such as schools and hospitals needs careful thought and planning.

Some children are drawn to construction sites as exciting places to play. You must do everything you can to keep them out of the site and away from danger.

The following specific steps are particularly relevant to child safety:

- Secure sites adequately when finishing work for the day.
- Barrier off or cover over excavations and pits.
- Isolate and immobilise vehicles and plant and if possible lock them in a compound.
- Store building materials (such as pipes, manhole rings, and cement bags) so that they cannot topple or roll over.
- Remove access ladders from excavations and scaffolds.
- Lock away hazardous substances.

SESSION 3&4

Learning Outcomes

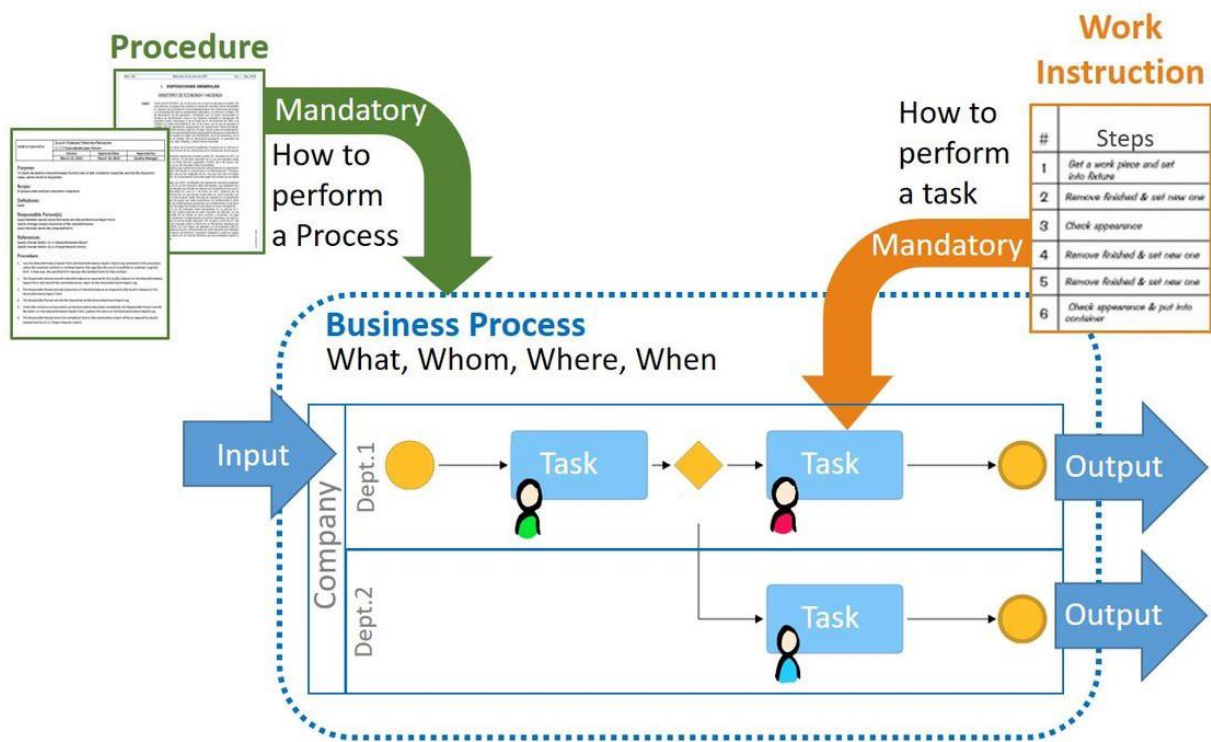
Inspect electrical circuits.

Test electrical circuits.

- Work instructions are interpreted, and a sequence of operation is determined according to worksite procedures, company standards and SABS safety requirements.
- The purpose of inspecting electrical circuits is explained in terms of legal and safety requirements.
- Tools, inspection equipment and materials selected are ensured to be appropriate for the requirements of the work activity. Reasons why specific tools and inspection equipment are selected are explained in terms of accuracy of work and required results.
- Personal protective equipment (PPE) is selected and used in accordance with statutory requirements. The consequences of not using personal protective equipment are explained in terms of personal risk and liability.
- Electrical circuits are inspected according to statutory requirements. The consequences of not inspecting all relevant components are explained in terms of potential risks and liability.
- Inspection tools and equipment are used in accordance with their design and manufacturer recommendations.
- Deviances from acceptable standards are identified and reported according to statutory requirements and manufacturer's specifications.

Work instructions are interpreted, and a sequence of operation is determined according to worksite procedures, company standards and SABS safety requirements.

Differences between Processes, Procedures and Work Instructions



Many people often confuse these three terms: business Process, Procedure, and Work Instruction. In fact, most people write work instructions and call them procedures, and define a procedure as a process. My objective in this paper is to provide the difference amongst the three terms to help you understand when to use each term in the Quality Management of your Business.

The ISO 9001:2015 standard (Quality management systems—Fundamentals and vocabulary) defines the concept of Business Process as: "A set of related or interacting activities, which transform inputs into outputs". It is possible to qualify something more by saying that a process is a sequence of activities with a realization order in time, which converts a given input into an output (result, a product). Any activity, or set of activities, that uses resources to transform input elements into results can be considered as a process.

Do not confuse procedure with process. The ISO 9001: 2015 says that a Procedure is: "specified way to carry out an activity or a process". When you have a process that must happen in a specific way, and you specify how it happens, you have a procedure. The procedure is mandatory and is necessary to develop any Quality Management System. Please pay your attention that not all processes need to have a procedure, and that the same process may have associated one or several procedures.

A procedure will be determined by the need to detail the specific way of doing the activities by legal issues, regulatory compliance or company policies. The procedures are used for those activities in which several operations are linked, and different people or departments of the company are involved (for example, Procedure for the presentation of Fiscal Models in the Treasury, Purchasing Procedure, Supplier Evaluation Procedure, Automation Procedure of Invoices, Procedure for Risk Work...

Main Differences between Process, Procedure and Work Instruction

PROCESS	PROCEDURE	WORK INSTRUCTION
Flow of sequences of activities that transform input elements into results	Specific way to carry out a process	Describe the correct steps to perform a specific task.
What we do By WHOM Where it takes place When it happens	How the work has to been performed and why	How accomplish a specific task within a process with very detailed directions
Orchestration the work	Mandatory method	Mandatory guidance
Can respond to 0, 1 or more procedures	It may consist of 0, 1 or more work instructions	Focus on the instructions of 1 task
Transversal by Business Units	Cross-Functional or only 1 business unit	Only 1 business unit
Participate more than one role	Participate more than one role	Participate only one role
Encapsulates activities	Explains how to do but doesn't get to the all details of how it is done	All the detail of all the steps to follow in an activity
Provides the workflow model in BPMN at a high level	Document with both narrative and images, usually in the form of use cases and workflow diagrams	Document with the maximum detail that explains step by step the instructions that must be carried out in an activity

The purpose of inspecting electrical circuits is explained in terms of legal and safety requirements.

What is an Electrical Inspection?

An electrical inspection refers to the thorough checkup of the electrical connections and wiring installed in your home.

Why is Electrical Inspection Important?

There are many reasons behind the necessity of getting an electrical inspection done. Some of the most common ones are:

Protection against possible surges: This is a common phenomenon in different regions, especially when there is a storm. Electrical surges can not only ruin your appliances, but can also create short circuits in your house. When the inspection is being done, the inspector will check for different points in the wiring that can cause this type of mishap.

Fire: Many a times it has been seen that an electrical short circuit has been the reason behind a fire breakout in a house. These short circuits are caused by different loose wires in the meter or in the cabling itself. During the inspection, the inspector can easily recognize these faults and ask you to rectify them.

Shocks: Imagine touching an electronic appliance and immediately experiencing a surging electric current in your body. Although the impact may not be that disastrous; however, this could lead to other mishaps easily. Apart from this, the body current in an electronic appliance can easily hamper its performance or can render it completely non-workable. An electrical inspector can catch these faults in the wiring in time to rectify them.

Electrical inspection is a must for every household to ensure the safety and security of the habitants. In fact, it is required, but it most certainly does not need to be a cause for high blood pressure. Knowing what to expect beforehand can be one of the best ways of dealing with electrical inspection. If you have the loopholes rectified as per the requirements of the state, then there is no way you will fail the inspection.

Why is Impedance Testing Important?

An earth loop impedance test is conducted to make sure that, if a fault occurs in an electrical circuit, the fault current will strong be enough to set off the circuit protection. If a fault current is left undetected, circuits can overheat and catch on fire.

Electricity takes the path of least resistance to the ground. Electrical wiring systems in buildings are often connected to the ground. This is referred to as an earth return circuit. The grounding wire is there for electrical current to pass through when a short circuit happens.

Resistance is a measure of how an electrical current is affected by its path. The resistance in the grounding wire needs to be low so that the fault current can travel down into the ground without causing damage to the system.

If the resistance in the earth return circuit is too high, the fault current may be too low to be detected, and the fault current will continue to travel around the main circuit – causing a short circuit. The circuit protection detects activity along the earth wiring and kicks in when the current is detected. If the resistance is too high, the circuit protection may not operate.

Well, if you care about the ongoing quality of the circuitry in your building, it's very important. In order to prevent overheating and fires, the loop impedance needs to operate at a certain level. The only way to maintain this optimum level is with regular testing.

Ensure Safety in Your Building

An impedance check is essential for ensuring safety in your building. Overheated circuitry and damaged wiring are a recipe for disaster. If left unchecked, you might be evacuating your workers during a fire soon, wondering where you went wrong.

The best way to run a safe business is to be thorough and invest in preventative measures. Consider all parts of electrical safety and don't forget your earth return circuits – they're in place to keep you safe.

Why is impedance testing important for business owners?

If you run a business, you are required to provide a safe working environment for your employees. Fires caused by short circuits will endanger your staff members and have many potential costly ramifications. Every year, the New Zealand Fire Service attend over 20,000 fires. Don't become part of the statistics – protect your property today.

Why is impedance testing important at home?

Short circuits can occur in any type of dwelling. If you're a homeowner, landlord, or property manager, we encourage thorough electrical testing throughout your domestic environment. Safety is nothing to mess around with – whether you're at home or at work.

What happens during a test?

During a resistance test, the technician will use an impedance tester to inspect the earth return circuit. The technician does this by plugging the test machine into a power socket or supply source. Two earth fault loop impedances will be measured. This will be the external impedance (the Z_e) and the total system impedance (the Z_s). The Z_s is a sum of the external resistance, the phase conductor resistance, and the earth conductor resistance.

Tools, inspection equipment and materials selected are ensured to be appropriate for the requirements of the work activity. Reasons why specific tools and inspection equipment are selected are explained in terms of accuracy of work and required results.

Tools for testing electrical wiring and panels

Power and distribution panels and their nearby wiring are a frequent starting point for technicians when electrical disruptions require troubleshooting.

Voltage and current are two of the most common electrical measurements taken at panels and the surrounding wiring. To isolate specific problems, a technician may also want to test for continuity or measure peak voltage, peak current, minimum/maximum voltage (or current) over time, temperature, harmonics, power factor, transient voltage, temperature or even analyze waveforms.

With the right equipment and use of appropriate protective gear, a technician can acquire a variety of measurements safely and efficiently in order to find and fix problems.

Voltage testers: These tools can quickly assess the presence of voltage. The best-selling Fluke T5-1000 speedily tests for voltage, continuity and current, and the Fluke T+Pro is a long-time for basic voltage testing.

Digital multimeters: Regarded as the Swiss Army Knife of measurement tools, DMMs can accommodate a long list of measurement or testing tasks: voltage, current, resistance, continuity, capacitance, frequency, diodes, duty cycle and potentially more, depending

on the model. The Fluke 87 V is a standard-bearer of such versatility, and the Fluke 3000 FC series has introduced wireless safety, efficiency and data sharing into the process.

Clamp meters: For wiring in tight spaces, panels that are beyond easy reach, and extra-large conductors, clamp meters greatly simplify measurement tasks. A top seller such as the

Insulation testers: The Fluke 1587 is an example of a tool that combines a digital insulation tester with a full-featured, true-rms digital multimeter into a single, cost-effective unit.

Infrared cameras and thermometers: Rugged, handheld IR noncontact cameras and thermometers quickly measure surface temperatures in hard-to-reach or hazardous areas. Infrared cameras are ideal for quick inspection of electrical panels in order to determine potential problem areas before further testing and troubleshooting.

TESTING ELECTRICAL CIRCUITS FOR POWER

The first step in almost any electrical project is to check for power to make sure the circuit or device is safe to work on. You can do this with a variety of inexpensive testers or even a multimeter.

Testers and How They Work

Standard probe-type circuit testers, such as neon circuit testers, voltmeters, and multimeters, have two wire leads with probes for checking circuit wiring or electrical devices. When you insert the leads into an outlet or touch them to a switch's screw terminals, a light or readout will indicate whether the device has voltage. An even simpler (and decidedly safer) type of tester is a non-contact voltage tester, which doesn't even need to be inserted into an outlet or to touch bare wire connections; merely bringing the sensor close to a power-carrying wire or device will cause the tool to light up or make an audible noise to indicate the presence of power.

There are also plug-in outlet testers with three small different-colored neon lights. These testers simply plug into an outlet and can check for an open neutral, lack of a ground, wires on the wrong terminals, or no power. A specific pattern of lights indicates each condition, and a chart on the top of the tester tells you how to interpret the light pattern.

While simple voltage testers can test only for the presence of voltage, multimeters have multiple testing functions and can measure voltage, ohms (for resistance), and amperage

(electrical current), indicating quantities on a digital readout or analog dial. Testing to see if the power is on is only one function of a multimeter.

Warning: Never touch the bare-metal probe ends of a tester during a test because they may have electricity running through them and can give you a dangerous shock. Also, never let the probes touch each other during a test.

Making Sure Your Tester Works

Always check to see if the tester is working properly before using it to check for voltage. The easiest way is to go to an outlet on a circuit that you know is live (has power). Insert the tester leads or sensor into the outlet slots. If the tester lights up, it's working fine. If it fails to light up, the tester is bad or needs new batteries.

How to Test Outlets for Power

A typical outlet receptacle has three holes in its face. The shorter straight slot is the "hot" lead and connects to the active hot wire in the outlet box. The longer straight slot is the "neutral" lead and connects to the neutral circuit wire in the electrical box. The slot that looks like a small D-shaped hole is the ground slot, and it is connected to the circuit ground wire.

To test an outlet for power, turn off the power to the circuit at the circuit breaker. Insert the two probes of the tester into the two straight vertical slots on the receptacle. If the power is on, the tester will light. Because there is a possibility that the outlet is "split-wired"—with the top and bottom halves of the outlet fed by different circuits—always check both halves for power before removing the receptacle to work on it.

You can also test to see if the ground system is properly connected to the receptacle. To test the ground, make sure the power to the circuit is on. Insert one tester probe into the hot (short, straight) slot and the other in the ground (D-shaped) slot. If the circuit is working and you have a good ground connection, the tester will light.

Testing Wall Switches

To test a switch for power, turn off the power to the circuit at the circuit breaker. Remove the switch's cover plate and flip the switch's toggle so the switch is on. Carefully touch one probe of the tester to one of the screws on the side of the switch. Touch the other probe

to the bare copper ground wire or the ground screw on the switch (you can also touch this probe to the electrical box if it is metal, but this test works only if the metal box is properly grounded; plastic boxes are not grounded). Next, touch one probe to the other screw terminal on the switch and touch the other probe to the ground wire or screw. Flip the switch's toggle to off and repeat the same tests. If the tester does not light for either test, the switch is not getting power.

Testing Light Fixtures for Power

When checking light fixture wiring for power, turn off the power to the circuit at the circuit breaker, then loosen the mounting straps holding the fixture to the ceiling box and pull the light fixture slightly away from the ceiling box for testing. Always test twice—with the fixture's wall switch on and with it off—because the fixture may get power in either position.

To test for power with a non-contact voltage tester, touch the sensor tip of the tester to each of the circuit wires. If the tester lights up when touching any of the wires, the circuit still has power.

To test a fixture for power using a probe-type tester, you need access to the fixture's screw terminals or, if the fixture has wire leads, to the ends of the wire leads. Touch one tester probe to the hot (black or red wire) screw terminal, and touch the other probe to the neutral (white wire) terminal. If the tester lights up, the fixture still has power.

If the fixture has wire leads connected to the circuit wiring with wire connectors (wire nuts), stick one probe into the connector for the black (or red) wires and the other probe into the white-wire connector. If the tester does not light up, confirm the test by carefully untwisting each wire connector—without touching the bare-metal wire ends or letting different-colored wires to touch—then touching each probe directly to the group of black (or red) and white wires.

Personal protective equipment (PPE) is selected and used in accordance with statutory requirements. The consequences of not using personal protective equipment are explained in terms of personal risk and liability.

Personal Protective Equipment (PPE) for Electrical Works

To protect the employee's body from injury Personal Protective Equipment (PPE) must be used.

Basic PPE consists of:

- Cotton protective clothing with long sleeves
- Helmet or hard hat
- Goggles for eye protection
- Gloves (leather or rubber)
- Hearing protectors
- Safety footwear



Helmets normally need be worn only when working on outdoor switchgear, where they serve to protect against falling objects and collisions with solid objects at head height. Outdoor substations should always be considered 'safety helmet' areas and helmet wear made compulsory.

Hearing protectors are only required when the in a noisy situation, that can occur during construction works.

Eye protection should be worn when working with hazardous fluids, particularly mineral oil, to prevent splashes into the eye. They should always be worn when washing down the internal parts of oil circuit breakers.

Safety footwear should be routinely worn in all working areas and the shoes or boots should incorporate steel toecap and non-slip soles.

Specific works require specific protective equipment.

Safety harness should be worn when working at any height greater than 1.5 m above ground level and a full harness equipped with a connector is preferred. Fall arrest equipment may be necessary when working at high level. Harnesses should be chosen that are suitable for their intended application and should be of a design that will support

the user in the correct position. A harness should be comfortable, allowing adequate movement of the user and the unhindered operation of other devices within the system.

Depending on the conditions of the work a sit harnesses may be necessary; sit harness has lateral and central attachment points and are designed primarily to be used for work in suspension, although they may also be used for work restraint purposes.

Electrical circuits are inspected according to statutory requirements. The consequences of not inspecting all relevant components are explained in terms of potential risks and liability.

Electrical Installation Regulations :

- Every user or lessor as the case may be shall have a valid Certificate of Compliance for the electrical installation he or she uses or leases
- The Certificate of Compliance shall be in the format of Annexure 1 and shall be accompanied by a test report in the format approved by the Chief Inspector
- Only a Registered Person shall issue a Certificate of Compliance after he/she has inspected and tested the electrical installation
- If at any time prior to the issuing of a certificate of compliance any fault or defect is detected in any part of the electrical installation, the registered person shall refuse to issue such certificate until that fault or defect has been rectified
- Any person who undertakes to do electrical installation work shall ensure that a valid certificate of compliance is issued for that work
- No person shall connect or permit the connection of any completed or partially completed electrical installation to the electricity supply unless it has been inspected and tested by a registered person and a certificate of compliance for that electrical installation has been issued
- Where any addition or alteration has been effected to an electrical installation for which a certificate of compliance was previously issued, the user or lessor of such electrical installation shall obtain a certificate of compliance for at least the addition or alteration
- Subject to the provisions of section 10(4) of the Act, the user or lessor may not allow a change of ownership if the certificate of compliance is older than two years

- The Certificate of Compliance is required by all insurance companies.

SESSION 5 Complete work activities.
Learning Outcomes
• The worksite is cleared, secured and restored to a safe and serviceable condition in accordance with statutory and worksite requirements. • The importance of returning the worksite to a safe and serviceable condition is explained in terms of workflow and minimising hazards. • Equipment and tools are checked, cleaned and stored according to company procedures. • Work is completed and reported to appropriate personnel in accordance with company procedures. The importance of accurately completing documentation is explained in terms of legal responsibility. • Work activities are completed within agreed timeframes. The importance of completing work within agreed timeframes is explained in terms of customer service and work interruptions.

The worksite is cleared, secured and restored to a safe and serviceable condition in accordance with statutory and worksite requirements.

Workplace Housekeeping - Basic Guide

Effective housekeeping can help control or eliminate workplace hazards. Poor housekeeping practices frequently contribute to incidents. If the sight of paper, debris, clutter and spills is accepted as normal, then other more serious hazards may be taken for granted.

Housekeeping is not just cleanliness. It includes keeping work areas neat and orderly, maintaining halls and floors free of slip and trip hazards, and removing of waste materials (e.g., paper, cardboard) and other fire hazards from work areas. It also requires paying attention to important details such as the layout of the whole workplace, aisle marking,

the adequacy of storage facilities, and maintenance. Good housekeeping is also a basic part of incident and fire prevention.

Effective housekeeping is an ongoing operation: it is not a one-time or hit-and-miss cleanup done occasionally. Periodic "panic" cleanups are costly and ineffective in reducing incidents.

What is the purpose of workplace housekeeping?

Poor housekeeping can be a cause of incidents, such as:

- tripping over loose objects on floors, stairs and platforms
- being hit by falling objects
- slipping on greasy, wet or dirty surfaces
- striking against projecting, poorly stacked items or misplaced material
- cutting, puncturing, or tearing the skin of hands or other parts of the body on projecting nails, wire or steel strapping

To avoid these hazards, a workplace must "maintain" order throughout a workday. Although this effort requires a great deal of management and planning, the benefits are many.

The importance of returning the worksite to a safe and serviceable condition is explained in terms of workflow and minimising hazards.

What are some benefits of good housekeeping practices?

Effective housekeeping results in:

- reduced handling to ease the flow of materials
- fewer tripping and slipping incidents in clutter-free and spill-free work areas
- decreased fire hazards
- lower worker exposures to hazardous products (e.g. dusts, vapours)
- better control of tools and materials, including inventory and supplies
- more efficient equipment cleanup and maintenance
- better hygienic conditions leading to improved health
- more effective use of space
- reduced property damage by improving preventive maintenance

- less janitorial work
- improved morale
- improved productivity (tools and materials will be easy to find)

How do I plan a good housekeeping program?

A good housekeeping program plans and manages the orderly storage and movement of materials from point of entry to exit. It includes a material flow plan to ensure minimal handling. The plan also makes sure that work areas are not used as storage areas by having workers move materials to and from work areas as needed. Part of the plan could include investing in extra bins and more frequent disposal.

The costs of this investment could be offset by the elimination of repeated handling of the same material and more effective use of the workers' time. Often, ineffective or insufficient storage planning results in materials being handled many times and being stored in hazardous ways. Knowing the workplace layout and the movement of materials throughout it will help when planning work procedures.

Worker training is an essential part of any good housekeeping program. Workers need to know how to work safely with the products they use. They also need to know how to protect other workers such as by posting signs (e.g., "Wet - Slippery Floor") and reporting any unusual conditions.

Housekeeping order is "maintained" not "achieved." Cleaning and organization must be done regularly, not just at the end of the shift. Integrating housekeeping into jobs can help ensure this is done. A good housekeeping program identifies and assigns responsibilities for the following:

- clean up during the shift
- day-to-day cleanup
- waste disposal
- removal of unused materials
- inspection to ensure cleanup is complete

Do not forget out-of-the-way places such as shelves, basements, sheds, and boiler rooms that would otherwise be overlooked.

The final step to any housekeeping program is inspection. It is the only way to check for deficiencies in the program so that changes can be made. Examples of checklists include inspecting offices and manufacturing facilities.

What are the elements of an effective housekeeping program?

Maintenance

The maintenance of buildings and equipment may be the most important element of good housekeeping. Maintenance involves keeping buildings, equipment and machinery in safe, efficient working order and in good repair. It includes maintaining sanitary facilities and regularly painting and cleaning walls. Broken windows, damaged doors, defective plumbing and broken floor surfaces can make a workplace look neglected; these conditions can cause incidents and affect work practices. So it is important to replace or fix broken or damaged items as quickly as possible. A good maintenance program provides for the inspection, maintenance, upkeep and repair of tools, equipment, machines and processes.

Dust and Dirt Removal

Enclosures and exhaust ventilation systems may fail to collect dust, dirt and chips adequately. Vacuum cleaners are suitable for removing light dust and dirt that is not otherwise hazardous. Industrial models have special fittings for cleaning walls, ceilings, ledges, machinery, and other hard-to-reach places where dust and dirt may accumulate.

Special-purpose vacuums are useful for removing hazardous products. For example, vacuum cleaners fitted with HEPA (high efficiency particulate air) filters may be used to capture fine particles of asbestos or fiberglass.

Dampening (wetting) floors or using sweeping compounds before sweeping reduces the amount of airborne dust. The dust and grime that collect in places like shelves, piping, conduits, light fixtures, reflectors, windows, cupboards and lockers may require manual cleaning.

Compressed air should not be used for removing dust, dirt or chips from equipment or work surfaces.

Employee Facilities

Employee facilities need to be adequate, clean and well maintained. Lockers may be necessary for storing employees' personal belongings. Washroom facilities require cleaning once or more each shift. They also need to have a good supply of soap, towels plus disinfectants, if needed.

If workers are using hazardous products, employee facilities should provide special precautions as needed such as showers, washing facilities and change rooms. Some facilities may require two locker rooms with showers between. Using such double locker rooms allows workers to shower off workplace contaminants and reduces the chance of contaminating their "street clothes" by keeping their work clothes separated from the clothing that they wear home.

Smoking, eating or drinking in the work area should be prohibited where hazardous products are handled. The eating area should be separate from the work area and should be cleaned properly each shift.

Surfaces

Floors: Poor floor conditions are a leading cause of incidents so cleaning up spilled oil and other liquids at once is important. Allowing chips, shavings and dust to accumulate can also cause incidents. Trapping chips, shavings and dust before they reach the floor or cleaning them up regularly can prevent their accumulation. Areas that cannot be cleaned continuously, such as entrance ways, should have anti-slip flooring. Keeping floors in good order also means replacing any worn, ripped, or damaged flooring that poses a tripping hazard.

Walls: Light-coloured walls reflect light while dirty or dark-coloured walls absorb light. Contrasting colours warn of physical hazards and mark obstructions such as pillars. Paint can highlight railings, guards and other safety equipment, but should never be used as a substitute for guarding. The program should outline the regulations and standards for colours.

Maintain Light Fixtures

Dirty light fixtures reduce essential light levels. Clean light fixtures can improve lighting efficiency significantly.

Aisles and Stairways

Aisles should be wide enough to accommodate people and vehicles comfortably and safely. Aisle space allows for the movement of people, products and materials. Warning signs and mirrors can improve sightlines in blind corners. Arranging aisles properly encourages people to use them so that they do not take shortcuts through hazardous areas.

Keeping aisles and stairways clear is important. They should not be used for temporary "overflow" or "bottleneck" storage. Stairways and aisles also require adequate lighting.

Spill Control

The best way to control spills is to stop them before they happen. Regularly cleaning and maintaining machines and equipment is one way. Another is to use drip pans and guards where possible spills might occur. When spills do occur, it is important to clean them up immediately. Absorbent materials are useful for wiping up greasy, oily or other liquid spills. Used absorbents must be disposed of properly and safely.

Tools and Equipment

Tool housekeeping is very important, whether in the tool room, on the rack, in the yard, or on the bench. Tools require suitable fixtures with marked locations to provide an orderly arrangement. Returning tools promptly after use reduces the chance of it being misplaced or lost. Workers should regularly inspect, clean and repair all tools and take any damaged or worn tools out of service.

Waste Disposal

The regular collection, grading and sorting of scrap contribute to good housekeeping practices. It also makes it possible to separate materials that can be recycled from those going to waste disposal facilities.

Allowing material to build up on the floor wastes time and energy since additional time is required for cleaning it up. Placing scrap containers near where the waste is produced encourages orderly waste disposal and makes collection easier. All waste receptacles should be clearly labelled (e.g., recyclable glass, plastic, scrap metal, etc.).

Storage

Good organization of stored materials is essential for overcoming material storage problems whether on a temporary or permanent basis. There will also be fewer strain

injuries if the amount of handling is reduced, especially if less manual material handling is required. The location of the stockpiles should not interfere with work but they should still be readily available when required. Stored materials should allow at least one metre (or about three feet) of clear space under sprinkler heads.

Stacking cartons and drums on a firm foundation and cross tying them, where necessary, reduces the chance of their movement. Stored materials should not obstruct aisles, stairs, exits, fire equipment, emergency eyewash fountains, emergency showers, or first aid stations. All storage areas should be clearly marked.

Flammable, combustible, toxic and other hazardous materials should be stored in approved containers in designated areas that are appropriate for the different hazards that they pose. Storage of materials should meet all requirements specified in the fire codes and the regulations of environmental and occupational health and safety agencies in your jurisdiction.

Work is completed and reported to appropriate personnel in accordance with company procedures. The importance of accurately completing documentation is explained in terms of legal responsibility.

Cost and Importance of Documentation in Engineering

Poorly written work instructions, standard operating procedures, incorrect drawings and blueprints, improperly constructed fixtures, inappropriate production equipment, and poorly designed workstation layout are just a few examples which contribute to human error. These poorly written documentations add addition trouble that could've been avoided. It is not unheard of to find components being fabricated according to out of date instructions because the correct information has been delayed in reaching the worker. Poor design and documentation quality are a major cause of product process inefficiencies, leading directly to delays, rework and variations, and contributing to increase project time and cost, for both the client and your company. But proper documentation is not just restricted to engineering paperwork, no, it involves every person working in that company. Let's see how poor documentation can delay even the best companies.

How It Works

The quality of the design and documentation produced has a major influence on the overall performance and efficiency of production projects. Designers provide the basic blueprints for representing the ideas and concepts that manufacturers need to make products a reality. However, it is the quality of the documentation provided which determines how effective and efficient turnaround time is for production. Inadequate and deficient design and documentation impacts the production process efficiency by leading to delays, rework and variations, contributing to increases in project time and cost. But there are levels before manufacturing can happen that can delay this process, or even bring production to a halt.

Work activities are completed within agreed timeframes. The importance of completing work within agreed timeframes is explained in terms of customer service and work interruptions.

HOW TO PLAN & ORGANIZE WORK ACTIVITIES

Business owners know that productivity is improved with employee efficiency. Helping team members plan and organize work activities set routines, providing direction on how to prioritize daily tasks and deal with potential obstacles. When employees have organized days, there is less time spent trying to figure out the next thing to do. The end result is less wasted time, improved efficiency and increased productivity. Keep planning and organizing work activities simple in order maximize effectiveness.

Determine Specific Tasks

Brainstorm all required tasks throughout the day. Don't worry about organizing at this stage. Think about everything that happens during the day from walking in the door, making the coffee and logging into the computer. If phones need to be turned on from a voice service, include this along with checking messages.

Think about filing, shredding and any inventory tasks that come up regularly. This brainstorming session will become the basis for the daily activities plan.

Prioritize and Sequence Tasks

Group tasks together. There may be tasks that need to be done in a specific sequence or together. For example, there may be a series of actions required in opening the office from unlocking the door, disarming the alarm, switching voicemail off, checking messages and making the first pot of coffee. Go through the entire brainstormed task list and group related tasks.

Once you have the grouped lists that can range from the opening, closing, sales and service tasks, establish the priority items. Re-write each grouped list based on the order of priority. For example, it might be a service priority to return all phone messages within the first hour of opening. Calling a supplier about inventory changes might not be a top priority first thing in the morning.

Set Realistic Timetables

There are some grouped tasks that need to be done at specific times of the day. For example, opening and closing tasks can't be done in the middle of the day. However, other tasks have timetable flexibility. Break the day down into sections to establish a routine. For example, the first hour of the day could be devoted to returning messages from voicemail and email. The next hour could be devoted to sales related tasks with the following hours before lunch reserved for client meetings.

Think about the best time to perform activities. For example, if you need more energy to write proposals, it might be better to place this task earlier in the day and reserve less energy-intensive tasks for later in the day when your energy is lower. Establish realistic timetables to accomplish tasks. For example, if you know it takes a minimum of two hours to process orders and get them out the same day, don't give yourself the last two hours before the shipping deadline to complete the task; set this priority earlier in the day.

Remove Potential Distractions

There are so many potential distractions in everyone's workday. Personal cell phones, text messages, instant messages, social media and internet surfing are huge distractions but aren't the only ones. Excessive coffee breaks and talking to co-workers greatly reduce productivity as well. Set break times and turn off all devices that could potentially distract from completing tasks.

Be realistic about checking devices and talking to co-workers but develop the discipline to limit it to certain times of the day. Some people even turn off email notifications, setting specific times to read and respond to email, focusing energy on immediate tasks rather than trying to multitask.

The Project Workplan will help you:

- to plan your project by outlining the activities to be accomplished, timeframes, and inputs needed, including how project results will be sustained and lessons learned disseminated. The Project Workplan will also help you to formulate your project and grant budget.
- to manage implementation of your project by enabling you to track implementation against expectations.
- to report on your progress in implementing the project by sharing information in the progress reporting column of the outcome workplan.
- to evaluate outcomes by identifying the indicators you will use to assess whether outcomes are achieved, the data collection methods and timeframe for collecting outcomes evaluation data, and by sharing evaluation results in the evaluation results column of the outcome evaluation plan.